A genome-scale census of the inositol 1,4,5-trisphosphate receptor family

Range, origin, retention, constraint and record quality across 503 genomes and 7,691 reference proteomes

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The analyses this document reports were carried out over 35 working sessions in the repository it is built from, and the document was written from their committed outputs. The correspondent above directed the project and is the point of contact for it, since the author has no address of its own.

Abstract

The inositol 1,4,5-trisphosphate receptor (IP_3R, gene family *ITPR*) is the endoplasmic reticulum's ligand-gated calcium-release channel. In vertebrates it exists as three paralogues, ITPR1, ITPR2 and ITPR3, whose origin, distribution and retention had never been measured against a declared search space. This thesis reports that measurement, and it gives equal space to the instruments that made it possible.

The family was enumerated across 7,691 reference proteomes and 503 genomes, of which 309 are vertebrate and 194 are not. The ryanodine receptors were searched alongside at every stage. That control is not decoration: RYR1 carries all four of the Pfam signatures that define an IP_3 receptor, is twice its length, and is returned by every query this project ran. Separating the two families is therefore treated as a positive test at every stage rather than as an assumption, and where an instrument could not make that test the result is reported as undecided rather than as an absence.

Three results follow. First, the family is ancestrally eukaryotic and has been lost repeatedly and independently. It is absent from all 384 land-plant and all 1,353 Dikarya reference proteomes, from Apicomplexa, Microsporidia, Rhodophyta, diatoms and four other fungal phyla, and from every archaeal and bacterial proteome swept. All 35 clade-level absences hold at assembly level in genomes where a measured positive control recovers a comparably long, deeply conserved gene. Second, the three vertebrate paralogues are not lost at all. Across 927 genome by paralogue cells, none reaches the absence state, Dollo parsimony places no loss anywhere on the tree, and what is reported instead is the set of analytical settings that would manufacture one. Third, the family's public record is substantially worse than the family itself. Most full-length *ITPR* protein records carry no usable gene symbol, and three quarters of the genes demonstrated here cannot be reached by any protein-database search.

The duplications that produced the three paralogues are dated and their asymmetry is explained. ITPR2 and ITPR3 are sisters. The first split sits on the vertebrate stem and the second on the gnathostome stem. The surviving two-round paralogon links run through ITPR1, which is also the only paralogue that was doubled and retained after the teleost genome duplication, and the only one held under roughly twice the purifying selection of its sisters. The three paralogues share most of their intron positions in every genome that carries them, an enrichment of nearly two orders of magnitude over a null drawn from the alignment itself, while the ryanodine receptors, which carry every diagnostic domain of the family, share one. Constraint mapped onto the cryo-EM channel identifies the gate and the selectivity filter as the least changeable elements, places the ten measured IP_3 contacts outside the Pfam domain named after the ligand, and finds the constrained unit at the ligand site to be a pocket rather than the contacts themselves.

Three things here are usable rather than only informative. The archive audit produces a list of 297 coordinate-level corrections with the evidence a curator would need to act on each. The per-residue constraint map is scored as a variant classifier and identifies which of four conservation layers a clinical resource for this family should quote. And five of the results are about method rather than about receptors, including a measured false-negative rate for genome-scale orthologue sweeps and a measurement of what a protein bait panel's breadth is actually worth.

How this document is organised, and why it is long

The same work is also reported as a paper of 17,807 words. That paper is the right length for its claims and the wrong length for its instruments, and the difference between the two is the reason this document exists.

Almost every number in the paper rests on a threshold, and almost every threshold in this project was measured rather than chosen. Examples include the intron length above which a gene reads as a fragment, the alignment margin that separates two families of the same architecture, the coverage bar a reassembled pseudogene has to clear, and the identity floor below which a locus is not a locus. Each of those measurements has a result of its own, and several of them changed the answer. In a paper each is a Methods sentence. Here each sits in the chapter that first depends on it, together with what it was measured against and what guessing would have cost.

The same applies to what did not work. A thesis is the only format in which a measured dead end is worth its page, and this one carries several: an exhaustive substitution-model scan that was started, timed and abandoned at 11 of up to 1,232 models; an attribution threshold inherited from a sister project and overturned because it came from a family half as similar to itself as this one; a duplication detector that reached a specificity of 0.16 before it was scoped correctly; and a synteny-caller threshold that optimising the obvious quantity would have set at the loosest value on offer. None of these is filed in a chapter of failures. Each sits in the chapter that owns the instrument it was measured on, because that is where it explains why a number can be believed.

Chapters 1 and 2 set out the receptor and audit what the literature could be trusted to say about it before any of this was measured. Chapters 3 to 5 build the search and report the family's range. Chapters 6 to 9 cover its history: the tree, the duplications, the gene's own architecture, and a retention result that required more care to state than any positive finding in the project. Chapters 10 to 12 cover the protein: selection, constraint, and the one part of it the ryanodine receptors do not share. Chapter 13 audits the archive. Chapter 14 is the methods, written as argument rather than as procedure, and it carries the project's decision log and its constructed negative controls. Chapter 15 discusses what the whole says and what it does not, and Chapter 16 describes how the project itself was carried out.

What is checked in this document, and by what

Nothing here is typed twice. Every figure is copied from the results directory that committed it and is never re-plotted. Every load-bearing number is declared with the committed table it comes from and the operation that recovers it, and the build re-reads those tables on every run. Every citation is a stable key resolved against one reference table, and the bibliography is rendered from that table rather than written. The claims ledger additionally requires each declared number to appear in the chapter that declares it, which is what stops a ledger being padded with checks the text never makes.

Running `python scripts/s25_assemble.py` rebuilds the whole document and exits non-zero on a missing figure, a missing chapter, a cited key with no reference row, a reference added without an audit, a glyph the document font cannot set, or a failed claim. Every one of those guards is broken on purpose on every build to confirm that it fires. Chapter 14 explains how.

Where the data, the code and the outputs are

Every number in this document is recovered from a committed table, and every one of those tables is in the repository this document is built from, under `results/`, one directory per analysis task. The analysis code is under `scripts/`, one file per task, and the search application it was built on is under `src/`. The exact version of every external program the project ran is recorded in `results/toolchain_manifest.txt`, and the versions of the public databases queried are recorded in each task's own statistics file.

Bulk inputs are deliberately not in the repository, because 503 genome assemblies and 7,691 reference proteomes are not a reviewable artefact. What is committed instead is the manifest that declares exactly which ones were used, with the rule that admitted each, so the download is reproducible from a table rather than from a description. `manuscript/deposit_notes.md` lists each excluded bulk class together with the command that regenerates it.

How this project was carried out

This project was carried out by Claude Code, which is Anthropic's Claude running as an agent in a terminal with access to the file system, the shell and the network. Claude Code wrote every line of the analysis software, chose and calibrated every threshold, ran every search, generated every table and figure, recorded every methodological decision, and wrote this document and the paper that accompanies it.

The human collaborator set the goal, provided the machine and the storage, ran the small number of commands that need an interactive login, and read and corrected the drafts. He did not write the code, select the genomes, choose the statistical tests or draft the text.

That division of labour is unusual enough to be worth describing rather than leaving for a reader to infer, and it is also relevant to how the work should be judged, because an agent that can run for hours unattended can produce a great deal of plausible output. **Nearly every methodological rule in this thesis exists because an autonomous worker needs to be stopped from convincing itself.** The measured thresholds, the positive controls carried through every stage, the negative controls that must fire, the reports rendered from tables rather than written, and the claims ledger that re-reads every number are all answers to the same question: how does anyone, including the agent, tell whether what it just produced is true?

Chapter 16 describes the whole arrangement in detail, covering how the work was organised into tasks, how the literature was surveyed and audited, how the decisions about what to download and what to measure were made, how the analyses ran, and how the results, the figures and these documents were produced.

1. The IP_3 receptor, and the questions this thesis answers

1.1 The receptor is a channel that reads a second messenger

Almost every eukaryotic cell keeps a store of calcium inside its endoplasmic reticulum at a concentration three or four orders of magnitude above the cytosol, and almost every eukaryotic cell has a way of opening that store on demand. In animals the principal way is the inositol 1,4,5-trisphosphate receptor, a ligand-gated channel in the endoplasmic reticulum membrane that opens when it binds the soluble second messenger IP_3 and releases calcium into the cytosol.

The pathway that produces the ligand is one of the best-characterised in cell biology. A receptor at the plasma membrane activates a phospholipase C, which cleaves phosphatidylinositol 4,5-bisphosphate into diacylglycerol and IP_3, and the IP_3 then diffuses to the endoplasmic reticulum and opens the receptor (Figure 1.1). What follows, meaning the amplitude, the timing and the spatial pattern of the calcium signal, is largely set by the channel itself. That is why the receptor has been studied for forty years as a signalling device rather than merely as a pore.

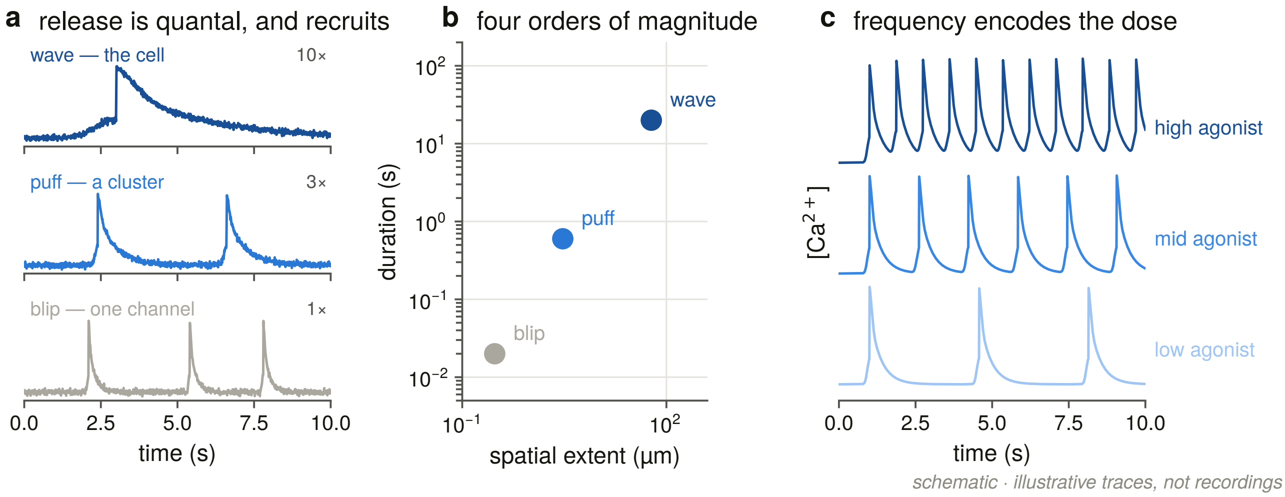


Figure 1.1. The receptor's place in the phosphoinositide pathway, drawn as a hierarchy of scales running from the messenger to the cellular output. The layer this thesis is about is a single one of these boxes, and the figure fixes the level at which a genome-scale census can and cannot speak. A census counts genes, so it can say whether the channel is present and how many copies a genome carries, and it cannot say what a cell does with the signal. Chapter 12's result about the enzyme that makes the ligand is the one place where the box above the channel is measured too.

The receptor was identified in the decade after IP_3 was shown to release calcium from a non-mitochondrial store in pancreatic acinar cells [1]. Purification and reconstitution demonstrated that a single protein was sufficient for the flux [2], and the primary structure, obtained from the cerebellar P400 protein, revealed a very large polypeptide of about 2,700 residues [3,4]. The title of one of those first reports named the problem this thesis had to solve before it could measure anything: *Putative receptor for inositol 1,4,5-trisphosphate similar to ryanodine receptor* [5].

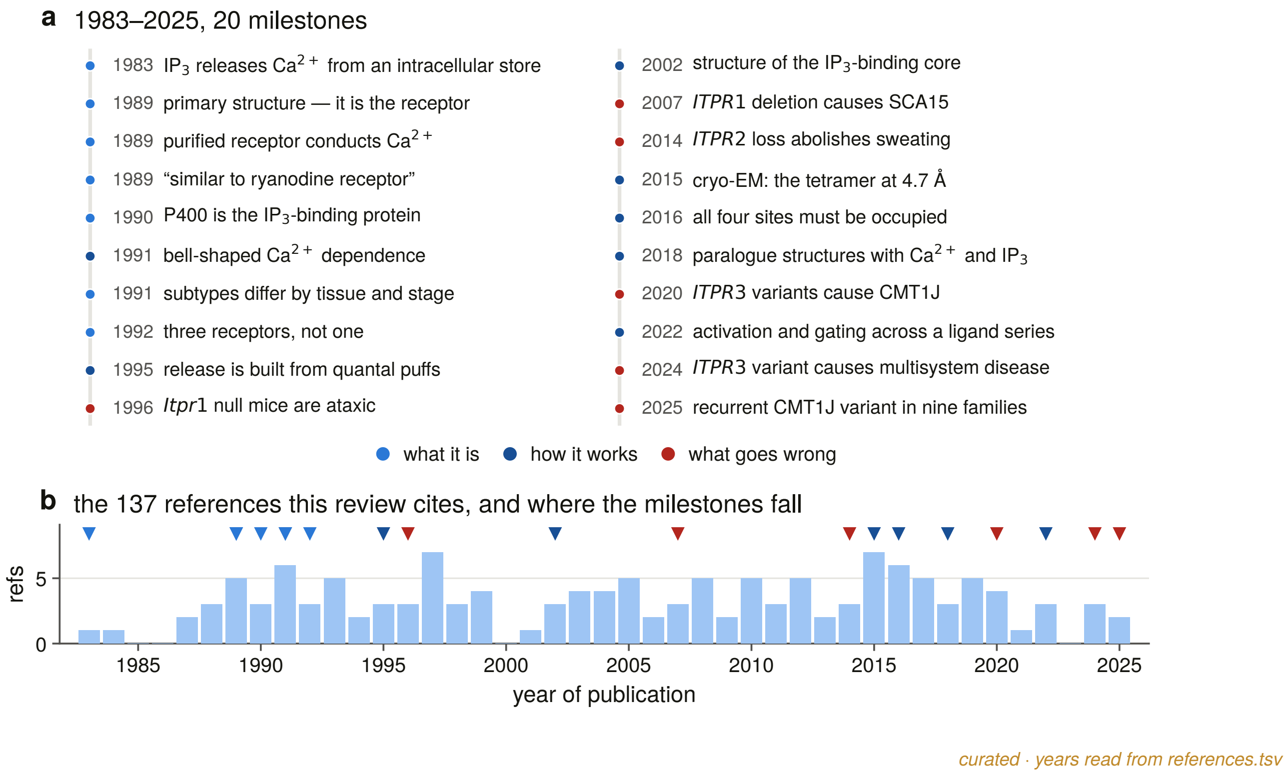
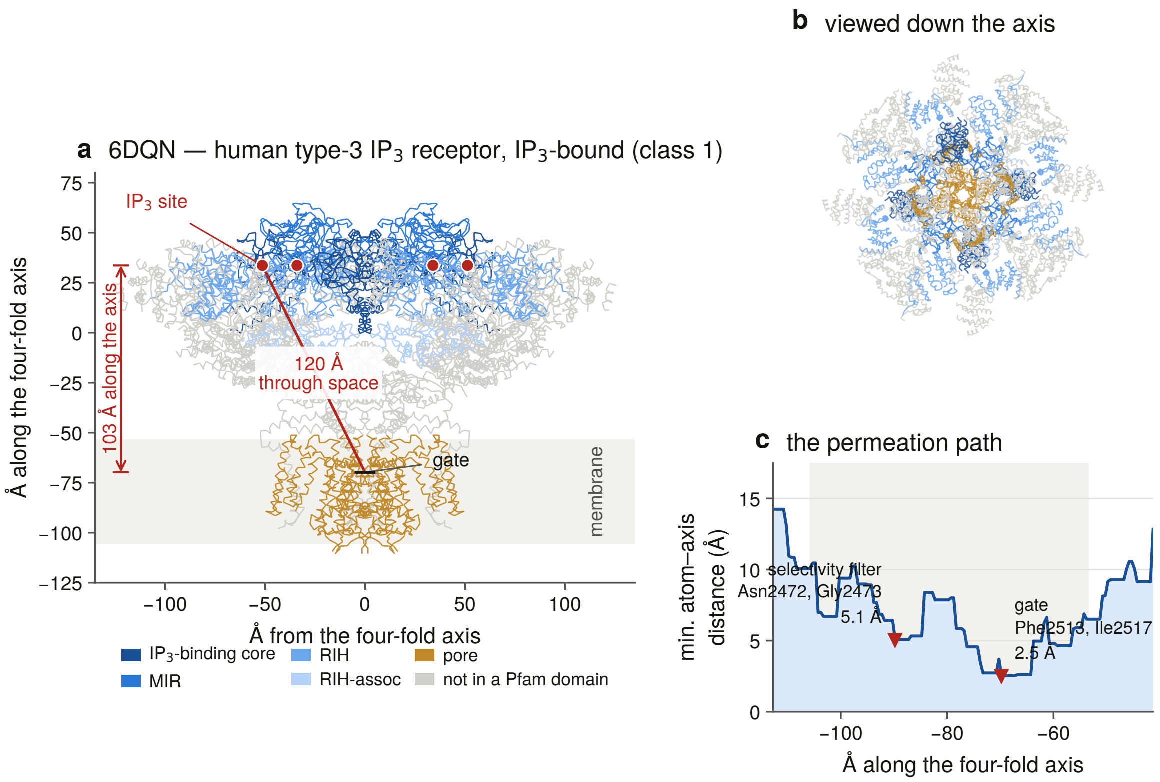


Figure 1.5. Four decades of work on the receptor as the literature records it, with each milestone drawn at the year of its published source. The figure is generated from a curated table in which no year is typed: each is read from the reference row it cites, so a milestone cannot be dated differently from the paper it rests on. What the shape shows is how recently the evidence this thesis depends on became available. The biochemistry and physiology were settled over forty years, but the near-atomic structures that make a residue-level constraint map interpretable are less than a decade old, and the genome assemblies behind a 503-genome census are more recent still, so the questions asked here could not have been asked before the right-hand end of this figure.

1.2 The receptor shares its entire domain architecture with a much larger sister family

An IP_3 receptor is a homotetramer. Each subunit is around 2,700 residues, of which roughly nine tenths are cytosolic. Reading from the N-terminus, a subunit consists of a β-trefoil suppressor domain, then the IP_3-binding core (a β-trefoil with an armadillo fold), then a long central solenoid of armadillo repeats, and then, near the C-terminus, a six-transmembrane pore module of the voltage-gated-channel superfamily, followed by a tail that folds back into the cytosolic mass (Figure 1.2). The ligand binds some 100 Å from the gate, and everything between the two is the machine that couples them.



measured · PDB 6DQN, 3.33 Å

Figure 1.2. The channel measured rather than drawn. The figure shows one C4-symmetric subunit's Cα trace from PDB 6DQN, a human IP_3R3 in the IP_3-bound state at 3.33 Å [6], together with the four-fold axis, the axial extent of the membrane, the pore radius profile and the two constrictions recovered from the coordinates. The narrowest luminal point lands on the GGGVGD selectivity-filter motif and the cytosolic constriction on the gate residues, neither of which the geometry was told about. That blind agreement is what makes the same geometry usable as a coordinate system later: it can define a luminal loop boundary no annotation carries, which Chapter 11 needs, and a ligand pocket measured by distance rather than taken from a contact list, which Chapter 12 needs. A structure drawn from a published residue list could do neither, because it would only return what was put into it.

In domain-annotation terms the diagnostic signatures of the family are the MIR domains (PF02815) in the suppressor region, the IP_3-binding core Ins145_P3_rec (PF08709), the RyR–IP_3R homology domain RIH (PF01365), the RIH-associated domain (PF08454), and the generic Ion_trans pore (PF00520).

Every one of the first four is also carried by every human ryanodine receptor (Figure 1.4). This is not an artefact of database annotation. The two families are one structural superfamily: their N-terminal regions are conserved to the point of direct comparison [7], and the near-atomic structures published for both in 2015 [8,9,10,11] show the same organisation, in which a vast cytosolic solenoid transduces ligand binding to a C-terminal pore. They differ in scale, the ryanodine receptors being nearly twice the size at around 4,900 to 5,000 residues.

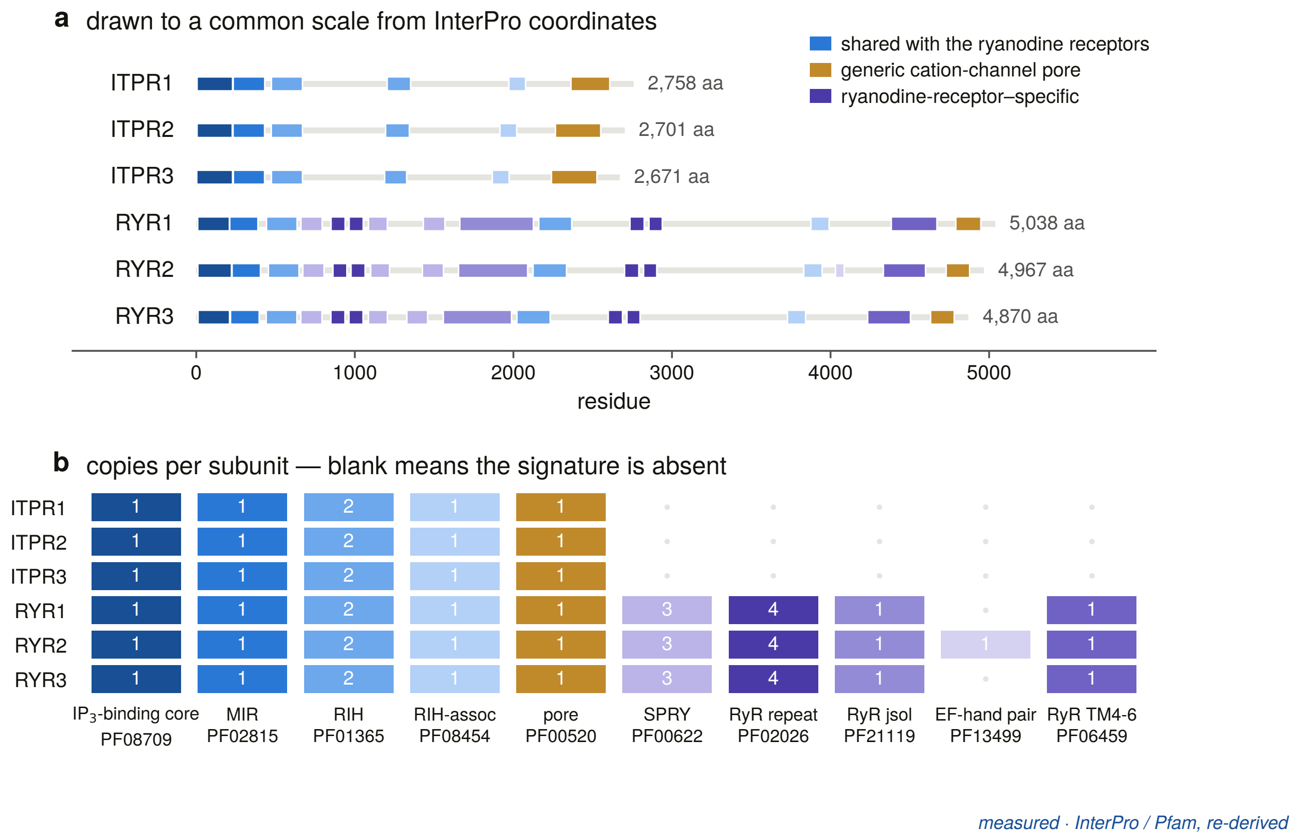


Figure 1.4. Every signature that defines an IP_3 receptor is also carried by a ryanodine receptor, in the same copy number, down to the two RIH domains. What separates the families is what the ryanodine receptors carry in addition: four further domains and some 2,200 extra residues. This figure is the reason the project has the shape it does. Because the diagnostic signatures cannot separate the two families, every stage from the first enumeration to the last structural comparison had to carry a positive family test, and the sister family had to be searched alongside rather than filtered out. Roughly half the methodological rules in Chapter 14 exist because of what this figure shows.

Three consequences of that shared architecture run through every chapter of this thesis.

Any similarity search for one family returns the other. The scale of the problem is easy to underestimate. A single query for the IP_3-binding-core signature in zebrafish returns 109 protein records across 10 gene symbols, of which 53, or 49 %, are ryanodine receptors. Seven of those records come from a 4,900-residue locus that carries no gene name at all. That number is measured in Chapter 2, in the exact query this project's own planning document had quoted as a clean result.

The separation between the two families must therefore be a positive test. It cannot be a filter that removes what looks wrong. It has to be evidence that a record is one thing rather than the other, in the form of best-profile assignment with a stated margin, or a labelled-bait identity margin, applied at every stage from the first enumeration to the last structural comparison. Where the evidence does not reach the required bar, the honest output is a refusal, and this project produces a great many of them.

Size is a filter and not evidence. The two families do separate cleanly by length in a well-annotated genome. That is a fact about annotation quality in that genome, and using it as the call would make every result downstream a restatement of how good the annotation already was, which is one of the things this thesis set out to measure.

1.3 What the channel is for, and why the three paralogues are not interchangeable

The channel's behaviour is not a simple function of ligand concentration. Its response to cytosolic calcium is biphasic, activating at low concentrations and inhibiting at high, which under IP_3 produces the bell-shaped calcium-response curve that has organised thinking about the receptor since it was measured [12]. IP_3 binding and calcium binding are cooperative rather than independent [13], and the stoichiometry required to initiate release has been resolved, in that more than one subunit must be occupied [14]. ATP modulates the channel at two sites [15], protein kinase A phosphorylates it [16], and it is a hub for protein interactions, including with Bcl-2 [17] and IRBIT [18].

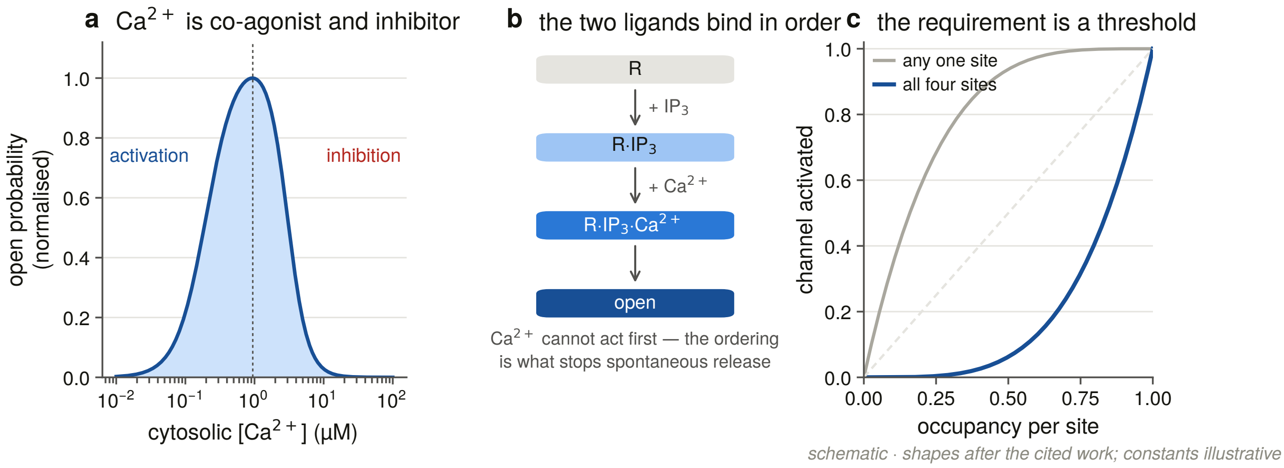


Figure 1.3. What opens the channel and what closes it, drawn as the logic rather than as a mechanism. The row that matters for a census is the last one: the same channel produces puffs, waves and oscillations, and the properties that distinguish those outputs belong to the individual paralogue. If the three vertebrate paralogues were interchangeable, their copy number would be a matter of dosage and their individual retention would carry little information. Because they differ in affinity, in calcium sensitivity and in downstream output, the finding in Chapter 9 that all three are retained in every vertebrate genome is a statement about three distinct functions rather than about three copies of one.

Those output differences are the reason the paralogues matter. The three vertebrate receptors differ in IP_3 affinity, in calcium sensitivity, in regulation and in tissue distribution [19,20,21], and they are not redundant. Calcium transfer from the endoplasmic reticulum to the mitochondrion, and the apoptotic decisions downstream of it, depend on which paralogue is present [22]. A cell's calcium signalling repertoire is in part a statement about which of ITPR1, ITPR2 and ITPR3 it expresses.

1.4 Four things about the family were not established

For a family this well studied, a surprising amount was not established at the level a genome-scale question requires.

Its taxonomic range had never been enumerated against a declared space. The receptor is described as animal machinery with scattered occurrences elsewhere. Those scattered occurrences had never been enumerated against a declared search space, and the conspicuous absences, meaning no IP_3 receptor in *Arabidopsis* and none in budding yeast, had never been tested at the level of an assembly rather than a gene set. An absence in a proteome is a statement about what a gene caller found. Whether it is also a statement about the genome is a different measurement, and nobody had made it.

Where the three vertebrate paralogues came from had no tested answer. That vertebrates carry three is a database fact. That the three arose in the two rounds of whole-genome duplication at the base of the vertebrates [23,24] is a reasonable inference from their number and their age, and it had not been tested against the genomic neighbourhoods that a duplication of that kind leaves behind. Which two of the three are sisters had no published answer at all. Worse, the proposition that the IP_3 and ryanodine receptor triplications happened independently, which this project's own background document asserted, turned out to have no primary source and had to be downgraded to an open question before any of the work could begin. Chapters 6 and 7 answer it.

What has happened to the paralogues since could not be asked without a false-negative rate. Gene families of this age normally lose copies [25]. Whether this one has, and where, is a question that cannot be answered without a false-negative rate, because a gene that a search fails to find looks exactly like a gene that is not there. That single sentence determined the structure of a third of this thesis.

How well the family is recorded had never been audited. Every one of the questions above is answered using public databases, and the quality of those databases for this family had never been audited. The audit turned out to be a result in its own right.

Those four gaps are the questions this thesis answers, and they can be stated in one sentence each. How far does the family reach, measured against a declared search space rather than against whatever records exist? Where did the three vertebrate paralogues come from, and which two of them are sisters? Has any vertebrate lineage lost one, and how would anybody know the difference between a loss and a search that failed? And how well is the family recorded by the public archives that every one of those answers depends on? Chapters 5, 6 and 7, 9 and 13 answer them in that order, and Chapters 3, 4, 8, 10, 11 and 12 build and test the instruments the answers rest on.

1.5 How every claim in this thesis is scoped

Every result here is scoped to a declared search space. The claim is never "all IP_3 receptors" but always a specific set: 7,691 UniProt reference proteomes and 503 NCBI genome assemblies, each set enumerated by a stated rule and committed as a manifest before any search ran. An absence is an absence from that space, measured with a positive control that demonstrates the search could have found the gene had it been there.

That discipline is why the negative results in this document can be read as results. The family is absent from all 384 land-plant reference proteomes, and it is also absent from those genomes, in assemblies where a measured control recovers a comparably long, deeply conserved gene. Those are two different claims and both are made separately, because the second is the one that means something biological and the first is what a database can support on its own.

The same discipline runs in the other direction. Chapter 9 reports that not one of the 927 genome by paralogue cells in the vertebrate scope reaches the state this project defines as a loss. A zero of that kind is worth nothing unless the instrument that produced it could have produced something else, so Chapter 4 measures the search's own false-negative rate on cells whose gene is independently known to be present, and Chapter 9 reports, cell by cell, the analytical settings that would manufacture a loss out of this data. The strongest form in which that result can be stated is the one that names its own escape routes.

1.6 What each chapter of this thesis does

Chapter 2 audits the literature this project started from, claim by claim, and reports the four statements that did not survive. Chapter 3 builds the instrument that separates the two receptor families and enumerates the search space. Chapter 4 takes that instrument to 309 vertebrate genomes and then measures what the search was worth. Chapter 5 takes it outside the vertebrates and reports the family's range and its absences.

Chapter 6 builds the alignment and the tree that everything downstream stands on. Chapter 7 asks where the three paralogues came from, using the genomic neighbourhood, the species tree and the duplication record. Chapter 8 measures the gene itself, meaning its exons, its introns, and what a fragmentary annotation actually is. Chapter 9 counts the losses.

Chapter 10 measures selection across the vertebrate family. Chapter 11 maps constraint onto the channel and asks whether it explains the clinical variants. Chapter 12 takes the one part of the receptor the ryanodine receptors do not share. Chapter 13 audits the archive. Chapter 14 sets out the methods as the argument behind each instrument rather than as a procedure, and Chapter 15 discusses what the whole says. Chapter 16 describes how the project itself was carried out, which is a question about the work rather than about the receptor and is answered separately for that reason.

2. Auditing what the literature could be trusted to say

2.1 Why this project began by auditing its own background document

Every genome-scale project begins with a background document, meaning the set of things taken as known against which the new measurements will be read. That document is normally written from reading, and it is normally the least scrutinised artefact in the project, because it is not a result.

This one was treated as a result. Nineteen atomic claims were extracted from the `[lit]`-tagged statements of the project's background document and checked against their primary sources. Every `[db]`-tagged number was re-queried against the live database it came from. The application that would do the searching was run against all three public interfaces to establish that it worked at all. The exercise took a session and changed four claims, one of which would have made a later chapter's conclusion into an assumption.

The tagging scheme is the point. A statement is `[db]` if it can be re-derived from a database with a stated query, `[lit]` if it rests on a published source, and `[open]` if it is a question this project answers. A background document that does not separate these three cannot be audited, because there is no way to say what would falsify any given sentence.

2.2 Four of the nineteen literature claims did not survive as written

Twelve of the nineteen were verified as written against their primary sources. Three were verified but qualified, meaning they were supported by wording that overstated the strength of the evidence and were re-written accordingly. The remaining four did not survive in the form they were stated.

One claim was promoted from literature to database. The cytogenetic locations of the three human genes, with ITPR1 at 3p26.1, ITPR2 at 12p11.23 and ITPR3 at 6p21.31, were carried as a literature claim. They were re-derived from Ensembl and confirmed exactly, so they became a `[db]` statement with a query attached rather than a citation.

One claim was demoted to an open question, and this is the important one. The background document asserted that the IP_3 and ryanodine receptor families had independently expanded to three vertebrate paralogues each. The three-paralogue state of each family is a database fact. The independence of the two triplications is not, because no primary source establishes it. Left in place, that sentence would have made Chapters 6 and 7 and the reconciliation that dates the duplications into an elaborate confirmation of something already assumed. It was retagged `[open]` before any tree was built.

Two claims were corrected. The first is discussed in §2.3. The second concerned Gillespie syndrome, which the background document described as caused by pore-proximal ITPR1 variants acting dominant-negatively in the tetramer. That is half the genetics. The syndrome arises from both biallelic recessive variants and de novo heterozygous ones, and only the latter act dominant-negatively through the channel domain. Stating only the dominant-negative mechanism would have misdirected the constraint analysis in Chapter 11, which asks where pathogenic variants sit relative to the constrained core, and would have had it looking for one signal where there are two.

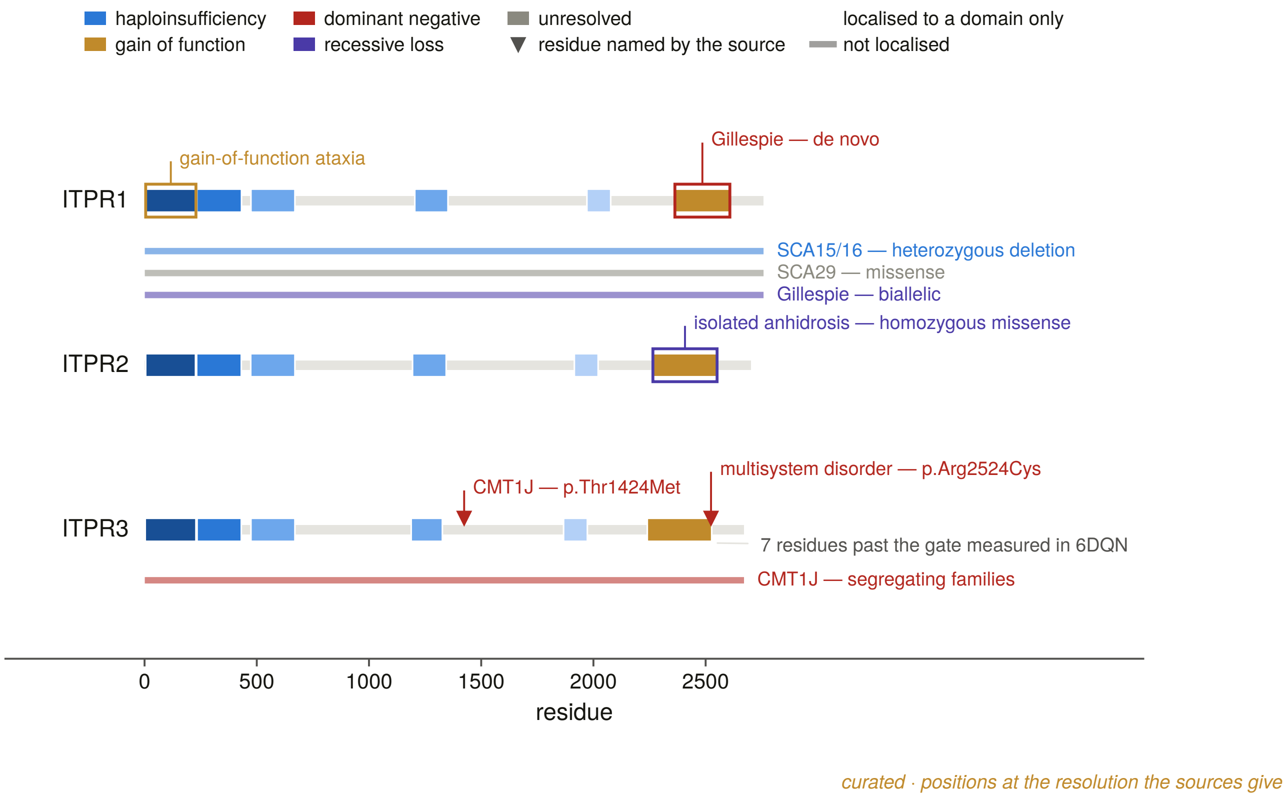


Figure 2.7. The curated disease-variant map, with each variant recorded as `point`, `domain` or `gene` according to how far its cited source actually localises it. That third column is the honesty of the figure: a source reporting a deletion of the whole gene cannot be drawn as a residue, and a map that drew it as one would assert a resolution the literature does not have. The distinction is what makes Chapter 11's variant analysis possible at all. A constraint map is scored against positions, and recording how far each source localises its variant is what showed that only two residue-level positions were available from the literature, which is why that chapter harvests a clinical database instead and uses those two as its positive control.

2.3 The background claim about gene size was false, and it became a measurement

The background document stated that each ITPR is "a ~58–60 exon gene spanning hundreds of kb". Ensembl's canonical transcripts say otherwise (Figure 2.1).

The exon counts are 62, 57 and 58, which is a range of 57 to 62 rather than 58 to 60. That is a small correction. The span is not: ITPR1 covers 354,174 bp, ITPR2 covers 497,888 bp, and ITPR3 covers **76,245 bp**. "Hundreds of kb" is wrong for one of the three, and the genomic span varies 6.5-fold across the three paralogues while the protein length varies by about 3 %.

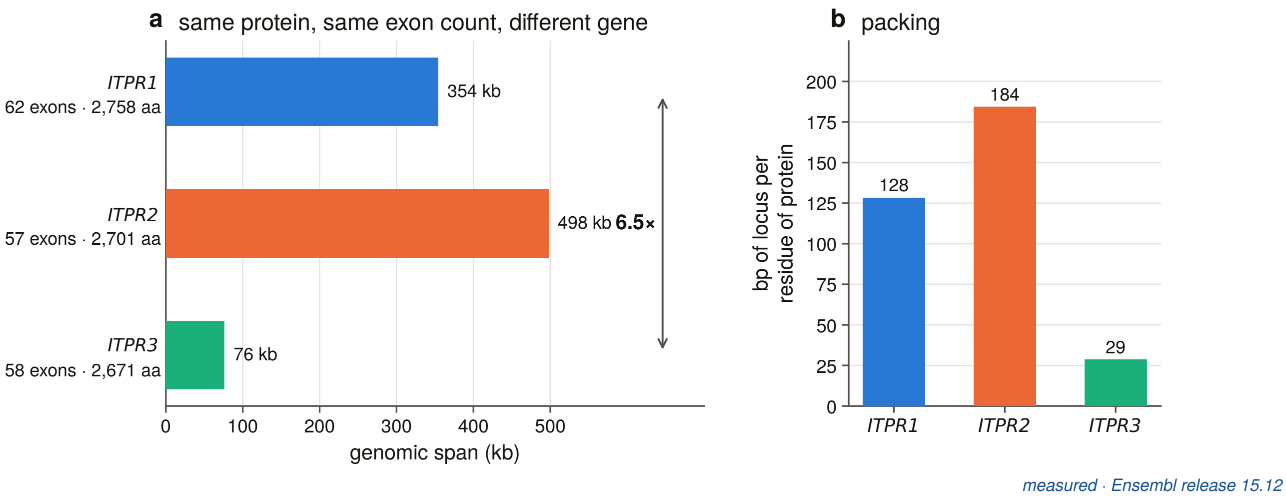


Figure 2.1. The three human genes drawn to one scale from Ensembl coordinates. Exon count is nearly identical across the three and genomic span is not, and the difference between them is almost entirely intron length. The figure is where a corrected sentence became a measurement. Three paralogues encoding proteins whose lengths differ by 3 % occupy genomic spans differing by a factor of six and a half, which must mean that either their exon structure or their intron content differs, and only the first would be a statement about how the coding sequence has evolved. Chapter 8 resolves it in favour of the second, and that chapter exists because of this figure.

A failed background claim is normally just a correction. This one became a measurement. If three paralogues of one gene, encoding proteins of the same length, occupy genomic spans that differ by a factor of six and a half, then either the exon structure differs between them or the introns do, and either answer says something about how the family has evolved. That question is Chapter 8, and it exists because a sentence taken from reading turned out to be false.

The correction also has an operational consequence that reaches into Chapter 4. A protein-to-genome aligner has a maximum intron length, and a gene whose largest intron exceeds it is split into pieces, so a split gene reads out of a genomic sweep as a fragment. Setting that parameter is not a performance choice, and it cannot be made from a claim about "hundreds of kb" that is wrong for a third of the family.

2.4 Every database number in the background document reproduced exactly

Every `[db]` number in the background document was re-queried on the day the audit ran. All of them reproduced exactly, which is the boring outcome and the one worth recording: the numbers were right, and they are now right with a query beside them.

Two of those numbers set up the whole project. The IP_3-binding core signature PF08709 was carried by 12,338 proteins, the RIH domain PF01365 by 13,177, the RIH-associated domain PF08454 by 12,062, the MIR domain PF02815 by 23,453, whose excess over the others is the O-mannosyltransferases that share it, and the generic pore PF00520 by 206,115, which is why the pore is not a family signature.

The taxonomic distribution carried an anomaly. Forty Viridiplantae proteins and forty-one fungal proteins carried PF08709, the IP_3-binding core, while *Arabidopsis thaliana* and *Saccharomyces cerevisiae* had none. Eighty-one records in kingdoms whose model organisms have no IP_3 receptor is either a real distribution or an artefact, and no amount of reading resolves which (Figure 2.2). Chapter 5 chases all of them individually.

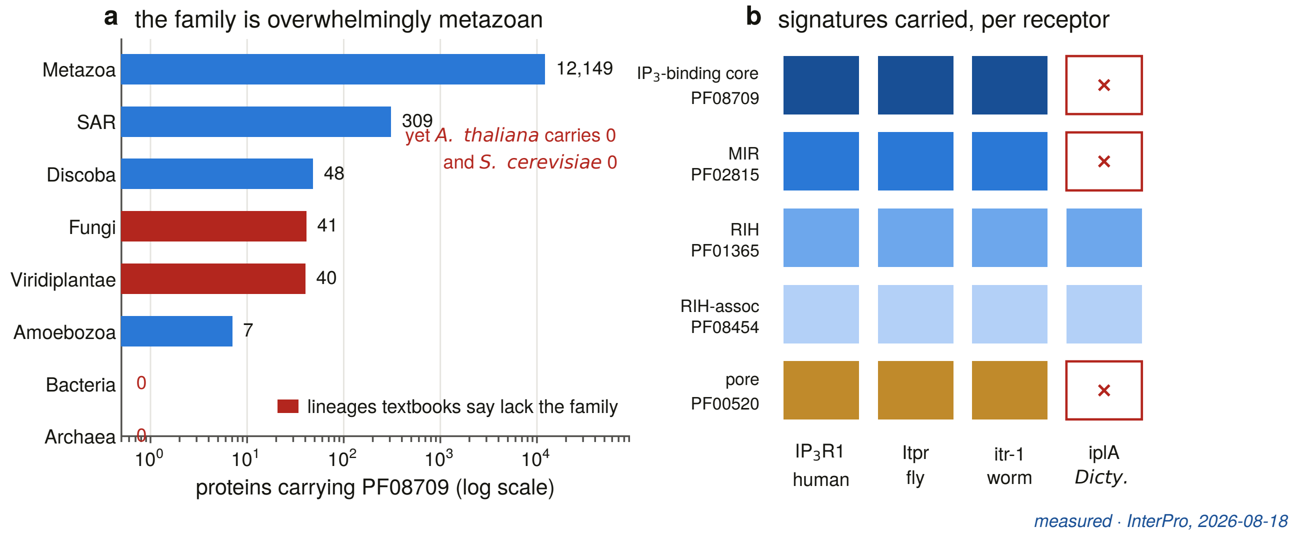


Figure 2.2. What the databases said the family's range was at the start of this project, shown as signature counts by taxon from a live InterPro query. The plant and fungal columns are the anomaly: 81 proteins in kingdoms whose model organisms have none. The figure is important as a statement of what was not known, and as an illustration of why a count of records is a weak instrument for a range question. A record enters here by carrying a domain annotation, and nothing in the count says whether the protein is real, whether it is in that organism's genome, or whether it is an IP_3 receptor at all. Turning this figure into a biological range claim took the two chapters and 194 genome searches that Chapter 5 reports.

2.5 The sister-family hazard was measured in the query that was supposed to be clean

The background document cited a UniProt query on taxonomy 7955 and Pfam PF08709 as evidence that zebrafish carries four IP_3 receptors. Re-running that query returns 109 protein records across ten gene symbols, and only four of those genes are IP_3 receptors. Fifty-three of the 109 records, or **49 %**, are ryanodine receptors.

This is the single most useful number the audit produced. It is a measured floor on the contamination that any signature-driven enumeration of this family inherits, obtained in the exact query that had been quoted as a clean result. It is why every later chapter separates the two families by positive evidence rather than by filtering. It also carries a second finding in passing: one of the six ryanodine-sized genes in that result, a 4,900-residue locus, carries no gene name at all. That is the first appearance of the unnamed-locus problem that Chapter 13 eventually audits across 503 genomes.

The length band happens to separate the two families cleanly in this particular query. That is a fact about zebrafish annotation quality rather than a rule, and treating it as one would import the quality of an annotation into the definition of a gene family.

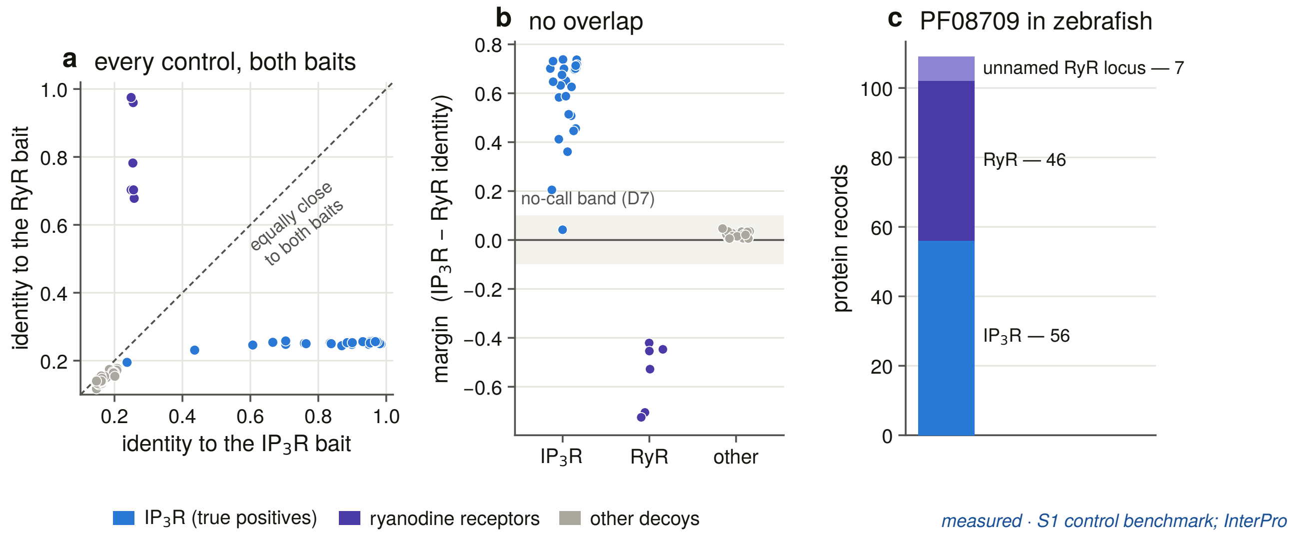


Figure 2.3. Separating the families by evidence rather than by name, using the control panel Chapter 3 builds. Panel **a** shows every control scored against a labelled IP_3 receptor bait and a labelled ryanodine receptor bait, and the two families fall on opposite sides of the diagonal with nothing between them. Panel **b** shows the same data as a margin, with the band inside which this project declines to call either way drawn rather than described. The one positive inside that band is *Dictyostelium* iplA, a characterised receptor that the family's own defining signature does not find. Panel **c** shows the zebrafish query and the 53 records in it that are the sister family. The importance of the gap in panel **b** is that it is what every later instrument had to reproduce. A margin with no gap would mean the two families cannot be separated by sequence, and every downstream count would then be a mixture of both. The one positive inside the no-call band is equally important in the other direction, because it shows that this instrument cannot reach the deepest branches, which is why the profile models of Chapter 3 exist.

2.6 Two sequence measurements become a coordinate system for later chapters

Two further measurements were made for the literature review, and the rest of this thesis uses them as a coordinate system rather than as a result.

The first is a per-column conservation profile over a family alignment, in human ITPR1 numbering (Figure 2.4). The second is a set of alignment windows anchored on sites measured in the structure, meaning the three stretches that contact the bound IP_3, the selectivity filter and the gate, rather than on a residue list taken from a paper (Figure 2.5).

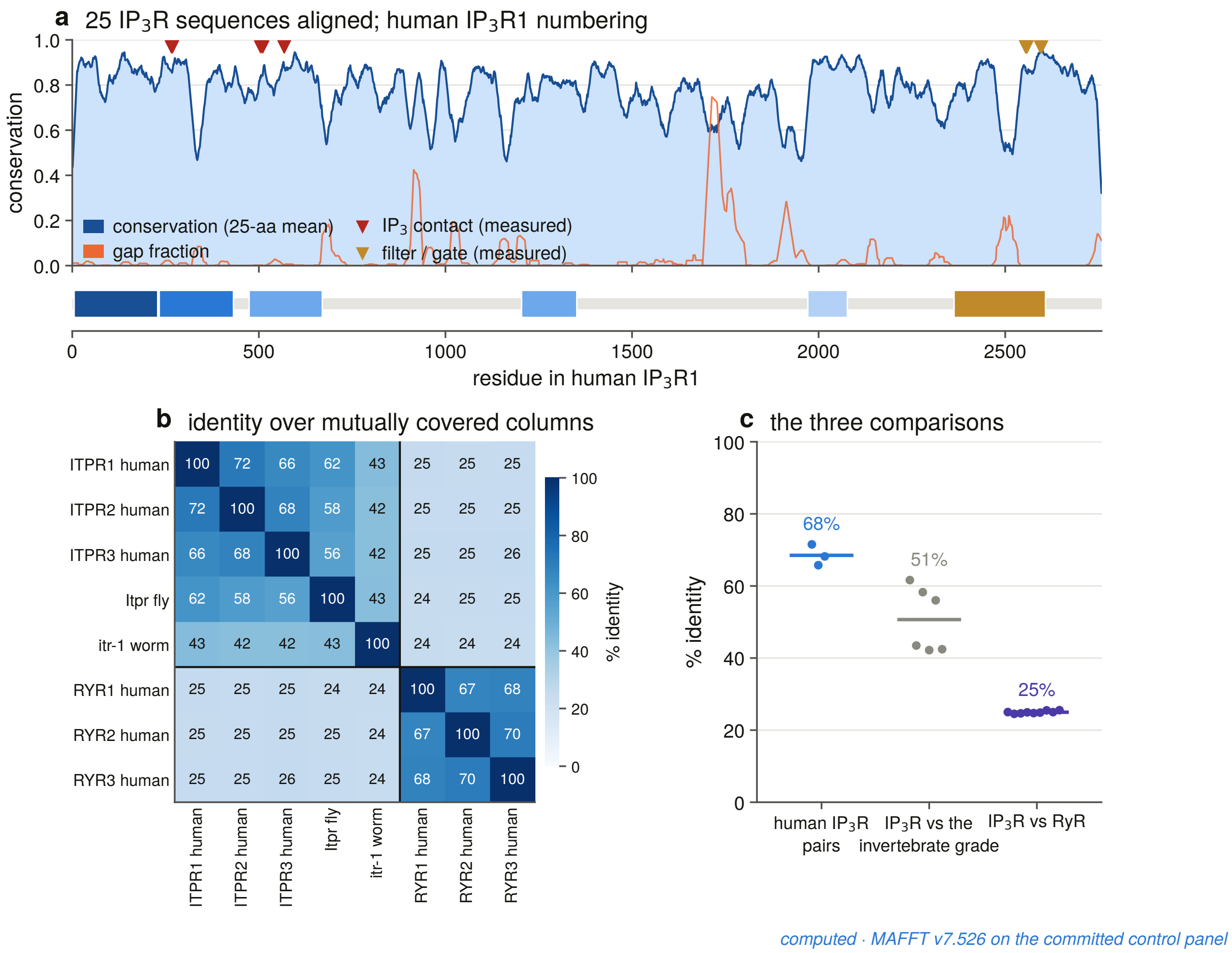


Figure 2.4. Per-column conservation across a 25-sequence family alignment, mapped onto human ITPR1 numbering, with the domain architecture beneath it. Chapter 11 rebuilds this at a very different depth, using 249 to 265 orthologues of each individual paralogue rather than 25 sequences of the whole family, and the two agree about where the peaks are. That agreement matters for a practical reason: if a shallow family alignment and a deep within-paralogue alignment disagreed about where the receptor is conserved, neither could be used to interpret a clinical variant. It is what licenses Chapter 11 to compare four conservation layers as classifiers rather than to pick one on faith.

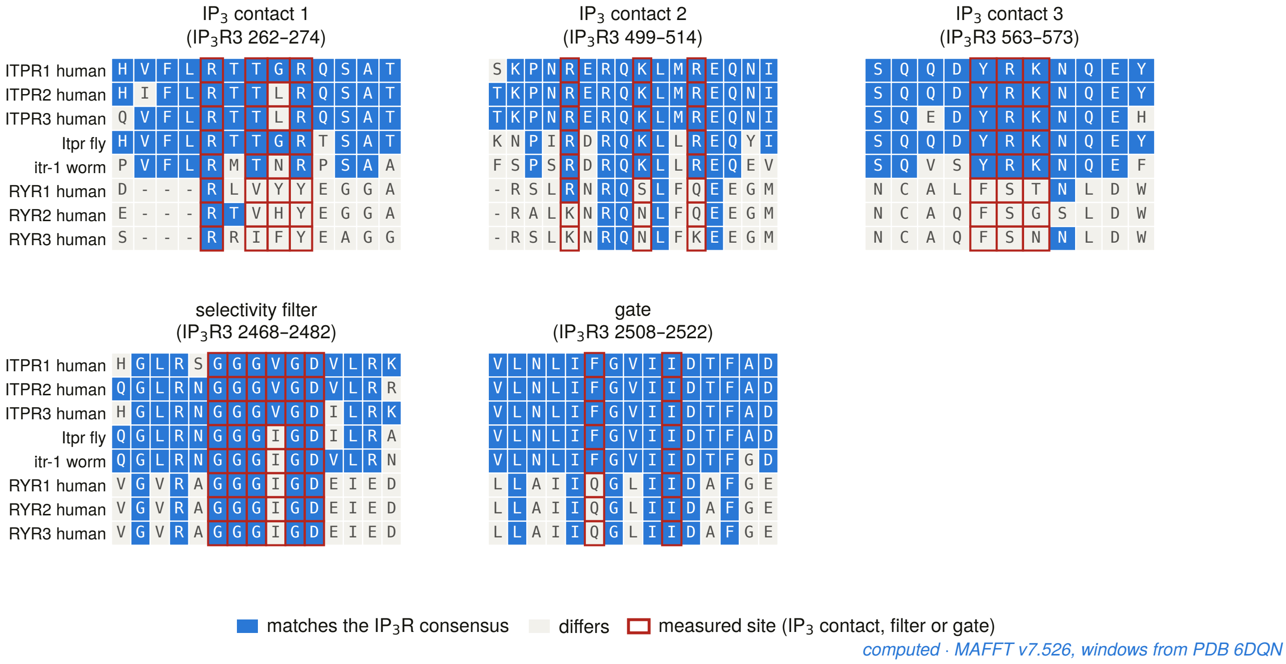


Figure 2.5. Where the two families agree, and where they stop agreeing. At the pore they are interchangeable, showing GGGVGD against GGGIGD in the filter and a gate that differs by conservative substitution. At the ligand site they are not: the arginines and lysines that grip the trisphosphate are absent from all three ryanodine receptors, and one window carries a three-residue deletion. This asymmetry is the biological reason Chapter 12 exists, because it identifies the one place in the protein where a functional difference between the two families is visible in sequence. Everything that chapter measures follows from it, including the result that the module named after the ligand is the less constrained of the two.

Measuring the structure for these figures produced three things the text did not have. The channel's two constrictions were recovered blind from the coordinates and landed on the GGGVGD filter motif and on Phe2513 and Ile2517, without the geometry being told where the filter was. The distance from the ligand to the gate resolved into 103 Å along the axis and 120 Å through space, two numbers that are usually quoted as one. And *Dictyostelium* iplA, a characterised IP_3-gated channel, carries none of PF08709, PF02815 or PF00520, making it a real receptor that the family's defining signature does not find. That last observation is why Chapter 3's profile seeds include it deliberately.

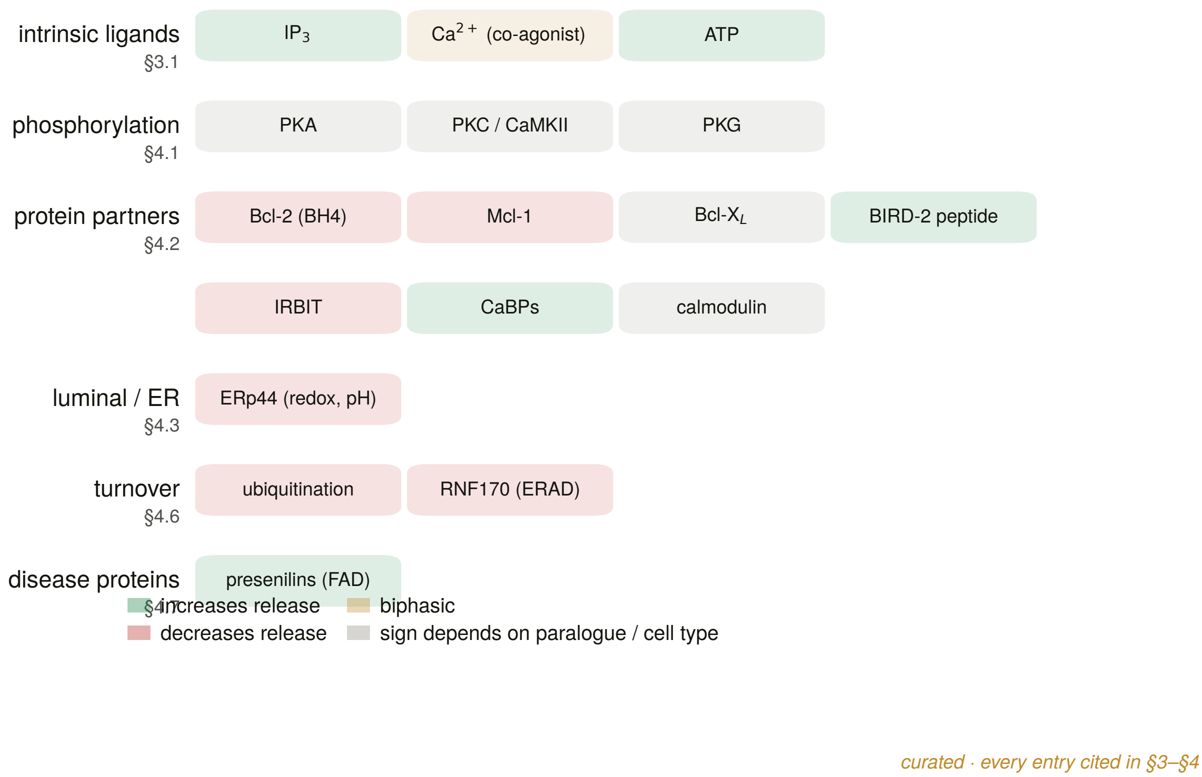


Figure 2.6. The curated regulator map, showing what binds the receptor, where, and with what effect. It is here as a statement of scope rather than as a result, because it is the part of the biology this thesis does not measure. Nothing in a genome-scale census can say whether a protein interaction is conserved, so the regulatory layer drawn here is untouched by every measurement in this document, and a reader deciding what the census settles should be able to see the size of what is left out.

2.7 A fault in one public interface, and why it was worth diagnosing

The audit ended by running the search application against all three public interfaces. Two of four runs returned all three sources, and the two failures were more useful than the successes.

Ensembl returned nothing at all for the human panel. This was not an outage and not a slow network. The endpoint the client used to resolve gene symbols, `/xrefs/symbol/{species}/{symbol}`, **stalls indefinitely for `homo_sapiens`**, and it does so for *BRCA2* as well as for *ITPR1*, so the fault is not gene-specific. The same endpoint answers in 0.6 s for `danio_rerio`. That is why the zebrafish run got three sources and the human run got none, and it is why a retry budget could never have fixed it, because there is nothing to retry against. The client now resolves through the database's other symbol endpoint [26], which works, and falls back to the old path only on a clean 404, never after a transport error, which would walk straight back into the stall. Human ITPR1 went from 0 variants to 24.

Fixing that fault exposed a second one underneath it. Ensembl's latency is unstable: the same 451-byte call measured 0.61 s, 7.61 s and 13.91 s inside one session, and the call that carries the transcript and exon payload costs about 12 s every time. A panel of eight species by three genes is 24 sequential pairs, which is somewhere between five and thirty-eight minutes. The headless run budget was raised accordingly. The instruction that came out of it is worth restating, because it applies to every latency figure in this thesis: do not treat a single measurement of a public interface as a rate, and treat the spread as the design constraint.

2.8 What Chapter 3 inherits from this audit

Chapter 3 inherits four things. A background document whose every statement is tagged by how far it can be trusted, with four claims corrected. A set of database numbers with queries attached, re-derived and reproducing exactly. A measured floor on sister-family contamination of 49 % in the query that had been quoted as clean, which turns the separation of the two receptor families from a caveat into a design requirement. And a gene-architecture question, raised by the one background claim that was simply false, which becomes Chapter 8.

3. Separating two receptor families that share every diagnostic domain

3.1 Why the separation has to be a positive test rather than a filter

Chapter 2 measured the hazard: 49 % of the records returned by a query on the IP_3-binding-core signature in one well-annotated genome are ryanodine receptors. This chapter builds the instrument that separates the two families, and it does so in three passes, because no single test reaches the whole family.

The requirement is stated as a rule rather than a preference. Every record in this project is assigned to one family or the other by a **positive test**, meaning evidence that it is one thing rather than evidence that it is not the other. Where no positive test reaches the required margin the record is left uncalled and counted as uncalled. The gene's own name is never an input to that test at any stage, and a length filter is never the call.

Three instruments follow, in the order they were built. The first is a labelled-bait identity margin, benchmarked against 56 controls. The second is a domain-architecture rule, applied to an exhaustive enumeration of the search space and audited against gene symbols it never sees. The third is a pair of profile hidden Markov models, one per family, calibrated against the architecture rule before being used.

3.2 A control benchmark was run before any search, and it failed on the first attempt

Nothing was searched until the discovery scorer had been run against a panel of known answers: 25 positive controls, comprising the three vertebrate paralogues across a species panel plus the invertebrate and non-metazoan single-receptor grade, together with 31 decoys.

The decoys were chosen to be hard rather than plausible. Six are the human and mouse ryanodine receptors, which are the sharp decoy because they carry all four family-diagnostic signatures. Two are the O-mannosyltransferases POMT1 and POMT2, which share the MIR domain. Fifteen are in-band channels and non-channels, meaning proteins of the right size and the wrong identity, from CACNA1A to talin. Four are out-of-band giants.

Recall was 24 of 25, and specificity 31 of 31.

Specificity was not 31 of 31 on the first run. It was 25 of 31, and every one of the six failures was a ryanodine receptor. Each carried all four family signatures, so each took the domain component, which also satisfies the evidence gate that stops size and novelty alone promoting a candidate, and each scored 45 against a threshold of 40. The instrument, run as designed, called the sister family the family.

What fixed it is the labelled-bait margin, and its three properties are the reason it is used unchanged for the rest of the project. It compares a candidate's identity to the nearest labelled IP_3 receptor bait against its identity to the nearest labelled ryanodine receptor bait, and assigns to whichever wins by more than a stated margin. It is a positive test on distances, so an unnamed 4,900-residue locus is called exactly as a named one is. The length band contributes only to the size component and never to the call. And the margin is recorded on every candidate, so every exclusion can be audited afterwards.

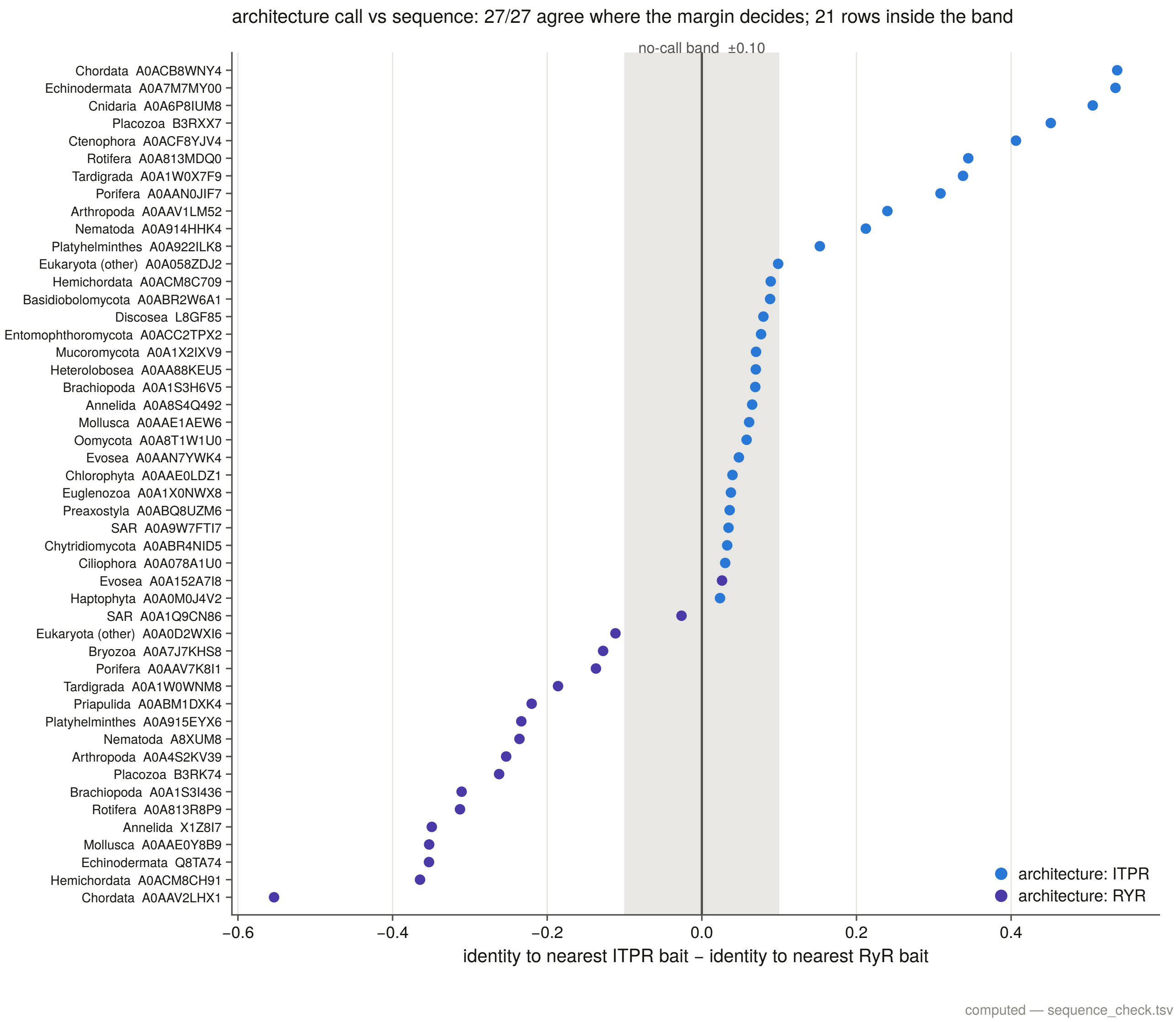


Figure 3.4. The labelled-bait margin measured across the census, inside and outside the band in which the project declines to call. The two families do not overlap: the narrowest true positive sits at +0.065 and the narrowest ryanodine receptor at (-)0.536, a gap of 0.601 with nothing in it. The importance of the gap is that it is a measured floor rather than a chosen threshold. Because the narrowest correct call and the narrowest correct rejection are separated by 0.601 with nothing between them, the 0.10 margin the project uses is not a compromise between sensitivity and specificity; it sits in an empty region six times its own width. That is what allows every later chapter to treat the family call as settled and spend its uncertainty budget elsewhere.

That gap is comfortable, and one true positive sits inside the no-call band anyway. *Dictyostelium* iplA, a characterised IP_3-gated channel, has a margin of +0.065, which has the right sign and is smaller than the 0.10 the rule requires for a call. The margin is not wrong about it; the margin simply cannot reach it. That is a scope limit on this instrument, measured on the first day, and it is why the profile route in §3.5 is not optional.

The obvious identity metric understated the risk. A ryanodine receptor is about 1.8 times the length of an IP_3 receptor, so identity computed over the full alignment dilutes every cross-family comparison toward zero. The ryanodine decoys sit at 0.105 to 0.110 identity to the nearest IP_3 receptor bait, which is below the floor at which the scorer's novelty component fires. Under a fragment-aware metric that scores only mutually covered columns, the same comparisons rise to 0.249 to 0.258, which is inside the twilight zone and worth a further 20 points. A later switch to the better metric, which the one recall failure argues for, would have promoted every ryanodine receptor to 65 without the sister test. Both columns are committed, which is how that is known rather than suspected.

The single missed positive is worth its line. *Drosophila* Itpr scored 35. It needs one non-known sibling at 0.35 identity or better, and its nearest is the worm receptor at 0.342 under the metric the scorer uses, short by 0.008. Under the fragment-aware metric the same pair scores 0.420 and the component fires. The honest worst case is therefore measured rather than assumed: **a lone true family member, 40 to 60 % diverged, with no sibling in the set, scores 35 and is missed.** In the census that follows, the invertebrate grade is represented by many lineages that corroborate one another, so the artefact does not recur, but it is the shape of this instrument's failure and it is on the record.

3.3 The search space was enumerated to exhaustion, and its own counts cannot be trusted

The census was built by walking three Pfam signatures [27] to exhaustion, through the domain database that serves them [28]: PF08709 (the IP_3-binding core), PF01365 (the RyR–IP_3R homology domain) and PF08454 (RIH-associated). The MIR domain PF02815 is deliberately not a seed, because it is carried by the O-mannosyltransferases as well as by both receptor families, so it widens the space without adding evidence. It is kept as an annotation column instead.

Three things about that walk are worth a thesis's space.

The database's own record count is not a completeness criterion, and it is wrong in both directions. PF08709 advertises 12,339 records and serves 12,507, an excess of 168. PF01365 advertises 13,177 and serves 13,233. PF08454 advertises 12,066 and serves 11,890, which is 176 fewer than advertised. The advertised values were stable across re-queries, so this is not an edit during the walk, and in every case what the interface serves matches UniProt's independent count for the same signature to within two records. It is the advertised number that is wrong.

Both directions bite. Stopping at an understated count drops records in silence. Trusting an overstated one declares a complete walk incomplete, which is what happened: this task's own completeness test failed PF08454 and exited non-zero on a walk that had run its cursor chain to the end. The recorded criterion is therefore cursor exhaustion, with both counts kept beside it.

The documented interface was down for the duration. The database answers the same queries on two hosts, the documented one and the one its own website uses. During this work the documented host returned HTTP 500 to 11 of 12 probe requests while the other served 12 of 12, with identical payloads and the same counts. The client now tries both before it sleeps, which is why the walk completed at all.

One signature would not have been enough. A census built on PF08709 alone, which is the signature that names the family, would have enumerated 12,507 of the 15,421 proteins here and missed 2,914, including 758 that this census calls IP_3 receptor across 385 taxa. Only 10,256 records, or 66.5 %, carry all three seeds. *Dictyostelium* iplA is among the records a single-signature census would have missed, because it carries

PF01365 and PF08454 and none of PF08709, PF02815 or PF00520. It is a characterised receptor that the family's defining domain does not annotate.

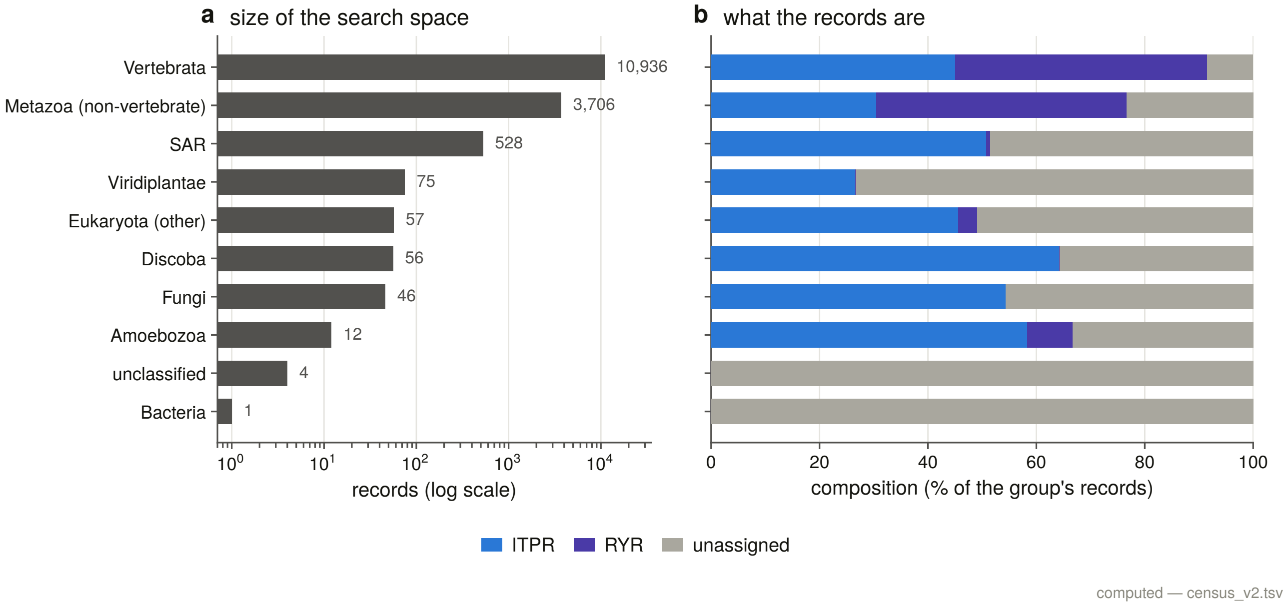
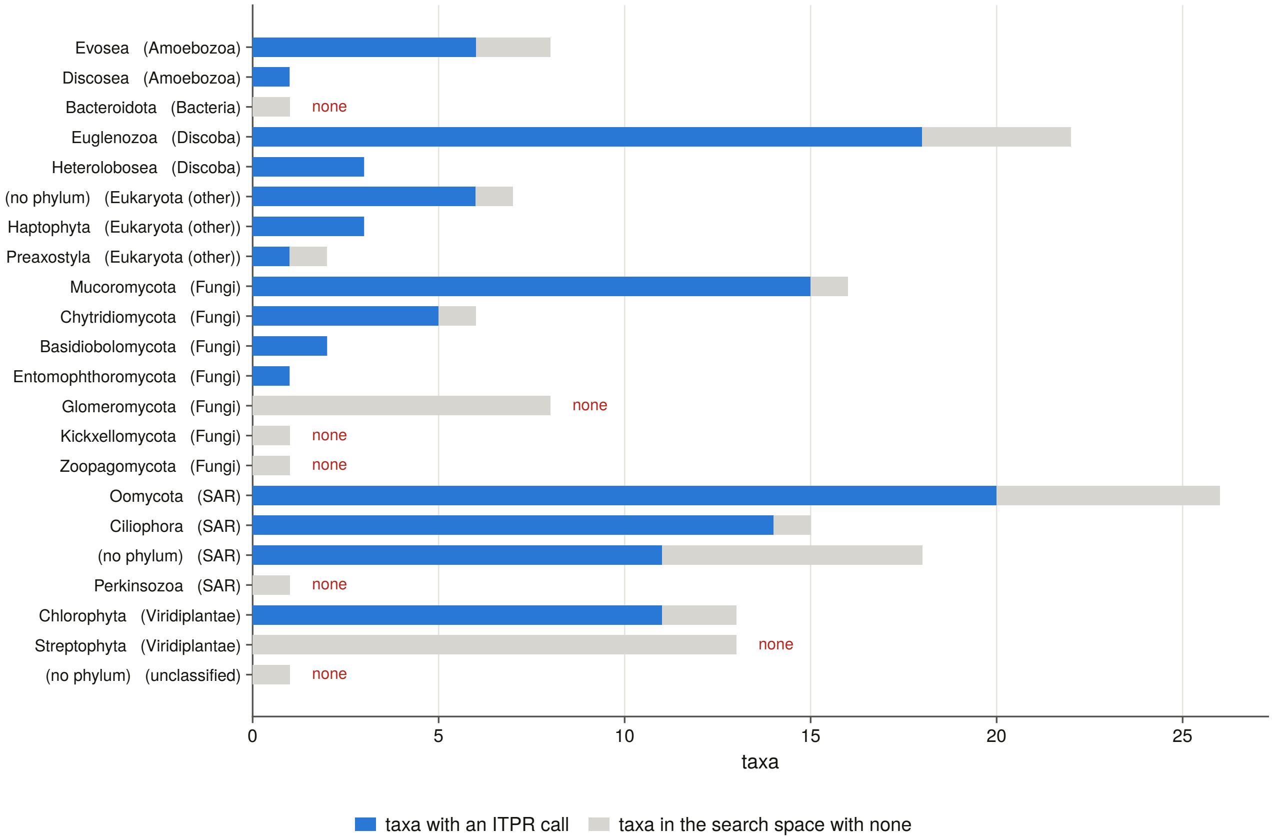


Figure 3.1. The enumerated search space by signature, with the overlaps drawn. The union is 15,421 proteins across 1,488 taxa and the intersection of all three seeds is 10,256. The difference between those two numbers quantifies what a conventional search would have missed: a census built on the signature that names the family, which is what a reader looking for IP_3 receptors would query, would have missed 2,914 proteins including 758 this census calls family across 385 taxa. Enumerating the union is the difference between a search that finds what is annotated and one that finds what is there.

3.4 A domain-architecture call is audited against gene symbols it never sees

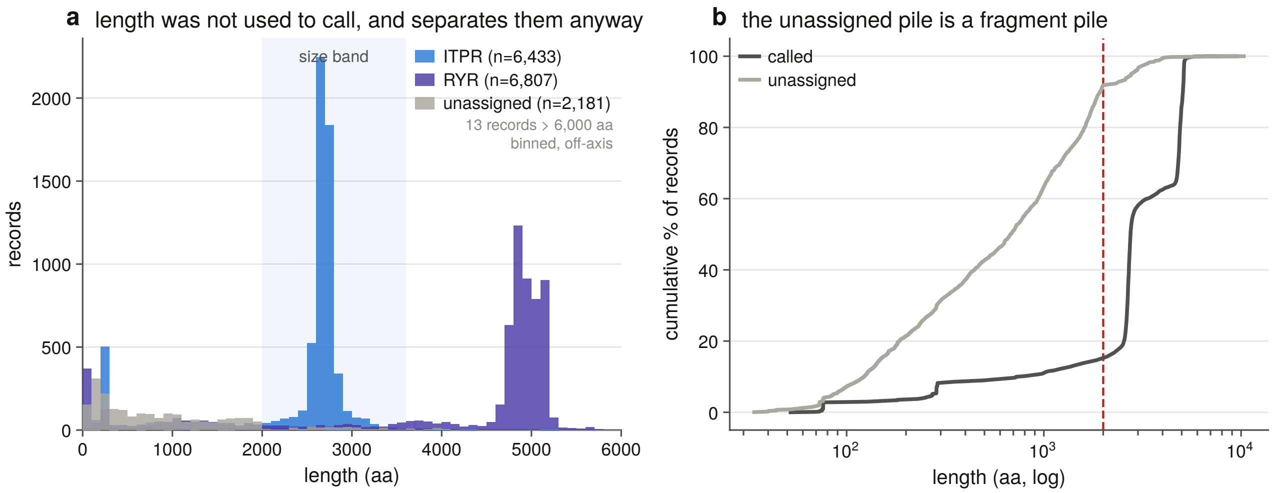
Every record is called by domain architecture. A record is a **ryanodine receptor** if it carries any of four RyR-specific signatures, namely PF02026, PF06459, PF21119 or PF00622. It is an **IP_3 receptor** if it carries the complete five-signature architecture of PF08709, PF01365, PF08454, PF02815 and PF00520 and none of those four. Length is recorded on every row and enters only as support for a medium-confidence call. The size band was chosen to exclude ryanodine receptors, so letting it decide would beg the question the call exists to answer.

The result is 6,433 IP_3 receptors, 6,807 ryanodine receptors and 2,181 records that satisfy neither positive test.



computed — lineage_calls.tsv

Figure 3.2. The census by lineage with both family calls, and with the sister family drawn beside the family everywhere. The figure is a warning about denominators. The vertebrate bar carries 10,936 of the 15,421 records, which is a fact about where sequencing effort has gone rather than about where the family lives, so any range claim read off record counts would report the history of genome sequencing. That is why Chapter 5 counts proteomes and taxa instead.



computed — census_v2.tsv

Figure 3.3. Length distributions of the two families as called. The medians are 2,671 and 4,856 residues and the distributions barely touch. The figure shows why a filter that works is still not evidence: length separates the two families almost perfectly here, and it does so on a population whose annotations are already good, which is precisely where no separation is needed. In the fragmentary and unannotated records where the

call actually matters, this separation is unavailable, and a project that had trusted it would have inherited annotation quality as a definition of the gene family.

The audit of that rule is possible because the rule never sees a gene symbol, which makes symbols an independent label to score it against. Every symbol-labelled record in the census is used rather than a sample. The rule agrees with the symbol on 2,479 of 2,479 decided IP_3 receptor cases and 2,492 of 2,492 decided ryanodine receptor cases, disagreeing on none. Four subsidiary claims about individual signatures are scored the same way and each is 100 % correct across two to three thousand records.

One of those subsidiary claims deserves suspicion and gets it. The rule leans on SPRY (PF00622) being diagnostic of ryanodine receptors, and SPRY sits in roughly 114,000 UniProt proteins and is in no sense specific to that family. The claim being made is conditional, in that SPRY is diagnostic *among proteins that already carry a family seed signature*, and the audit tests exactly that conditional claim on 2,335 records, where it holds. It is stated here as the row of the audit table a reader should check first, because it is the one that is a claim about this search space rather than about a domain.

A note on the 1,220 calls that rest on a gene symbol. Some records have an architecture too partial to decide and a name that is not. Those are called from the name at explicitly low confidence, with the reason written into every affected row and the pure architecture call kept in its own column, so the audit above can score the rule with no symbols anywhere in its input. Section 3.5 revisits all of them with an instrument that reads residues, and overturns one.

The unassigned pile is a pile of fragments. The 2,181 records that satisfy neither test have a median length of 678 residues against 2,671 for a called receptor, and only 143 of them fall inside the size band at all. A rule that reads the absence of a domain as evidence cannot call a partial annotation, and refusing to is the intended behaviour rather than a shortfall.

Because length never decides a call, the size band was free to catch something else. Seven records carry the complete five-signature architecture inside 2,000 residues, from 1,528 to 1,993, against 2,000 for the shortest real family member. An intact architecture in two-thirds of the length is a truncated gene model. None of the seven is flagged as a fragment by UniProt, because a truncated model submitted as a whole protein is not marked as one, and all seven are unnamed locus tags from three species. They are the first evidence in this project for the annotation failures Chapter 13 audits.

3.5 Two profile models are calibrated against the architecture rule before they are used

The architecture rule reads annotation and a profile reads residues. Building both and requiring them to agree is what makes the census defensible, because they can fail in different ways.

Two profile hidden Markov models [29] were built, one of 2,684 match states from 34 seeds and one of 4,930 from 22, each seed drawn from the census and carrying that census's call, so the profiles are labelled by a rule rather than by a gene name. Five selection rules are enforced in code rather than trusted: a seed whose census call, architecture count or length has drifted since the manifest was written aborts the build, and on the first run it did.

Five seeds carry an incomplete architecture and are in the set deliberately, each with its reason written into the manifest row. The most important is *Dictyostelium* iplA, at two signatures of five, which is in the IP_3 receptor seed set precisely so that the profile can find its relatives, since the architecture rule cannot.

Two build decisions are measurements rather than settings. MAFFT is run **single-threaded**, because at automatic thread count it is not reproducible: the same ryanodine seeds aligned twice gave 8,510 and 8,468 columns and profiles of 4,933 and 4,908 match states. MAFFT's return code is checked and a ragged alignment aborts the build, because the benchmark in §3.2 established that a silent MAFFT failure degrades to a star alignment with no other symptom at all. The SHA-256 of every seed set, alignment and profile is recorded, so a rebuild that drifts is visible in the data rather than only in a count.

The instrument is calibrated before it is used. Both profiles were run over the archived sequence set, which holds 15,381 of the census's 15,421 records, every one of them called by a rule that read annotation rather than residues, and the two instruments were scored against each other. With the seed sequences removed, because they are in that set by construction, the profile assignment agrees with the architecture call on 11,875 records and disagrees on 1.

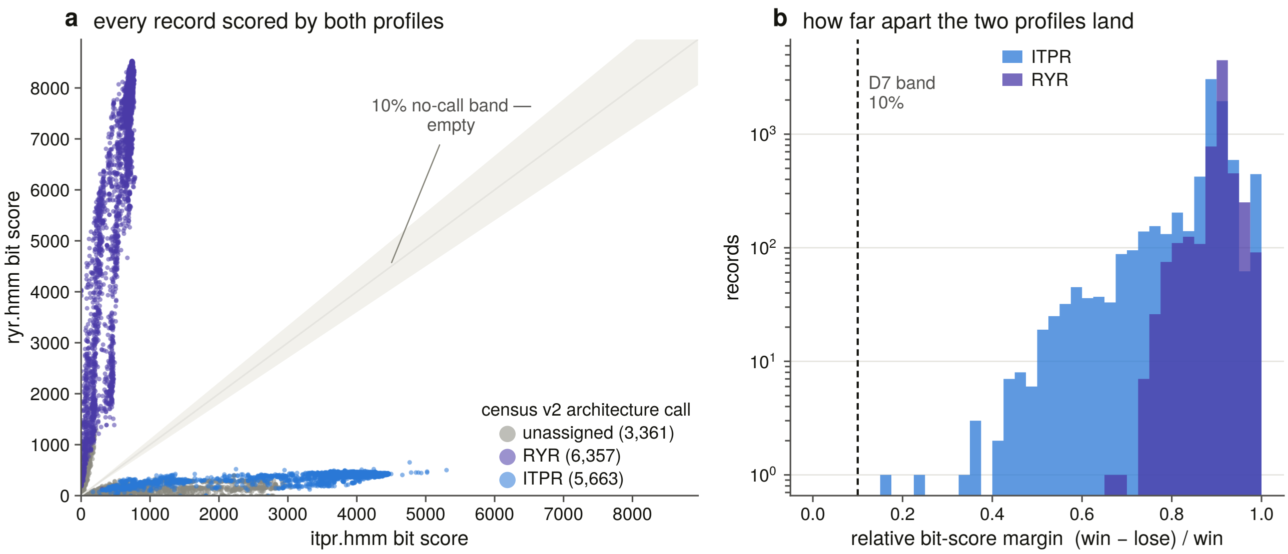


Figure 3.5. The two profiles' scores against each other, with the band inside which no call is made drawn rather than described. A call requires the winning profile to clear 30 bits, to span at least 200 match states, and to beat the loser by more than 10 % of its own score. That margin is relative rather than absolute because bit scores scale with alignable length, so a fixed gap would confidently call every full-length protein and refuse every fragment, restricting the census to the sequences that were never in doubt. The relative form is what allows 2,314 records the architecture rule could not decide to be called at all.

The span floor is measured rather than tuned. In this project's own structural domain coordinates the shortest observed PF08709, which is the IP_3-binding core that names the family, is 200 residues, and the longest observed SPRY is 137, so the floor sits in the gap between them. What it costs is on the record: 89 records that the architecture rule could call lose their profile verdict to it.

The second instrument earns its place exactly where the first fails. **2,314 of the 3,360 records the architecture rule leaves unassigned get a call from the profiles**, because a partial architecture defeats a rule that reads absence as evidence and does not defeat a sequence profile. Of the 1,220 calls that had rested on a gene symbol, the profiles scored 1,219 and overturned exactly one. The symbol fallback was sound, and it is now checked rather than assumed.

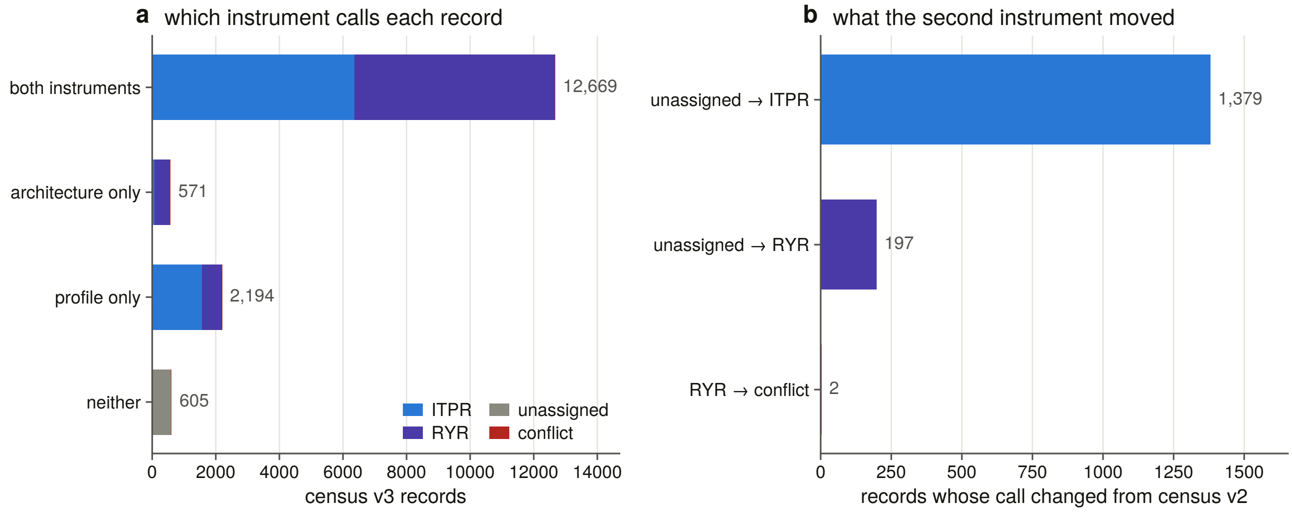


Figure 3.6. Which instrument calls each record. The census keeps the two verdicts side by side and merges them by a stated rule: both agreeing gives a high-confidence call, one speaking gives that one's, a disagreement is kept and reported as a conflict rather than resolved by preference, and neither speaking leaves the record unassigned. Two instruments reading different evidence disagree on two records in 16,039, which is the number that justifies treating the merged call as reliable, and keeping both verdicts is what makes a disagreement visible instead of assigning it to whichever instrument the author trusted more.

3.6 A sweep of 763 vertebrate proteomes required a span gate to be usable

The profiles were then run over 763 vertebrate reference proteomes [30], comprising 14,414,821 canonical proteins, one per gene, because isoform sets add isoforms of genes already counted and this census counts genes.

The union of the two searches is 18,501 targets. Before that number means anything, **12,609 hits are declined as module-only matches.** Both profiles are full-length channel models, so a protein sharing one small domain with either scores against it, and the ryanodine profile carries SPRY, already flagged as sitting in roughly 114,000 proteins. Without the span gate this sweep called 14,981 vertebrate proteins ryanodine receptors, headed by troponin, DEAD-box helicases and calcium-binding proteins. That is what a sensitive profile search returns if nothing stops it, and it is the single clearest demonstration in this project that a threshold on score alone is not an instrument.

Of the declined hits, 1,794 carry a gene symbol from this family or its sister. They are not domain-sharing proteins that happen to score. They are pieces of split or truncated gene models, sitting under the gate because there is not enough of the protein left to span a family domain. They are kept as an annotation lead rather than discarded, and Chapter 13 takes them up.

618 sweep hits are absent from the enumeration entirely, meaning proteins in a vertebrate reference proteome that the exhaustive InterPro walk never returned, 188 of them called IP_3 receptor. A signature-based enumeration and a profile sweep of the same organisms do not return the same set, and the difference is not small.

3.7 Three iterative searches forced the kill criterion to be rewritten

The completeness argument needs a third line of evidence: iterate a model from a single sequence until it stops finding new things, and see whether the family it converges on is the one the census holds.

Three iterative searches [31] were run to convergence, from a human IP_3R1, a *Drosophila* Itpr and an *Acanthamoeba* receptor, which is to say from a vertebrate, an insect and an amoebozoan. Only the fly run converged. The other two reached the ten-round ceiling and were killed.

Killing them is the point. The failure mode for this family is specific: an IP_3-seeded iterative search drifts across the shared architecture into the ryanodine receptors and reports a converged, confident, wrong answer. So the criterion is coded and evaluated per round on that round's own inclusion list, rather than judged by eye afterwards.

The first version of that criterion was wrong, and how it was wrong is a result. It was coded as a flat ceiling of 5 % sister-family content, and it fired on every run at round one, because a single IP_3 receptor sequence searched at a sensitive threshold already returns 32 % to 37 % ryanodine receptors before any iteration has happened. That is not contamination. The two families are genuine homologues sharing the entire pore, and a search sensitive enough to reach *Acanthamoeba* is necessarily sensitive enough to reach RYR1. The level is a fact about shared ancestry, and only the change in it can be attributed to iterating the model. The rule measures the rise.

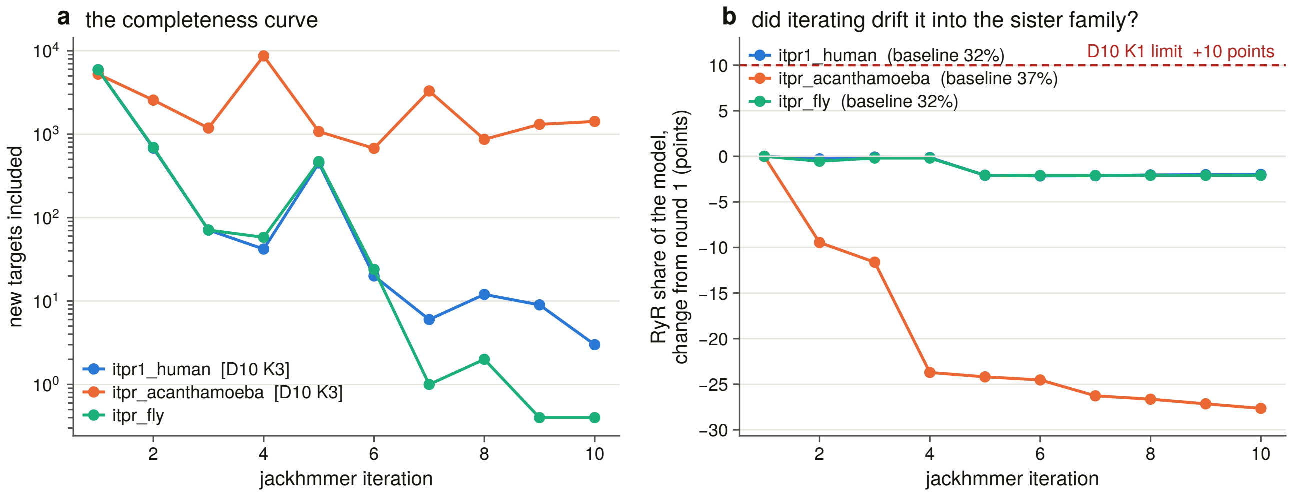


Figure 3.7. Convergence for each seed, with the sister-family trace beside it. Two runs never converge and are excluded from any completeness claim, and the third does. The curves decay to an asymptote of a few new targets a round rather than to zero, and the tail is proteins sharing one small domain rather than family members. That is why convergence alone cannot be a completeness criterion here, and why the argument in §3.8 rests on three distant seeds recovering the same family core rather than on any single run converging.

3.8 The family core is the same whichever seed finds it

A convergence curve says a search has stopped finding things. It does not say what it found. Every target supporting each run's final model was therefore classified by the sweep's own two-profile verdict.

The result is the strongest completeness statement in this chapter, and it does not depend on convergence at all. **The family core is the same whichever seed finds it.** Three starting points as far apart as a human, a fly and an amoeba produce final models carrying identical family-called content: 1,656 architecture-complete IP_3

receptors in all three, 1,565 to 1,569 architecture-complete ryanodine receptors, and 952 partial IP_3 receptors in all three.

What differs between them is everything that is *not* the family. The *Acanthamoeba* run's final model rests on 26,266 targets against the human run's 7,222, and 18,670 of those are proteins that neither profile scores at all. The rise-based kill criterion **cannot see that drift**, because off-family accretion dilutes the sister-family share rather than raising it. Across that run the share falls from 36.6 % to 8.9 % while the model grows five-fold, so the rule that catches the failure it was designed for reads this one backwards. The ten-round ceiling is what caught it, and the composition table is why the run is reported rather than quietly dropped. Chapter 4 returns to this and scores all three rules as classifiers of an outcome measured on the finished model.

3.9 Completeness checked in the expensive direction: which known records the profiles missed

A sweep is a completeness argument only if it is also checked the other way round. How many records that the enumeration called IP_3 receptor, in a proteome that was swept, did the profiles fail to find?

The answer is 2,787. Each was then looked up in the 8.8-gigabyte search database itself, in one streaming pass, rather than explained away by its gene name. Of those, 1,987 were absent from the database though another entry of the same gene was hit, and 800 were absent and never searched.

Not one was in the database and missed. Every apparent miss is a UniProtKB entry that its taxon's reference proteome does not contain, because a reference proteome holds one canonical protein per gene and the enumeration walked all of UniProtKB, so it was never searched at all. The profiles' sensitivity failures over 763 vertebrate reference proteomes number zero.

Fifteen of those 763 proteomes carry no IP_3 receptor record. That is a list of leads for the genome sweep rather than a list of losses, because a reference proteome is an annotation, and Chapter 4 tests all fifteen against the genomes themselves.

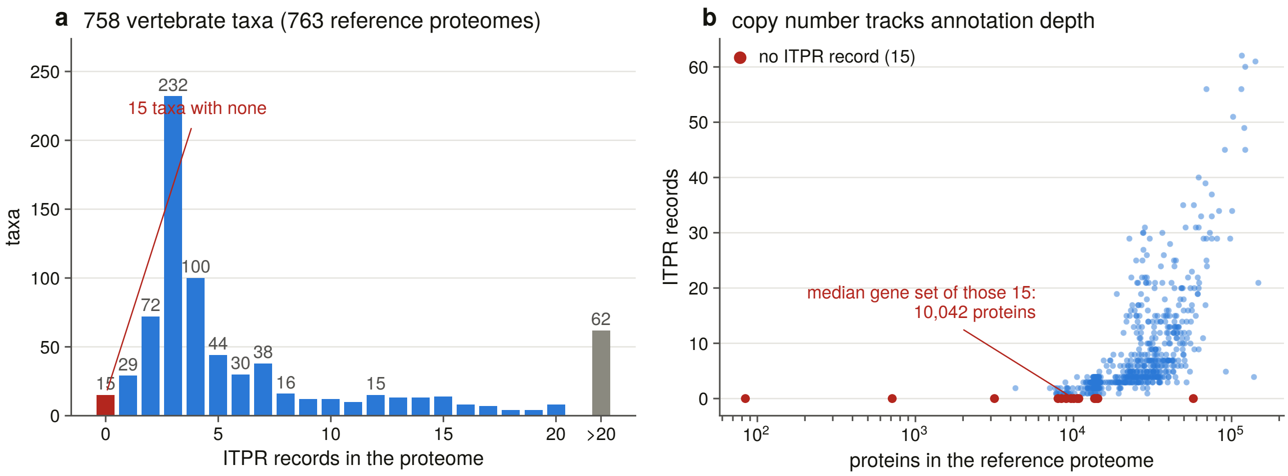


Figure 3.8. Copy number per proteome against annotation depth. This relationship is the single strongest argument in the thesis for searching genomes rather than gene sets: a census built on reference proteomes measures how thoroughly each organism has been annotated at least as much as it measures how many receptors that organism has, and the correlation drawn here is that confound made visible. Every absence claim in this thesis is taken to an assembly for this reason, which is what Chapter 4 begins.

3.10 What census v3 holds, and the three limits it carries forward

The merged census is 16,039 records: 8,000 IP_3 receptors, 7,432 ryanodine receptors, 605 unassigned and 2 conflicts. Both instruments speak for 79.0 % of records, the profiles alone for 13.7 %, and the architecture rule alone for 3.6 %.

Three limits carry forward, and each becomes a later chapter.

The sweep is vertebrates only. Every count here is a statement about 763 vertebrate reference proteomes, and nothing in it bears on the plant, fungal or protist questions the enumeration opened. That is Chapter 5.

A reference proteome is a gene set, not a genome. A gene missing from one may be missing from the annotation rather than from the DNA. That is Chapter 4.

Profile assignment inherits its seeds' breadth. The deep branches are represented by seven non-metazoan seeds, and a lineage more diverged than any of them is outside what this instrument has been shown to call. Chapter 5 builds a second panel by different rules for exactly that reason.

4. Searching 309 vertebrate genomes, and measuring what that search was worth

4.1 Why the reference proteomes were not enough

Chapter 3 ended with two limits. A reference proteome is a gene set rather than a genome, so a gene missing from one may be missing from an annotation rather than from the DNA, and fifteen of 763 vertebrate reference proteomes carried no IP_3 receptor record at all. Neither of those is a biological statement. Turning them into one requires searching the assemblies.

This chapter does that over a declared scope of 309 vertebrate genomes, and then does something less usual: it measures what the search was worth. The second half is not a methods appendix. The numbers it produces are the error bars on Chapters 5, 9 and 13, and a census that reports zero losses, as Chapter 9 does, is worth nothing unless somebody has measured how often the same search fails to find a gene that is demonstrably there. That measurement belongs with the instrument rather than with the result it qualifies.

4.2 The denominator was declared before the search ran

The scope is 309 assemblies, and it is the union of two rules.

The first rule is taxonomic. It takes one best assembly per vertebrate order [32], 161 of them, ranked by a stated function that prefers annotated over unannotated, RefSeq over GenBank, then assembly level, then scaffold N50, which is the length such that half the assembly sits in pieces at least that long and is the standard summary of how contiguous an assembly is.

The second rule is a margin rule, and it is computed from the census rather than hand-listed. Four positive tests put a species in scope: its reference proteome returned no family record at all, or it carries fewer than three paralogues, or every record it has is below the family's minimum length, or it is an anchor species the project needs for another reason. That produces 169 margin species, and the union with the order representatives is 309 genomes and 552.5 Gbp.

The point of computing the margin set from the census is that the denominator rebuilds itself. If the census changes, the scope changes with it, and the scope document is rendered from the manifest rather than written.

That design has a consequence discovered later and worth stating here, because it shapes every absence claim in the thesis. The margin species were selected for what their *proteomes* lack. They turn out to be very largely the same genomes whose *assemblies* cannot hold the gene: 68 % of margin species fall below the contiguity bar against 12 % of order representatives. The two signals are confounded by construction, and separating them is what §4.7 and Chapter 9 exist to do.

4.3 Seven rules chose the bait panel, and six slots stayed empty

The sweep aligns proteins to genomes, so it needs baits. There are 38 of them, comprising 30 IP_3 receptors and 8 ryanodine receptors as an internal positive control, 121,294 residues in total, derived from the census by seven stated rules rather than assembled by hand.

The rules are worth listing because each is a positive test. A bait's paralogue label must come from the census rather than from its own gene symbol. It must be full length and carry the complete architecture. There is one bait per paralogue per clade band, and two in the bands that dominate the scope. Every candidate passes a chimera screen. The ryanodine baits are a presence control. The existing profile seeds are reused rather than re-derived. And the panel is scoped to the vertebrate genomes of this sweep and no other.

Six slots stay empty, and that is a finding rather than an oversight. They are ITPR1 and ITPR2 in cartilaginous fish, ITPR2 in the coelacanth grade, and all three in cyclostomes. The census holds no labelled, full-length, complete-architecture record for those combinations at all, because the paralogue labels simply run out below the well-annotated clades. Rather than promote an unlabelled record into a labelled slot, three unlabelled baits cover those genomes without making a paralogue claim, and Chapter 6 settles the assignment.

The chimera screen has its own negative control, run on every build. The screen rejected none of the 57 candidates, and a screen that cannot fire looks identical from outside to a shortlist that is clean. So every build constructs three synthetic failures from real panel baits and requires each to be rejected by the rule responsible: an IP_3 receptor N-terminus joined to a ryanodine receptor C-terminus, caught by the family assignment; a mis-joined gene model with 435 unaligned residues inside its envelope, caught by the shape rule; and a truncated model, caught by the length band.

That test earned its place on its first run by failing. It originally asked the envelope rule to reject the truncation, which the envelope cannot and should not do, because a protein truncated to 40 % of its length aligns 100 % of itself to the profile in one clean segment. Truncation is the length band's job and the envelope's job is fusion.

4.4 The aligner's intron parameter is a threshold on the call, not a performance knob

The protein-to-genome aligner [33] takes a maximum intron length. Its default is 200 kb, and it is not a performance knob: a gene whose largest intron exceeds it is split into pieces, and a split IP_3 receptor reads out of this ledger as a fragment. Setting it wrongly manufactures exactly the observation the thesis is trying to measure.

Chapter 2 supplied the wrong number for this. It measured genomic *spans*, with human ITPR2 at 498 kb, and a span over 57 exons is not an intron. So the largest single intron of every IP_3 and ryanodine receptor gene was measured directly across an eleven-species panel.

No IP_3 receptor gene in the panel has an intron over the default. The widest is 152,216 bp, in human ITPR1. The ryanodine control exceeds 200 kb twice, the widest being human RYR2 at 227,927 bp. So the default is adequate for the deliverable and marginal for the control, which is the opposite of what the span figures suggested.

The rule the sweep uses takes the widest measured intron, doubles it, scales it by genome size, floors it at the aligner's own default and caps it for tractability. It is deliberately not the intron-per-gigabase ratio, because that statistic is largest in the smallest genomes measured, and extrapolating it linearly asks for a 7.4 Mbp setting on the lungfish, which is an artefact of a small denominator rather than a measurement. The cap binds on two genomes and the ledger flags both, because a fragment there could be a real intron the sweep declined to span.

4.5 What the sweep found across 309 genomes

The sweep covered 309 genomes of the declared 309, recorded 2,144 loci, and carried a ryanodine receptor bait with every one of them.

The control fired in 309 of 309 genomes. Ryanodine receptors are present in three copies in every vertebrate, so a genome where the control finds nothing has an assembly or a pipeline problem rather than a biological result, and until it fires that genome's IP_3 receptor cells say nothing. None failed, so no result below is excluded on control grounds.

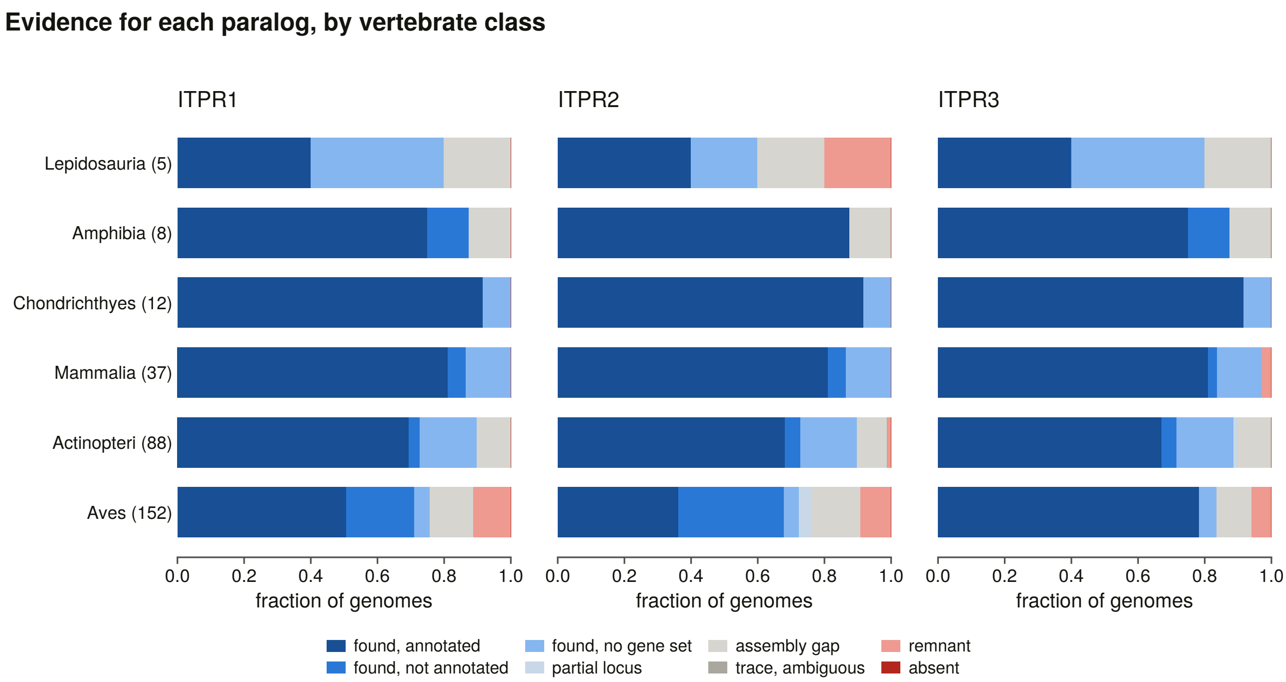


Figure 4.1. The three-paralogue ledger across 309 vertebrate genomes, by class. This is the raw ledger before any of Chapter 9's restatement, so it shows the problem that chapter had to solve. The family is found nearly everywhere it is looked for, and what red there is falls in particular vertebrate classes rather than scattered across them. Those classes turn out to be the ones whose assemblies are worst rather than the ones whose biology is different, so reading this figure directly as a map of gene loss is the mistake the rest of the thesis is built to avoid.

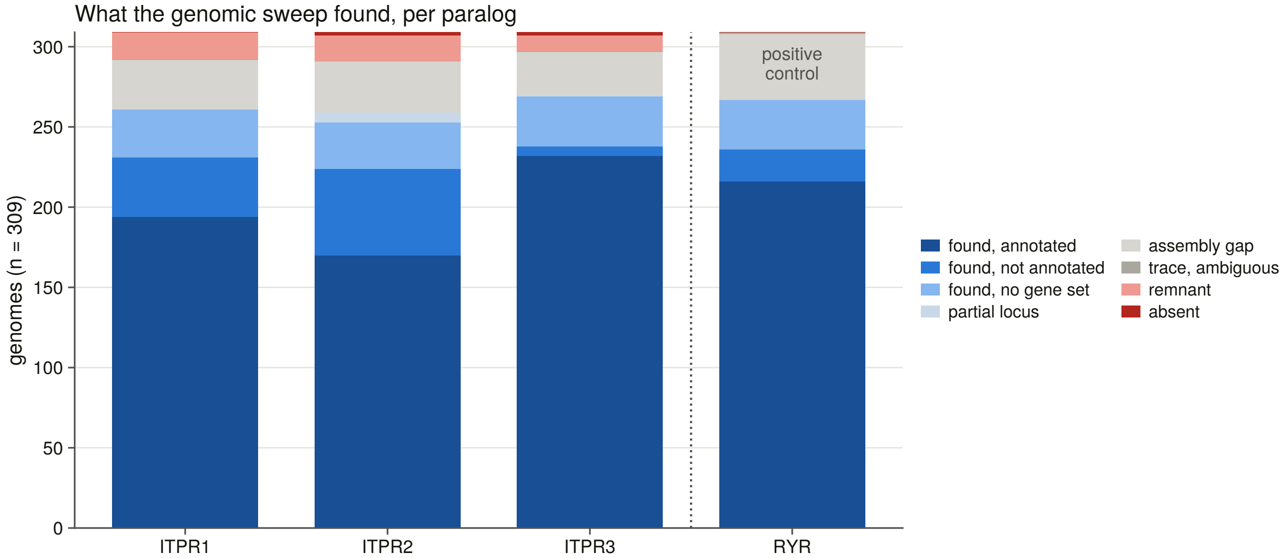


Figure 4.2. Cell status across the sweep, by paralogue. Four cells of 927 are called absent, meaning no spliced-alignment locus and no remnant from the rescue search. Those four are the strongest negative the search itself can produce, and they are still not a loss claim: Chapter 9 shows that every one sits in an assembly that could not hold the gene or in a genome whose paralogue labels the bait panel cannot resolve. The distance between this figure and that conclusion is the distance between a search result and a biological result.

The two families never contested a locus. At all 2,144 loci, only one family's baits aligned at all. That is a far sharper separation than the protein level afforded, where, before the sister test existed, all six ryanodine decoys were promoted as candidate family members, and it holds from the six-genome pilot through to full scale. Spliced alignment against a labelled panel settles the family before any margin has to be applied.

4.6 A threshold inherited from another project was measured and overturned

The rescue step [34] attributes a fragment to a paralogue when its best bait beats the runner-up by a stated relative margin. That margin arrived in this project as an inheritance: 0.333, a 1.5-fold bit-score ratio, from a sister project on a gene family whose paralogues are 40 to 50 % identical. These are 61 to 68 % identical.

The question is whether the threshold transfers, and it can be answered without any new search. Loci whose paralogue identity the assembly's own annotation establishes carry a margin measured on evidence the bait scores did not produce. Across 542 such loci the margin runs from 0.136 to 0.411 with a median of 0.314, and **368 of the 542 fall below the inherited threshold.**

A complete locus bounds a fragment's separation from above, so a threshold that complete evidence fails cannot be met by a fragment. Under the inherited value every rescue trace in this project would have been reported ambiguous and no absence claim could ever have been attributed to a paralogue at all. The threshold is now 0.22, which is the highest value that rejects none of those correct calls, and its limitation is stated rather than hidden: derived from complete loci and applied to fragments, it is a ceiling on the right answer rather than the right answer.

The full sweep then produced the fragments themselves. Fifty-three rescue regions overlap a gene the assembly names for a paralogue, so their identity is established independently of the bait scores. The attribution agrees with the annotation on 53 of 53, and none falls below the threshold in force. That bounds the threshold from below, meaning it establishes how low the threshold must be to keep correct calls, and not from above. That limitation is said plainly because no region was attributed against its annotation, so nothing measures where wrong calls would start.

A second inherited constant failed the same way. Rescue attribution originally ranked every bait clade against every other, including the unlabelled baits. Those are the gar, chimaera and lamprey seeds, which are evidence that a region is an IP_3 receptor rather than a rival hypothesis about which one. Ranking them as competitors collapsed real margins. In one bird genome an ITPR1 trace scoring 916.1 against ITPR2's 700.6, a margin of 0.235, was reported at 0.027 because an unlabelled bait sat between them at 891.6. Every ITPR1 and ITPR2 trace in that genome was deflated the same way.

4.7 Assembly contiguity is why four absences are not four losses

An assembly whose contigs are shorter than the gene cannot carry the gene on one contig, so an empty cell in such an assembly is evidence about the assembly.

The bar this project uses is the median measured IP_3 receptor genomic span, 142,212 bp of contig N50, taken from gene geometry with no error rate in its derivation. **120 of 309 genomes fall below it, which is 39 %, and they are not a random 39 %.** Assembly quality across the vertebrates is uneven by clade and by sequencing era, which is the problem the reference genome consortia were set up to address [35], and this family's genes are long enough to feel it. Two thirds of birds fail it against 11 % of ray-finned fish, and 68 % of margin species against 12 % of order representatives.

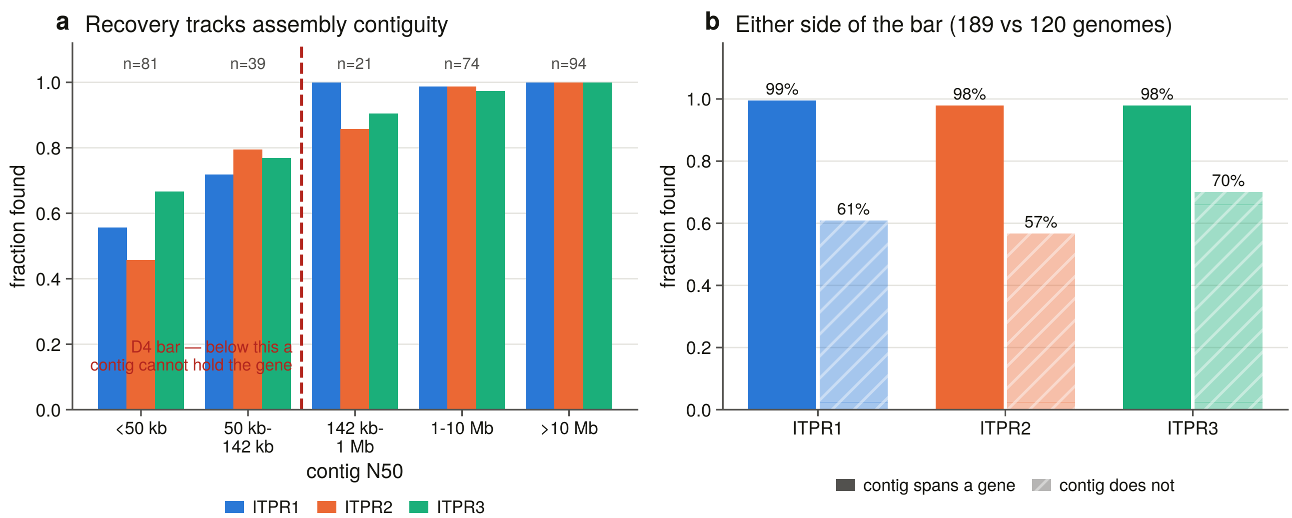


Figure 4.3. Recovery against assembly contiguity, binned on the bar rather than across it. A sliding window straddling the threshold would report a recovery rate that no genome in the window has, smoothing away the very discontinuity the panel exists to show. The relationship drawn here is the confounder every absence claim in this thesis has to survive, and it is why the contiguity floor is applied before any cell is read as evidence about biology.

The pilot had predicted, from six genomes, that a fragmented assembly loses the long paralogues first, because a 231 kb ITPR2 needs a contig that an 82 kb ITPR3 does not. At 309 genomes that is confirmed: above the bar all three paralogues are found in 98 to 99 % of genomes, while below it ITPR1 is recovered in 61 %, ITPR2 in 57 % and ITPR3 in 70 %.

This matters more than it looks. The detection bias runs in the same direction as the loss signal the margin species were selected for. A per-paralogue absence in a fragmented assembly cannot be read as a loss, and the confound is structural rather than incidental.

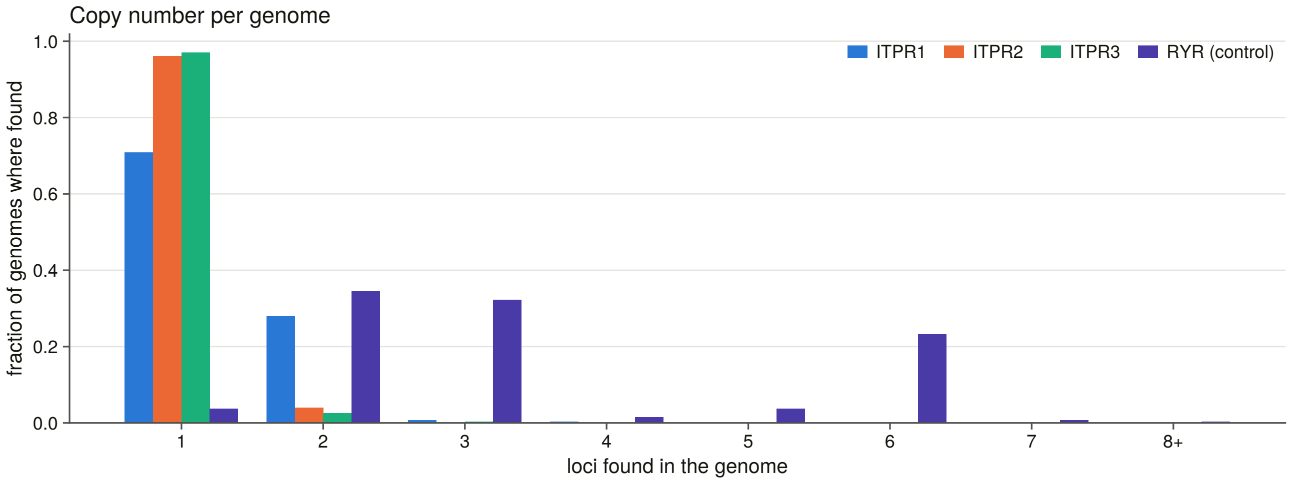


Figure 4.4. Copy number per genome across the sweep. The excess above three copies is not spread across the vertebrates but sits almost entirely in the ray-finned fish, which is what a whole-genome duplication confined to one lineage looks like from a copy count. Chapter 7 turns that observation into a tested claim about which paralogue was doubled and retained.

4.8 The three paralogues are not annotated equally well

The sweep also produces something the protein databases cannot: for every gene it finds, whether an annotation is there and whether that annotation names the right paralogue. The measurement is restricted to loci the sweep found in an assembly that carries a gene set, so the denominator is genes that exist and are annotatable, and it then additionally holds contiguity constant, because otherwise it measures assembly quality and calls it annotation quality.

Above the bar, a gene model is present at 96 % of found loci for all three paralogues. It names the paralogue correctly at 65 % for ITPR1, 87 % for ITPR2 and 88 % for ITPR3.

A 23-point gap between genes of near-identical protein length in the same genomes is not a biological difference. A census built on gene symbols inherits that gap as an apparent difference in copy number, which is why Chapter 13 exists.

The uncontrolled version of that table is instructive about the control. Without holding contiguity constant, ITPR3 appears to be 21 points better annotated than ITPR2. Above the bar the two are within one point, so the apparent annotation gap between those two paralogues was assembly quality.

4.9 The sweep found 318 genes that exist only as DNA

The sweep contributes 1,058 gene models from 224 genomes, each scored against both profiles so that a new census row rests on the instrument that called the old ones. The census becomes 17,097 records.

The interesting column is not how many but **how a database holds each locus**. 318 IP_3 receptor gene models exist only as DNA, meaning there is no gene model at the locus, or the assembly has no gene set at all. A further 167 sit inside an annotated gene that carries no family name, so no search by name can reach them however complete the protein databases are.

Three loci are claimed by a paralogue cell other than the one the assembly's annotation names, and one of those sits in a genome that also carries a separate locus for the paralogue the annotation names, so the two cannot both be that paralogue and the disagreement is internally coherent. They are handed to Chapter 13 rather than adjudicated here, because one instrument does not overturn a public annotation and this family's paralogue margins are narrow.

4.10 How the search can be used as its own control series

Everything above is a search. What follows is a measurement of that search, and it is placed here rather than in a methods appendix because the numbers it produces are the error bars on Chapters 5, 9 and 13.

The design is what makes it possible, and it depends on a result that has not been reported yet. Chapter 9 reconstructs no losses anywhere in this scope. If no gene in the scope is absent, then every cell whose gene is independently known to be present and which the ledger did not find is a **false negative of the method**. It is not a biological absence and not an ambiguous case, but a miss with a known right answer. That turns the whole sweep into its own control series.

Two independent series are carried, because one of them would be circular on its own. The first is every cell that Chapter 9's state rules call present. The second is the ryanodine receptor sister cell, which uses none of those rules at all: every vertebrate has three ryanodine receptors, so every genome's control cell has a known answer that no part of this project's state logic produced.

Before any of it is written, 32 constructed negative controls run and refuse the build if they fail. Their character is worth noting, because it is the character of every test suite in this project: they are checks on refusal and on reachability. One requires that a false negative be **constructible** at all, because a control that can only ever return zero misses is not a measurement. Another requires the full-panel simulation to reproduce the committed ledger cell for cell. A third requires the suite itself to alter no committed table, and it exists because an earlier build's test called the real routine and overwrote a committed table with its two constructed rows, after which the report read two runs where there are seven.

4.11 The search misses about one cell in seven, and every miss is an assembly

Over the whole 309-genome scope the search misses **140 of 923 control cells, 15.2 %**.

The misses are not distributed evenly. A missed cell's assembly has a median contig N50 of 23,460 bp against 3,396,515 bp for a found one, and the odds of finding the gene rise 8.10-fold per tenfold of contig N50 in the IP_3 receptor series and 19.99-fold in the ryanodine series. On chromosome-level assemblies the IP_3 series misses 3 of 512 cells and the ryanodine series misses none of 172.

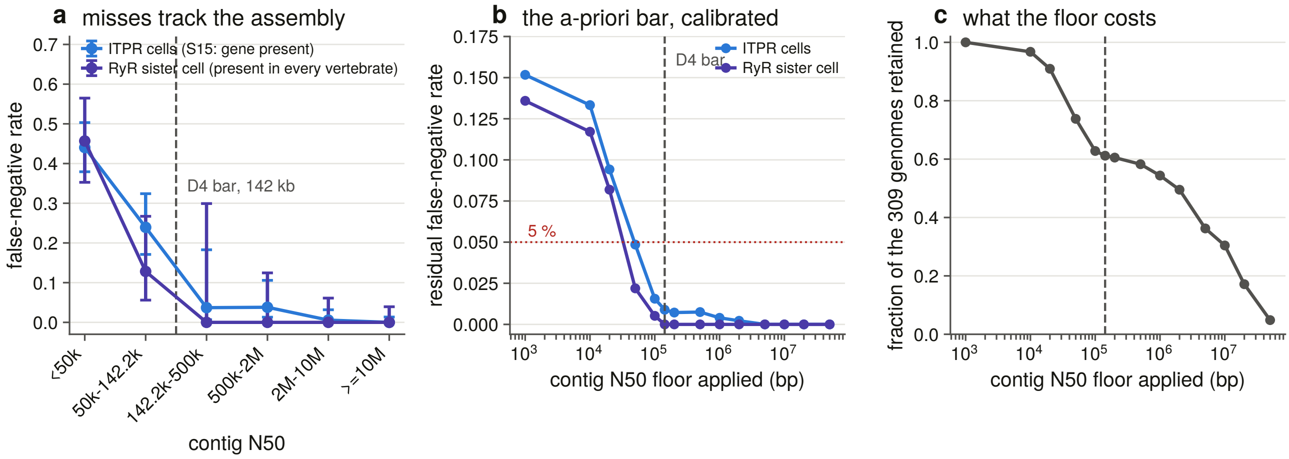


Figure 4.5. The measured false-negative rate against assembly contiguity, with the bar drawn. This is the figure that licenses the retention result, because a survey reporting zero losses is worth nothing unless somebody has measured how often the same search fails to find a gene that is demonstrably present. The two series are independent, since the ryanodine sister series uses none of the state rules the family series depends on, and they agree, which is what stops the number being read as the search agreeing with itself.

The contiguity bar was chosen a priori and survives calibration. The 142,212 bp floor came from gene geometry, meaning the median measured genomic span of the gene, with no error rate anywhere in its derivation, and it had never been scored against one. Scored now, the IP_3 series has a residual miss rate of 0.9 % over 563 cells above the bar and the ryanodine series 0.0 % over 189. That is conservative against a conventional five-per-cent target, which the scan reaches at a floor of 100,000 bp, and about right against a one-per-cent target.

The bar is not free. It removes 38.8 % of the genomes, and it removes them disproportionately from exactly the clades a loss survey is most interested in. That cost is quantified rather than absorbed: the clade composition of what each candidate floor retains is committed, so a reader can see which question each floor makes unanswerable.

The residual is also diagnosed locally rather than dismissed. Contig N50 is a genome-wide statistic and a regional assembly defect is invisible to it, so each remaining miss is checked against the flanking-gene consensus of its own paralogue to ask whether the region survived at all.

4.12 The bait panel was ablated exactly rather than modelled

How much of the panel was doing work? The question is normally answered by argument. Here it is answered exactly, because the aligner aligns each bait independently: dropping baits from a retained alignment and re-clustering reproduces exactly what the sweep would have reported with a smaller panel.

The full chain is reproduced rather than approximated. It runs cluster, then the identity floor, then the cell assignment where the family call lives, because scoring the best coverage over every alignment instead would let an ITPR3 bait's hit at the ITPR1 gene stand in as ITPR3 evidence and make every panel look alike. The simulation reproduces the committed ledger 1,236 cells out of 1,236, so every delta below is measured against the sweep itself rather than against a model of it.

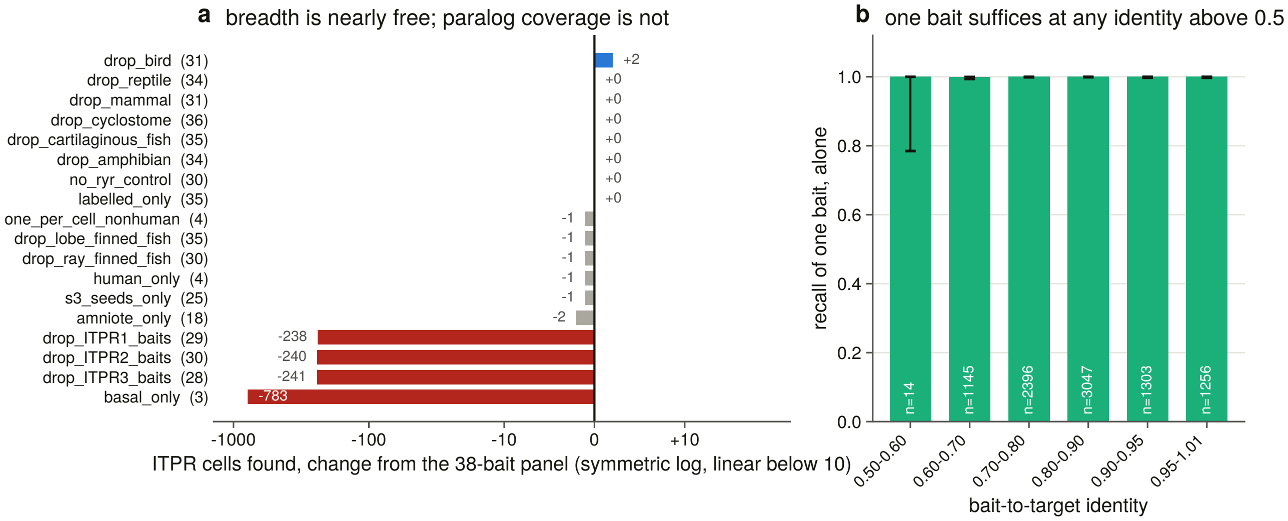


Figure 4.6. Nineteen panels, drawn as change from the full panel on a symmetric-log axis. Every panel scores between 0.58 and 0.85 of the cells, so on an absolute linear axis the entire breadth result would be one pixel, and the one ablation that removes every labelled bait sits 783 cells away from the rest. The importance of this figure is that it overturns the usual intuition about how to build a bait panel. Phylogenetic breadth, which is what a panel is normally padded with, buys almost nothing inside the vertebrates, while paralogue coverage buys everything. Anyone designing a comparable survey should spend their panel budget on paralogues and on clades where no labelled record exists, and this figure is the measurement that says so.

Two readings fall out, and they point in opposite directions.

Phylogenetic breadth buys almost nothing. Four human baits, one per cell, recover 782 of the 783 cells that the 38-bait panel recovers. Dropping any single clade band costs at most two cells. The three unlabelled baits cost nothing when removed and recover zero cells on their own, because the sweep offers an unresolved-clade locus to whichever labelled paralogue scores highest there, and with no labelled bait present there is nobody to offer it to. Their function is to keep a real shark or lamprey gene inside the family when it outranks every labelled bait, not to place it in a cell.

Paralogue coverage buys everything. Removing one paralogue's own baits costs 238, 240 and 241 cells, which is about a quarter of the recovery each, and no amount of breadth in the other two replaces it.

Removing the sister-family control changes no call. The positive family test at genome scale costs nothing in recall, which is the cheapest possible price for it.

An ablation can gain a cell, and that is a property of the rule. Sixty-eight of the 2,179 call changes across all panels are gains, by the same mechanism every time. A cell's coverage is read off its top-scoring bait rather than its best-covering one, so removing a competitor can promote a bait whose coverage is fractionally higher and push a marginal cell across the coverage bar. The two gains under one ablation are a chicken bait at 0.6956 coverage giving way to a lizard bait at 0.7022, which is seven thousandths either side of the threshold. They are reported rather than folded away, because an ablation table with unexplained positive deltas invites the reading that a smaller panel searches better.

4.13 What each search channel actually contributed, on three separate denominators

Three denominators are kept apart, because conflating them is the standard way this question gets answered wrongly.

How the census accumulated. The targeted database search that started the project returned 1,571 records. Exhaustive signature enumeration took it to 15,421. The vertebrate profile sweep added 618 and the genome sweep added 1,058. The non-vertebrate proteome sweep added 785 and the non-vertebrate genome sweep 183, giving 18,065.

Profile models against domain annotation, inside one database. Both channels are counted by what they call family rather than by raw hits, and both are restricted to accessions the swept files actually hold. Comparing a curated set against a raw hit list would credit the vertebrate sweep with the 13,371 targets its own gate declined.

At gene scale the two return nearly the same set. In the vertebrates the enumeration returns 3,135 gene-scale records and the profile sweep 3,136, of which 3,135 are shared, so the profile adds one. In the non-vertebrate metazoa it adds two to 1,021. Its entire gain is in the fragment tail. The one exception is worth naming: in the protists it adds 89 gene-scale records the enumeration never returned, which is where the family's architecture is least well annotated.

The other side of that statement is the same measurement seen from the other direction: 766 vertebrate records carry a family signature and are declined by the profile pair, all of them under 1,000 residues, which is the span gate doing exactly what it was added for.

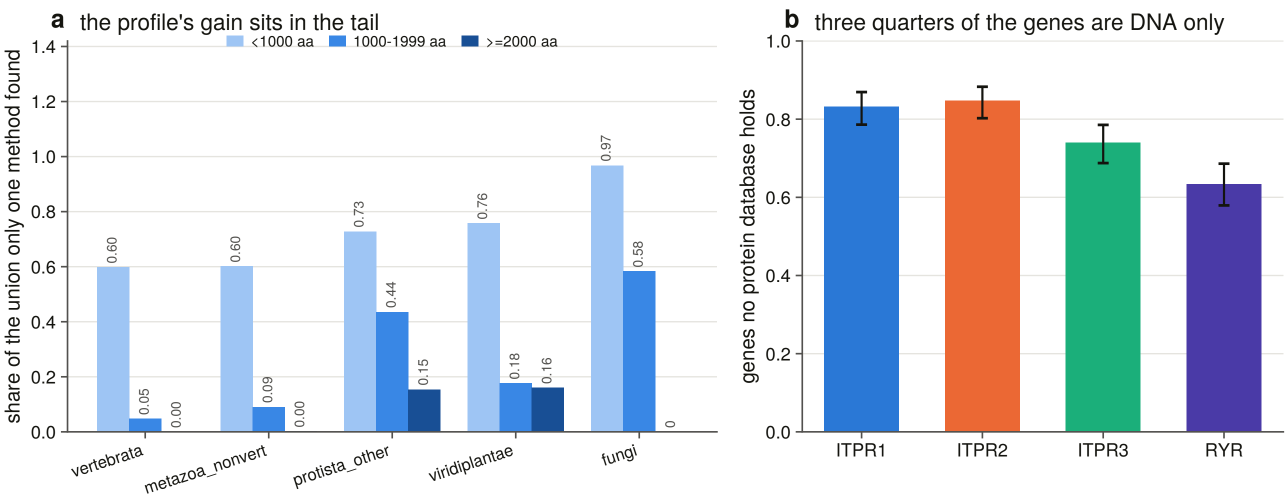


Figure 4.7. What each channel contributed, drawn as disagreement rates rather than as stacked counts. The claim is a proportion, and stacking proportions on a logarithmic axis misreads them by construction. The figure matters because it prices the standard approach. A family profile searched over reference proteomes, which is how most gene-family surveys are done, returns at gene scale almost exactly what domain annotation already returns. Its entire gain is fragments, and the only place it finds whole genes that annotation missed is the clades where annotation is worst.

Per gene, what a protein-database search would have missed. Asked per genome by cell rather than per record, with the genome sweep as ground truth for where the genes are, **940 of 1,232 demonstrated genes, or 76.3 %, are not reachable by any protein-database search.** They are not hard to find. No protein record of them exists.

That is not an artefact of the margin species. Split by why each genome is in scope, the rate is 74.3 % for order representatives against 78.5 % for margin species. Chapter 13 is what that number turns into.

4.14 The rule written to catch iterative-search drift never fires

Chapter 3 left a question open: two of three iterative searches were killed by a round ceiling rather than by the rule written for the hazard. With seven such runs now available across the whole project, the rules can be scored as classifiers.

The outcome is measured on the finished model rather than taken from any session's prose, using the share of a run's final included set that neither profile scores at all. The seven runs separate cleanly, with three at 0.81, 0.97 and 0.99 against four at 0.23 to 0.34 and nothing between 0.34 and 0.81, so the labels are not a judgement call.

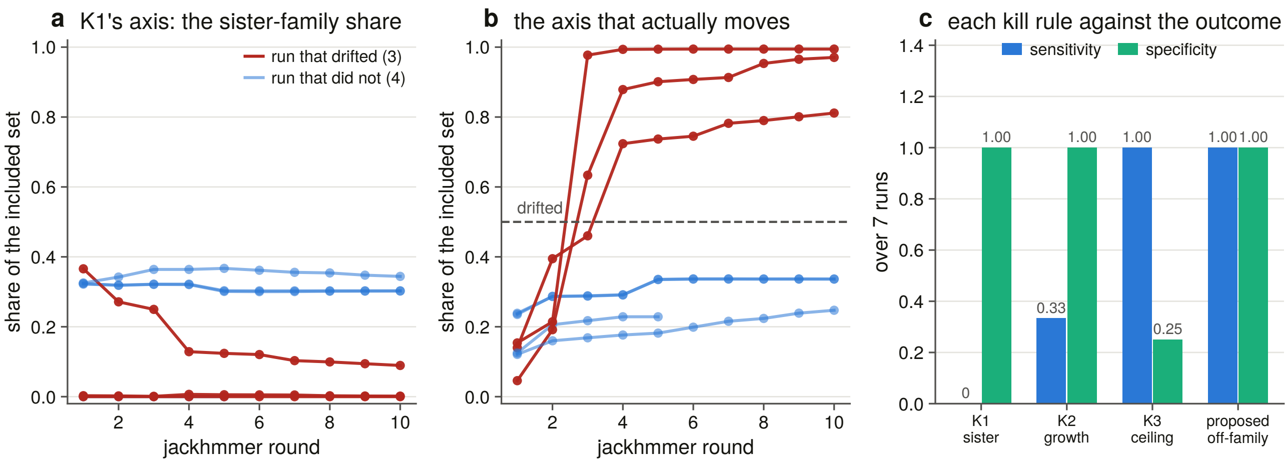


Figure 4.8. Each kill rule scored as a classifier of a drift outcome measured on the finished model. The figure records an instrument failing at the one job it was built for: the rule written to catch sister-family drift has a sensitivity of zero across seven runs, including all three that drifted, because the drift dilutes the very quantity the rule measures. The correction is a one-line change to a different axis, and it is reported rather than applied, because it was validated after the fact on the runs it would reclassify.

The sister-family rule fires on none of the seven runs, including all three that drifted. The reason is now measured rather than described: off-family accretion dilutes the sister-family share, so the rule's own statistic moves the wrong way while a run drifts. The sister share fell over the run in four of the seven.

The round ceiling catches all three drifted runs and also flags three that did not. It counts rounds rather than content, which is right for a budget and useless as a diagnosis.

One change fixes it. Moving the same threshold from the sister-family share to the off-family share separates all seven runs completely, firing at round 2 or 3 in each of the three that drifted and never in the other four, at sensitivity 1.00 and specificity 1.00. It is reported and **not applied**, because it is validated after the fact on seven runs with an inherited threshold, and applying it would overturn committed verdicts. That is a decision for a session that owns those tables rather than for the one that noticed.

Seed choice does not change the family, only the debris. The three vertebrate runs were seeded from a human paralogue, a fly gene and an amoebozoan gene. Their family content is the same set three times over:

they intersect on 4,785 family records and differ by at most 162. What the seed changes is everything that is not the family, in that the amoebozoan run carries 19,949 non-family targets against roughly 2,439 for each of the other two. A distant seed does not reach further into the family. It reaches further out of it.

4.15 The inference ran out in four places, rather than the search

Four limits were met while doing something else, and each is a general result about the method rather than about this family.

A reconciliation's loss count is mostly sampling. Losses implied by a gene tree over a 134-tip alignment, checked cell by cell against the 309-genome ledger, give 47 implied cells, 26 of them the paralogue present in the genome and absent only from the sample, and none corroborated.

Synteny disambiguation is accurate and unavailable. The neighbourhood caller built in Chapter 7 is right wherever it acts and almost never acts on the cells that need it: of 432 trace regions it reaches 8, because a fragment's contig carries no neighbours to read. Accuracy and reach are different numbers, and quoting only the first would report contig lengths as method performance.

Per-site codon models are past their power at these divergences. The synonymous rate is unidentifiable at 671 of 7,368 sites, and 373 of 420 pairwise comparisons are flagged saturated. Any per-site rate formed as a ratio of sums is dominated by sites whose denominator is not estimable.

A likelihood tree need not resolve the question asked of it. The test that compares the three sister arrangements leaves two topologies inside its 95 % confidence set, so the sister question is answered by what the data excludes rather than by reading the best tree. Chapter 6 reports both.

4.16 Five results here apply to anyone doing the same thing

1. **A genome-scale orthologue sweep misses about one cell in seven, and every miss is an assembly.** A survey that reports absences without a contiguity floor is reporting assembly quality.
2. **A contiguity bar can be set from gene geometry before any error is measured**, and it lands where a calibration would have put it, at a cost of 38.8 % of the genomes, disproportionately in the clades a loss survey most cares about.
3. **Phylogenetic breadth in a protein bait panel is nearly worthless within the vertebrates, and paralogue coverage is everything.** Spend the budget on paralogues and on clades where no labelled record exists.
4. **A family profile over reference proteomes is not a discovery method for whole genes.** At gene scale it returns what domain annotation already returns, its gain is fragments, and it adds gene-scale records only where the annotation is worst.
5. **The obvious rule for catching iterative-search drift measures the wrong thing.** Sister-family contamination is the quantity this project and its predecessor both chose to monitor, and it is diluted by the very drift it is meant to detect.

5. How far the family reaches across the eukaryotes

5.1 Two questions that look like one, and have to be answered separately

"Where does this gene family occur?" is normally answered from a database and reported as a range. That answer conflates two questions that have to be kept apart, and this chapter answers them separately.

The first question is about the record: in how many reference proteomes does a family member appear? That is a statement about what gene callers have found, and it is the one a database can support on its own.

The second question is about the DNA: in genomes where no gene caller found one, is there one? Answering it requires searching assemblies and, crucially, a **positive control that demonstrates the search could have found the gene had it been there**. Without that control, "no IP_3 receptor in *Arabidopsis*" is indistinguishable from "the search did not run properly on *Arabidopsis*". The distinction is not academic in this family. Plant cells release calcium in response to IP_3, no plant receptor gene has been identified, and the standing review of the subject asks in its title whether the receptor is real [36], so a plant absence is a claim somebody has been waiting for.

Both questions are answered here, over 6,928 reference proteomes and 194 genomes.

5.2 The proteome sweep covers 6,928 proteomes and partitions the eukaryotes

The sweep covers six groups: non-vertebrate metazoa, fungi, green plants, other protists, archaea and a genus-stratified bacterial sample. Together they comprise **6,928 reference proteomes, 63,144,898 proteins and 24.93 gigabases of residue.** The four eukaryotic groups partition Eukaryota with Chapter 3's vertebrate sweep, so no proteome is swept twice and the two denominators add.

Two sampling decisions are stated rather than assumed. Bacteria are sampled at one proteome per genus, taking the one with the most proteins, which is the most sensitive member of each genus, so a negative claim is made in the places most likely to break it. Archaea's reference set is small enough to take whole, so no sampling caveat attaches to it.

Twenty-three proteomes that UniProt lists are not published in the release tree and return a permanent 404. They are excluded from the denominator and recorded, together with what they took out of it, rather than retried. A listed but unpublished proteome is not a transient failure and retrying cannot fix it.

The instrument is Chapter 3's, unchanged: the same two profile hidden Markov models [29] over the same reference proteome collection [30], with the same 30-bit floor, the same 200-match-state span gate and the same relative margin. Reusing it rather than building a new one is what makes the vertebrate and non-vertebrate numbers comparable at all.

5.3 The family is present in 662 of 6,928 proteomes, and in no prokaryote

662 of 6,928 reference proteomes carry an IP_3 receptor call, and 45 of the 135 clades swept do.

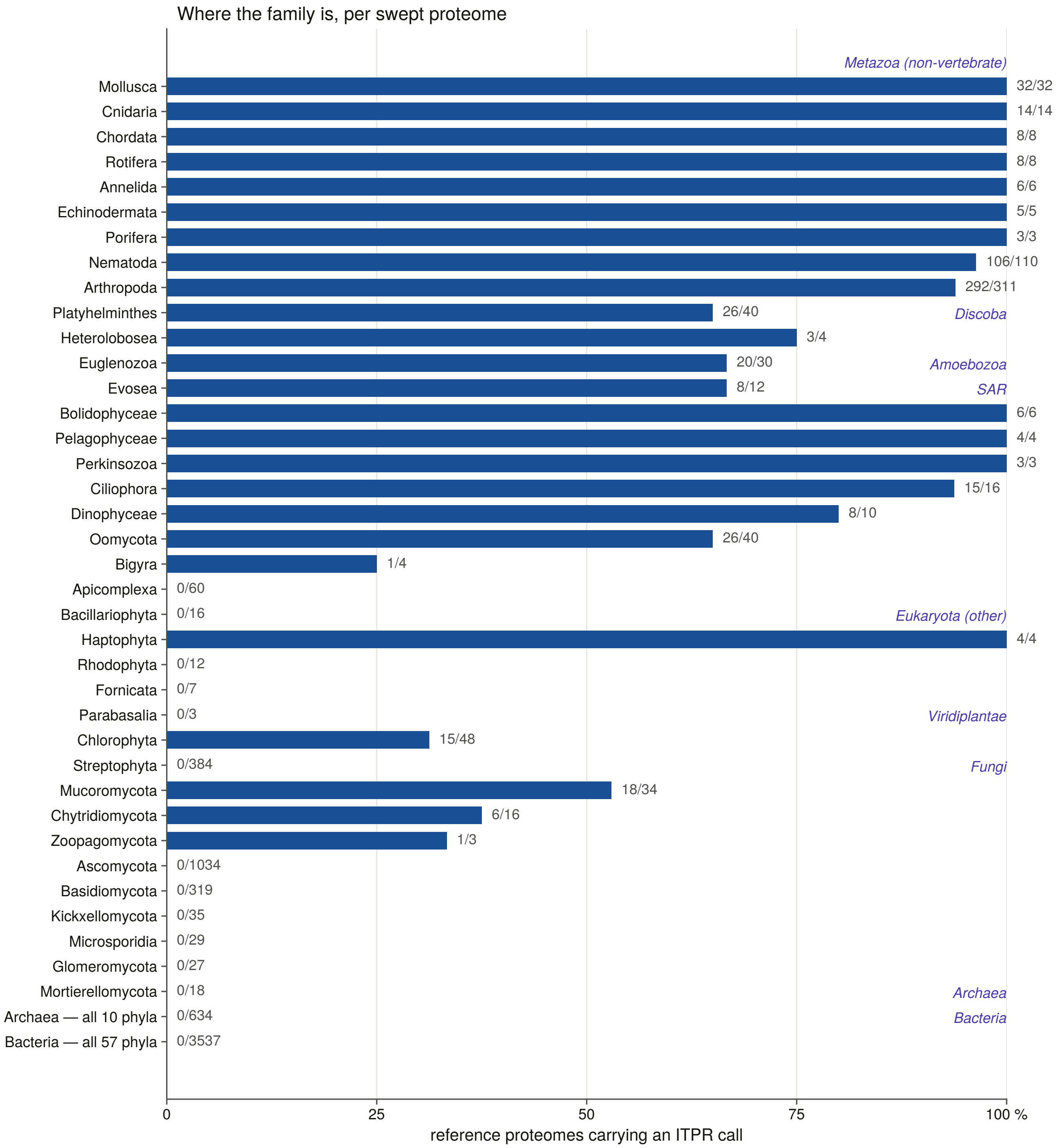


Figure 5.1. The family across the eukaryotes, drawn as a fraction of swept proteomes rather than of records. That denominator is the whole point: counted by records, a single well-sequenced alga outvotes a sparsely sampled phylum, and the figure would report sequencing effort. The importance of this figure is the shape rather than any single bar. A family present across the metazoa, in several protist lineages, in the green algae

and in the early-diverging fungi, and absent from land plants and from the yeasts and moulds, is a family that was present in the eukaryotic ancestor and has been lost repeatedly since. That is a different claim from an animal innovation, and it is what the figure supports.

The shape is a family that is ancestrally eukaryotic and has been lost repeatedly, which is what earlier surveys of the calcium-signalling toolkit had inferred from smaller samples [37,38,39] and what repeated loss is now understood to look like as an ordinary evolutionary mechanism [25]. It is in 94 % of arthropod proteomes, 96 % of nematode, and 100 % of molluscan, cnidarian and sponge. It is in 65 % of oomycetes, 67 % of euglenozoans, 94 % of ciliates and 69 % of amoebozoans. It is in none of 634 archaeal and none of 3,537 bacterial proteomes.

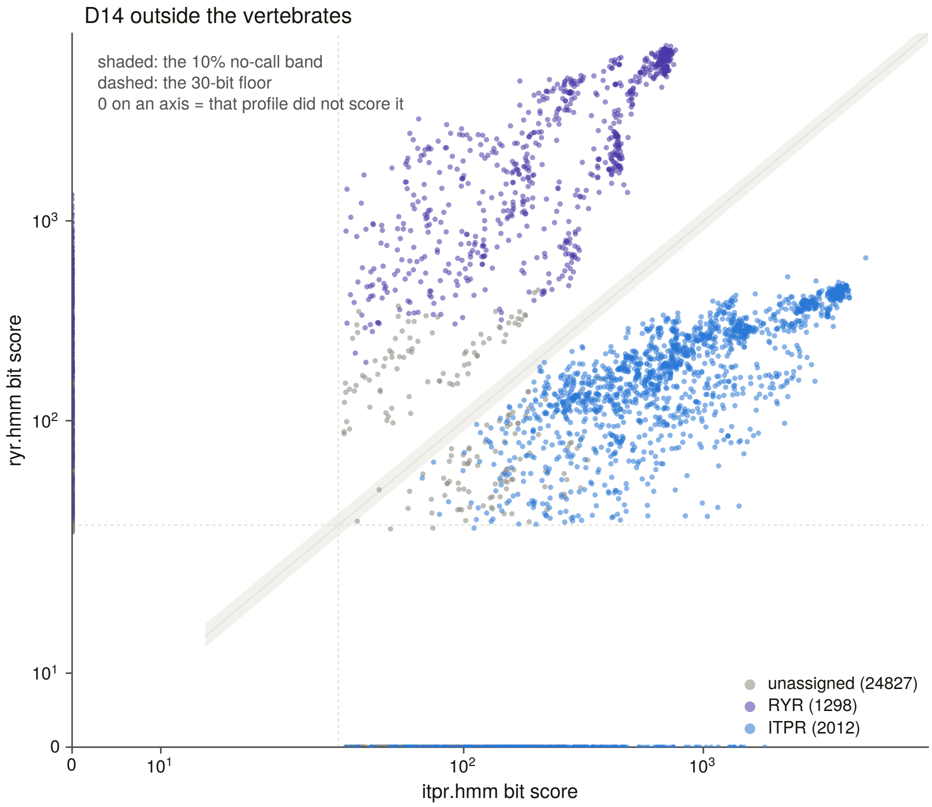


Figure 5.2. The same two-profile separation outside the vertebrates, with the no-call band drawn. 28,137 targets were scored by at least one profile and 2,769 by both above the floor, which are the only ones where the two families can be said to compete at all, and exactly one of those falls inside the band where the instrument declines to choose. The family separation therefore transfers intact to organisms a billion years of divergence away from where it was calibrated. Without that, every non-vertebrate count in Chapter 5 would be a mixture of two families rather than a measurement of one.

5.4 Land plants and Dikarya have no IP_3 receptor, and their early-diverging relatives do

Chapter 2 recorded an anomaly: forty plant and forty-one fungal proteins carry the IP_3-binding-core signature while *Arabidopsis* and budding yeast have none. Chapter 3 sharpened it. In the enumerated space every green-plant call was in Chlorophyta and **Streptophyta, the land-plant lineage, had none**, while every fungal call was in an early-diverging phylum and **Dikarya contributed no records at all**.

Those were statements about a space a protein enters only by already carrying a family domain annotation.

Sweeping the proteomes themselves asks the question without that filter, and both statements hold.

For land plants, **0 of 384 Streptophyta reference proteomes carry a call**, against 15 of 48 Chlorophyta. For the fungi, **0 of 1,034 Ascomycota and 0 of 319 Basidiomycota** carry one, against 18 of 34 Mucoromycota, 6 of 16 Chytridiomycota and both Basidiobolomycota.

That is the shape of a loss rather than of an absence: the family is present in the early-diverging lineage of both kingdoms and gone from the derived one, with the fungal split read against the accepted backbone of the kingdom [40]. In the plants it puts a genomic answer under a question the plant calcium literature has kept open, because the green algae these lineages descend from carry the gene [41] and the land plants that followed do not, so whatever performs the physiology there is built from other channel families [42,43].

Every plant and fungal record was then chased individually, 64 and 35 of them, through four independent lines of evidence, because a handful of records in a kingdom whose model organisms have none is exactly the shape of a contamination artefact.

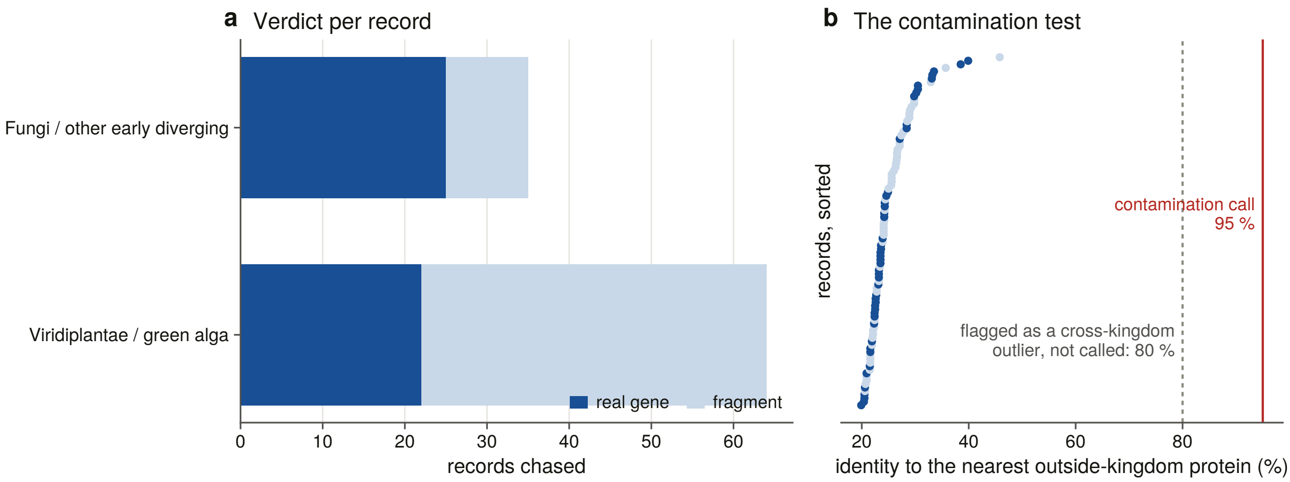


Figure 5.3. Every plant and fungal record chased, with the cross-kingdom identities the contamination test rests on. That axis decides whether these records are biology or contamination, which is the question the whole range claim turns on: a genuine deep homologue is 20 to 40 % identical to its metazoan relatives, while an assembly contaminant is 95 to 100 % identical to one particular animal. Nothing here is above 40 %, so the surviving records are real green-algal and early-diverging fungal genes, which is what makes the land-plant and Dikarya absences losses rather than the family's boundary.

Twenty-two plant records and 25 fungal records survive every test, and the rest are fragments of real genes. The surviving plant records run from 19.9 % to 39.9 % identity to anything outside their kingdom, and the fungal ones from 20.5 % to 33.5 %. None is a contaminant.

They are also not evenly spread. *Cymbomonas tetramitiformis*, a green alga, contributes twelve of the twenty-two plant records against a median of one per species. Whether that is a real copy-number expansion or a duplicated assembly is not a question a proteome can answer, and §5.8 answers it.

5.5 Every negative claim was made twice, at two stated sensitivities

Every negative claim in this chapter was made twice: once with the two full-length family profiles at a strict threshold, and once at a much looser one with those profiles plus the family's four individual Pfam domain models. A 213-state domain model can reach something a 2,684-state channel model cannot, and a negative claim should be made with the most sensitive instrument available rather than the most specific.

The design behind that is worth stating. The same search is run once and filtered twice: every search runs at the loose threshold and the strict call is taken by filtering the same output, so the strict set is exactly what a strict run would have produced and the loose set is a superset from the same search rather than a second experiment.

That equivalence was tested rather than assumed, and **the assumption behind it turned out to be wrong.** The reporting threshold is applied to the conditional expectation value, which is normalised by how many sequences passed, so a looser setting inflates every conditional value by a constant factor and pushes marginal domain rows out of the report. Measured across 798 protist targets, the sequence sets and all 798 assignments agree and no gate decision moves, but 11 domain rows differ. A second effect appears only at scale: the sequence expectation value is printed to two significant figures, so a target whose true value sits just above the cut prints as if it were at the cut, and 42 of 13,770 plant targets on the ryanodine profile are admitted by a less-than-or-equal filter that a strict run would never have reported. All 42 are declined by the span gate anyway.

The pass criterion is therefore about the call rather than about set equality: no assignment moves, no gate decision moves, and no one-sided target is called. The equivalence test also refuses to pass vacuously. Its first run went green on the archaeal group, which has no hit at either threshold, so a comparison of fewer than 25 targets is now a failure with its own message.

Each group was also searched again iteratively, from a seed native to that group, under the same coded kill criterion Chapter 3 used. The completeness question this answers is precise. A model seeded in one lineage and iterated to convergence either reaches a neighbouring lineage or it does not, and the targets that only iteration found are the ones that matter. If they sit in the same lineage as the seed, the absence next door is not a sensitivity artefact.

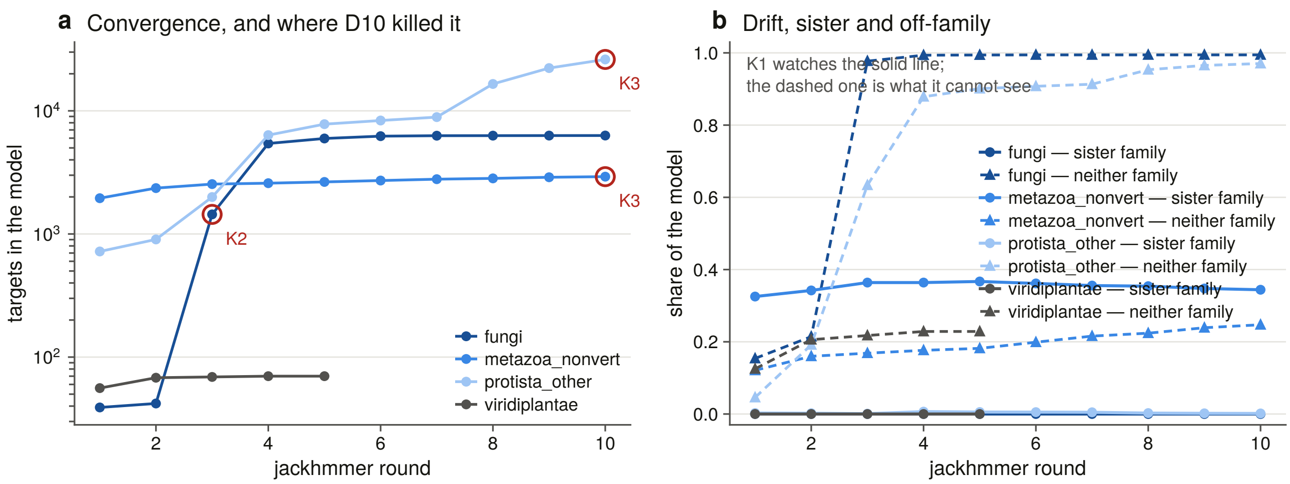


Figure 5.4. Per-group convergence with the sister-family trace beside it. One of four groups converges and the other three hit the kill criterion, and Chapter 4 explains why the rule that catches them is the wrong one. The figure matters as a sensitivity check on the absences this chapter reports: an iterative search that reached into an empty lineage and found a receptor would overturn them, and what the trace shows instead is a search running until it meets the sister family and stopping there.

Nine targets entered an accepted model that the single pass never reported. They are named rather than counted, and two of them sit in a lineage this chapter calls empty, so the absence there is not quite absolute and what those two are decides whether that matters. **Both are mannosyltransferases**, meaning the MIR-domain sharer that Chapter 3's decoy panel was built around rather than a receptor. The iterated search reaches these lineages exactly far enough to pick up the known false positive and no further, which is the most informative result the iteration could have produced.

5.6 The absences were taken from the proteome to the genome

A proteome absence is a fact about a gene caller. The 194-genome sweep turns the ones that matter into facts about genomes.

The scope is chosen to sample most finely where the negative claim is. Five rules were applied to 24,596 NCBI eukaryotic reference assemblies [32], minus the 6,216 vertebrate ones: one assembly per non-vertebrate eukaryotic phylum; one per class in the five phyla with more than 100 swept proteomes; one per clade with at least ten swept proteomes and zero calls; the anchor organisms whose answer is known from the literature; and the highest-copy species per class. That gives 194 genomes and 100.4 Gbp.

The third rule re-derives the target list from the presence table rather than repeating a summary, and in doing so it **found three absences that summary never named**: diatoms at 0 of 16, red algae at 0 of 12, and **Cestoda at 0 of 11, a metazoan clade with no record at all.** The first two sit in lineages whose calcium-channel repertoires have been surveyed and found to differ sharply from the animal one [44], and the third is a clade that has shed gene families across its whole biology as an adaptation to parasitism [45]. The apicomplexan and microsporidian absences belong with the severe genome reduction those lineages have undergone [46,47].

Both genomic thresholds had to be re-measured, and neither transferred. A vertebrate IP_3 receptor gene spans 76 to 498 kb and a *Drosophila* one spans 22 kb. Measured across sixteen genes in sixteen species from NCBI's own gene annotations, and deliberately not from the aligner, whose own intron setting shapes the loci it reports so that a measurement taken that way cannot falsify the setting it calibrates, spans run from 3,739 to 324,840 bp with a median of 19,435. That is a hundred-fold range against the 6.5-fold measured inside the vertebrates, so **the contiguity bar is set per group**. Metazoan genes are about nine times longer than protist and fungal ones, at 83 kb against 7 to 9 kb, and the genomes carrying this chapter's negative claims are judged against the smaller bar, which almost any modern assembly clears.

One gap is stated in the output rather than hidden. No green-plant gene span could be measured, because the chlorophyte census records carry locus tags with no gene record, so every plant genome is judged against the global median. The calibration function returns that fallback in its own return value so that a caller cannot fail to notice.

5.7 The positive control had to be replaced twice

Chapter 4 could write "the ryanodine control fired in all 309 genomes" because every vertebrate has three ryanodine receptors. Outside the metazoa that sentence is not available, because the proteome sweep found architecture-level ryanodine receptors in **2 of 6,928** non-metazoan proteomes. A land plant with no ryanodine locus is the correct answer, so it cannot be that genome's proof of search.

The first replacement was the MIR-domain sharer, meaning the O-mannosyltransferase family, which is in the family's own signature set, present in the clade each claim is about, and simultaneously the sharpest available decoy. Where a query on the IP_3-core signature returns nothing in land plants, a query on MIR returns 633.

It was still one profile fixed in advance for every clade, and that is what failed. Both apicomplexan classes carry a single MIR protein each across 60 swept proteomes, and the pilot's *Toxoplasma gondii* came back with neither a receptor nor a control, making it the one genome in the pilot whose absence claim therefore rested on nothing.

The fix is not a different profile. It is not fixing the profile in advance. A control's job is to fire in the clade whose absence is the claim, so which family makes the best control is a property of the clade and is measurable. Six candidate profiles, all large, deeply conserved, multi-exon eukaryotic families, were run over each clade's own swept proteomes, and each clade takes the one that is actually there, with every candidate's coverage committed rather than only the winner's. The MIR bait stays in the panel whatever the measurement says, because dropping it would buy a proof of search and sell the family's negative control.

193 of 194 genomes are controlled, and 115 of them across a kingdom boundary, meaning that a control bait from a different kingdom-level group aligns across the same locus. That is the tier an absence claim needs, because it shows the search reaches across the divergence any receptor here would have to be found across rather than merely that the assembly is readable. In the pilot, 1 of 14 genomes was uncontrolled. Here, none.

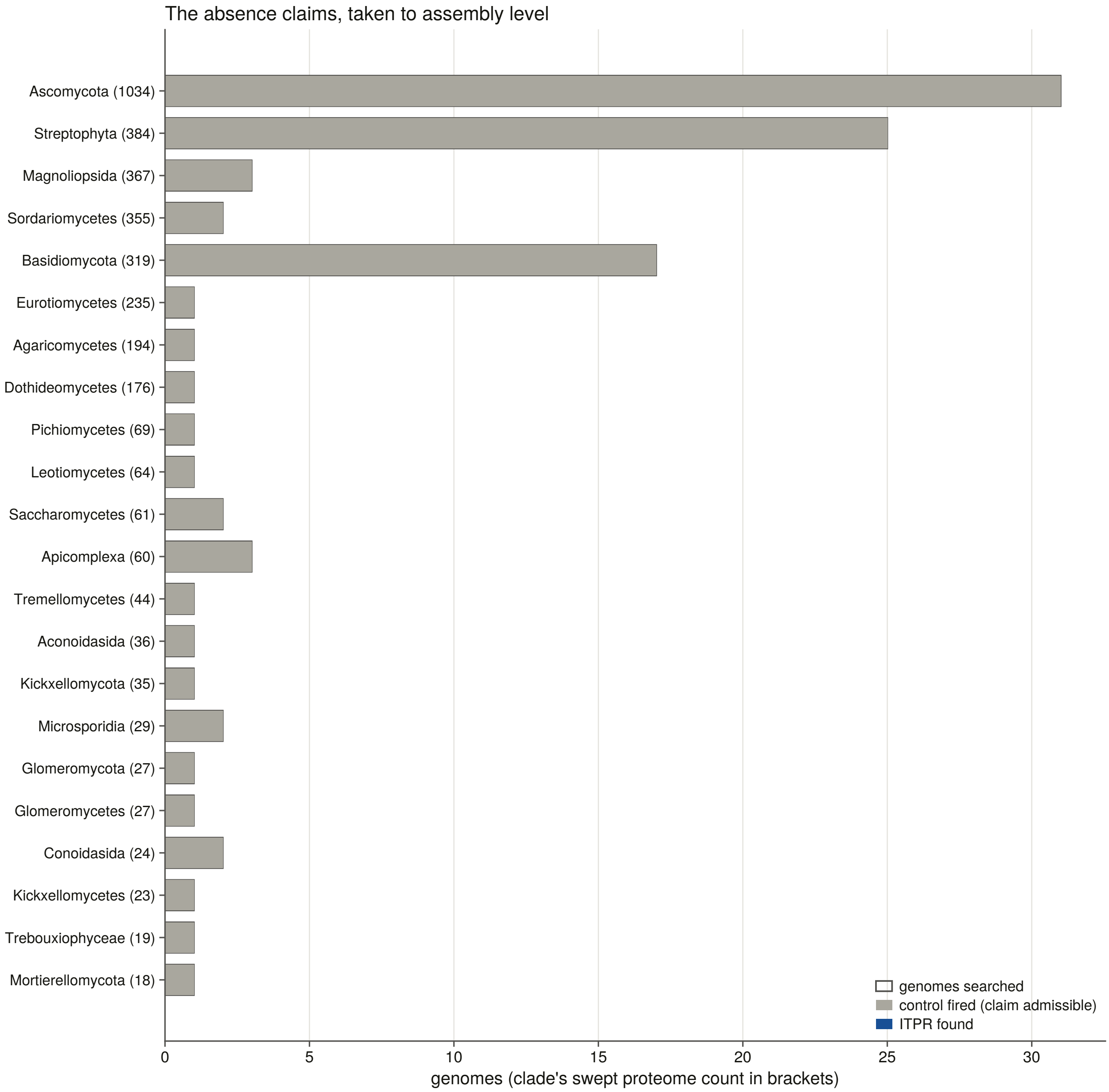


Figure 5.6. Each absence claim with the number of controlled genomes drawn beside the number searched. The two bars are the difference between a database observation and a biological claim. A clade with no records in any proteome may simply not have been searched properly, so what licenses an absence claim is the number of genomes in which a measured positive control recovered a comparably long, deeply conserved gene. A claim whose control bar is short is standing on nothing, and drawing the two counts together is the only honest way to present a table of zeros.

Every one of the 35 clade-level absences holds at assembly level, and none fails. Ascomycota has none in 31 controlled assemblies, Streptophyta none in 25, and Basidiomycota none in 17. Apicomplexa, Microsporidia, Glomeromycota, diatoms, red algae and Cestoda likewise.

5.8 Copy number outside the vertebrates reaches eighteen

Outside the vertebrates the three paralogue cells do not exist, because ITPR1, ITPR2 and ITPR3 are a vertebrate whole-genome-duplication product and Chapter 4 found the trio absent below the cyclostomes. The question is simply how many receptors a genome has.

Of 193 controlled genomes, 116 carry none, 43 carry one, and 34 carry more than one. The top of the distribution is a flatworm, *Macrostomum lignano*, with 18 complete gene models, then a ciliate, *Stentor coeruleus*, with 13, then two sponges at 8 and 6.

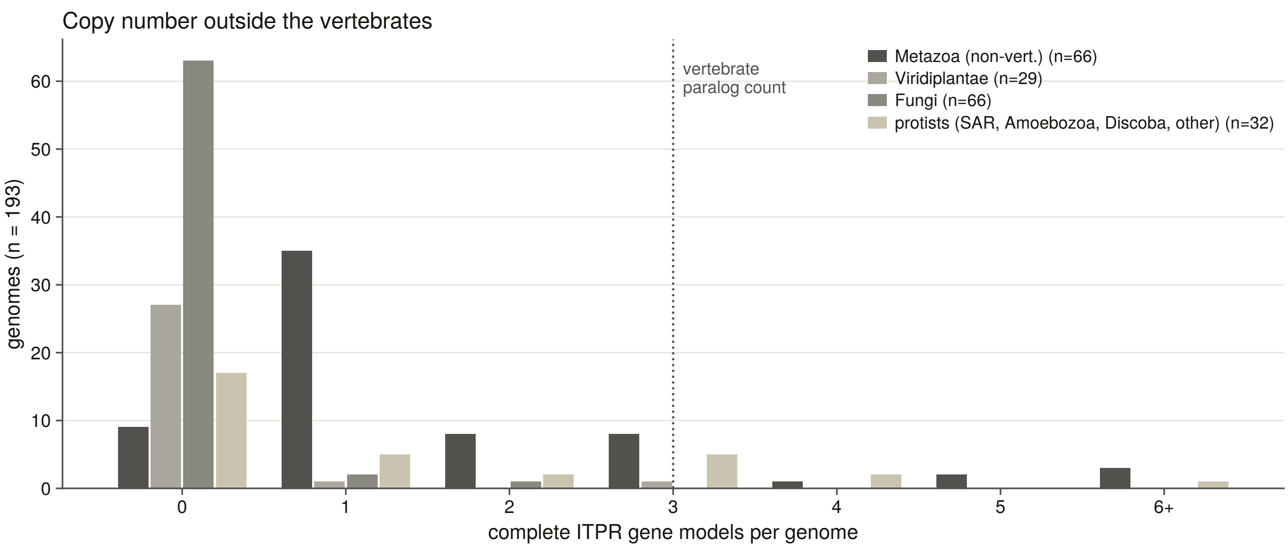


Figure 5.5. Copy number outside the vertebrates, with the vertebrate paralogue count drawn as a reference line rather than as a category. Drawing it that way shows the vertebrate three to be unremarkable: a flatworm carries eighteen receptors and a ciliate thirteen, so copy-number expansion in this family is not a vertebrate story and three is not the family's high-water mark. What is specific to the vertebrates is that their three copies are ancient, distinct and universally retained.

Cymbomonas, the green alga that contributed twelve of the twenty-two surviving plant protein records, carries **three** complete gene models across two clusters. Twelve records and three genes is what a duplicated assembly and a multi-isoform gene set look like from the protein side, and it is the answer the proteome could not give.

5.9 Two further thresholds had to be measured for this scope

Two thresholds in this sweep are measurements rather than settings, and both had to be made here because neither transfers from the vertebrates.

A cluster of alignments is not a gene. With the intron parameter set high, as §5.6 requires, a cluster is a large object, and two rules pull in opposite directions. Neighbouring same-strand clusters whose bait spans are complementary are one gene the aligner broke, and one cluster containing two distinct complete models is two genes. Both were implemented, both record the numbers they fired on, and their measured effect over the sweep is seven genes recovered in seven genomes.

The identity floor below which a locus is not a locus was measured against loci whose identity the assembly's own annotation establishes, which is evidence the alignment score did not produce. The sweep deliberately records every cluster down to a much lower floor precisely so that the floor is not measured from the population it has already filtered.

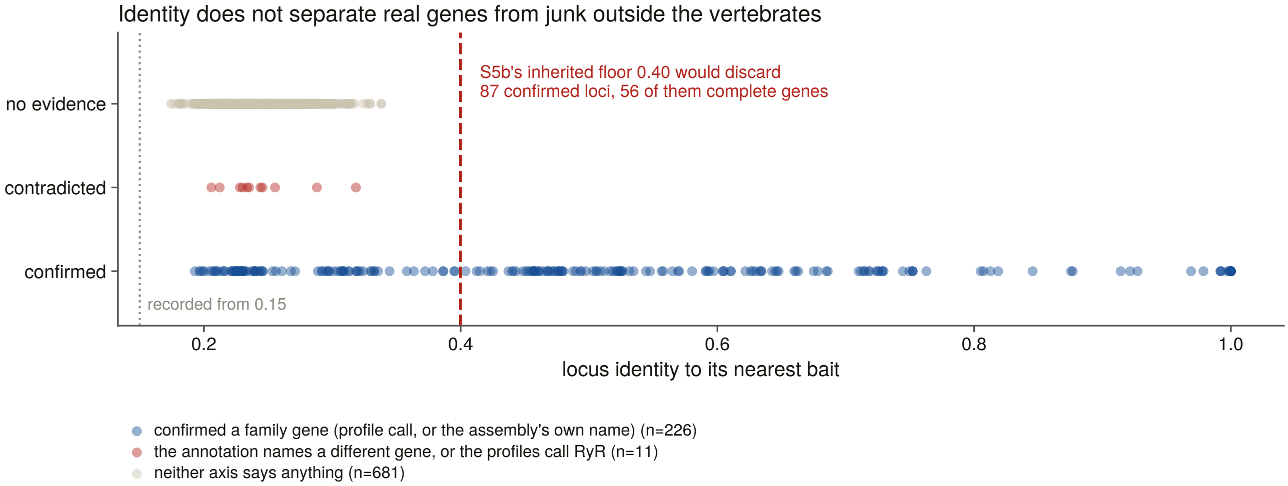


Figure 5.7. The measured identity floor and the two populations it separates. The importance of the figure is a negative result that changed the instrument. Outside the vertebrates the annotation axis is nearly empty, with 21 of 917 clusters carrying an informative gene name, and neither identity nor coverage separates the confirmed from the contradicted loci cleanly, so a threshold placed on either would have been a number with no population behind it. A second, independent axis was added for that reason, scoring every recorded cluster against both family profiles, and the calibration reports both axes rather than the one that looked better.

That calibration also refuses to write below a floor of genomes and confirmed loci, and the reason is an incident. A smoke-test run over one genome produced a threshold of 0.25 from two loci, wrote it, and the next sweep read it back and moved three genomes from one status to another. A calibration that can pass vacuously is worse than no calibration.

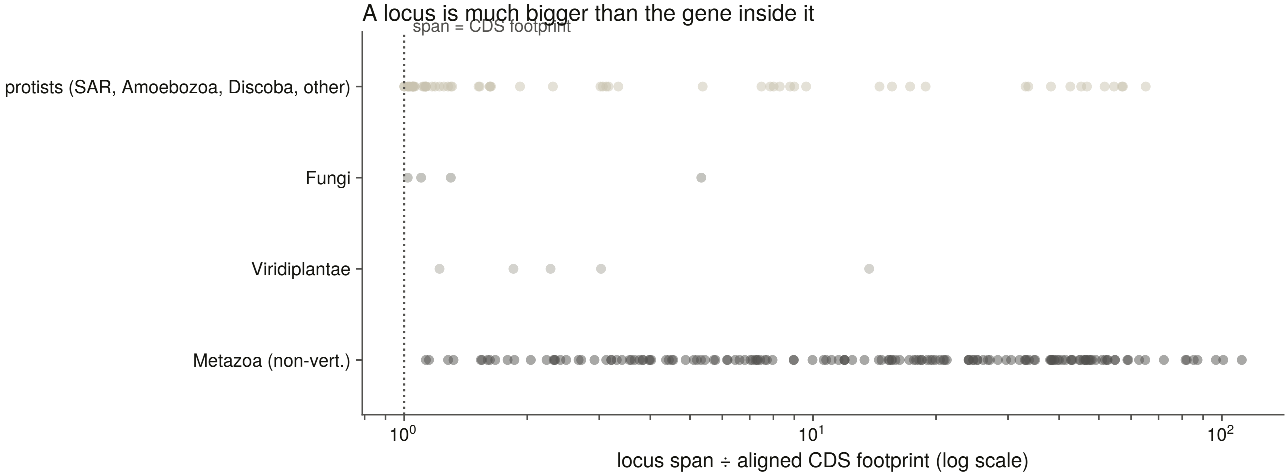


Figure 5.8. Locus span against coding footprint, by group. The figure shows why a cluster of alignments is not a gene: a locus is much larger than the coding sequence inside it, and by a factor that varies roughly nine-fold between metazoan and protist genomes, so a single intron setting and a single contiguity bar would be far too lax for one group and far too strict for the other. This is the measurement behind the decision to set both per group, and behind the rule that a locus is not a copy.

5.10 The family's range is stated with its scope attached

The family is **ancestrally eukaryotic**. It is present across the metazoa, across several protist lineages including ciliates, oomycetes, euglenozoans and amoebozoans, in the green algae, and in the early-diverging fungi. It is absent from land plants, from Dikarya, and from every archaeal and bacterial proteome swept.

Every one of those absences is a claim about a declared space. The proteome-level claims are about 6,928 reference proteomes, and the genome-level claims are about 194 assemblies in which a measured positive control recovered a comparably long, deeply conserved gene. Where the two disagree, which they do not in any of the 35 clades tested, the genome would win.

The merged census is 18,065 records, of which 8,990 are called IP_3 receptor across 1,402 taxa. Chapter 12 returns to the absences with a different question: whether the enzyme that makes the ligand went with the receptor.

6. Building the alignment and the tree that later chapters stand on

6.1 What this chapter has to produce, and why the care is warranted

Chapters 7, 10, 11 and 12 all rest on a single multiple sequence alignment. It is worth the care that goes into it, and it is worth saying which decisions in building it are measurements and which are choices.

The requirement is a set of representatives spanning the whole family, meaning every clade and every kingdom in which Chapter 5 found the receptor, with the ryanodine receptors included to root the tree.

6.2 Eight rules choose the representatives, and none of them uses length or identity

Eight rules choose the set, and the first thing to say about them is what none of them does. **No rule ranks or filters on sequence identity, and no rule takes the longest record per species.**

Longest-per-species is the obvious rule and it reliably selects chimeric gene models, because a mis-joined model is longer than the real gene, so a rule that prefers length prefers the error. Identity was retired as a call gate outside the vertebrates in Chapter 5, after it was measured and found to separate neither confirmed from contradicted loci, and a selection rule stated in a statistic that does not separate the populations is not a rule.

What the eight rules do select is the following. They take the vertebrate paralogue grid of clade band by paralogue, with human forced in. They take the teleost co-orthologue pairs as a stress test. They take the deep vertebrate grade, meaning cyclostomes, cartilaginous fish and the coelacanth grade, entered **unlabelled**, because a paralogue label is a vertebrate concept and assigning one here would assert what the tree is being run to test. They take non-vertebrate metazoa by phylum. They take non-metazoan eukaryotes, gated on Chapter 5's per-record plant and fungal verdicts, so that a suspected contaminant cannot become the plant branch's tip. They take the copy-number expansions, because Chapter 5 found up to 18 copies in one genome and a single tip would misrepresent that. They take the novel gene models no database annotates. And they take the ryanodine outgroup.

The quality key is **lexicographic rather than a weighted sum**, precisely so the audit can name the component that decided each pick. Every chosen row carries its rule, the cell it filled, the runner-up it beat and which component separated them.

Length targets are measured rather than borrowed. Each group is scored against its own median over its non-fragment records, because the family's size varies far more outside the vertebrates than within, and one target would score the plant and ciliate grades against a vertebrate ruler. The bar for a record to count toward that median is four of five signatures rather than five, and the record floor is three rather than five. Both were loosened for a measured reason. At the strict setting two groups had too few qualifying records and fell back to the global median: the green plants had two and the amoebozoa one, so both were scored at 2,694 residues, a metazoan number, and in the plant grade that decided which tip was chosen. At the looser setting the well-populated groups barely move and the starved ones gain a real ruler, in that the green plants go from two qualifying records to thirteen and their target from a metazoan 2,694 to 3,182. The tier and record count behind every target is committed, so a number computed from four records is visible as such.

One deviation from the brief is stated rather than absorbed. The brief asks for every novel gene model from both sweeps. There are 424 full-length models with no database record, and including all of them would triple the alignment, put it past the size where the chosen aligner is affordable single-threaded, and, the real objection, fill the tree with vertebrate gene models, since 206 of the 241 vertebrate ones are birds and ray-finned fish. The rule instead takes one per clade band and paralogue cell and one per non-vertebrate group, which is what the requirement is for: that sequences no public database contains are in the tree, and that each clade where they are the only evidence has one.

The result is 134 representatives, 375,137 residues, the longest of them a 5,317-residue ryanodine receptor.

Six of the selection rules have constructed negative controls that run on every build, covering a fragment, a ryanodine-length record offered as a family member, a suspected contaminant, a protist whose UniProt name reads “type 2” by annotation transfer, one species under two spellings, and a score-ranked single-clade pool. Two further checks cover properties the step before the aligner must have for reproducibility to mean anything downstream: determinism, and Newick-legal unique tip labels.

6.3 The alignment is pinned to one thread because otherwise it is not reproducible

The aligner is MAFFT in its most accurate iterative mode [48]: 61 minutes, producing 11,777 columns at 76.23 % gaps.

It is run single-threaded, by decision. At automatic thread count it is not reproducible on this machine, because the same seeds aligned twice in Chapter 3 gave 8,510 and 8,468 columns, since the iterative refinement stage combines partial results in whatever order the threads finish. Everything downstream is built from this file, so it is pinned and the SHA-256 of input and output are recorded. A ragged alignment is a hard failure rather than a warning, because a silent aligner failure degrades to a star alignment with no other symptom.

An automated trimming heuristic [49] then keeps **1,797 of 11,777 columns, 15.3 %**. The choice of heuristic is recorded rather than tuned, because a trimming threshold picked by looking at the resulting tree is a threshold fitted to the answer.

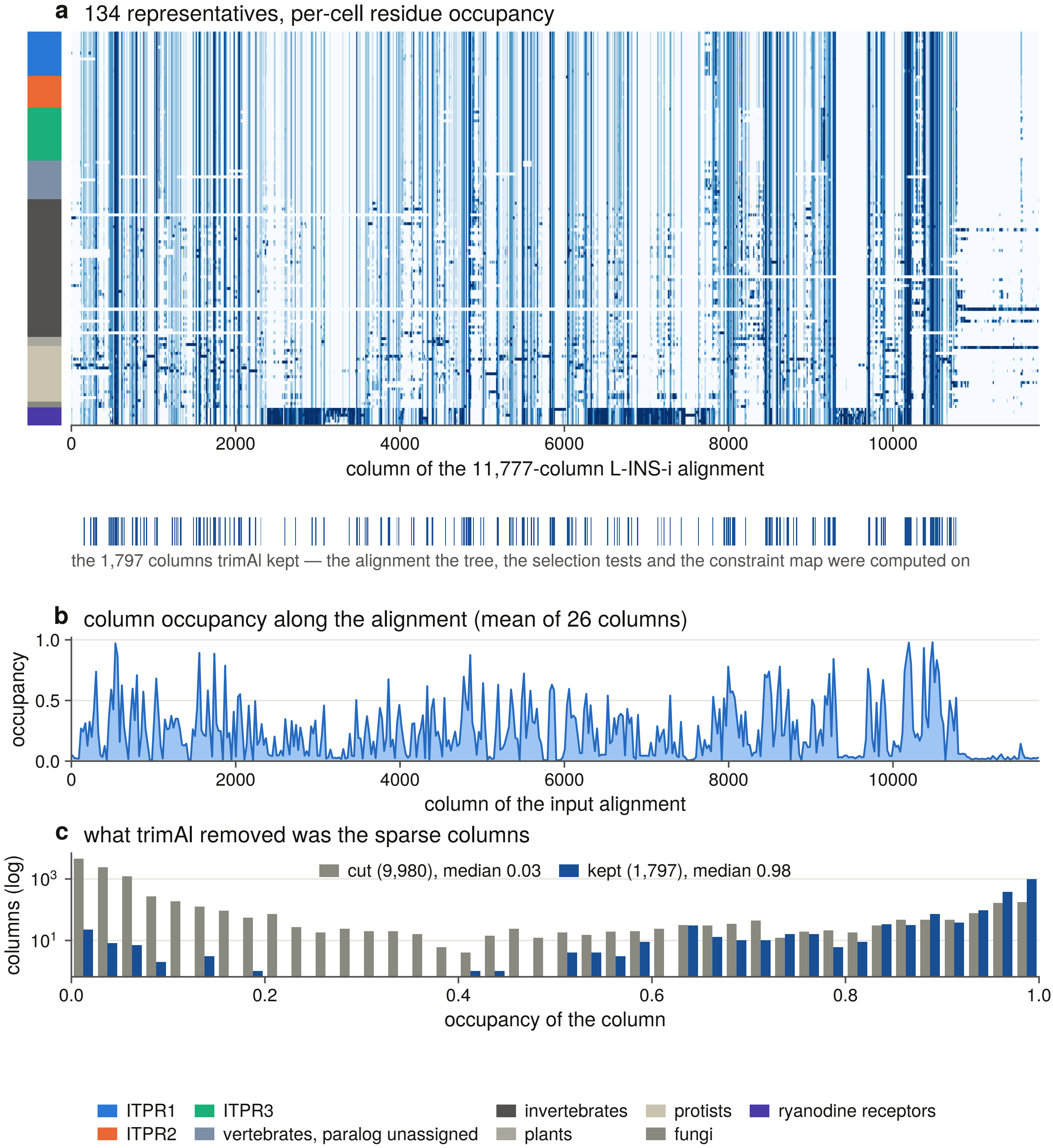


Figure 6.5. The alignment, with the columns the tree actually saw marked in the input's own coordinates. The trimmed alignment is the input with 9,980 columns deleted and the two share no x-axis, so only one raster can honestly be drawn and the cuts marked beneath it. The raster bins columns and plots occupancy rather than residue identity, because at 11,777 columns one printed pixel is nine columns and a residue palette would draw whichever residue happened to land on it. The importance of this figure is that it lets a reader see what the tree was actually computed on. Trimming removed 85 % of the columns, and the ones it removed were the sparse ones, at a median occupancy of 0.03 against 0.98 for those it kept. Every tree, selection estimate and constraint layer downstream rests on the kept set, and a reader who wants to know how much of the alignment that is can read it here rather than take it on trust.

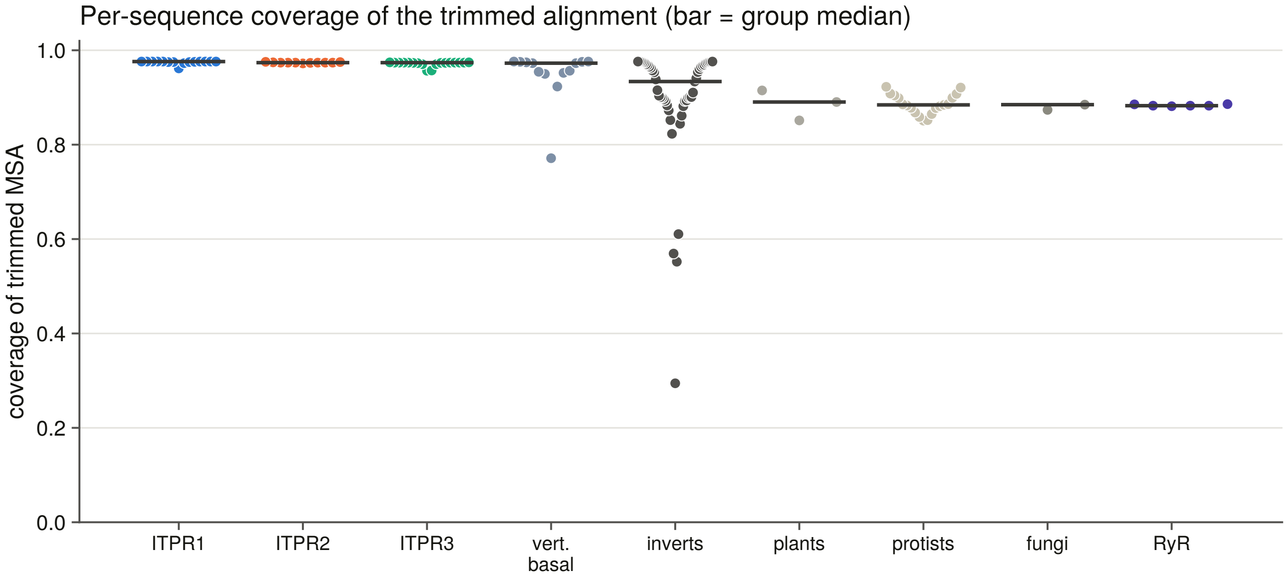


Figure 6.4. Per-sequence coverage of the trimmed alignment. The median is 0.96 and one tip of 134 covers less than half. Coverage matters because a tip that covers half the alignment contributes gaps to every column the tree is inferred from, and gaps are not neutral for a likelihood model. Naming that one tip, rather than reporting only the median, is what allows a reader to judge whether any deep node depends on it.

6.4 What the alignment says before any tree is built

The family separation, re-measured. The ryanodine receptors are in this alignment on purpose, and their separation from the family is a positive test at every stage rather than an assumption. Measured here, mean identity within a vertebrate paralogue group is 0.910 and from each paralogue group to the outgroup 0.253, giving a separation of 0.656 identity units. Chapter 3's benchmark, using a different estimator on a different panel, measured 0.828 and 0.249. The comparison is made on the separation rather than on either absolute value, and it is confirmed.

That separation is what every stage of this project works inside. A 5,000-residue ryanodine receptor and a 2,700-residue IP_3 receptor are 25 % identical over the columns they share, which is not far enough apart to trust a heuristic with.

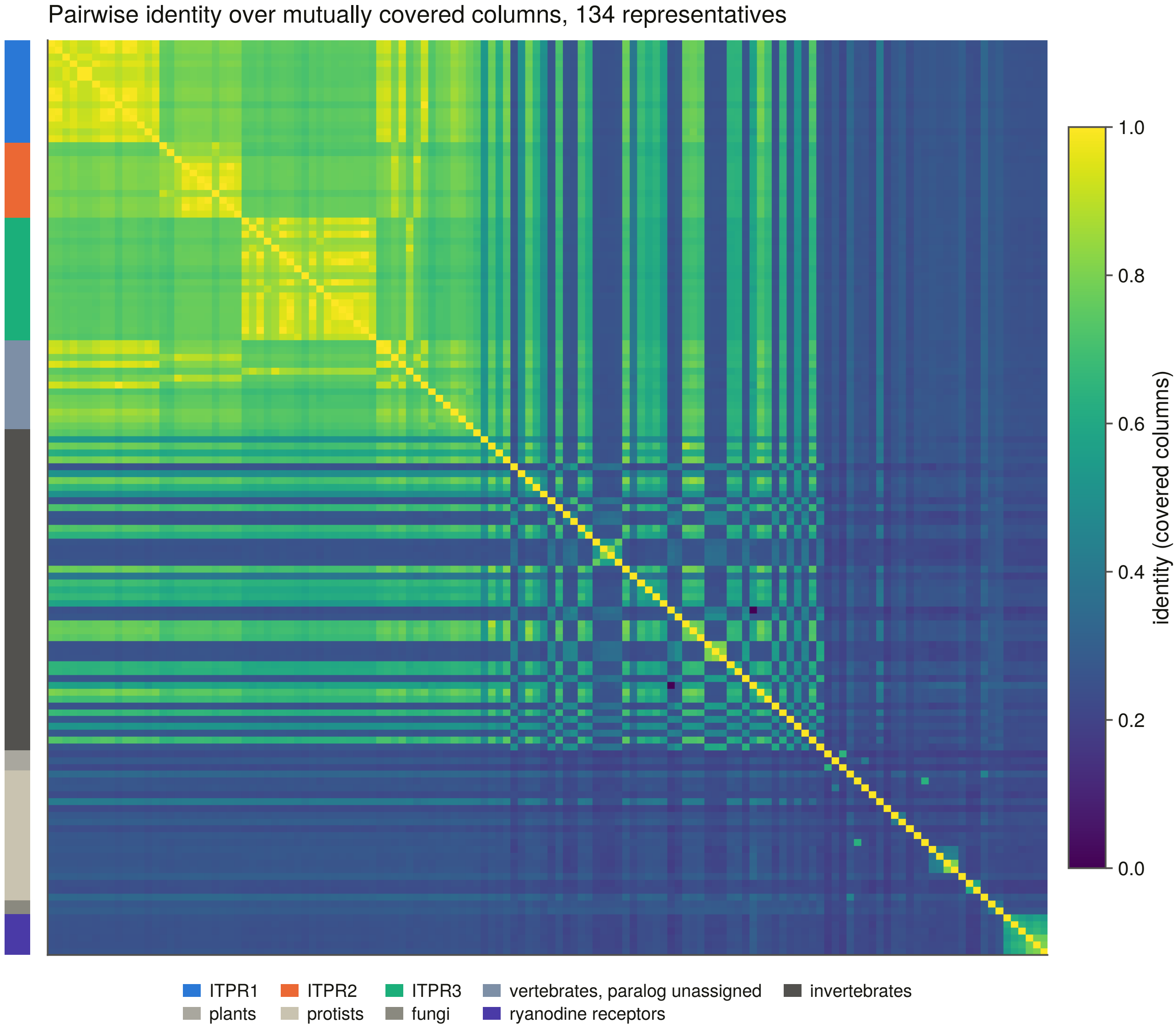


Figure 6.2. All-pairs identity across the representative set. Two identity matrices are committed rather than one, because they answer different questions and disagree systematically wherever a fragment is involved. A metric that counts gaps as mismatches makes every partial sequence look divergent, which would confound sequence divergence with assembly quality, and that confound runs through this entire project.

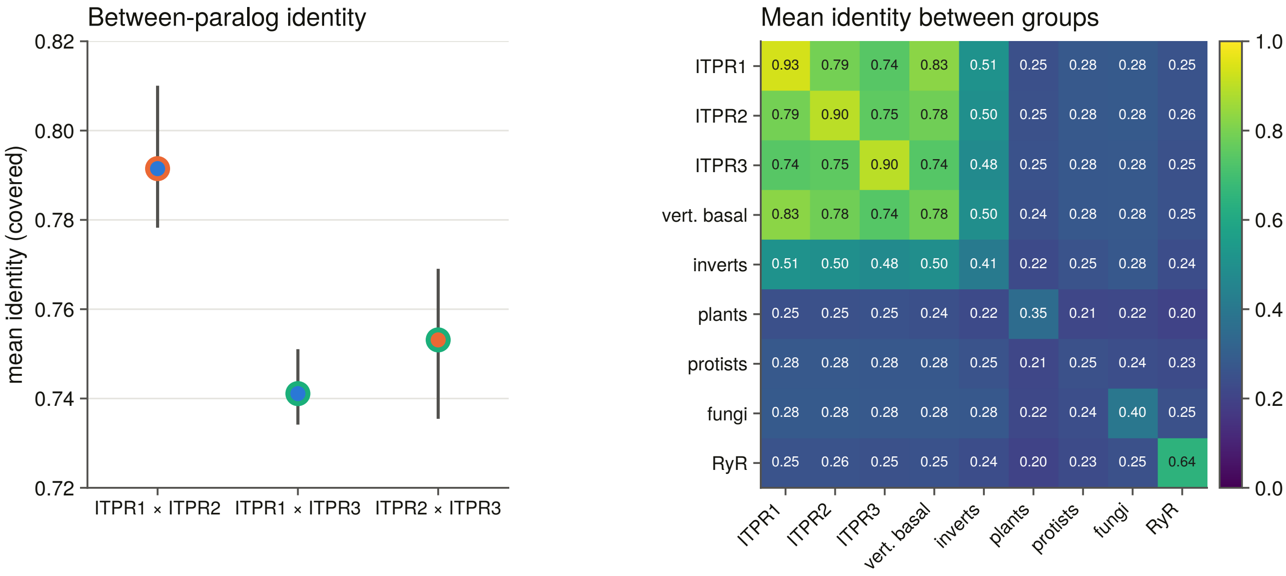


Figure 6.3. Identity within and between groups. This figure fixes the scale every later comparison is read against: the three vertebrate paralogues sit at 0.74 to 0.79 to one another and at 0.91 within themselves, which is close enough that a bait from one aligns at all three genes. That single fact is why the duplication detector of Chapter 8 needed the family call before its geometry meant anything, and why the attribution margin inherited from a less similar family had to be overturned.

The alignment's preview of the sister question turns out to be wrong. The alignment can rank the three between-paralogue identities: ITPR1 with ITPR2 leads at 0.791, against 0.753 and 0.741, and the leading pair's interquartile range does not overlap either of the others. That is a real ranking and it is not a phylogenetic estimate, because it ignores the outgroup, the rate variation and the branch lengths, so it was reported as a preview with the tree named as the answer. Section 6.8 reports that the tree contradicts it.

The alignment's preview of the cyclostome loci is wrong in a different way. Every cyclostome in this set carries three loci, all won by the same bait, so nothing before the alignment could say which was which. On identity all six fall nearest ITPR1, at margins of 0.014 to 0.066.

That table needs a control, because leaning towards one paralogue could be a fact about the cyclostomes or a fact about the metric. If ITPR1 is simply the slowest-evolving of the three, everything deep is nearest it. The same statistic over groups that are certainly not vertebrate paralogues gives the no-signal baseline: invertebrate metazoa lean toward ITPR1 at a median margin of 0.008, protists at 0.002, and the ryanodine receptors at 0.002. The cyclostome margin is five times that, so the lean is real. What it means is a different question, and §6.9 gives an answer neither reading predicted.

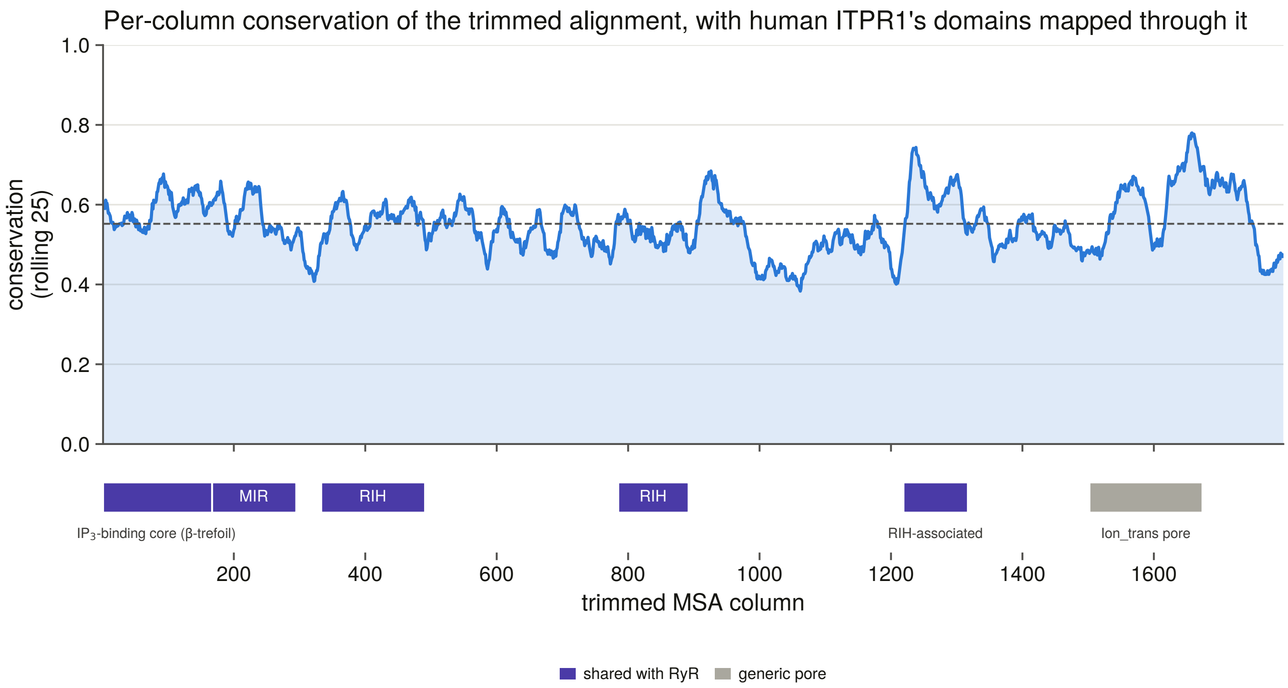


Figure 6.1. Per-column conservation with human ITPR1's domain architecture mapped onto it through the alignment, by walking the human row and counting ungapped positions, rather than by scaling residue coordinates onto column coordinates. The second method places every domain boundary wrong by exactly the gap content of the sequence. Since Chapters 11 and 12 both make claims about which element a residue belongs to, a systematic offset here would have propagated into every one of them.

6.5 The exhaustive model scan was measured and abandoned

The obvious command is an exhaustive model scan [50]. It was started, timed and abandoned, and the measurement is the reason: **11 models of up to 1,232 in 672 seconds, projecting about 20.9 hours.**

The cost is the free-rate models, and they are not optional on this alignment, because the best free-rate model beats the best gamma model by 682.3 units of the Bayesian information criterion, the fit statistic the selection is scored on. Dropping them to buy the time back would have returned a measurably worse model.

The replacement is greedy in two stages, with both tables committed: every exchangeability matrix at fixed rate heterogeneity, then every rate model on the winning matrix. Stage A chose its matrix by 614.6 BIC units over the runner-up, and stage B bought a further 1,109.4 by changing only the rate model.

What that costs is stated rather than buried. The matrix that wins under a gamma model need not be the matrix that would win under the final rate model. That is the bet a greedy selection makes, and the margin it was made on is in a committed table rather than in a sentence, so how close the second-best matrix came is checkable.

6.6 The tree search is pinned and guarded, and its rooting doubles as a control

The analysis used one maximum-likelihood search [51], taking 71 minutes, with 1,000 ultrafast bootstrap replicates [52] and 1,000 approximate likelihood-ratio replicates [53]. **Threads are pinned rather than automatic**, because automatic thread count reads the machine's current load and the search is reproducible only at a fixed seed and a fixed thread count. The number pinned is what the program's own benchmark returned for this alignment, so pinning costs nothing.

Support is reported as both branch tests together, and a node counts as well supported only when it clears both thresholds. One alone is not enough, because the bootstrap is optimistic under model violation, which is exactly the condition a 134-tip alignment spanning four kingdoms is in.

Two guards on the search itself came out of one incident. Every stage resumes on the report file rather than the tree file, because the program writes a tree at checkpoints and a search killed part-way leaves a tree of an unfinished topology that the next run would silently reuse. And a run refuses a prefix that a live process already owns. Both exist because a previous session's constrained searches were interrupted, kept running orphaned for fifty minutes, and a fresh run started three more on the same prefixes, after which the topology test was handed whichever checkpoint tree happened to be on disk.

Rooting doubles as a control on the whole alignment. The tree is rooted on the ryanodine receptors, and if they did not come back as a clade the alignment underneath every downstream result would be the thing to doubt. They do, at maximal support.

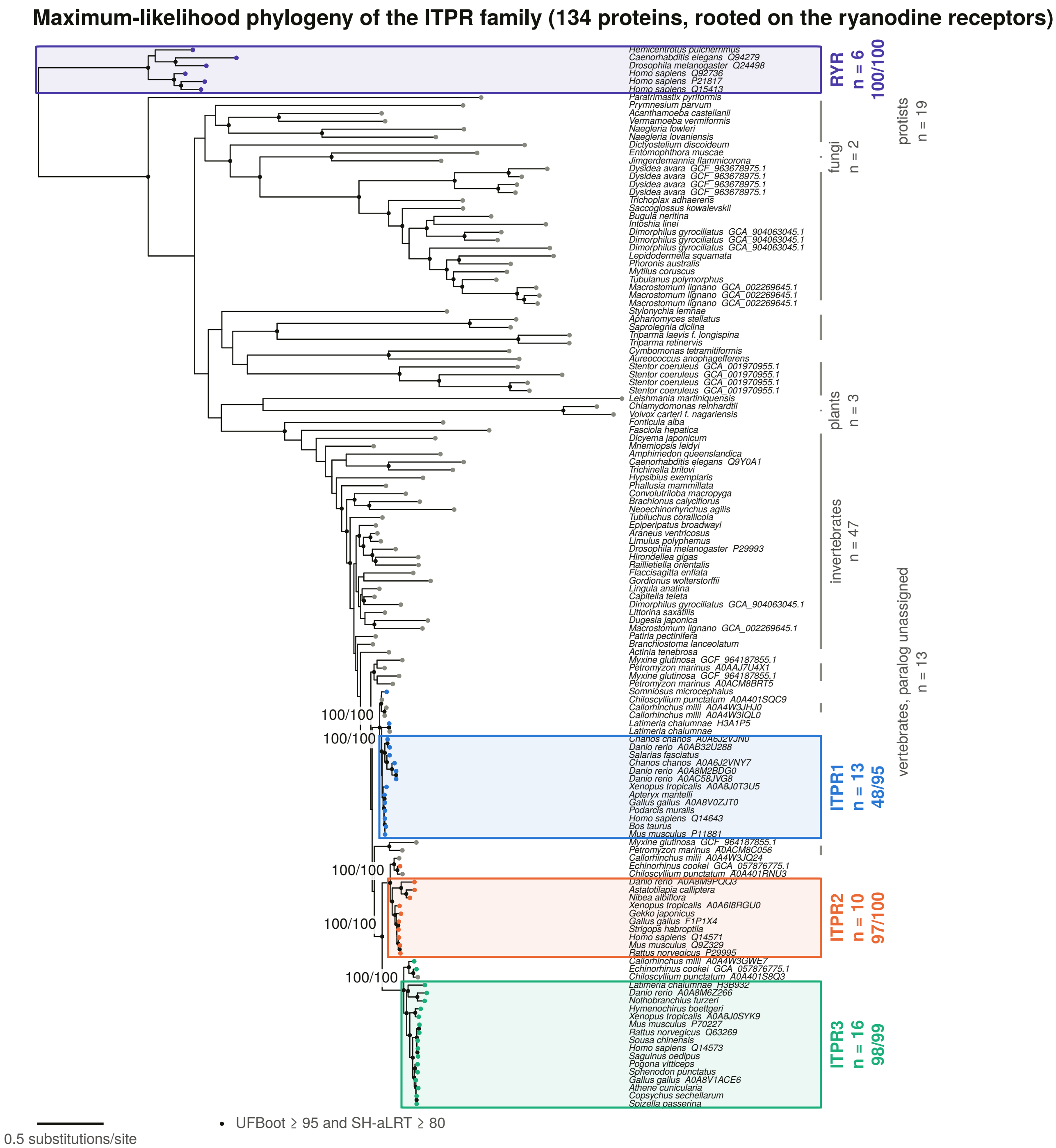


Figure 6.6. The rooted maximum-likelihood phylogram. Branches are in neutral ink, the three paralogue clades and the outgroup are boxed, and a filled dot marks every node clearing both support thresholds. No tip is ringed, because the relabelling rule described in §6.7 fires on none of them, and the legend entry for a ring appears only when a ring does, since a key naming a marker the figure does not carry asserts a correction that was never made. The tree matters because it answers a question the literature had left open: the review's own audit found no published, support-annotated maximum-likelihood analysis with a ryanodine outgroup that fixes which two of the three paralogues are sisters.

6.7 The three paralogues are clades, but only after the tree's own corrections

Every stage since Chapter 3 has assumed they are clades: the architecture call assigns a paralogue per record, the genome ledger has one cell per paralogue per genome, and the representative grid is built on clade band by paralogue.

Asked on the census's own labels with nothing corrected, **none of the three is monophyletic.** ITPR1's largest pure clade holds 13 of 15 tips, ITPR2's 10 of 11, and ITPR3's 16 of 18.

A label is not evidence, and a single mis-annotated tip breaks a clade of forty. So the same question is asked after the tree's own corrections, derived by coded rule and never by hand. A tip inside its group's largest pure clade keeps its label. A tip outside it whose smallest containing clade holds at least three members of exactly one other paralogue, at a well-supported node, is moved. Anything else is left out of all three sets rather than assumed either way.

Under tree-corrected membership **all three are monophyletic.** The difference between the two tables is the size of the family's naming problem rather than a change in the tree.

The correction rule fires on nothing. 39 tips agree with their label, 0 are reassigned, and 5 are left unplaced. That is worth saying plainly: the tree does not overturn a single census name in this set.

The rule that produces that result was itself wrong on its first version, and the way it was wrong is instructive. It climbed the tree from a mislabelled tip until some paralogue dominated and then assigned the tip there, which hands a tip to whichever clade is largest three nodes up, and that is a fact about clade sizes rather than about the tip. It now stops at the tip's own nearest neighbourhood. Two constructed controls, a positive and a negative half, caught it: a weakly supported placement must not overrule a census label, and a strongly supported one must.

Every naming call the tree disputes is checked by a different instrument. Neither the alignment nor the model can tell a mis-annotated record from a tree error, because both produced the placement in question. So each of the five unplaced tips is put through reciprocal best hits against the human reference proteome and back, using an instrument that knows nothing about this alignment, this model or this tree. **All five uphold the census name.** For each of them the tree offered no alternative, only a refusal to place, so what is upheld is the name: the record is correctly named, and the conflict is this tree's own uncertainty about where to hang it.

That verification is set up so that it can fire. An earlier version filtered to the reassigned tips only, which made it report that the tree contradicted no census label while the table it had just read held five tips the tree declined to place. A verification that cannot fire is not a verification.

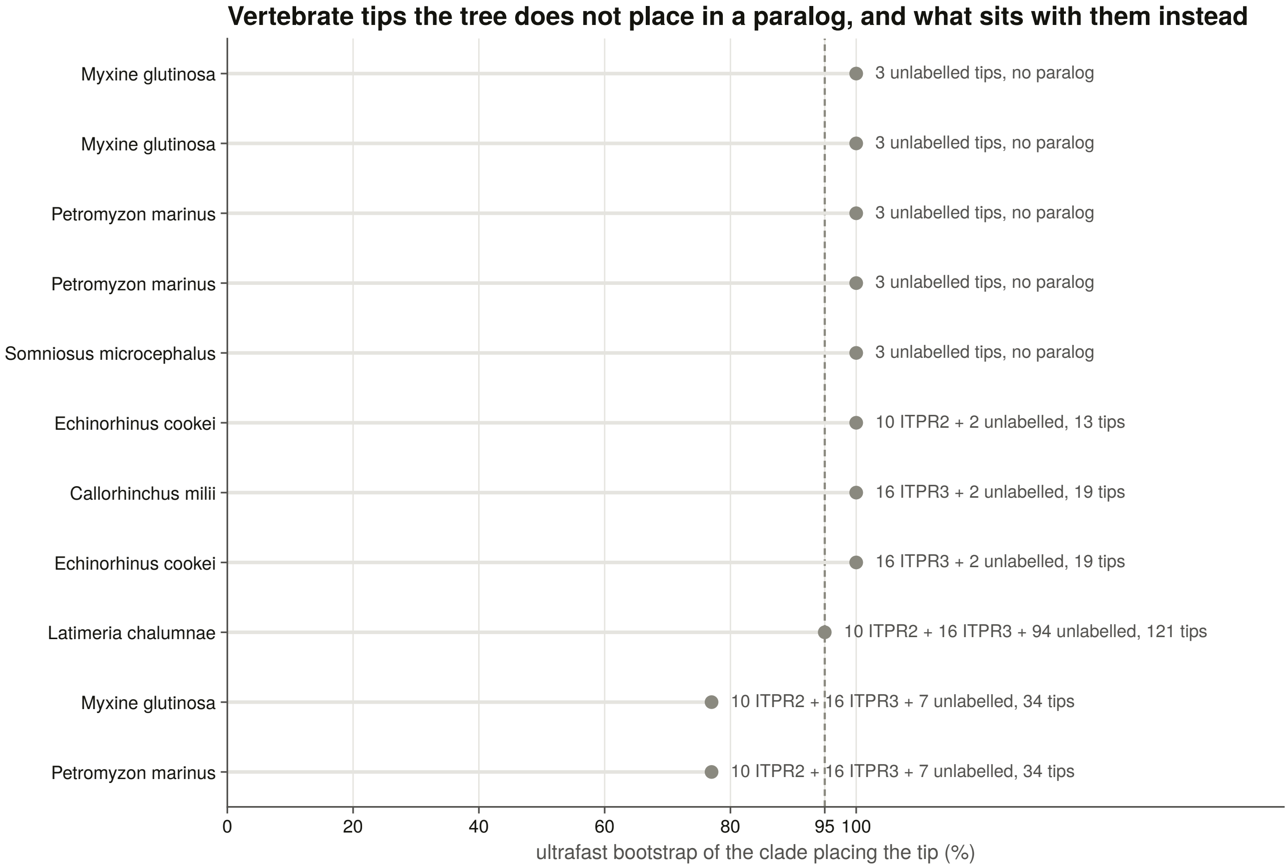


Figure 6.9. Where the tree places each tip against its census label. Thirty-nine agree, none is reassigned, and the five the tree declines to place are named, with reciprocal best hits against an independent database upholding the name for every one of them. The panel matters because a gene tree disagreeing with the annotation would have cast doubt on the paralogue cells every later chapter is built on.

6.8 ITPR2 and ITPR3 are sisters, and the other two arrangements are rejected

Three rooted hypotheses were tested, one per way of pairing the three paralogues, each realised as a constrained search under the same model, and each making all three paralogue clades monophyletic, so that the test compares the sister arrangement and nothing else.

Two traps had to be avoided and the second was walked into first.

Membership must be **tree-corrected**, because a constraint built from census labels asks the test about a topology the data rejects for reasons unrelated to the sister question, and all three hypotheses then fail together. That trap is documented, and the correction here is derived from the tree rather than hand-listed.

The second trap is that the constraint mechanism places freely exactly the taxa a constraint omits, so a tip that is listed, even in a top-level polytomy, is pinned outside every group the constraint declares. The first constraint files named all 134 tips and thereby forced the unlabelled vertebrate tips out of the paralogue clades this tree nests them in, identically in all three hypotheses. The test then rejected every arrangement at a log-likelihood difference of about 1,500, **including the one the tree itself holds at maximal support.** Each constraint now names the three paralogue cores and the outgroup and nothing else, and a constructed control fails any constraint that names a free tip.

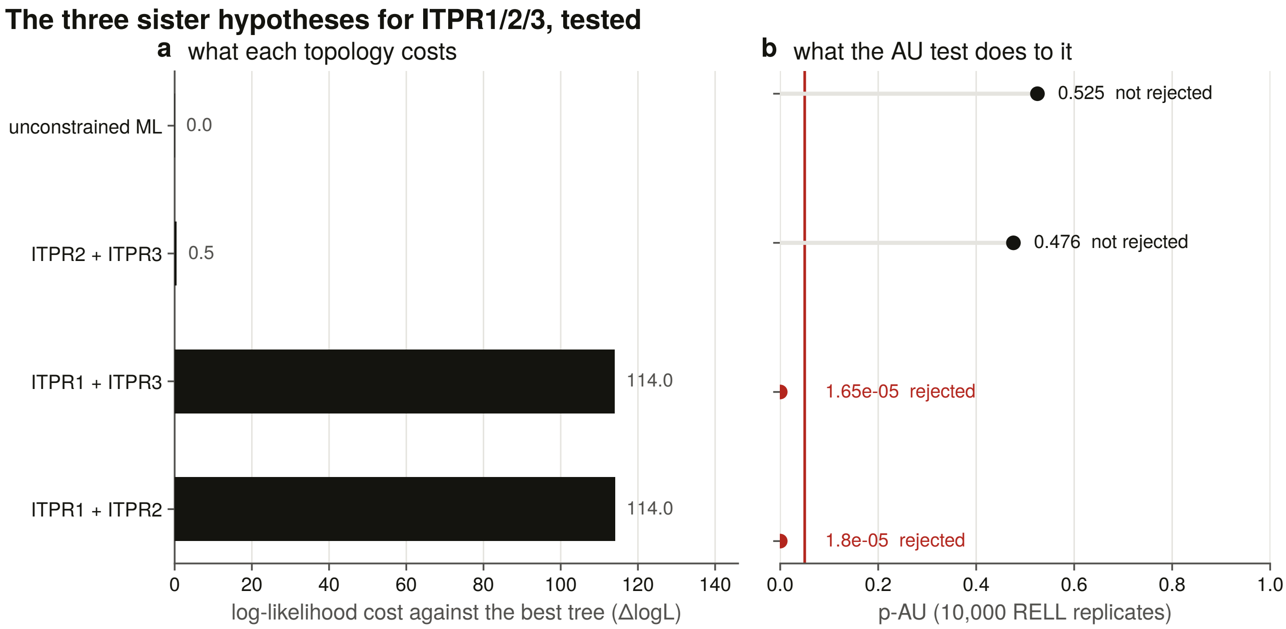


Figure 6.7. The three hypotheses under the approximately unbiased test [54], with log-likelihood difference and p-value on two panels sharing one row of categories rather than on two y-axes. They are different measures on different scales, and a twin axis invites a comparison that has no meaning. This figure is the answer to Chapter 6's central question, and its importance is that it excludes rather than merely prefers. Two of the three possible sister arrangements fall outside the 95 % confidence set of topologies for this alignment, at p below 2×10^{-5}, so the result is not a ranking that a slightly different alignment could reverse. Chapter 7 then shows the genomic neighbourhood pointing the other way, and the two are reported as a disagreement rather than reconciled.

The unconstrained tree groups ITPR2 and ITPR3 at maximal support, and the test agrees. The arrangement pairing ITPR1 with ITPR2 is rejected at $p = 1.8 \times 10^{-5}$, ITPR1 with ITPR3 at 1.65×10^{-5}, and ITPR2 with ITPR3 is not rejected at $p = 0.476$.

The alignment's identity preview predicted ITPR1 with ITPR2 and the test returns ITPR2 with ITPR3, so **the preview is contradicted**, which is the reason it was reported as a preview.

This is an answer the literature did not have. The review's own audit found no published, support-annotated maximum-likelihood analysis with a ryanodine outgroup that fixes the pair.

6.9 The cyclostome loci are two ancient lineages, which neither earlier reading predicted

The alignment left two hypotheses. Three one-to-one orthologues from the vertebrate duplications would put each cyclostome locus inside a different paralogue clade, while a lineage-specific expansion would put all of them together, outside all three.

The tree gives a third answer. **Every one of the six loci sits in a cyclostome-only clade.** There are two such clades, each well supported, and both hold hagfish and lamprey together. So the loci are neither one expansion nor three one-to-one orthologues. They are two anciently separate cyclostome lineages, and a clade spanning the two species is a duplication older than the hagfish and lamprey split rather than a copy either genome made on its own.

The identity preview is contradicted as a reading rather than as a measurement. Identity ranked all six against the same paralogue because it can only measure distance to the labelled clades, and these loci are outside all of them. Which side of the vertebrate duplication each lineage attaches to is Chapter 7's question, and the neighbourhood is what would settle it.

6.10 A stress test on the naming failed, for the right reason

The teleost co-orthologue pairs are in the representative set as a stress test on the naming: if the tree does not recover a species' two same-paralogue copies as sisters, either the naming is wrong or the alignment is.

Neither set comes back as a clade. Both are broken only by other tips of the same paralogue, in that the copies sit in a small clade made entirely of that paralogue, with each copy nearer another species' copy than its own genome's.

That is exactly what a duplication older than the species looks like, and the teleost genome duplication is precisely that. The naming passes. What fails is the expectation that co-orthologues of a shared duplication should be sisters, which was the wrong prior rather than the wrong tree.

6.11 How much of the tree is resolved, and what the model-violation guard says

A sister question answered on a tree whose deep nodes are unsupported is not answered. Of 131 internal nodes, **91 clear both thresholds, which is 69.5 %**, with a median bootstrap of 100 and a median branch-test value of 99.

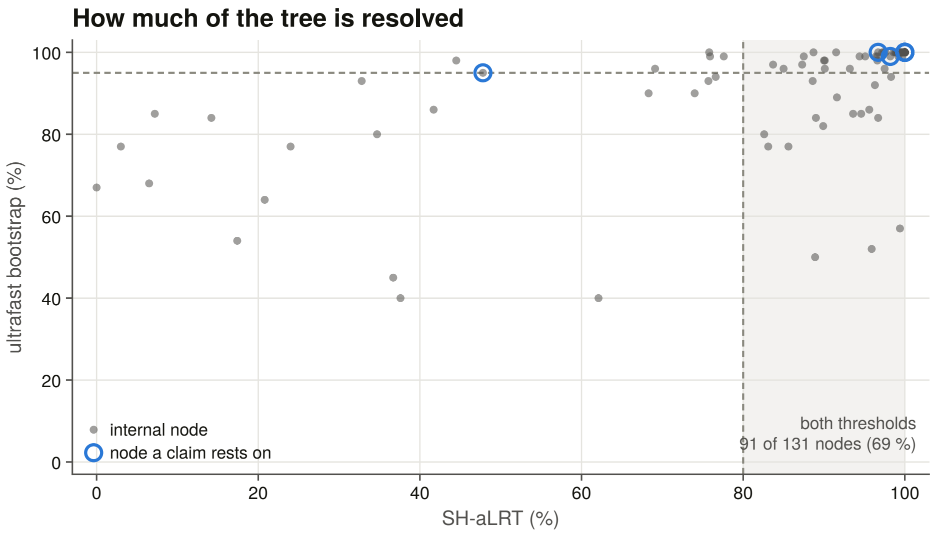


Figure 6.8. Node support drawn as a scatter rather than as two histograms, because the claim is about the joint condition and a pair of marginals cannot show it. A node is trustworthy here only when both branch tests agree. The bootstrap alone is known to be optimistic under model violation, which a 134-tip alignment spanning four kingdoms guarantees, so every well-supported claim in this thesis requires both.

Every node a claim rests on is listed with its own support. The ITPR2 with ITPR3 clade the sister result depends on is at maximal support. The one weakly supported claim node is ITPR1's core clade, meaning the census's labelled set taken bare, at a branch-test value of 47.8, which the tree prefers to group slightly differently. The same clade including the unlabelled tips is at maximal support.

A model-violation guard was run over the whole search. Because the bootstrap is optimistic under model violation, the whole search was re-run with an extra optimisation round on every bootstrap tree, and the same clade questions were asked of that tree, with membership read from a committed table rather than re-derived so that a bookkeeping difference cannot surface as a topology change.

No claim is weakened or lost. Nine of ten clear both thresholds in the guarded tree as well. The tenth was not well supported in the main tree either, so the guard took nothing away. It is listed rather than folded into the count, and the guard does push it further down, which is the honest reading.

The tree reported is also the best tree found, because no constrained search reached a higher likelihood than the unconstrained one. That check exists because a constrained search can reach a better optimum than the free one, and if it did, the tree every table is built from would not be the global optimum.

6.12 What the tree settles, and what it hands to Chapter 7

The tree settles that the three vertebrate paralogues are clades, that ITPR2 and ITPR3 are sisters with the other two arrangements outside the 95 % confidence set, and that the cyclostome loci are an ancient cyclostome-specific pair of lineages rather than orthologues of the three.

It does not settle when the duplications happened, where on the vertebrate tree they sit, or whether they are the two rounds of whole-genome duplication at all. A gene tree gives branching order, it does not give dates, and it cannot distinguish a genome duplication from a tandem one. That is Chapter 7, and it needs a different kind of evidence entirely: the genomic neighbourhood.

7. Where ITPR1, ITPR2 and ITPR3 came from

7.1 A gene tree cannot answer the questions this chapter asks

Chapter 6 established that the three vertebrate paralogues are clades, that ITPR2 and ITPR3 are sisters, and that the cyclostome loci are something else again. It could not say when the duplications happened, where on the vertebrate tree they sit, or whether they are whole-genome duplications at all.

Those are three different questions and they need three different kinds of evidence. Where a gene sits relative to its neighbours is a statement about the genome that no alignment contains. When a duplication happened requires a species tree with dates on it, which is an input rather than a result. And whether a duplication was part of a whole-genome event requires showing that the neighbourhood was duplicated too, which needs a paralogy map rather than a synteny map.

This chapter does all three, and the answers do not entirely agree. Where they disagree, the disagreement is reported rather than resolved by preference, and §7.13 says what it is.

7.2 Reading the genomic neighbourhood needs three decisions and a matched control

Every IP_3 and ryanodine receptor locus of every swept genome was taken with its flanking genes, and every measurement below is a comparison between two such neighbourhoods.

Three decisions make that comparison mean anything.

Loci come from the per-genome sweep output rather than from the ledger. The ledger holds one row per genome by cell and therefore only each cell's best locus. In a teleost a cell holds two genes, so a ledger-driven comparison would score one species' first copy against the next species' second copy and report the mismatch as a synteny result.

Two window rules are committed rather than one, because naming density is not constant: human annotation names 97 % of coding genes and sea lamprey 27 %. One rule takes a fixed number of coding genes each side, and the other takes the nearest informative symbols each side, so that a difference in neighbourhood similarity cannot be a difference in annotation depth.

Three key vocabularies are used. The first is a strict symbol, the second a relaxed one with teleost duplicate suffixes stripped, and the third a **root** key with the trailing digit run removed. The last is the only level at which a whole-genome-duplication ohnologue pair is visible at all, because after 500 million years the two copies almost never carry the same symbol.

Every real comparison is scored against a matched random control. A neighbourhood similarity of 0.21 says nothing on its own, so every pair of loci is scored against control pairs of random coding genes in the same two genomes, under the same window and key rules, which holds the two genomes, their annotation depth, their naming conventions and the normalisation constant fixed. Sampling is seeded per accession, and a contig too short to hold a window is refused rather than allowed to degrade the null.

Eleven constructed negative controls run on every build. The most instructive requires that **two empty neighbourhoods score zero rather than one**, because otherwise two unannotated genomes are a perfect synteny match, and a survey across 309 genomes of very unequal annotation depth would find its strongest signal in its worst data. Another caught a real hash-seeded nondeterminism in a ranked output table.

7.3 The neighbourhood confirms the paralogue cells at two orders of magnitude

Every stage since Chapter 3 has treated a cell's paralogue label as orthology. Nothing before this had tested that with evidence outside the gene itself.

It holds, by two orders of magnitude. Within-paralogue neighbourhood similarity runs at 216 to 413 times the matched null, with 98 to 99.8 % of individual pairs beating their own control.

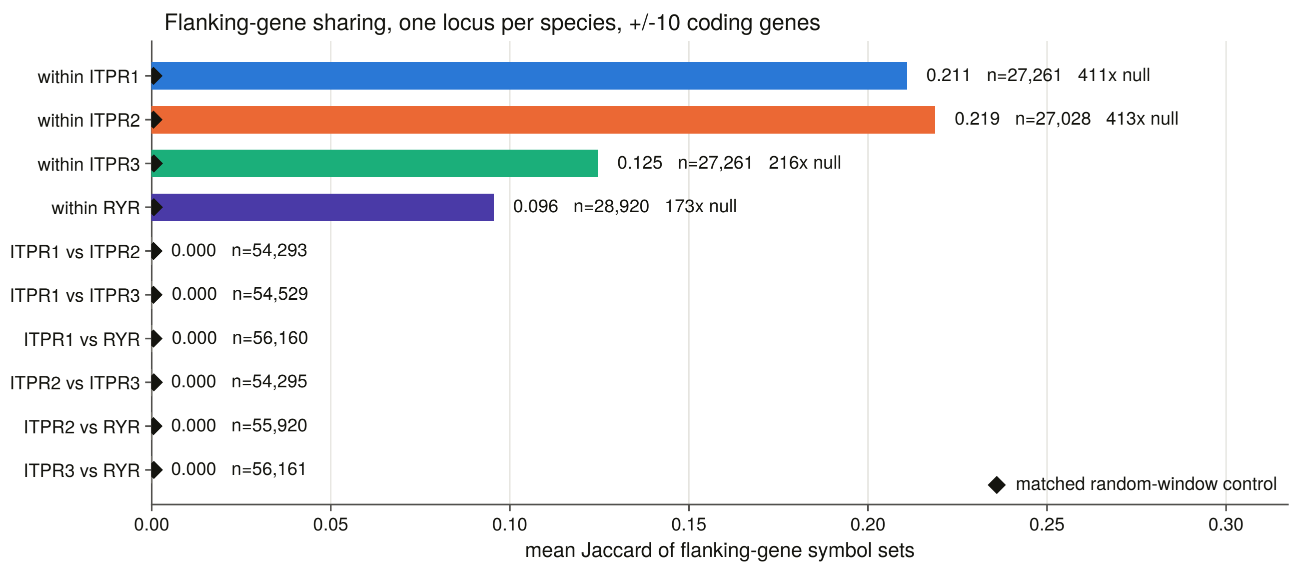


Figure 7.2. Every pair class against its matched random-window control. This is the first test in the thesis of an assumption every earlier chapter had made, since the paralogue cells were assigned from sequence and nothing before this had checked them against evidence outside the gene itself. Neighbourhood similarity within a paralogue runs at two to four hundred times a matched null while every cross-paralogue class sits at or below it, so the cells are orthology groups rather than similarity bins.

Across the family boundary there is nothing. Over 168,241 pairs of IP_3 and ryanodine loci the highest mean similarity of any class is 0.0002, which is below the random-window control itself. An instrument that shares nothing with the sequence evidence returns the same family separation.

The three paralogues are not equal, and pooling hides it. A pooled within-paralogue mean answers two questions at once: whether two mammals share a neighbourhood, which they nearly trivially do, and whether a mammal shares one with a teleost. Only the second is about the locus.

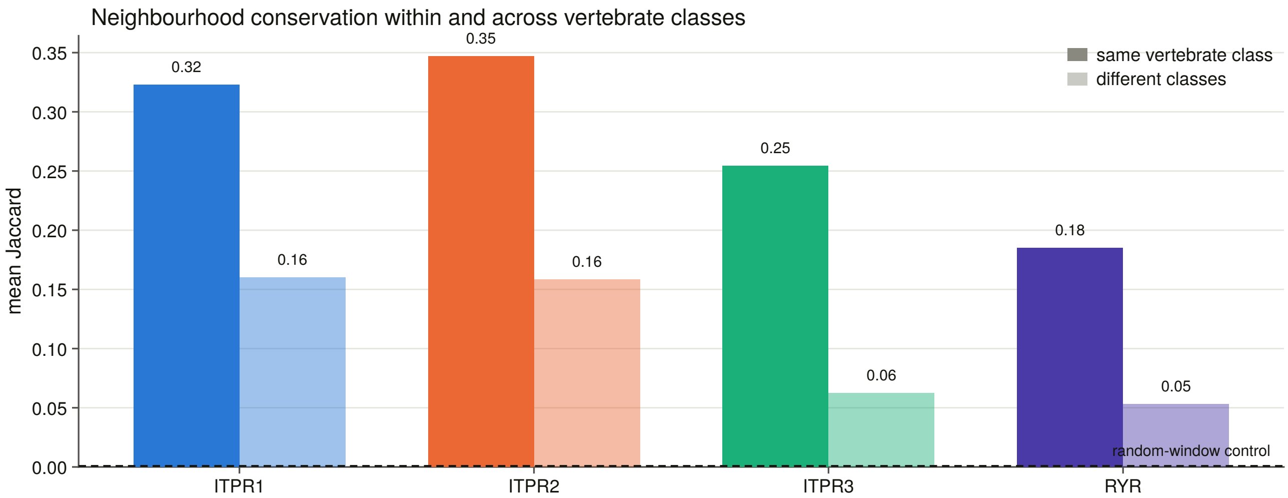


Figure 7.4. How far a neighbourhood travels, split into within-class and cross-class comparisons. Within a vertebrate class the three paralogues span 0.254 to 0.347. Across classes, ITPR3 retains 0.062 against 0.160 and 0.158, which is a 2.5-fold gap and the one asymmetry in this panel that a pooled mean would have hidden.

ITPR3's neighbourhood is the one that does not travel. It is still 157 times its own null and 96.9 % of its cross-class pairs still beat their control, so this is decay rather than absence. It is also worth noting what it is not. ITPR3 is the best-recovered paralogue in the genome sweep and the best-annotated. Ease of finding a gene and stability of the ground it sits on are different properties, and in human that ground is the major histocompatibility region at 6p21.31, one of the most rearranged and most polymorphic neighbourhoods in the genome.

7.4 The surviving paralogon links run through ITPR1

If the three paralogues are the product of whole-genome duplication [23,24], their neighbourhoods should be paralogous rather than identical, meaning the flanking genes should be surviving copies of the same ancestral families under different names. That is exactly what a same-symbol test cannot see, and it is why cross-paralogue symbol similarity above is a flat zero.

At the root-key level two links survive, and each was scored against the rate at which random windows in the same genomes share a root family.

ITPR1 with ITPR3 share a metabotropic glutamate receptor family, present in 42 % of one side and 53 % of the other against a background of 0.45 %, which is an enrichment of 93-fold. **ITPR1 with ITPR2 share a basic helix-loop-helix family**, at 62 % and 85 % against 0.74 %, which is 84-fold.

ITPR2 with ITPR3 share nothing, at any threshold, and nor does any pair of IP_3 and ryanodine neighbourhoods.

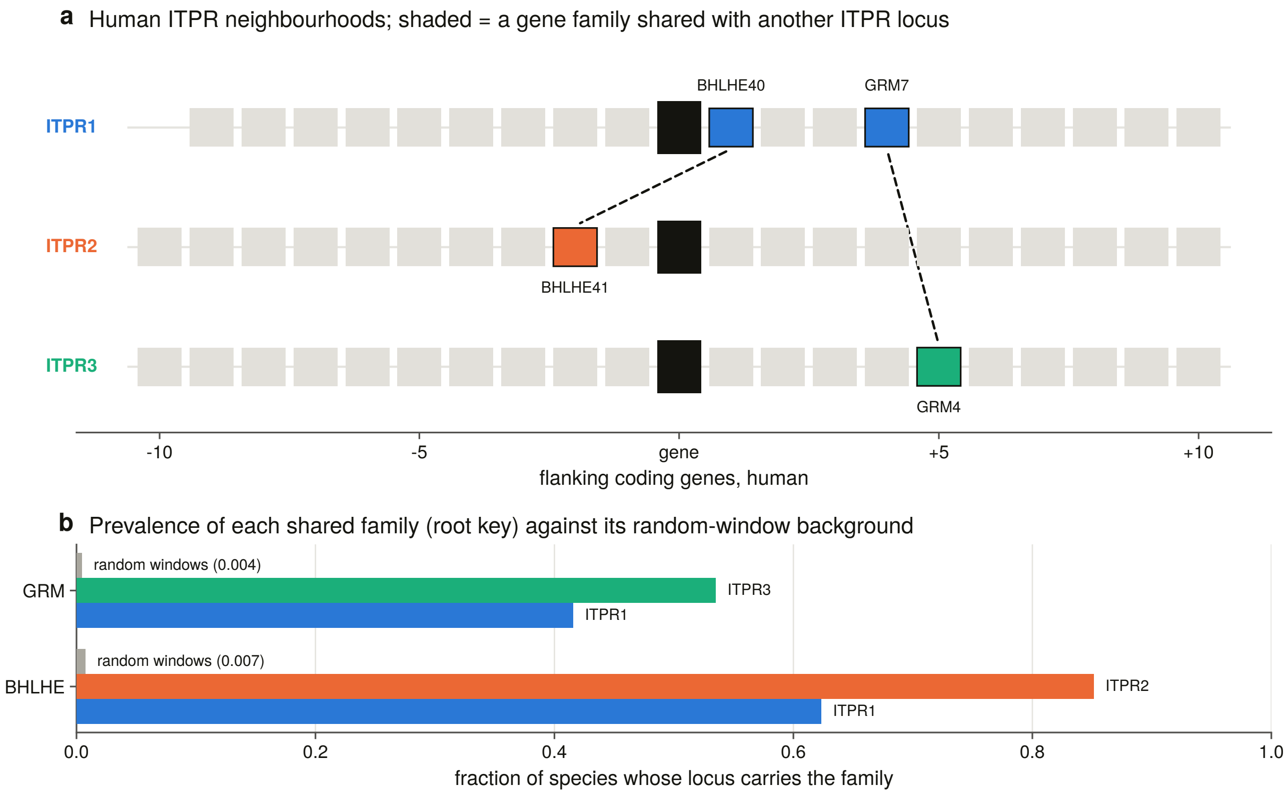


Figure 7.1. The three human neighbourhoods drawn as gene tracks with the shared ohnologue families linked. The tracks are read out of a committed table rather than named in the figure code, so the figure cannot show a gene the data does not have. This figure carries the neighbourhood half of the duplication argument. If the three paralogues arose in whole-genome duplications, their neighbourhoods should be paralogous rather than identical, and the surviving shared families are what remains of that. The importance of the asymmetry is that both surviving links run through ITPR1 and none connects ITPR2 to ITPR3, which is the first of five independent measurements that put ITPR1 on its own side.

That is a clean measurement, since both families are under 1 % of random windows at every threshold from 10 % to 50 %, and it is the first result in this thesis that does not corroborate an earlier one. Chapter 6's tree makes ITPR2 and ITPR3 sisters, and ITPR2 and ITPR3 are the one pair whose neighbourhoods share nothing.

That is not a contradiction, and §7.13 says why at length. In short, a tree estimates the order of duplication, while a retained flanking ohnologue records which copies survived deletion beside each gene, and the quartets left by whole-genome duplication lose flank copies lineage by lineage in a way that carries no memory of the duplication order [55,56]. What can be said is that the synteny does not corroborate the tree's pair, and that whichever pair is sister, the ITPR1 neighbourhood is the one that kept its ohnologues.

7.5 A caller built only from the neighbours places 131 unlabelled loci

The neighbourhood can also be used forwards. If each paralogue has a characteristic set of neighbours, an unlabelled locus can be assigned by its neighbours alone.

A consensus is built per paralogue from the keys carried by a stated fraction of the species holding it, **with the query's own species dropped from every consensus first**, so a locus cannot be scored against evidence it supplied. The caller is calibrated on loci whose paralogue their own assembly's annotation establishes, which is evidence the caller never sees, since it reads the symbols of the neighbours and the truth is the symbol of the gene itself.

The threshold is measured, and the obvious way to measure it is wrong. The sweep reports what each setting buys, meaning the call rate on the confirmed loci, and what it costs, meaning the rate at which random neighbourhoods in the same genomes are called a paralogue. Optimising call rate alone selects the loosest setting on offer, which is how an instrument gets tuned into agreeing with itself. The rule is to maximise the difference, with ties broken toward the stricter setting.

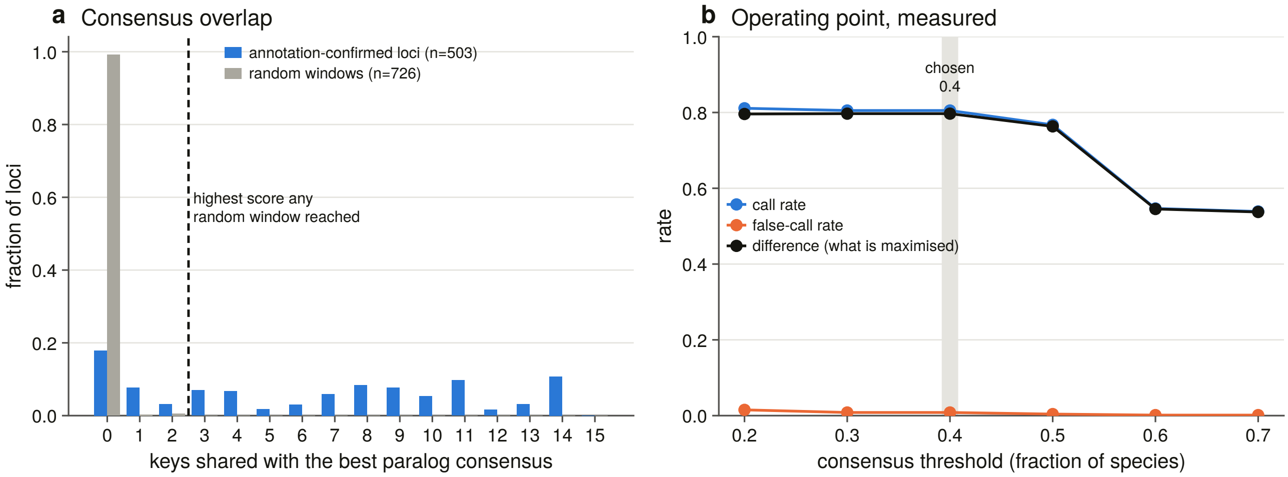


Figure 7.3. The caller calibrated and swept across its threshold range, with the random-window false-call rate drawn beside the call rate. At the chosen setting it is correct on 405 of 405 calls over 503 annotation-confirmed loci, calls 80.5 % of them, and calls 6 of 726 random control windows. Drawing both curves is what makes the operating point defensible, since the two move together and only their difference says what a setting is worth. The caller then places 131 loci that the alignment could not label, each with a null tail probability attached.

Accuracy is 1.000 across the whole range, so it separates nothing and is not what is optimised. It is reported rather than used.

The reporting bar is the **null's own maximum** rather than a probability cut. The highest consensus overlap any random window reached is 2, so a call at or below that score is reported as inside the null however clean it looks. With 726 control windows, a one-in-726 tail is not a rate to build a claim on.

131 loci gain a paralogue assignment that no random neighbourhood could have produced, comprising 73 ITPR1, 42 ITPR2 and 16 ITPR3, each carrying its score, its margin and its null tail probability. Thirty-six of them are second and later copies inside one paralogue cell, and 36 of 36 are placed in the same paralogue as the cell they were filed under. A cell holding two loci holds two copies of one gene rather than a misfiled second gene.

7.6 The neighbourhood cannot reach the cyclostome question

Chapter 6 handed this chapter a specific question: the six cyclostome loci sit in two ancient cyclostome-only lineages, and which side of the vertebrate duplication each attaches to is undecided.

The neighbourhood cannot answer it. Of the six loci, four get a call at all, and **every one of those sits inside its own null.** The other two are no-calls.

That is not a negative result, it is an absence of measurement, and the distinction matters enough to have its own verdict in this project's vocabulary. A neighbourhood comparison taken where 27 to 29 % of the flanking genes carry a symbol is a measurement that did not happen. The corroboration this chapter would most like to have does not exist, and saying so is better than reporting the four inside-null calls as evidence.

7.7 Dating the duplications requires a species tree, which is an input

A reconciliation maps a gene tree onto a species tree and reads duplications off the mapping. The species tree is therefore an input, and it is declared as one rather than estimated here.

It is a hand-curated topology for the 31 vertebrate species the alignment samples, with 29 named internal nodes, every one carrying a literature age, the spread of published estimates, and its source [57,58,59,60,61]. A validator fails the build on a gene-tree species missing from the tree, a taxon with no gene-tree tip, an uncalibrated node, a point age outside its own spread, or any age inversion.

Two things it does that a cladogram with ages written on it does not.

Where the literature does not resolve an arrangement, the node is left a polytomy rather than given a shape the sources do not carry.

Every node carries a stem age beside its crown age. The sampled tree omits every unsequenced lineage, so a node's parent in this tree is usually older than the split that created it. Without the stem age, a duplication on the placental mammal branch would be bracketed at 96 to 319 Ma instead of 96 to 99.

The reconciliation itself uses the **non-binary** duplication rule [62] rather than the familiar binary one, and that is load-bearing rather than fastidious. The binary rule is valid only on a binary tree, and on the support-collapsed root, which is a four-way polytomy, it reads a speciation, which would have reported the collapse as the duplications disappearing. Losses are counted by the standard method [63] and the mapping is the standard least-common-ancestor one [64].

Fourteen groups of constructed negative controls run before anything is written, and two of them caught real errors: the two halves of the non-binary duplication rule, and the support-collapse routine refusing a tree with no support labels. That last exists because a constrained search writes no support values, and the first run dissolved all three constrained topologies into one 134-tip polytomy.

7.8 The two duplications sit on different branches

Twelve reconciliations were run. Five topologies, comprising the maximum-likelihood tree, the three constrained sister hypotheses and the model-violation re-search, were crossed with three variants: all tips, the cyclostome loci pruned, and every node below the support bar collapsed. Three combinations are refused with their reason.

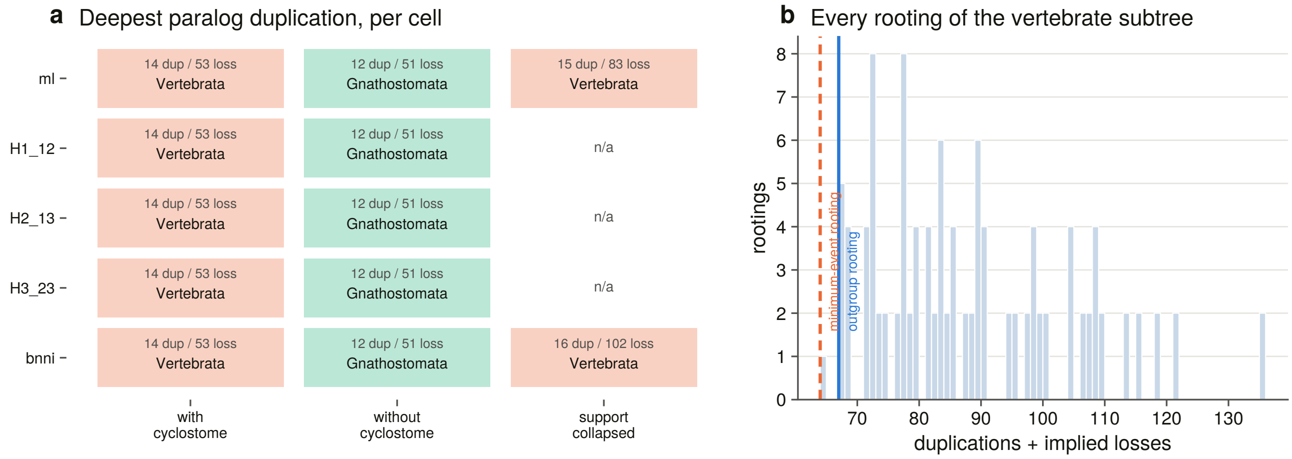


Figure 7.10. The topology by variant matrix, with the deepest paralogue duplication in every cell, and beside it every rooting of the vertebrate subtree scored by total events, with the outgroup rooting and the minimum-event rooting marked. The matrix matters because it separates the placement from the topology it was read off. Twelve reconciliations across five topologies and three taxon treatments put the younger duplication on the gnathostome stem in every cell, so that placement does not depend on which sister arrangement is correct. The rooting panel beside it shows the same for the choice of root.

Read plainly, on the tree with every tip in, two results follow.

The split that separated ITPR1 from ITPR2 and ITPR3 is on the vertebrate stem, meaning older than crown Vertebrata [65,66], whose published estimates run 480 to 615 Ma. Nothing in this tree bounds it from above, and the bracket is reported open rather than closed silently.

The split that separated ITPR2 from ITPR3 is on the gnathostome stem, meaning between crown Gnathostomata at 462 Ma and crown Vertebrata at 563 Ma. It maps there in every cell of the matrix, cyclostomes in or out.

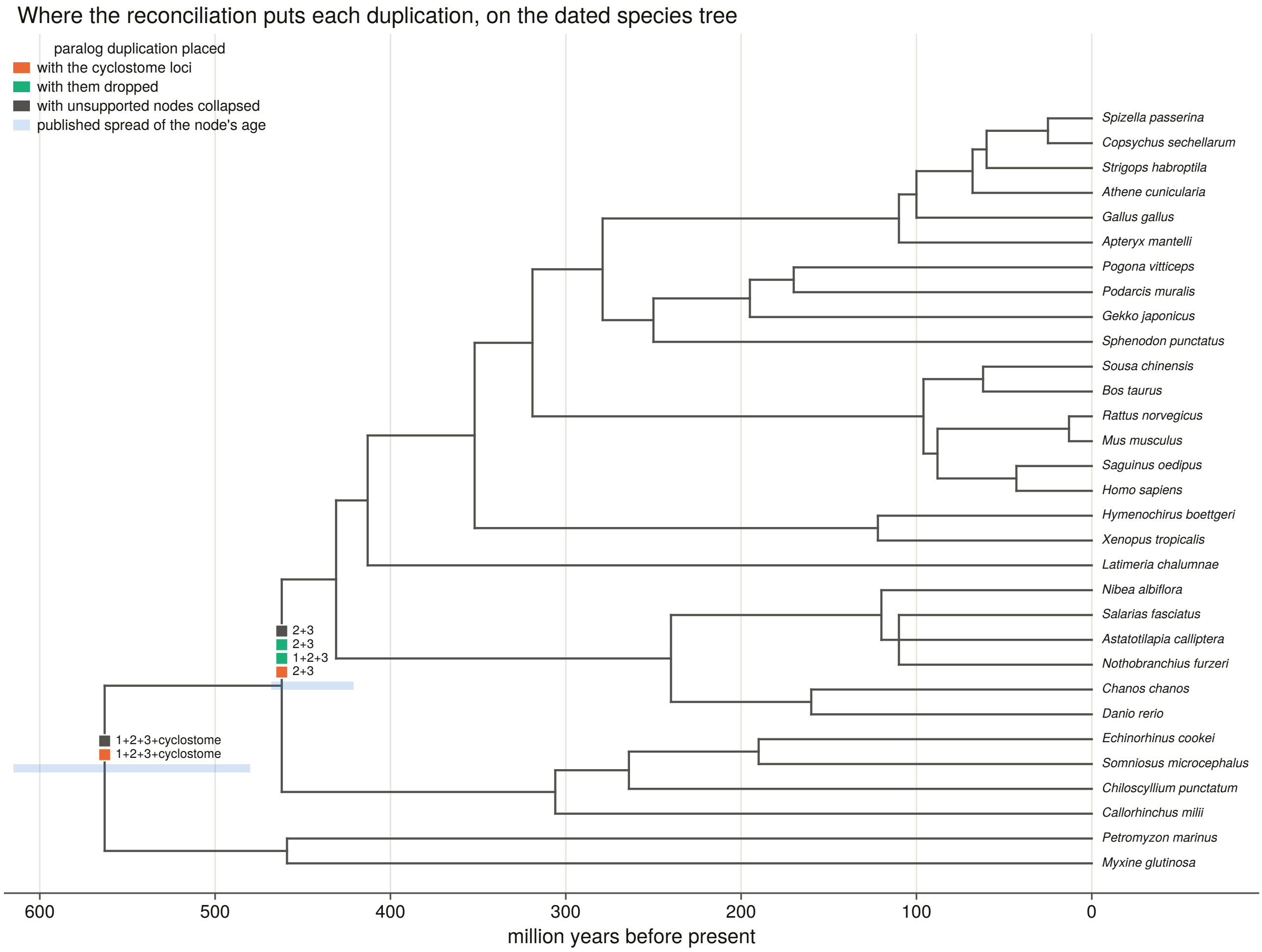


Figure 7.9. The dated backbone on a linear time axis with each calibration's spread drawn as a band, rather than a cladogram with ages written on it, because the result is an interval and a cladogram cannot show one. Placements are deduplicated to one marker per distinct arrangement, since the matrix repeats the same placement across cells and drawing each would make agreement look like weight. The deepest bracket is left open at its old end because nothing in this tree closes it, and the widest disagreement in the whole calibration set sits at exactly the node the older duplication maps to.

The two duplications are not on the same branch. That is the finding, and it is what makes ITPR1 the earlier-diverging copy.

7.9 Two objections to the deep placement were measured rather than argued

Two objections a reader should raise are measured rather than argued.

Does the placement rest on the weakest node in the tree? The gene-tree node separating ITPR1 from ITPR2 and ITPR3 sits at a branch-test value of 17.4 and a bootstrap of 54, which is the weakest node in the whole vertebrate subtree and apparently the one carrying the headline. It is not. Collapsing every node below the support bar dissolves that arrangement and the placement survives, because the root becomes a polytomy in which more than one child still maps into the gnathostome subtree, and the non-binary rule reads a duplication at Vertebrata from that alone. What the collapse costs is resolution, in that two separate vertebrate-stem duplications merge into one, rather than the placement.

What forces the older placement is one hagfish and lamprey pair. Of the six cyclostome loci, two sit in a cyclostome-only clade that is itself nested among the gnathostome paralogues, sister to ITPR2 and ITPR3 at maximal support. A lineage holding both cyclostome and gnathostome copies forces every duplication above it to predate the split between cyclostomes and gnathostomes. The same tip is what keeps the younger event on the gnathostome stem, because that pair is sister to ITPR2 and ITPR3 rather than inside either, so the node uniting ITPR2 with ITPR3 holds gnathostomes only. Had it fallen inside either, that split too would have been pushed onto the vertebrate stem. **The two answers are decided by where one two-tip clade attaches**, and saying so is the honest form of the result.

Are those tips long branches? Cyclostomes are the classic long-branch attraction risk in vertebrate phylogeny, and long branches are attracted to the root, which is exactly where this answer sits. So the objection is measured. Root-to-tip distance for all 57 vertebrate tips puts the six cyclostome loci at **0.96 to 1.04 times the median, ranking 18th to 55th of 57.** Not one is an outlier and two sit in the shorter half.

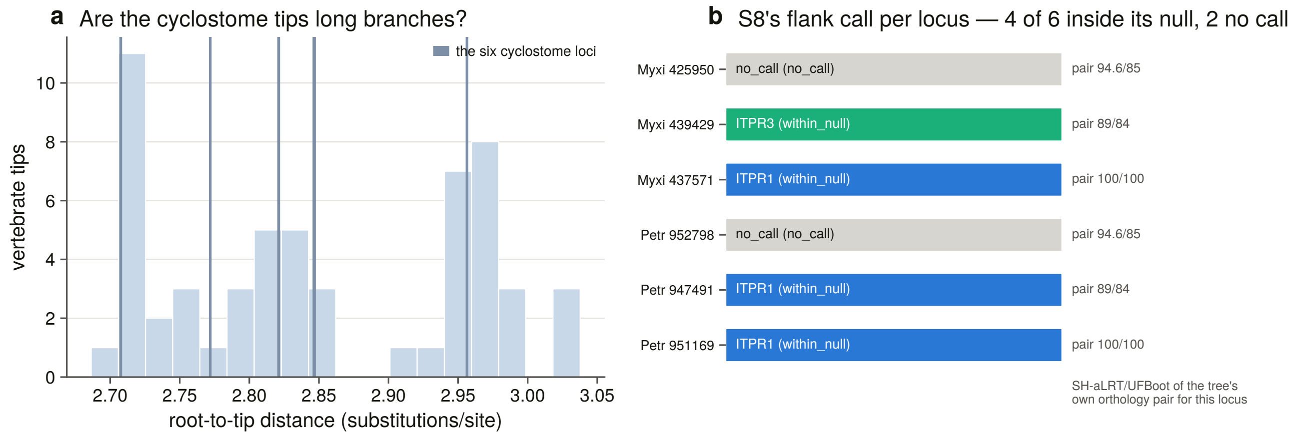


Figure 7.12. The six cyclostome loci marked on the distribution of root-to-tip distance across the tree, and beside it the independent neighbourhood call for each locus with the pair support the tree gives it. The first panel removes the standard objection to this placement by measurement rather than by argument. The second matters because the neighbourhood knows nothing about the alignment, so its agreement is corroboration from an instrument that could have disagreed.

That negative result is the one that matters most here. It is the reason the placement is offered as a finding rather than as a caveat.

A loss count from this analysis is not a loss count. The reconciliation implies 47 lost cells. Checked one at a time against the genome ledger, 26 of them are the paralogue present in the genome and absent only from the 134-tip sample, and none is corroborated.

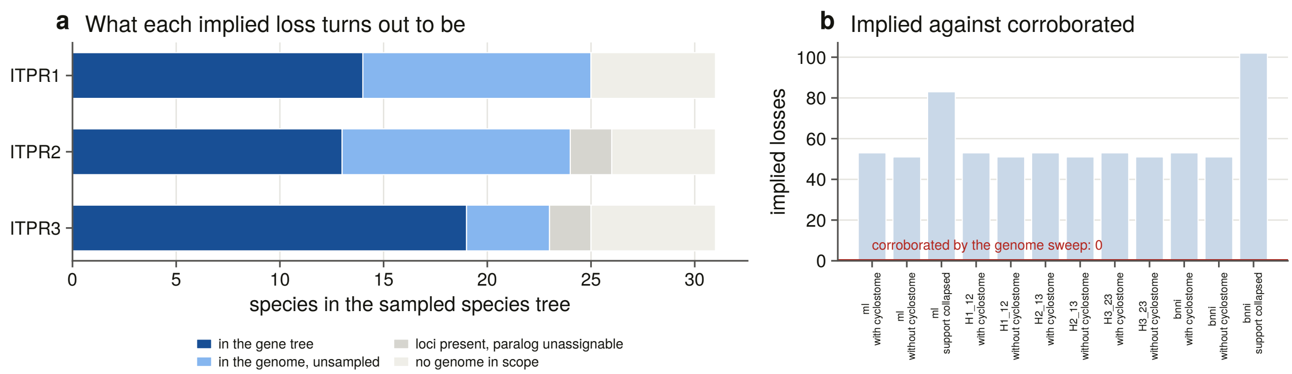


Figure 7.11. What each implied loss turns out to be once it is asked of the genome ledger, and the implied count in every cell of the matrix against the number the genomes corroborate. The figure is the reason a loss count is not read off a reconciliation: a reconciliation over a representative sample counts sampling, not biology. That is why Chapter 9 counts losses from 309 genomes instead, and why the two chapters' numbers are not alternative estimates of the same quantity.

7.10 Testing whole-genome duplication needs dated paralogy, not synteny

Neighbourhood similarity can show that two blocks are related. It cannot show when they became related, and that distinction is what separates an ohnologue pair from the two rounds of duplication from an older duplication whose two copies happen to sit beside these genes.

So the human paralogy map was taken from a comparative-genomics database [67] through a **pinned dated archive** rather than the rolling one [68]. That is the same reproducibility discipline applied to a database release, since the rolling host follows the release cycle and a re-run would score the same windows against a different tree. The archive's own registry is committed beside the map, so the release is evidence in the results directory rather than a sentence in a report.

Three rules make it a test rather than a description.

Every IP_3 and ryanodine receptor gene leaves every window. Counting the ITPR1 and ITPR2 paralogy as evidence that their blocks are paralogous is circular, and leaving a ryanodine receptor in an IP_3 window would import the control's answer into the test.

The null is drawn from real genomic windows at the matched gene count, so the clustering of gene families that a shuffled gene set throws away is kept, and it inflates the real window and the null alike.

The links are dated. The database's duplication node is what separates an ohnologue pair from the two rounds from an older duplication, and it is read in two nested vocabularies.

Dating the links changed the answer. Two paralogue links survive between the three human neighbourhoods at every window size, and they are exactly the two families the neighbourhood analysis found by a completely different instrument. But one of them, the basic helix-loop-helix pair linking ITPR1 and ITPR2, is dated to Opisthokonta. It is a pair far older than the vertebrates whose two copies happen to sit beside these genes. It is real paralogy and it is not an ohnologue pair from the two rounds. The metabotropic glutamate receptor pair linking ITPR1 and ITPR3 is dated to Vertebrata, which is what the two rounds mean.

The single human genome is underpowered, and the report says so twice. At the primary window each family retains exactly one vertebrate-dated ohnologue pair between two of its three neighbourhoods, each clears its own permutation null when asked once per family rather than three times per pair, and neither survives correction across all 135 tests the stage ran. The ryanodine trio's two-round origin is not in question, so what the human test measures is the instrument's ceiling on one genome rather than a difference between the families.

The replication is where the answer is. The same human map, asked of the flank sets in every swept genome, against matched random neighbourhoods in the same genomes drawn by the same sampler, gives the following. The ITPR1 neighbourhood carries a paralogue of the ITPR2 neighbourhood in 141 of 175 genomes, which is 80.6 %, and of the ITPR3 neighbourhood in 89 of 152, against 2.6 % of matched random windows. **ITPR2 and ITPR3 sit at exactly the background rate, at 2.7 % against 2.6 %, $p = 0.554$.**

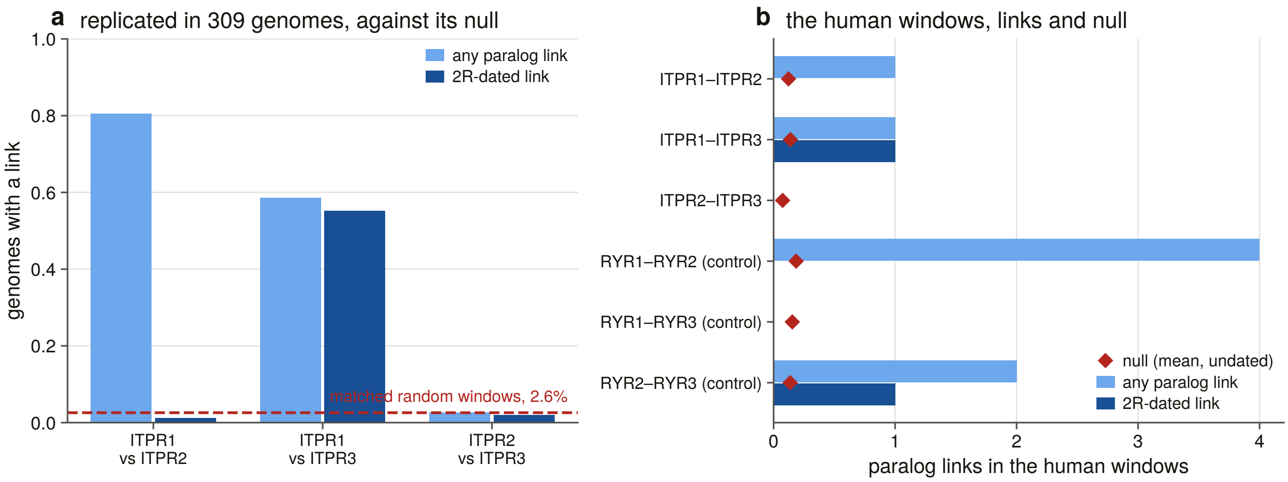


Figure 7.6. The two-round test with its null drawn across the bars rather than quoted in a caption, and the ryanodine trio run through the identical instrument in the same genomes beside it. What carries the claim is not the human window but the replication across 309 genomes against matched random neighbourhoods, and what makes it readable is the sister family, whose two-round origin is not in question and which behaves the same way through the same instrument.

The dated column splits the two links cleanly: the ITPR1 to ITPR3 link is vertebrate-dated in 84 genomes and the ITPR1 to ITPR2 link in 2.

The fourth slot cannot be found, and current reconstructions of what the vertebrate duplications left behind [69,56] are what a positive answer would have to be read against. A genome-wide block scan returns no block that both carries no family gene and looks like these blocks' missing sibling. With one dated ohnologue pair surviving between the blocks that do carry a gene, a block that lost the gene too has nothing left to be recognised by. That is a limit of the evidence rather than a claim that no fourth slot existed. The quartet test additionally flags its circular pairs in the data, because a window tested against its own top-scoring block tests the selection rather than the quartet, and all six such pairs are significant by construction.

7.11 The teleost duplication doubled ITPR1 and only ITPR1

Ray-finned fish underwent a further whole-genome duplication [70,71], and the copy-number landscape says that is where all this family's variation is.

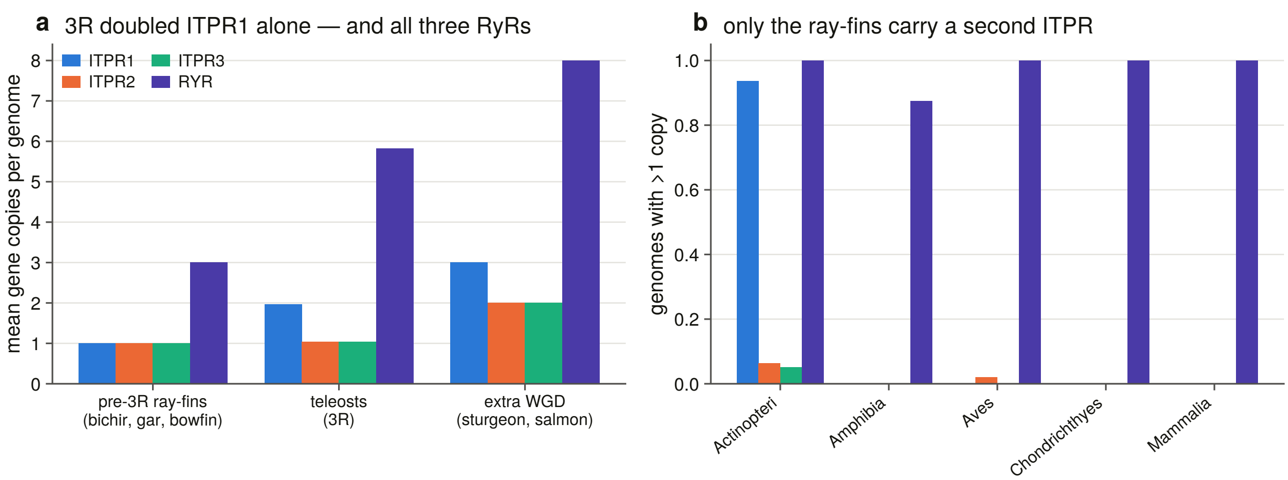


Figure 7.5. Copy number grouped by whole-genome-duplication status rather than by taxonomy, because the result is a contrast between lineages defined by which duplications they have been through, and a per-class bar buries it inside the ray-finned fish. Grouping by duplication history rather than by taxonomy is what makes this figure a test. The prediction is specific: lineages that diverged before the teleost duplication should carry one copy of each paralogue, teleosts two of one, and lineages with a further duplication more again. All three hold, and a second copy in the outgroup or an undoubled sister family would each have falsified the reading.

Above the contiguity bar, every non-teleost gnathostome genome in the sweep carries exactly one of each paralogue and three ryanodine receptors.

Five predictions were checked.

Pre-duplication lineages carry one copy. The bichir, gar and bowfin carry 1.00 of each and 3.00 ryanodine receptors, which is the gnathostome state.

Teleosts carry two of one. They average 1.97 ITPR1, 1.04 ITPR2, 1.04 ITPR3 and 5.82 ryanodine receptors, with 97.3 % of 73 genomes carrying more than one ITPR1.

Lineages with a further duplication carry more again [72]. Sturgeon and salmon carry 3.00, 2.00, 2.00 and 8.00.

A second copy appearing in the outgroup, or a teleost ryanodine count that had not doubled, would each have killed the reading, and neither does. **The ryanodine control is what makes the singletons interpretable**: the duplication duplicated ITPR2 and ITPR3 as surely as ITPR1, and the sister family in the same genomes kept all six of its copies. So the teleost ITPR2 and ITPR3 singletons are a statement about retention rather than about the sweep's ability to find a duplicate.

The copies are not tandem, and this is measured rather than asserted. Of 71 teleost genomes with two copies above the bar, 20 put them on different contigs and the rest sit a median 8,870,578 bp apart, with the closest pair anywhere at 535,374 bp.

Both copies keep part of the ancestral neighbourhood and between them account for it. Of 49 two-copy genomes with both copies flanked, 46 keep ancestral symbols on both copies against a tetrapod reference and 47 against a pre-duplication ray-finned one, and in 45 and 46 respectively **the two sets are disjoint**. Disjoint is the load-bearing word: the copies do not merely each resemble the ancestor, they partition it, which is what reciprocal gene loss after one duplication produces and what a pair of independent later duplications would not.

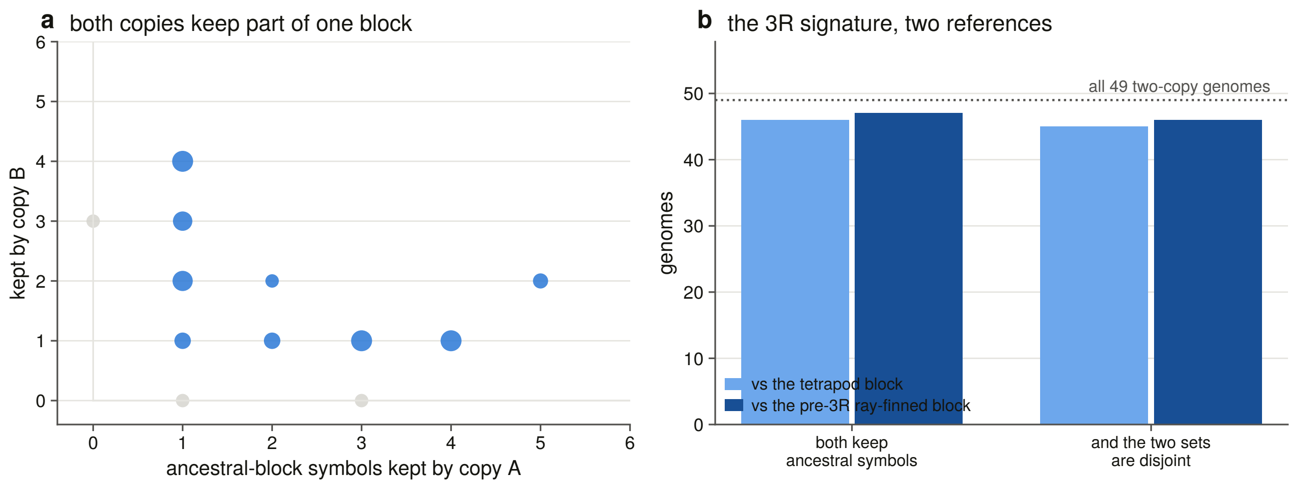


Figure 7.7. Double-conserved synteny, with the two copies plotted against each other so that disjointness is a geometric fact on the figure rather than a number in a table. The load-bearing word in this figure is disjoint. Two copies that each resemble the ancestral neighbourhood could be two independent later duplications, but two copies that partition it between them are what reciprocal gene loss after a single duplication produces. That distinction is what turns a copy count into a claim about one shared event.

Two references were used rather than one, because teleost gene symbols diverge from tetrapod ones even after normalisation, so a tetrapod consensus under-counts. Both give the same answer.

7.12 The two teleost copies are one ancestral duplication, not many

The five predictions are all consistent with a single duplication in the teleost ancestor. They are also consistent with a series of independent lineage-specific duplications, and separating those is the sixth check.

The obvious statistic has no power, because each genome's copy labels are arbitrary, so a matched-versus-crossed comparison measures nothing. Instead, anchors from different orders assign every genome independently, and the question is whether the anchors agree.

705 of 705 assignments agree, across six anchors from six orders. Under independent lineage-specific duplications the anchors carry no shared information and the statistic sits at 0.5, and a constructed control builds exactly that case and requires the statistic to land there, so the perfect agreement is a measurement rather than a property of the routine.

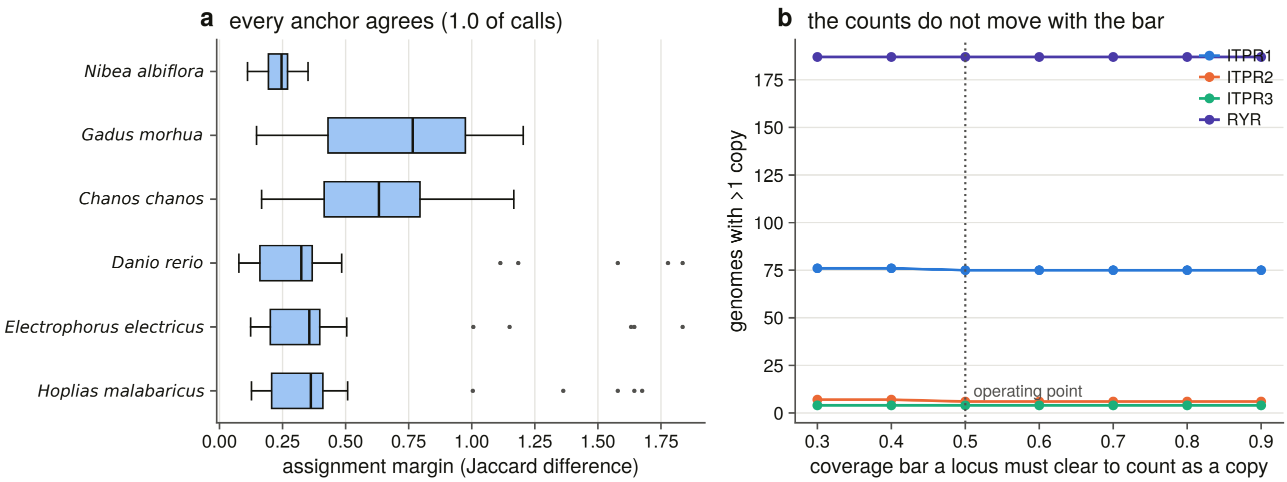


Figure 7.8. Cross-anchor block identity, and the sensitivity of every count to the coverage bar. Seven bars are scanned, because a duplication claim that survives only one bar is a claim about the bar. The anchor panel is what separates one ancestral duplication from a series of lineage-specific ones, which the copy counts alone cannot do, because anchors from six different orders assign every genome independently and would carry no shared information if each lineage had duplicated on its own.

It is corroborated by evidence of a different kind. Which bait won each copy is a sequence call made with no neighbourhood input at all, and in the six genomes whose two copies won different baits the sequence call and the neighbourhood block agree six times out of six. The reference for that check is chosen as the first anchor whose own two copies won different baits, because anchors are ranked on flank richness and taking the first one blindly makes the check unrunnable.

7.13 What this chapter settles, and the disagreement it leaves standing

Six things are settled. The three paralogue neighbourhoods are shared within a paralogue at 216 to 413 times a matched null and share nothing across the family boundary. The blocks are paralogous and the paralogy runs through ITPR1. One of the two surviving links is vertebrate-dated and the other is far older. The ITPR1 against ITPR2-plus-ITPR3 split sits on the vertebrate stem and the ITPR2 against ITPR3 split on the gnathostome stem, and the placement survives collapsing every weakly supported node. The teleost duplication doubled ITPR1 and only ITPR1, in 97 % of teleost genomes, while doubling all three ryanodine receptors in the same genomes, and the two copies are one ancestral duplication.

One disagreement is left standing. The tree makes ITPR2 and ITPR3 sisters. The neighbourhood makes ITPR1 the block that kept its ohnologues with both of the others, and ITPR2 with ITPR3 the one pair that retains nothing above background. Two instruments give two answers, and the project's verdict is *not corroborated* rather than *contradicted*.

The reason is that they are not the same quantity. A tree estimates the order in which copies diverged. A retained flanking ohnologue records which copies survived deletion beside each gene, hundreds of millions of years later, and quartets from whole-genome duplication lose flank copies independently of the duplication order. The reconciliation adds a third view that is consistent with both, because it makes ITPR1 the earlier-diverging copy, which is the copy whose neighbourhood kept a shared family with each of the other two. That is suggestive rather than a test.

Stating that as a disagreement rather than resolving it is a decision. The alternative, which is quietly preferring whichever instrument agreed with the headline, is how a project of this size talks itself into a wrong answer.

Three things are not settled. The chapter does not settle which of the two duplication rounds made which split, because the dated link is dated to Vertebrata, which is both rounds, and separating them needs the cyclostome side of the quartet, which §7.6 found underpowered. It does not settle whether the quartet had a fourth slot. And it does not settle **why ITPR1 alone kept its teleost duplicate**, because the observation is clean and the cause is not in this chapter's evidence. Chapters 10 and 11 are where a dosage or subfunctionalisation argument would have to be made.

8. The gene itself: exons, introns, and what a fragmentary annotation is

8.1 A failed background claim left a question about gene size

Chapter 2 struck a claim: that each IP_3 receptor is a 58-to-60-exon gene spanning hundreds of kilobases. The exon counts turned out to be 62, 57 and 58, and the spans 354, 498 and **76** kb. Three paralogues encoding proteins that differ in length by 3 % occupy genomic spans that differ by a factor of six and a half.

That leaves a question with two possible answers. Either the exon structure differs between the paralogues, or the introns do, and only one of those is a statement about how the gene has evolved. Answering it across 309 genomes also answers a second question the annotation audit needs: when a database calls a locus fragmentary, is that a real gene boundary or an annotation that stopped somewhere nothing splices?

8.2 Reading exons out of a genomic alignment has four traps

The measurement is made on the sweep's own alignments rather than on annotations, because Chapter 4 found that most of these genes have no usable annotation. Four things a naive reader of that output gets wrong, each with a constructed control that fires on it.

The model keys collide. The aligner numbers its alignments from one per run, so the two assemblies large enough to be processed in chunks repeat every identifier. A reader that keys models the way the aligner names them merges the blocks of two different genes, and the merged gene has more exons and longer introns than either.

Gene order is not coordinate order. Blocks come in target order, and an intron computed as the next block's start minus the previous block's end reverses every minus-strand gene.

Splice dinucleotides must be read strand-corrected. Reading an intron's two ends without reverse-complementing both turns every canonical minus-strand intron into a non-canonical one, which is a systematic signal that would discredit exactly the alignments this chapter is defending.

A block is not an exon. The aligner emits blocks, and some adjacent pairs are separated by a gap too small to be an intron.

Measurement also shows that **frameshifts sit inside blocks rather than between them**, which means the merge this chapter needs is a sub-intron-gap merge rather than a frameshift merge.

8.3 What counts as an intron was measured against the genome, not chosen

Rather than choose a minimum intron length, every block pair was binned by gap size and scored by its splice dinucleotides, which is evidence the alignment score did not produce.

The calibration then **refuses to hand out the threshold it was written to produce**. The sub-intron population is empty, meaning there is no population of small gaps that fail the splice test. So the floor goes at the smallest gap the genome calls spliceable, at 10 bp, rather than at a declared 30 bp fallback, which would have merged the ten junctions between 10 and 29 bp that read as canonical splice pairs.

8.4 The aligner places a canonical splice pair at 99.89 % of junctions

Before measuring exon structure with an aligner, it is worth knowing how often the aligner puts a boundary where no splice site is. **99.89 % of 112,254 junctions read as a canonical splice pair.**

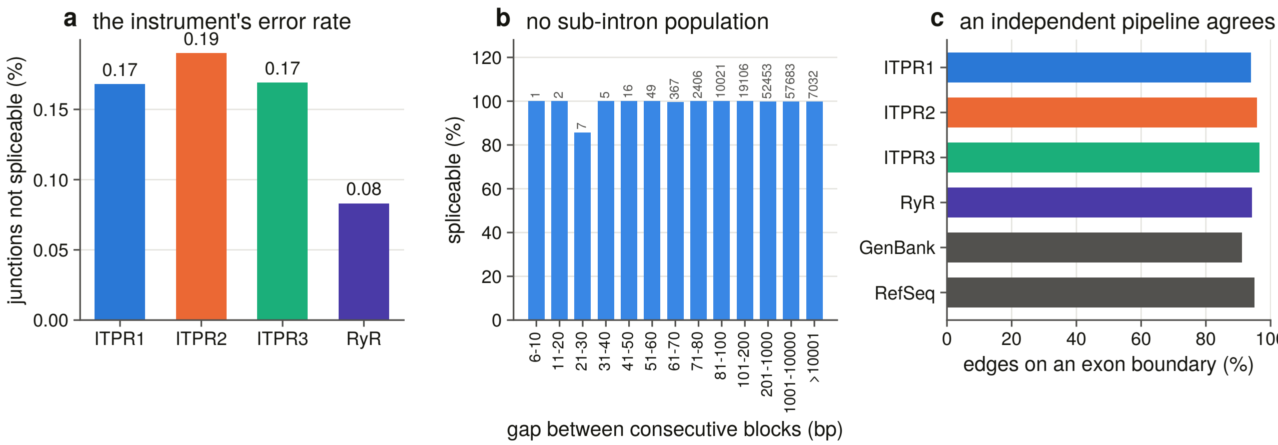


Figure 8.3. The aligner's junction error rate, drawn on the complement, because at 99.9 % agreement a bar of the agreement is four full bars and shows nothing. Each gap bin carries the count it rests on. This figure is the instrument's own error rate, and it has to be measured before exon structure inferred from alignments can be used for anything. At 99.89 % of junctions reading as a canonical splice pair, the boundaries the shared-intron test depends on are the aligner's rather than an artefact, and the independent check against 188,146 annotated edges from 164 genomes is what rules out a systematic bias the dinucleotides alone could not see.

That is corroborated by a second, independent pipeline: the sweep's boundaries against **188,146 annotated coding-sequence edges from 164 genomes**, at 94.5 % exact agreement. An earlier chapter made the same comparison for two loci against 35 and 40 genomes, and this is that check generalised.

8.5 One coordinate frame lets the three genes be compared, and it is checked

An exon boundary is a position in the bait's numbering, and the sweep used 38 baits, so comparing an ITPR1 gene's boundaries with an ITPR2 gene's needs a frame both can reach. That frame is Chapter 6's alignment, because all three human paralogues and a human ryanodine receptor are tips of it, so a boundary travels from the bait to its cell's human reference and then to an alignment column.

Ninety-eight frames were built and none refused.

The frame is checked rather than assumed. All fourteen residues measured on the structure, meaning the ten IP_3 contacts, the two selectivity-filter residues and the two gate residues, each in its own paralogue's numbering, land in the **same alignment column in all three paralogues**. A frame that had slipped anywhere in the pore or the ligand core would fail there, and a slipped frame is exactly the error that makes a shared-intron count look like a result. A constructed control shifts one paralogue's aligned row by a single column and requires the test to refuse.

Twenty-one constructed controls run before anything is written, and the suite was mutation-tested on six deliberate rule breakages, all six caught. Four of this chapter's rules return a good-looking number when they are wrong, which is why the suite is as large as it is: the model-key collision, the strand branch, an inverted intron-phase convention that swaps two phases everywhere without changing a single count, and the duplication detector discussed in §8.9.

8.6 The exon count is conserved and the packaging is not

Across 1,378 genes in 189 vertebrate genomes, the three paralogues sit at **57 to 58 coding exons** with median genomic spans differing by **4.2-fold**, and coding sequence lengths of 8,250, 8,100 and 8,008 bp. These are *coding* exon counts, which is why they sit below the 62, 57 and 58 that Chapter 2 read off the human canonical transcripts: a protein-to-genome alignment sees only the translated part of the gene, so a purely untranslated first or last exon is invisible to it and counted by the transcript annotation. What differs between them is how much intron is wrapped around nearly the same protein.

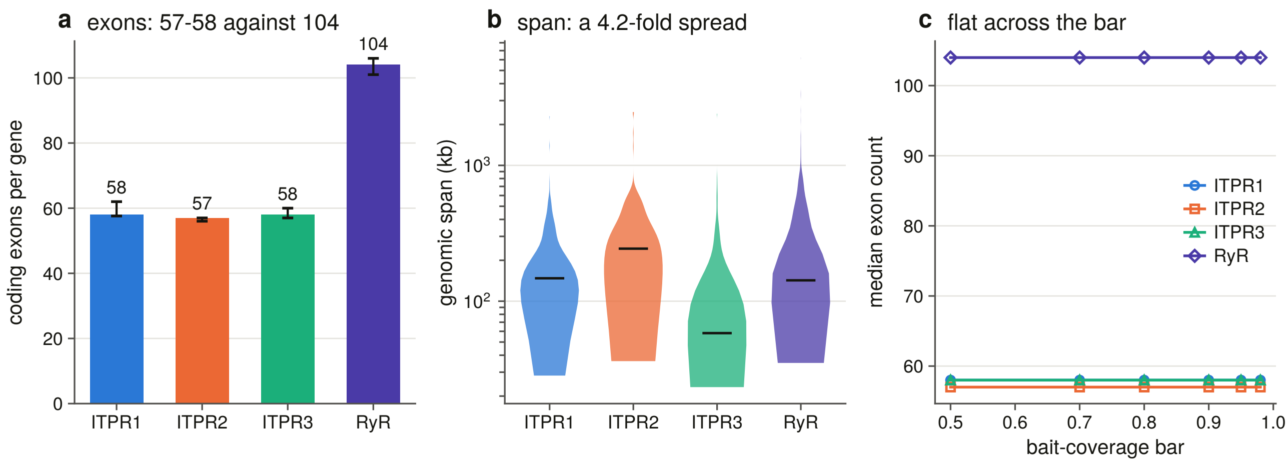


Figure 8.1. Exon count and genomic span on separate axes, with span logarithmic and count not. The separation matters because the result is a contrast between two quantities measured on the same genes, one of which is conserved and the other of which is not, and a shared scale would hide it. Fifty-seven to fifty-eight coding exons across 189 genomes, wrapped in genomic spans differing four-fold, means the paralogues inherited one coding architecture and then diverged in intron content alone.

The ryanodine receptors, measured through the identical instrument in the same assemblies, carry **104 exons over 14,910 bp** of coding sequence, which is nearly twice the gene in both dimensions, at almost exactly the same mean exon length of 144 bp against 138 to 142. The control tells us the instrument is measuring gene structure rather than a property of this family's alignments.

Every count is recomputed at six coverage bars, because a count that survives only one bar is a claim about the bar.

The comparison is paired inside one genome. Intron size, assembly quality and annotation completeness all scale with the assembly, so every cross-paralogue comparison uses one locus per genome per gene, sign-tested with ties dropped and counted, and the whole family of tests corrected together.

Two things fall out. The exon counts differ by one to three exons and the spans by tens of kilobases, in the same genomes. That intron content, rather than coding structure, is what varies is worth holding against the evidence that intron length is itself under selection, with shorter introns in highly expressed genes [73], although the explanation usually offered for that pattern has been contested [74]. And the ordering of the spans is consistent gene by gene rather than an artefact of averaging, in that ITPR3 is the shorter gene than ITPR1 in 161 of the 182 genomes carrying both.

8.7 The three paralogues share the same introns in the same places

An intron's position here is a pair: the alignment column its upstream exon ends in, and its phase. Intron positions are among the most durable characters a gene has, conserved across kingdoms and gained and lost slowly enough to carry information about deep relationships [75,76], which is what makes this comparison worth making at all. Both halves of the pair are needed. Two paralogues can carry an intron between the same two residues in different frames, which is not one ancestral intron, and a column alone would call any two introns in the same region shared.

Within a paralogue, each carries a core of about 48 to 49 intron positions present in at least 90 % of the genomes that have the gene, out of 175 to 185 positions seen anywhere. The ryanodine receptors have their own core of 91, on the same scale relative to their 101 introns per gene. So the architecture is not merely a count that happens to be stable. It is the same introns, in the same places, across the vertebrates.

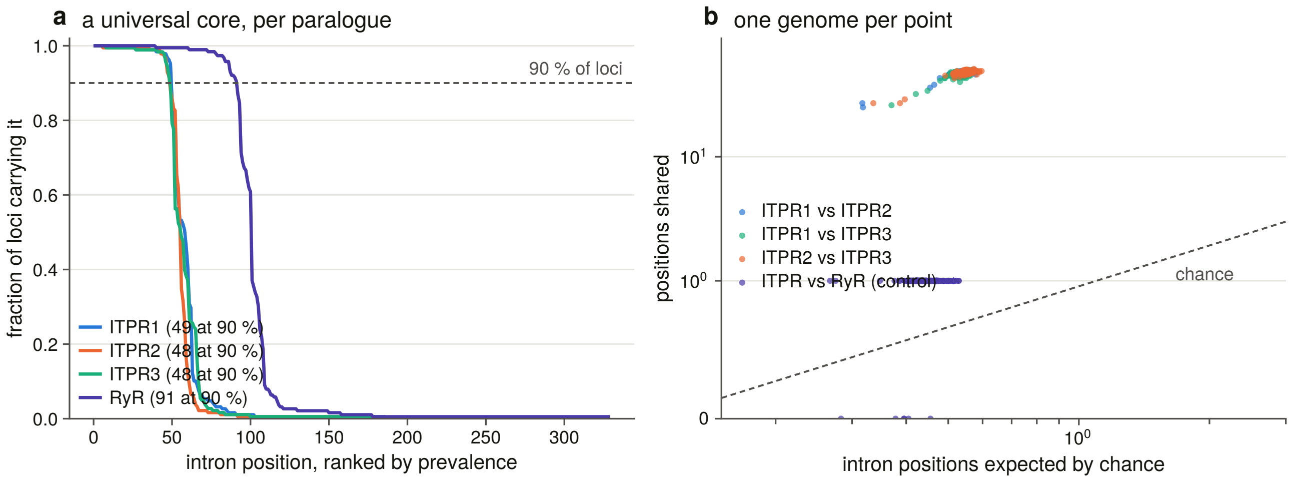


Figure 8.2. Intron positions by alignment column and phase, per paralogue. The columns line up, and that alignment is what the enrichment in §8.7 is measured on. Plotting phase as well as column is what keeps two introns that fall between the same two residues in different reading frames from being counted as one inherited intron.

Between paralogues is the test this chapter exists for. Two genes with about 58 introns each spread over about 2,700 aligned residues will share some positions by chance, so every comparison is paired within a genome and scored against a null in which each of one gene's introns is placed independently and uniformly on the columns both loci have residues in, keeping its own phase. That makes the match count a Poisson-binomial whose upper tail is exact, with no random number generator and no replicate count, and a seeded permutation without replacement is run beside it as a cross-check.

The three paralogues share 46 to 49 of their roughly 58 intron positions in every genome that carries them, at an enrichment of 85 to 88-fold, significant in every one of 181 to 183 genomes.

The ryanodine receptors share one. That is an enrichment of 2.1 to 2.3-fold, significant in none of 188 genomes.

That is the sharpest single result in this chapter and it needs the control to be readable. The ryanodine receptors carry every diagnostic domain of this family, they are the same superfamily, and they have been in every search. Their intron positions have nothing to do with the IP_3 receptors', while the three IP_3 receptors' have almost everything to do with each other. The three paralogues did not converge on a shared architecture. They inherited one, from a common ancestor much younger than the split from the ryanodine receptors.

8.8 A fragmentary annotation usually stops where nothing splices

Chapter 13's audit calls 264 loci fragmentary and 27 split, meaning coding sequence is present and no single model covers the gene. That is a statement about coverage and says nothing about **where** the annotation stopped, and the difference is the whole claim. A model that stops at a genuine junction has produced a plausible short gene. One that stops in the middle of an exon has produced a boundary no splicing machinery could make.

So the unit is the annotated model's own **terminus** rather than its internal exon edges, because a model's internal boundaries are its own splice sites and are canonical by construction, so scoring them would answer §8.4 a second time. Each terminus is scored on two axes that know nothing about each other: against the alignment's exon boundaries, and against the two genomic bases immediately outside it read in gene orientation. A terminus coinciding with the gene's own end is excluded by rule, because a real gene legitimately starts and stops inside an exon.

The question is asked of all 291 such loci rather than only those in the architecture scope, because those loci sit in the poorer assemblies by construction and restricting the question would have answered it on 50 of them and dropped the hardest.

Of 1,448 internal termini, 406 sit inside an exon of the gene model and 353 inside one of its introns, while 689 land on a boundary the model has. **228 of the 291 loci carry at least one mid-exon terminus**, which falsifies the reading that the annotation stopped at a real gene boundary. Three are broken entirely at junctions the gene has.

The verdict is one-sided by design. A mid-exon terminus falsifies. A terminus landing on a boundary does not prove the pieces are separate genes, so that verdict is named for what it is and claims nothing more.

8.9 A duplication detector had to be scoped before it measured duplication

An exon count is only a gene's if the locus is one gene. The aligner aligns each bait independently, so a genome encoding the same part of a protein twice has the same bait aligning twice at two disjoint places.

Run naively, that geometry measures **paralogy** rather than duplication, because the three receptors are 61 to 68 % identical, every bait aligns at all three genes, and the detector's first version reached a specificity of

0.16. The pair has to be inside the cell's own loci, with the sweep's clustering and family attribution imported unchanged.

Scored as a classifier of a copy count it never sees, which is decided by coverage, identity and aligned length and by no pairwise geometry at all, the detector reaches **99.7 % sensitivity and 97.7 % specificity over 1,236 cells.**

The ryanodine control's specificity is lower on purpose, because that cell holds three genes in every tetrapod, so the detector should fire there, and it does.

No locus in the sweep encodes the same part of the protein twice. The within-locus class, meaning two alignments of one bait inside one cluster, which would be an internal partial duplication, fires on zero pairs, and a constructed control builds such a pair and requires the branch to fire, so the zero is a measurement rather than an unreachable code path.

Figure 8.4. The verdict on every split and fragmentary locus, where the annotation's internal termini sit relative to the gene model, and the duplication detector scored against an independent copy call. The first panel converts a database label into a testable claim: a locus called fragmentary might have been annotated as a plausible short gene, or the annotation might have stopped where no splicing machinery could stop, and only the second is an error. 228 of 291 such loci carry at least one terminus in the middle of an exon, which falsifies the innocent reading. The second panel makes the same point about an instrument, since scoring the detector against a copy call it never sees moves its specificity from 0.16 to 0.977 once the family assignment is applied first.

8.10 What this chapter settles about the gene

The three vertebrate paralogues inherited one gene architecture and have kept it: 57 to 58 coding exons, of which 46 to 49 positions are shared between any two of them in every genome that carries both, at nearly two orders of magnitude above a null drawn from the alignment itself. The ryanodine receptors, which are the same superfamily, share one.

What varies is intron length, and it varies by a factor of four in the median and consistently gene by gene. The 6.5-fold span difference that Chapter 2's failed background claim exposed is entirely intronic.

And a database's "fragmentary" is usually not a gene boundary, because 228 of 291 such loci stop somewhere nothing splices.

9. Counting a loss that never happened

9.1 A zero is the hardest result in this thesis to state

Gene families of this age normally lose copies [25,77]. The prior for a three-paralogue family present since the origin of the vertebrates is that somewhere in 309 genomes at least one lineage has dropped one.

This chapter reports that **not one of the 927 genome by paralogue cells in the vertebrate scope reaches the state this project defines as a loss**, that Dollo parsimony [78] places no loss anywhere on the tree, and that no rate can be fitted to a character that never changes.

A zero is the easiest result to obtain by accident and the hardest to defend. It is what a broken pipeline produces, what an over-conservative rule produces, and what a search too insensitive to find anything produces. So most of this chapter is not the count. It is the machinery that makes the count mean something: a state vocabulary in which exactly one state can be counted as a loss and every other explanation must be positively excluded first; an instrument for the cells the sweep could not decide; a false-negative rate, measured in Chapter 4, that says how often the search misses a gene that is there; and a sensitivity matrix that names, cell by cell, the analytical settings that would manufacture a loss out of this data.

9.2 Eight states are assigned, and only one of them can be counted as a loss

The genome ledger's own statuses were not built to license a loss count. In that ledger "absent" means the rescue search attributed no region, which in a shattered assembly is a statement about contig lengths. So every cell is restated under a chain of ordered positive tests, where the first rule that fires wins and each writes the number it fired on into its own row.

The design constraint is that **exactly one state may be counted as a loss**, reachable only by passing every test that could explain the absence otherwise.

The chain runs as follows. The genome's positive control did not fire, so no cell in it supports any claim. Or there is a locus at or above the coverage bar. Or a locus below it, truncated at a contig edge or a run of ambiguous bases. Or a locus below it with no assembly excuse. Or no locus, but the reference reassembles from sequence outside every locus the aligner found. Or no locus and no reassembly, but the genome carries family loci that no cell claimed. Or nothing found, and the assembly cannot hold the gene on one contig. And finally, nothing found in a controlled assembly that could have held it.

Two of those rules exist because of specific incidents.

The sixth is the cyclostome rule. Both cyclostome genomes carry three family loci apiece, all filed in the ITPR1 cell because the bait panel has no cyclostome-labelled bait, so their ITPR2 and ITPR3 cells read as absent in the ledger. A count taking that at face value would score two independent losses per cyclostome that never happened.

The seventh is the contiguity rule. An assembly whose contigs are shorter than the gene cannot represent it as one locus, so it cannot be evidence that the gene is missing, and two thirds of the bird assemblies in this scope are in that position.

The rule order is itself tested. A constructed control moves the cyclostome rule after the absence rule and requires the suite to fail.

9.3 Reassembling a reference across contigs decides the undecided cells

The sweep's undecided cells are the ones a loss count can neither ignore nor use. The aligner placed no locus, so the ledger records no coverage, and the rescue search attributed regions, so they are not nothing either.

What is measured is the **non-redundant coverage of the reference protein, reassembled across contigs**. Three properties make that a positive test rather than a hopeful one.

It is computed outside every locus the aligner found, passing through the sweep's own exclusion set of every clustered locus in the genome, any bait, padded. For a family whose paralogues are 61 to 68 % identical that is the failure mode that matters, because without the filter a missing gene reassembles out of its own paralogues and reports 0.95 coverage. A constructed control removes the filter and requires the suite to fail.

It uses one reference at a time, never a union over baits, because a union over three orthologous baits counts a residue covered in any of them and reports a coverage no single protein achieves.

The bar is measured against a gene that is accounted for, with the separation statistic [79] reported beside the threshold. The negative population needed no new search: it is the regions the full bait panel attributes to a paralogue whose gene the aligner already placed at a locus in the same genome. That paralogue is accounted for, so the fragment set cannot be that gene, and what it recovers is what cross-paralogue similarity delivers on its own, at a median of 0.030 against a candidate median of 0.795.

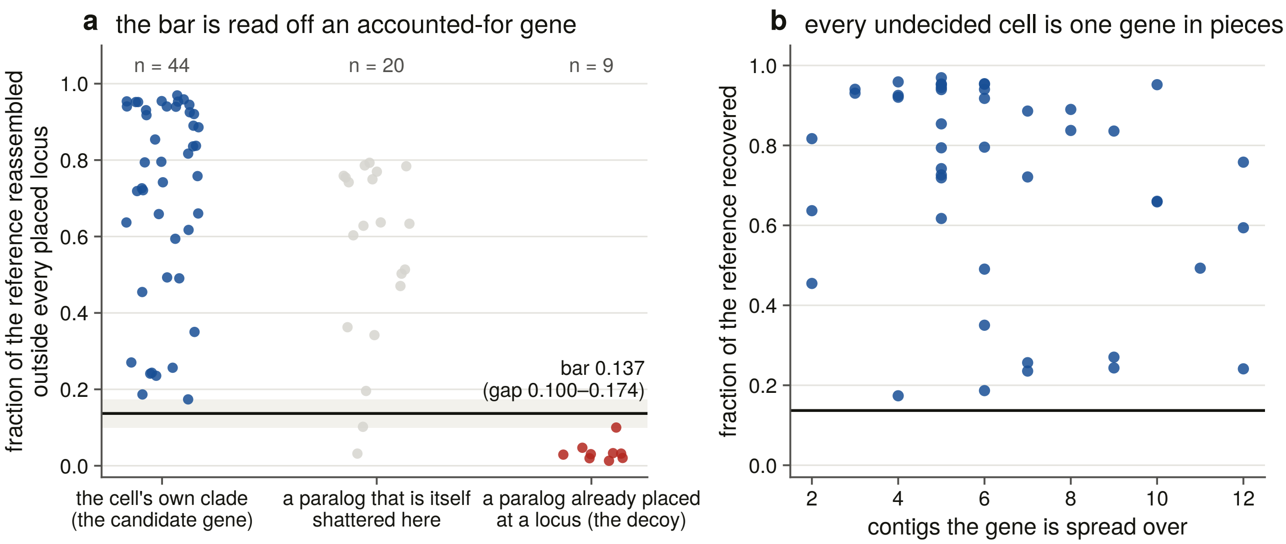


Figure 9.2. The three populations the bar is read off, with the gap shaded and the operating point drawn, and every undecided cell against the number of contigs its gene is spread over. A calibration figure that asked to be believed would not be one, so both edges of the gap are marks rather than a caption. The third population

is the one worth looking at: it lies between the other two, it was not anticipated, and the paragraph below explains what separating it out did to the threshold.

The first version of this calibration did not separate, at a Youden index of 0.52, and why is a result in itself. The decoy was every region attributed elsewhere, but in a genome where two paralogues are both shattered, a fragment attributed to the other one is a piece of a real gene. Those 20 regions are now reported as their own population, with a median of 0.631 sitting squarely between the two, and they are the **measured size of the paralogue-attribution problem in a shattered assembly**. A constructed control folds them back into the decoy and requires the suite to fail.

The operating point is the **midpoint of the gap** rather than the Youden threshold, which on a perfectly separated pair lands on the lowest positive and is the most permissive bar the data allow. Both edges are committed, so a later run whose populations have drifted into contact is visible in the table, and the calibration refuses to hand out a threshold it has marked unusable.

9.4 Synteny could not reach the undecided cells, and that is the result

The task was asked to disambiguate the undecided cells **by synteny**, with measured accuracy. Chapter 7's caller already existed and was already calibrated, so the work was to point it at these regions and measure whether it arrives.

It does not, and that is the result. Of 432 rescue regions, **273 sit on a contig carrying no annotated gene at all**, 151 have too few informative flanking symbols, and 8 reach the caller's floor. The median region has zero informative neighbours and zero coding genes on its own contig, which extends a median of 39.9 kb, shorter than a single IP_3 receptor gene.

That is not a failure of the caller. Binned by the number of keys available, its accuracy on its own calibration set is 100 % at every key count at which it acts.

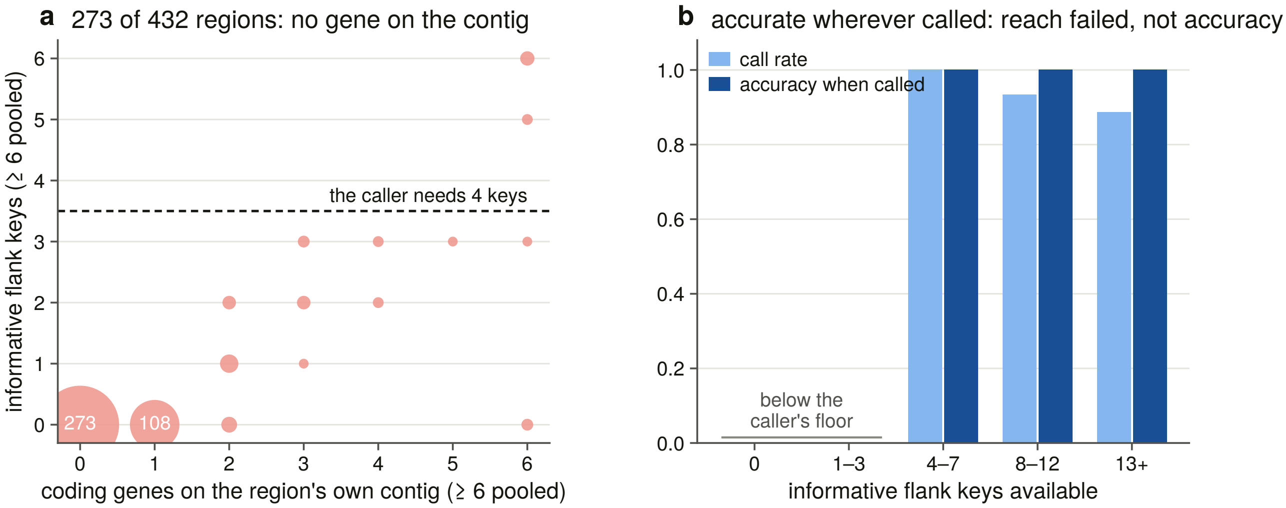


Figure 9.4. Why synteny could not answer, showing the joint distribution of what a trace region has to work with, with the caller's floor drawn, beside the caller's own accuracy on the same axis. The neighbourhood caller is right at every key count at which it acts, and it acts on 8 of 432 regions, because a fragment's contig carries no neighbours to read. Putting reach and accuracy on one axis is the only way to show that the method's failure here is a property of the data rather than of the method, which is what makes accurate and unavailable a readable verdict rather than an excuse.

On the 8 regions it can reach the caller returns a call for 6 and agrees with the alignment's own attribution on all 6. That is an independent instrument corroborating the attribution, on six regions, stated for what it is.

9.5 No cell in 927 reaches the absence state

Of the 927 cells, 783 are a single locus at full coverage, 90 a locus truncated by the assembly, 7 a partial locus with no assembly excuse, 43 reassembled across contigs, and 4 family loci the panel cannot file. Zero fall in each of the three states that would license a loss claim or invalidate a genome.

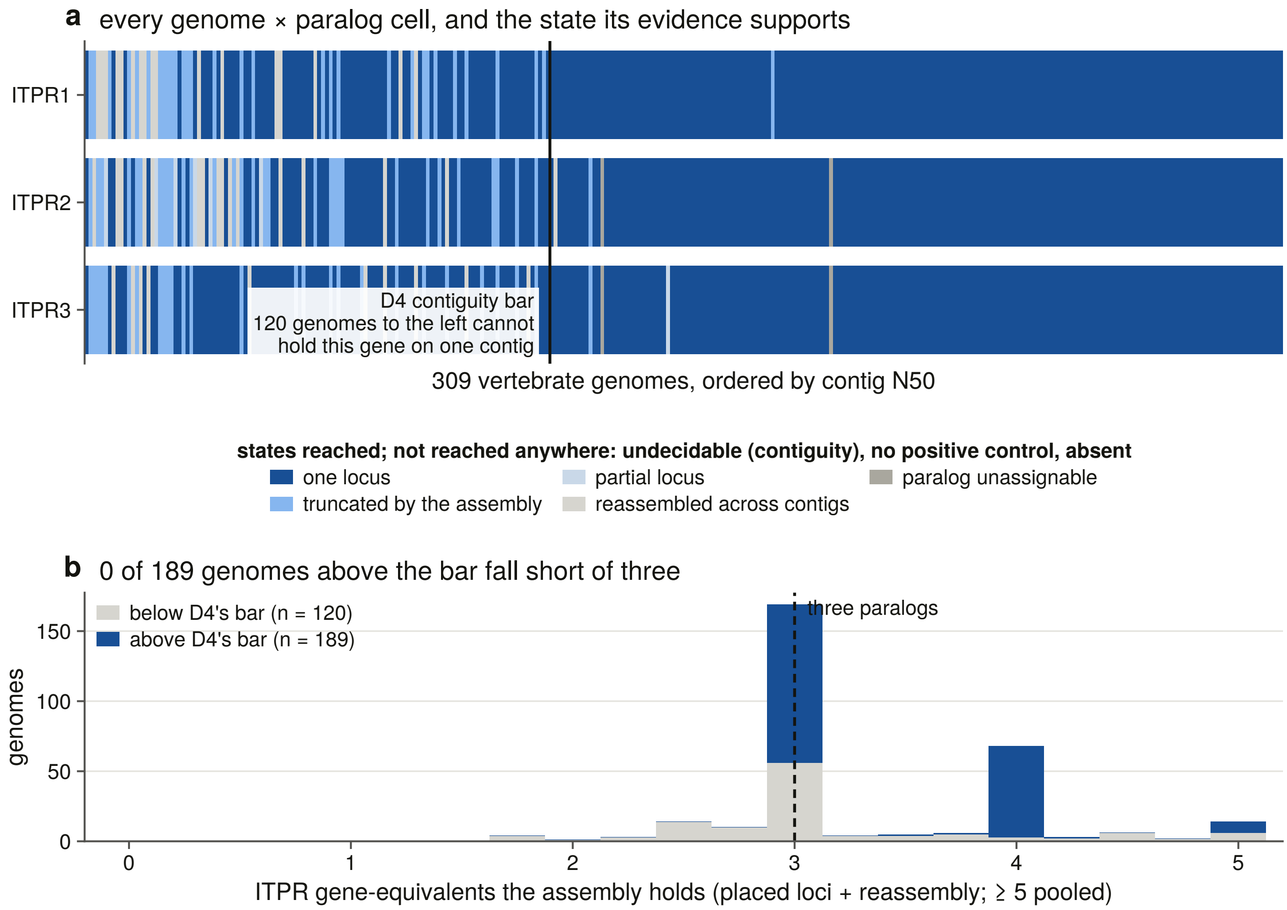


Figure 9.1. Every genome by paralogue cell, ordered by assembly contiguity with the bar drawn, and the gene-equivalents each assembly holds. The panel exists so a reader can see that the red the genome ledger showed is gone, and see where it went. This figure is the whole of Chapter 9's argument in one panel. Ordering the genomes by contiguity and drawing the bar shows that the apparent absences in the raw ledger are concentrated below it, and the second panel shows that in every one of the 189 assemblies contiguous enough to carry the gene, all three paralogues are present. What a reader should take from it is where the red went, not merely that it is gone.

The 43 reassembled cells are the substantive change this chapter makes to the ledger, because those were the cells a naive count would have read as candidate absences. Every one of them holds a gene.

The four unfilable cells are ITPR2 and ITPR3 in the two cyclostomes, which are exactly the four the reconciliation flagged, recovered here by a rule that reads the sweep's own per-genome locus counts rather than that chapter's table.

Reference coverage cannot count copies, and genomic sequence can. The paralogues are similar enough that a genome holding only ITPR1 recovers most of the ITPR2 reference, which is why the calibration needed a decoy at all. A placed locus and a reassembly occupy different places, so the per-cell contributions add.

In every one of the 189 assemblies contiguous enough to carry this gene, all three paralogues are there, at a median of 3.00 gene-equivalents and a minimum of 3.00. Below the bar, 11 of 120 fall short, and the shortfall tracks contig N50 rather than taxonomy.

9.6 Lesion counts are mostly an alignment statistic, and the confounder is not the obvious one

The sweep records how many frameshifts and in-frame stops each locus required. Read naively that is a pseudogene screen. Read honestly it is mostly a sequencing and alignment statistic, and the family's own control makes the point: the ryanodine receptors, which are a 5,000-residue gene nobody claims is dead, have the lowest share of lesion-free loci of the four cells.

Lesions are counted per kilo-aligned-residue rather than per gene, or the 1.8-times-longer ryanodine reference leads every ranking by construction. The bar is the upper quantile of a population the screen never scores, comprising loci at full coverage, in contiguous assemblies, whose own annotation names them as family, which are genes a second pipeline independently calls functional. 72 % of them carry no lesion at all.

The confounder that matters is not the one anybody expects. Measured over every scored locus, assembly contiguity barely moves the lesion count (ρ = (-)0.077) and the locus's identity to its bait moves it a great deal (ρ = (-)0.397). A lesion count is substantially a measure of how far the reference is from the gene, because a poorly matched bait buys alignment with frameshifts.

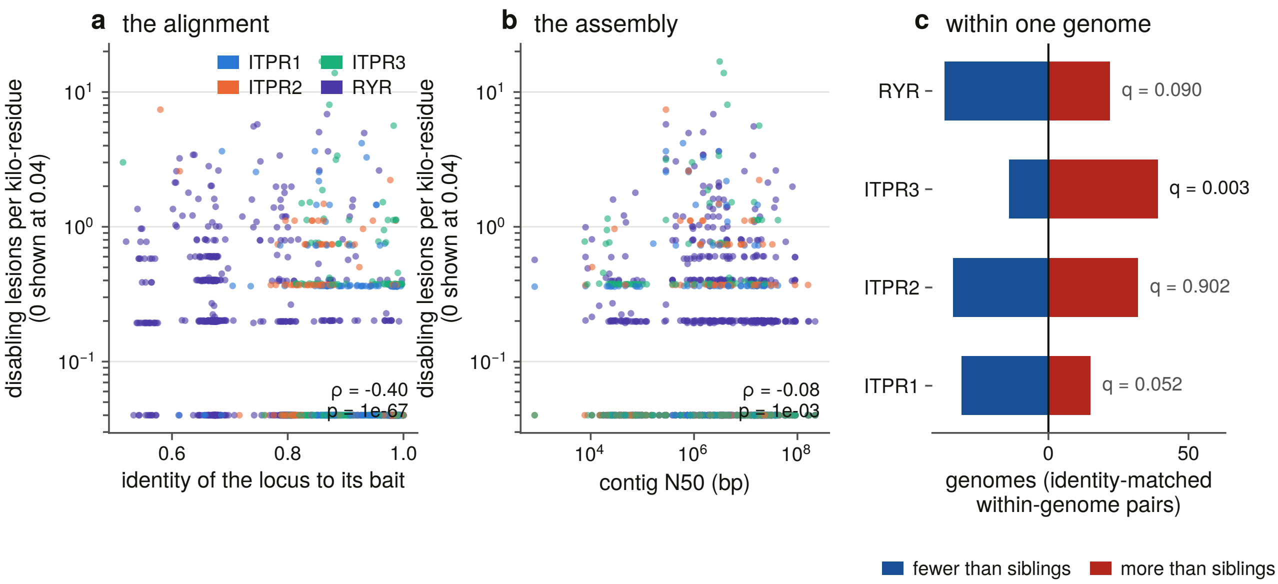


Figure 9.3. Lesion density against the two confounders that could produce it without a gene being dead, and the paired within-genome test that removes both. The identity panel is drawn first because contiguity is the confounder everyone expects and identity is the one that turned out to be real. Density is logarithmic with an explicit zero band, since 72 % of intact loci carry no lesion and a linear axis puts the whole calibration population on one pixel.

So the paired within-genome test, in which each cell is compared against the same genome's other family loci and which removes the assembly entirely, is run twice, once identity-matched, with ties dropped and counted and the four tests corrected together by the false-discovery-rate procedure this project uses throughout [80].

One result survives, and it is paralogue-specific. ITPR3 carries more disabling lesions than its own genome's identity-matched sibling loci, in 39 genomes to 14. ITPR2's excess in the unmatched test disappears once identity is matched, so it was an alignment artefact. ITPR1's deficit does not survive correction, and the ryanodine control shows no excess, so the ITPR3 signal is not a property of the family's gene structure as a whole.

What that is **not** is a pseudogene finding. All 44 loci above the bar in contiguous assemblies are at full coverage with an intact gene model, and the one sound direction here was already established: zero stops falsifies a pseudogene call, and a handful does not establish one. The verdict column stays one-sided.

9.7 The tree the count is placed on is a taxonomy, declared as an input

A reconciliation needs dates and a hand-curated tree, and a loss count needs coverage. Chapter 7's 31-species tree cannot serve here, because 278 of the species whose cells this matrix holds are not in it.

So the topology is NCBI taxonomy over all 309 assemblies, taken from the same archived dumps the manifest was built from: 309 genomes placed, 468 internal nodes, **49 of them polytomies**. There is one tip per accession rather than per species, because the sweep's unit of observation is an assembly.

It is an input rather than a result. Its polytomies are real and are not resolved, which for parsimony makes a placement less confident and never wrong, but a count of independent losses under a polytomy is bounded by the resolution, so the degree distribution is committed. The tree's compatibility with the curated one is checked rather than assumed, and the check is asked so that the two trees' different sampling cannot register as a disagreement. The first version counted assemblies of species the curated tree never sampled as intruders and reported 21 of 29 clades as unrecovered while every matched node carried the right name.

9.8 Dollo parsimony places no loss, and the gain nodes agree with Chapter 7

Under the operating point, on both codings, the family character and all three paralogue characters are present in every genome that can be scored, absent in none, and Dollo places **zero losses**, at both the maximum and the minimum.

The count is zero because no cell reaches the state that would license it, rather than because the routine cannot return anything else. A constructed control puts a loss on a known edge and requires it found, another requires two sister losses merged into their parent, and a third constructs a contiguous, controlled, spare-free empty cell and requires it to come back as an absence. **A count that can only ever return zero is not a measurement**, and these are what stop this one being that.

The gain nodes are a consistency check nobody asked for. Dollo places the single gain at the most recent common ancestor of the tips carrying the character, and for the paralogue characters that node is not pinned. The three answers are Vertebrata, Gnathostomata and Gnathostomata, which is exactly where Chapter 7 placed the two duplications, recovered by a method that reads no gene tree, no alignment and no reconciliation.

It is worth printing and it is **not independent evidence**, because the reason ITPR2 and ITPR3 have no cyclostome tip is that those cells are unfilable for want of a cyclostome-labelled bait, and the reason the reconciliation placed the duplication there is a reconciliation.

9.9 What it would take to manufacture a loss out of this data

With no loss to place, what is worth reporting is the shape of the zero: how far the rules must move before a loss appears, which rule has to move, and what each move buys.

Thirty-two settings were walked, comprising an ordered eight-rung evidence ladder, each rung named after what it refuses, crossed with the two protective rules on and off, on both codings, under three branch-length schemes. The first three rungs of the ladder are the calibration's own measured gap edges and midpoint rather than round numbers, so the first sensitivity question asked is whether moving the bar across its own uncertainty changes anything.

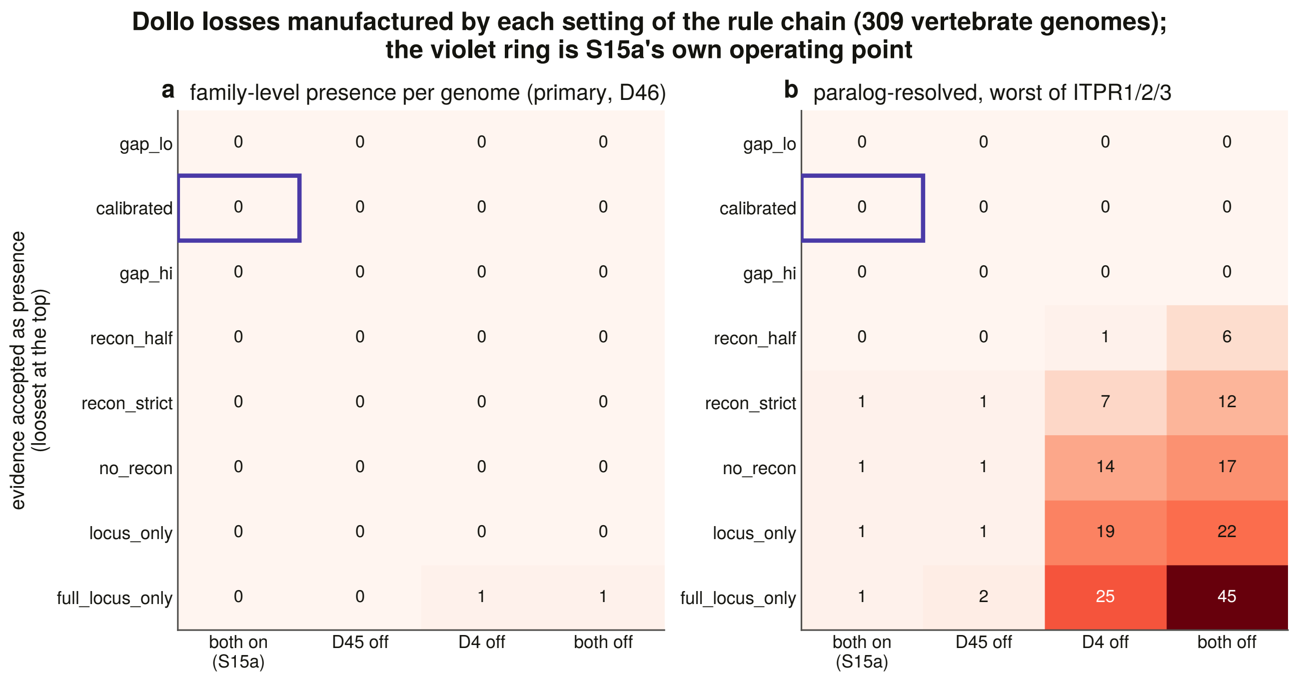


Figure 9.6. The loss count in every cell of the grid, for the family-level coding and the paralogue-resolved one, with the operating point marked. This is what a zero looks like when it is defended properly. Rather than assert robustness, the count is recomputed under every combination of four analytical axes, and the asymmetry that reveals is the result: a family-level absence is stable across 30 of 32 settings while a paralogue-resolved one is fragile to every knob. A zero is drawn as an explicit zero and never as an empty cell, because an empty cell reads as not measured and the zero is the result.

Moving the reconstruction bar across the whole gap the calibration measured manufactures no loss on either coding. The bar's position inside its own uncertainty is not what any result here rests on.

Read across the two codings, the result is an asymmetry rather than a number. The **family-level coding manufactures a loss in 2 of 32 settings**, and its worst case is one genome in 309, reached only by refusing everything except a complete locus and ignoring the contiguity bar at the same time. The **paralogue-resolved coding manufactures one in 18 of 32**, up to 45 loss edges. A per-paralogue absence is fragile to every one of these knobs and a family-level absence is not.

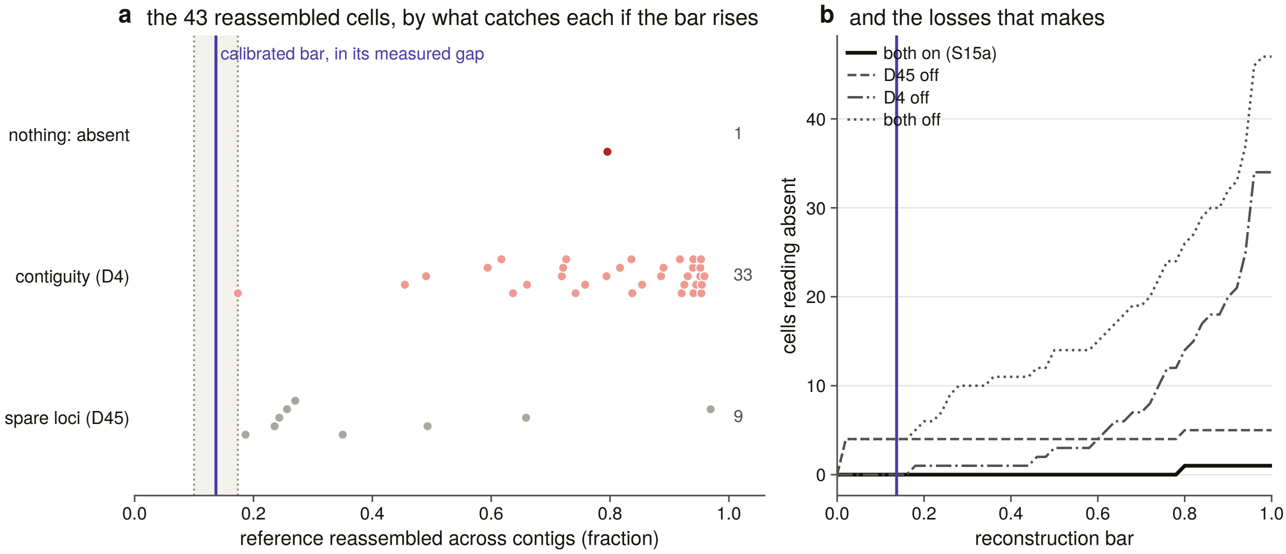


Figure 9.5. The calibrated bar with the gap's two edges drawn. The within-row offset of each point is its rank in its own row, using no hash and no random number generator, so the figure is reproducible. Drawing both edges is what makes the threshold's own uncertainty visible, and the answer to the first sensitivity question is that moving the bar anywhere inside it changes no cell in 927.

The branch-length axis changes nothing, and that is reported rather than omitted. Across all 32 settings and all three schemes, the number of settings at which a branch-length scheme changes a parsimony count is zero. Parsimony counts edges and does not read a length, so this could not have come out any other way, but the brief names branch lengths as an axis and a matrix that quietly dropped one axis would be indistinguishable from one that had tested it.

9.10 The rate models are refused, and the refusal is measured

The brief asks for equal-rates, all-rates-different and irreversible Markov models [81]. On the primary character all three are **refused**, and the refusal is measured rather than asserted.

An invariant character contains no transition to estimate. Profiling each likelihood along a rate grid over eight orders of magnitude gives, at every one of the twelve model by branch-length by axis combinations, a monotone curve with its maximum on the grid's boundary. The loss axis falls to the floor and the all-rates-different model's gain axis rises to the ceiling, which is what unidentifiability looks like when it is drawn rather than argued.

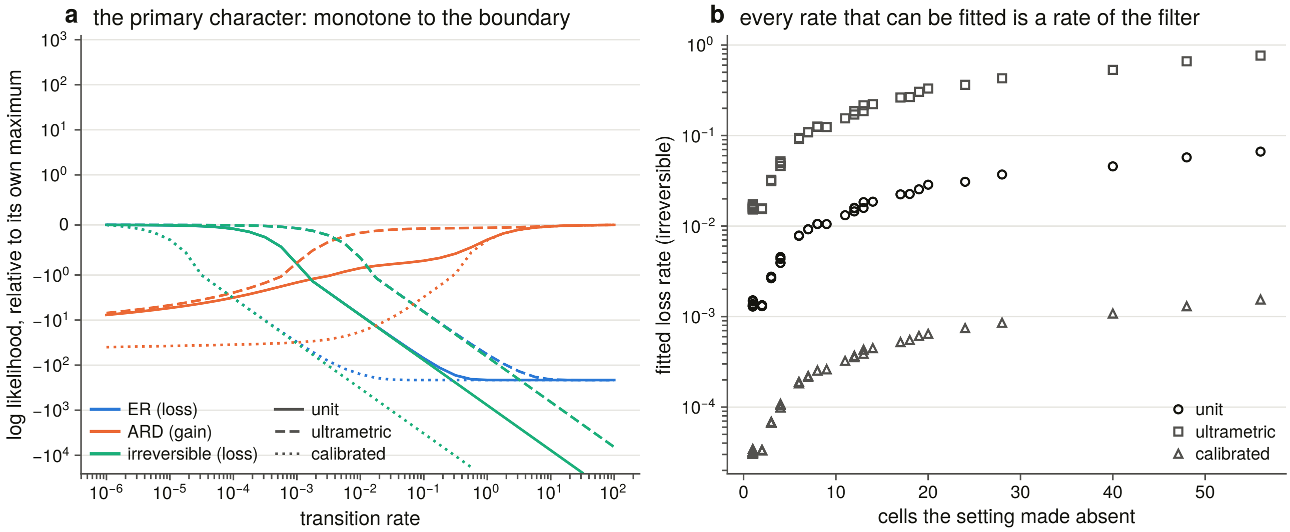


Figure 9.7. The likelihood along a rate grid for every model, axis and branch-length scheme. The all-rates-different model is profiled on its gain axis rather than on its diagonal, because the diagonal is the equal-rates model by construction and would put the same curve on the figure twice under two names. The figure is a refusal turned into a measurement: rather than assert that an invariant character cannot be fitted, it shows what the likelihood surface actually does, which is to run to the edge of the grid in every one of the twelve combinations.

So no rate is reported for the primary character, because a fitter run on it would return its own starting point.

Where the sensitivity matrix does produce variation the fits are real, and they are reported for what they are: a property of the filter rather than of the family. In the most extreme cell of the matrix the three branch-length schemes give rates spanning a factor of 495 while describing the same character, which is the branch-length axis doing the only thing it can do here, namely setting the units a rate is quoted in.

9.11 The fossil test has no dead loci to run on, and it hands on one lead

If a paralogue died in an ancestor, its descendants should share lesions more often than independent decay would give. That test needs dead loci.

Of 1,760 scored loci, 44 clear the measured lesion bar, and **all 44 are at full coverage, delivering a complete gene model.** The screen was made deliberately generous, counting a locus as a candidate if any of three readings fires, because a test made hard to pass would make the zero uninformative. Even so, seven family loci in 874 fire any reading at all, and all seven do so on one or two internal stops in a complete model.

One or two stops in a 2,700-residue model at full coverage is not the signature of decay. **The shared-lesion test has no dead loci to run on, and that is reported with its denominator** rather than omitted, because an omitted section is indistinguishable from a section nobody ran.

The one lead this chapter hands on is a bird result. The ITPR3 lesion excess of §9.6, stratified by vertebrate class and corrected across the strata, is 25 genomes to 2 in birds and 7 to 6 in ray-finned fish, which is no signal at all on a sample that is smaller but not small.

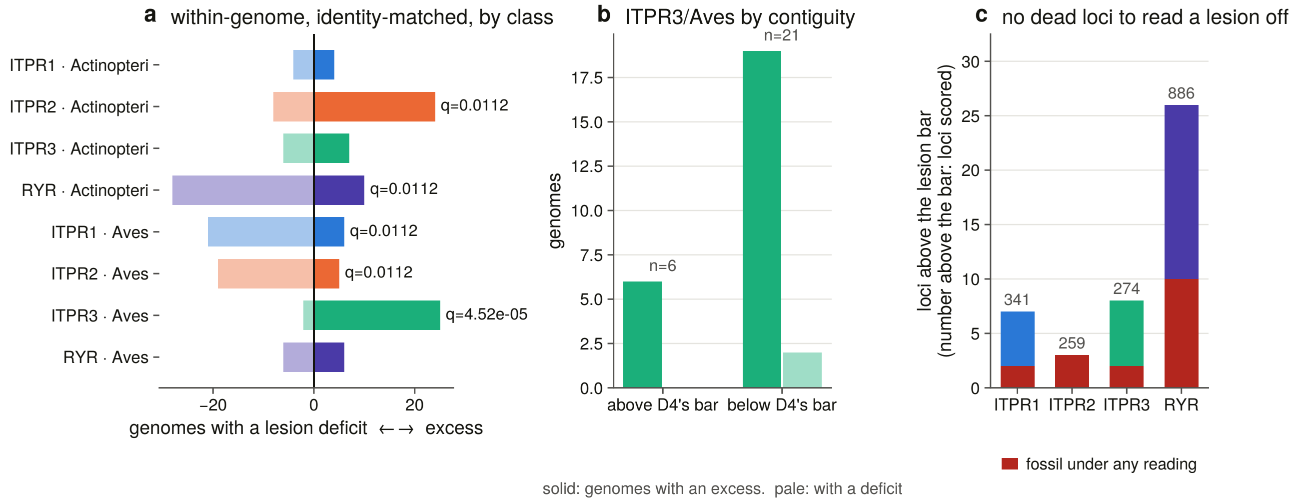


Figure 9.8. The identity-matched sign test stratified by vertebrate class, with the contiguity control drawn beside it. The stratification names a lineage and the control immediately removes the ground from under it, which is why both belong on one figure. The verdict this panel supports is that the mechanism behind the excess is unsettled rather than found.

The design makes a cell and its siblings the same observation twice, because the comparison is within one genome against that genome's other family loci, so a bird ITPR3 that is elevated makes its own ITPR1 and ITPR2 look deficient by construction. The three bird rows are one result rather than three.

The control has to be printed. Twenty-one of the 27 bird pairs are in assemblies below the contiguity bar, and that is where the test has its power. Above the bar all six pairs point the same way and none points against, but six pairs cannot carry a test. Two thirds of bird assemblies are below the bar, the worst of any class, so this is precisely the class where an indel signal is hardest to separate from an assembly signal. The identity control is clean, so the confounder that mattered in §9.6 is not what this is. **The lineage is now named and the mechanism is not, and the contiguous half of the evidence is too small to settle it.**

9.12 What the zero means, and the two ways to escape it

Every vertebrate genome in a declared scope of 309, searched with an instrument whose false-negative rate is measured at 15.2 % overall and 0.9 % in contiguous assemblies, carries all three IP_3 receptors. No lineage has lost one. The count is robust to moving every threshold across its measured uncertainty, and the settings that would manufacture a loss are named and counted rather than avoided.

That is a strong claim, and the strongest form in which it can be made is the one that names its own escape routes. There are two. A per-paralogue absence is fragile to the analytical settings in a way a family-level absence is not, and a reader who wants a loss can have one by refusing all reconstruction evidence and ignoring assembly contiguity, which is a description of the setting rather than a defence of it. And the scope is 309 vertebrate genomes, so nothing here extends to a lineage with no assembly.

10. Measuring selection across the vertebrate family

10.1 An identity is a distance, and this chapter measures a rate

Chapter 6 measured identity within a paralogue at 0.910 and concluded the family is conserved. That is a distance. A conserved protein and a slowly evolving one are not the same statement: identity says how far two sequences have moved apart, and a selection test says whether the moves that did happen were the ones selection tolerates.

This chapter measures the rate. It is the shortest analytical chapter in the thesis and the one with the most traps in it, because a codon model returns a perfectly plausible number when its input is wrong, when its optimiser has found a local peak, or when its denominator has stopped existing.

10.2 Every coding sequence had to be proved to encode the aligned protein

A ratio of non-synonymous to synonymous substitution is a statement about codons, so every number here rests on a nucleotide sequence that provably encodes the exact protein the alignment holds. Nothing is a translated database record taken on trust.

Three routes supply candidates, comprising a genome database, cross-references from the protein record, and the genomic gene models the sweep produced, and each is a **generator of candidates** rather than an answer. A protein entry lists every transcript of its gene and the entry's own sequence is one particular isoform. Human ITPR1 lists five and ITPR2 two, and in **both** the first is not the isoform the alignment holds. Returning the first coding sequence that downloads would silently substitute a different isoform for the protein the tree was built on, which is a wrong codon alignment rather than a missing one.

Every candidate is translated against the aligned protein and **every** disagreement is masked, not only internal stops. A codon that encodes something else is not that protein's codon at that site whatever the cause, and the model must see missing data rather than a residue that is not there.

Fifty-seven of 57 vertebrate tips have a validated coding sequence, from two routes, with 24 codons masked across the whole set.

The genomic route has its own trap. The sweep ran the aligner in a mode whose emitted protein skips frameshift residues, so the alignment's coding blocks run out of frame against it and splicing them is not offered at all. The locus is realigned in a mode that keeps those residues, and the reconstruction is accepted only if it reproduces the sweep's protein. The one concession is measured rather than assumed: a locus re-run indexes a fraction of the genome the sweep indexed, so a terminal exon can be placed differently, by nine residues of 2,666 in one frog. Same length and within 1 % is the same gene model and its disagreeing codons are masked, and anything beyond that is a different model and is refused.

The codon alignment is then produced by a protein-guided aligner [82] and **cross-checked nucleotide by nucleotide against an independent in-house mapping**, with a disagreement aborting the build. A residue masked in step one is written as an unknown in the protein rows too, so the aligner is never asked to reconcile a pair that disagrees, which it does by dropping the sequence silently. Trimming is chosen on the protein and applied codon-aware, whole triplets only.

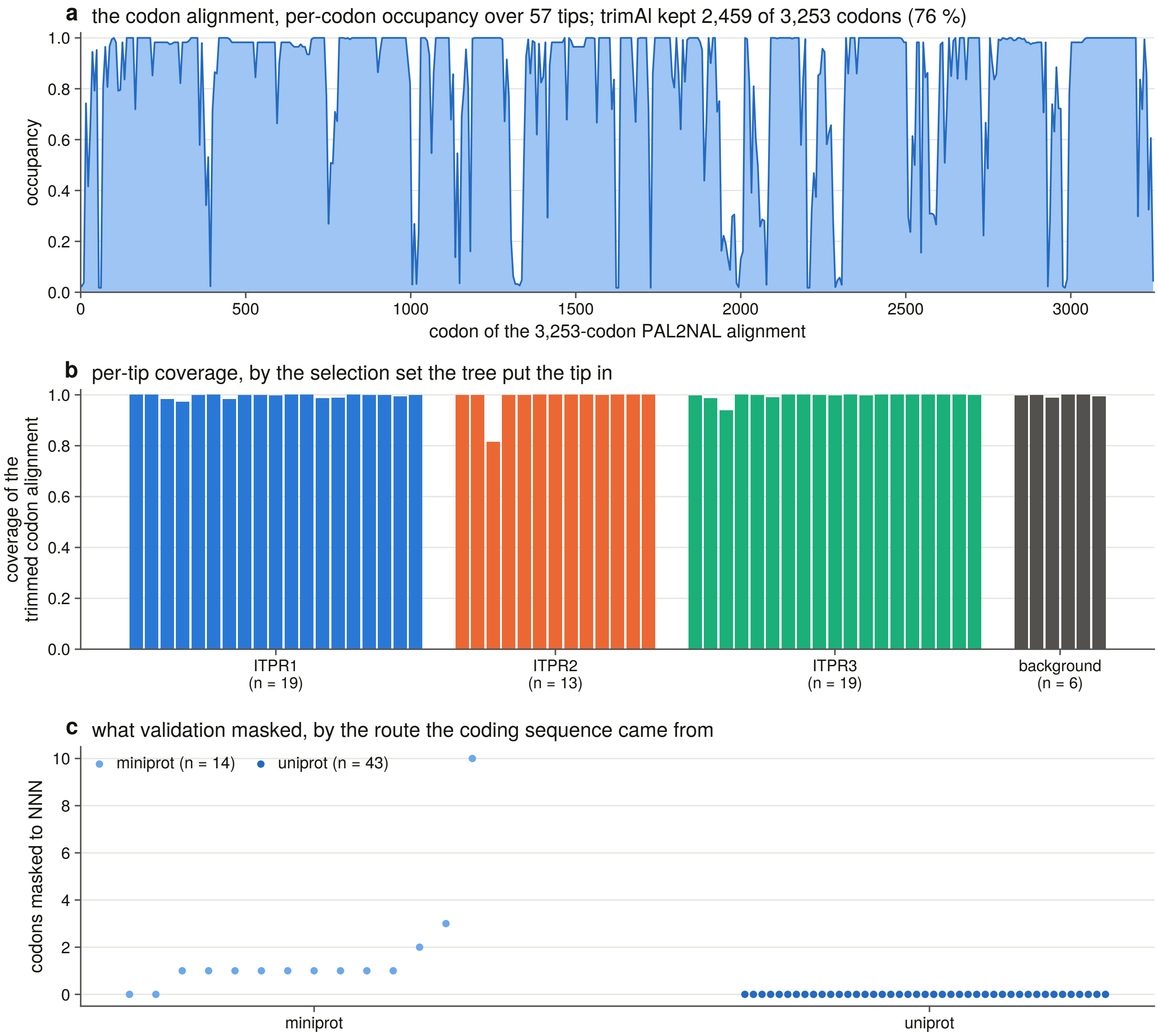


Figure 10.5. The trimmed codon alignment behind every estimate in this chapter, plotted in codons, because an axis in nucleotides would make a three-fold difference look like a property of the data. The panel exists so a reader can see the size of what every rate in Chapter 10 was estimated on, which is 2,459 codons of 3,253 after trimming, across 57 tips whose coding sequences were each proved to encode the aligned protein.

Twelve constructed negative controls run on every build, and their character is the point. A codon alignment is the one artefact in this project where a silent error is invisible downstream, because the model will fit a frame-shifted alignment and return a perfectly plausible rate. So the controls are checks on refusal: a frame-shifted sequence, one encoding a different protein, an internal stop that must be masked and not refused, whole-triplet gap stripping, and a non-monophyletic foreground being refused.

10.3 The tree decides which tips are in which selection set

A branch model marks a node. A foreground that is not a clade does not fail. It silently marks a larger one and returns a well-formed rate for a hypothesis nobody asked.

So the three selection sets are Chapter 6's extended paralogue clades, re-derived here from the rooted tree with that chapter's own rule and cross-checked against its committed table, with a disagreement in size or membership treated as a hard failure.

Two consequences follow. A tip the tree recorded as unplaced is placed by the tree or by nothing, never by a database symbol. And the seven unlabelled tips the tree nests inside a paralogue clade join that set, because outside the vertebrates a paralogue label means nothing, but inside a maximally supported vertebrate paralogue clade the tree has just supplied one.

Six tips sit in no paralogue clade. They stay in the whole-tree analyses, because dropping them would change the branch lengths every other estimate is made on, and they are in no foreground, because a locus the tree could not place is not evidence about a paralogue.

They are the six cyclostome loci. Chapter 6 placed them in two cyclostome-only clades and handed the question on, Chapter 7 reported the neighbourhood underpowered because after 550 million years there is no shared flank vocabulary left, and a third instrument now declines the same question for a third reason: a codon model can only ask about a paralogue whose clade the tree defines, and for these six it defines none.

10.4 All three paralogues are under strong purifying selection

One-ratio estimates [83] give ITPR1 at 0.0238, ITPR2 at 0.0430 and ITPR3 at 0.0415. The highest of the three is **23 times below neutrality**.

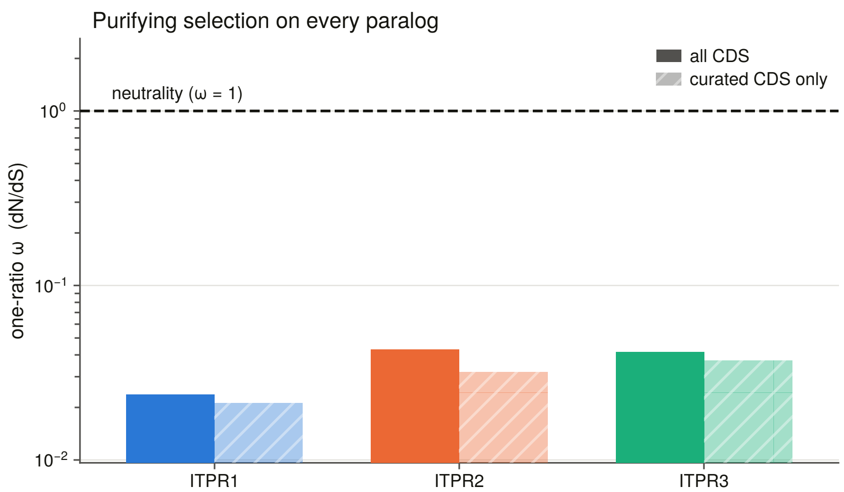


Figure 10.1. Per-paralogue rate with the curated-sequence sensitivity estimate beside it, and neutrality drawn rather than described. The axis is logarithmic because the result is how far below neutrality these rates sit, and on a linear axis every bar is on the floor. The sensitivity estimate beside each bar is what shows the rates are not an artefact of the reconstructed gene models, since it is computed with every one of them removed.

The identity Chapter 6 measured says the same thing far less sharply. This is a 2,700-residue channel accumulating one non-synonymous change per 23 synonymous ones.

The sensitivity subsets, meaning the same paralogues with every genomic gene model removed so that no masked codon contributes, move the estimate by at most 0.0112. The estimates are not an artefact of the reconstructed models.

10.5 Synonymous saturation qualifies every other number in this chapter

Synonymous sites are **saturated across the vertebrate span, inside a single paralogue and not only between them.** In the worst set, 94.2 % of within-paralogue pairs exceed the conventional saturation bar, and the medians run from 4.6 to 13.5.

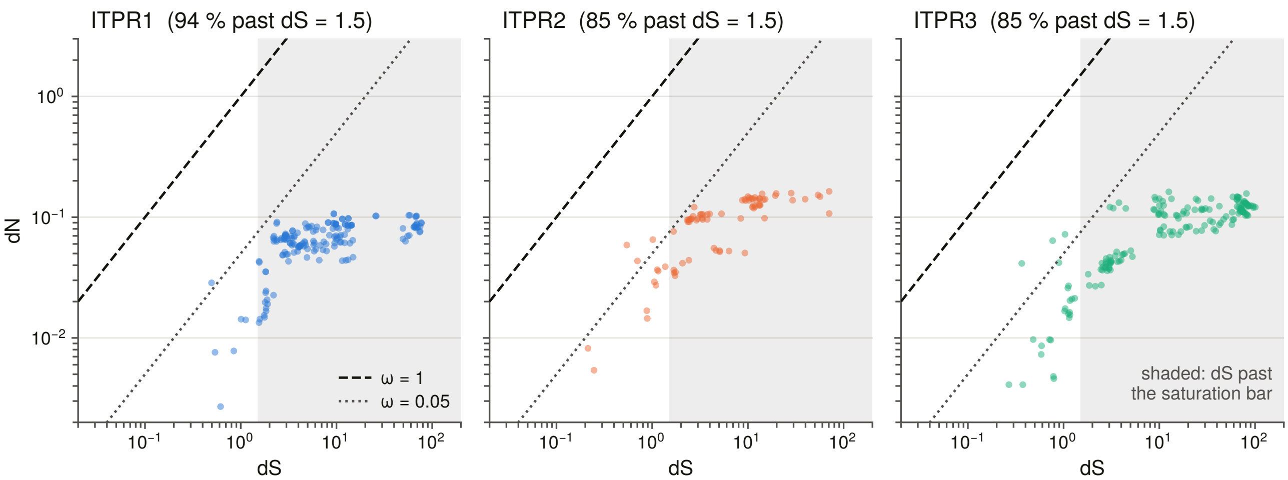


Figure 10.3. Pairwise rates within each paralogue, drawn log-log with the neutral diagonal and the saturation bar. On a logarithmic-x, linear-y plot the neutral diagonal is not a line at all: the first draft's neutrality ran off the panel within the first pixel and left the saturation bar as the only line on the figure, which reads as neutrality. What the panel shows is that saturation is reached inside a single paralogue and not only between the three, which is the reason no rate in this chapter is taken from a pairwise matrix.

The expectation going in was saturation between the paralogues, which are older than 500 million years. It is already reached within them, because a single paralogue set spans shark to teleost to mammal.

That has one immediate consequence and one that runs through the rest of the chapter. **The pairwise rate matrix is a diagnostic here rather than an estimate**, so every rate quoted comes from a tree-based model, which distributes substitutions over branches instead of asking one pair to carry 450 million years. And every rate should be read as a lower bound on precision rather than a point estimate with a small error, because a ratio whose denominator is soft is soft.

10.6 ITPR1 is the most constrained of the three

Ranked from most constrained to least, the order is ITPR1, then ITPR3, then ITPR2.

Each paralogue clade tested against the rest of the family as a two-ratio branch model confirms it, with every test corrected across the family. ITPR1's foreground rate is 0.0241 against a background of 0.0432, ITPR2's is 0.0435 against 0.0317, and ITPR3's is 0.0455 against 0.0308.

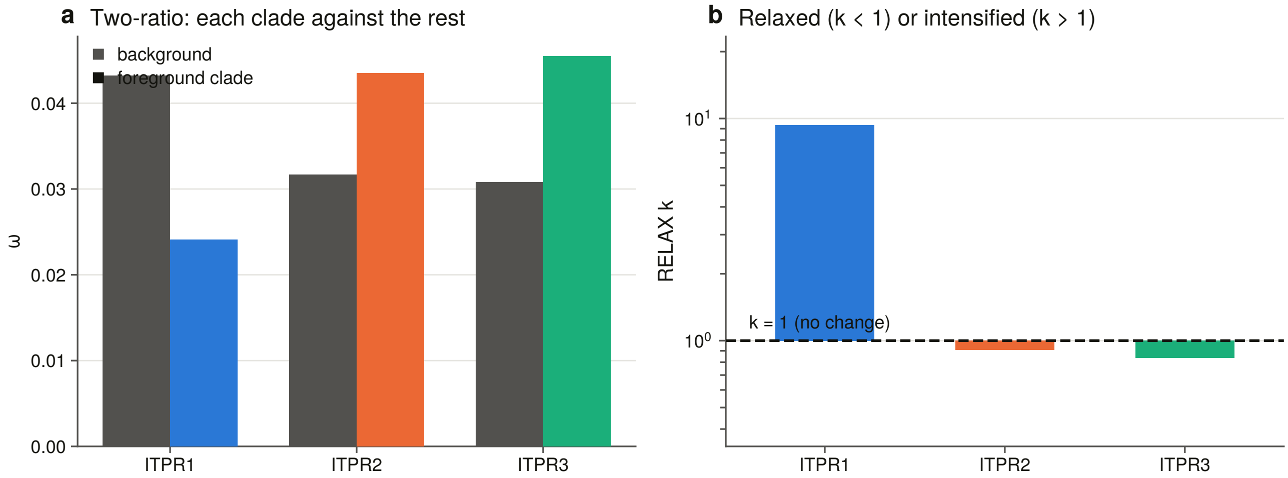


Figure 10.2. Background against foreground rate per paralogue clade, with the relaxation coefficient beside it. The importance of this figure is that two different statistics on two different models give the same ordering. A point-estimate contrast and a test on the whole rate distribution both put ITPR1 under roughly twice the purifying selection of its sisters, and both make its selection intensifying relative to theirs. That agreement is what turns a ranking into a claim about the three copies having been held to different standards.

ITPR1 is the paralogue that carries the family's dominant missense disease burden, and it is the most constrained. It is also the paralogue whose neighbourhood Chapter 7 found best preserved and the one that alone kept its teleost duplicate.

ITPR3's neighbourhood is the one that does not travel, while its coding sequence is not the least constrained. Those are different quantities, because neighbourhood conservation is rearrangement history and this is coding sequence rate, so the mismatch is not a disagreement, and the project's verdict vocabulary has a word for that rather than forcing a choice.

10.7 The stem branches carry three optimiser failures a single run would have hidden

A duplicate's fate is decided on its stem, meaning the interval between the duplication and the first surviving split of the new copy. If a paralogue was ever free to change, that is when. Branch-site model A [84] asks whether a class of sites on that one branch has a rate above one while the rest of the tree does not.

Model A is restarted from four initial rates by construction here rather than repaired afterwards. A nested alternative cannot have a lower optimum than its own null, and yet **3 of the twelve restarts converged below their own null, one on every stem.** Each of those is a local-optimum failure that a single-start run would have reported as its answer.

A different starting value fails on each stem, so no single initial value would have been safe. That is the case for running several rather than for choosing a better one.

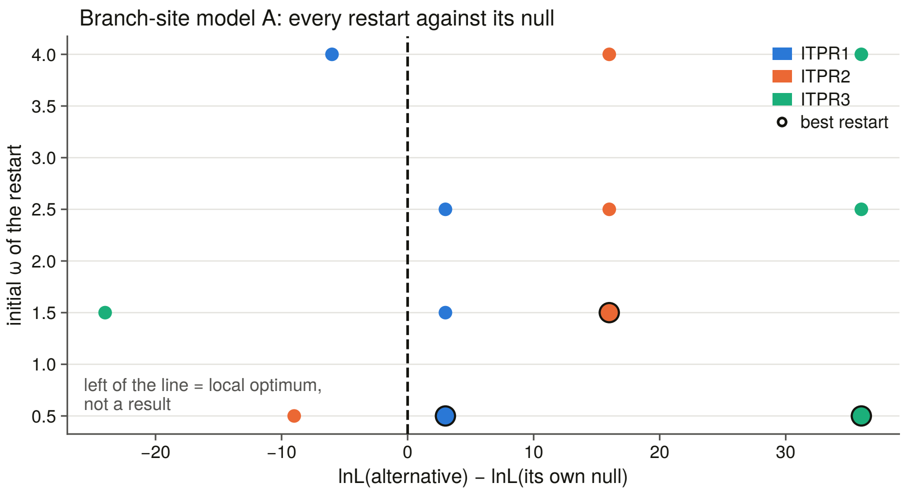


Figure 10.4. Every restart against its own null. A point below the line is a local optimum rather than a result, and three of twelve sit there, one on every stem, with a different starting value failing each time. The importance is general rather than local: a branch-site result from a single-start run is a result whose optimiser has not been checked, and nothing in its output says so.

All three stems are significant after correction, and one of the three carries a rate the data actually determine. That distinction is the substance of this section, and it comes from a parameter printed outside the test rather than from the test.

On the ITPR2 and ITPR3 stems the foreground rate is pinned at the optimiser's upper bound of 999. That is not an estimate of 999. It is the optimiser reporting that the foreground has no synonymous signal left to normalise a rate against, which is exactly what §10.5 predicts for a branch this old. Restarts reaching the same likelihood put one of them at 162 and at 999, which is a six-fold spread at an unchanged likelihood and is the definition of an unidentified parameter. On one of them, no site clears the posterior bar at all.

On the ITPR1 stem the rate is **5.53 on 11 % of sites**, well inside the estimable range, stable across restarts, with 8 sites clearing the posterior bar [85] and 5 clearing the stricter one.

So the reportable branch-site finding is ITPR1 alone: a class of sites on its stem evolving several times faster than neutrally while the rest of the tree sits near 0.03. The other two stems' tests are significant and their effect size is unmeasurable, and those are different sentences. A pipeline that printed the three p-values would have reported its strongest signal on the stem whose parameter is least determined.

10.8 A significant site-model test can mean something quieter than its p-value

Site models ask whether any site in a paralogue has a rate above one across the whole clade, which is a different question from the stem, and the one that would find recurrent positive selection anywhere in the vertebrate history of that copy.

The pattern that recurs is a significant likelihood-ratio test whose fitted parameter says something quieter than the p-value suggests: an extra rate class at exactly 1.000 on 0.3 % of sites, or a class above 1 with a share of zero, and no site clearing the posterior bar. A significant test with no site above the posterior bar is the signature of a fit driven by saturation rather than by identifiable substitutions, so every significant test in this chapter is printed beside its posterior column.

That discipline is the reason three of this chapter's significant tests are reported as meaning something other than what their p-values suggest, and in every case the tell was a parameter printed outside the test.

10.9 Selection on ITPR1 is intensified, and on its sisters relaxed

A branch model asks whether a foreground's rate differs. The relaxation test [86], run locally in the same framework as the per-site model [87], asks whether the whole distribution on the test branches is pulled towards neutrality or away from it, which in a family where every rate is far below one is the sharper question.

ITPR1 is intensified at $k = 9.36$, ITPR2 relaxed at $k = 0.908$, and ITPR3 relaxed at $k = 0.836$.

That is the same ordering the one-ratio estimates give, arrived at by a different statistic on a different model, in that one compares point estimates and the other compares whole distributions, so the two are a check on each other rather than one number told twice.

The three copies have not been held to the same standard since they were made. ITPR1 is under roughly twice the purifying selection of its sisters and is intensifying relative to them, and it is the copy the teleost duplication kept, the copy whose neighbourhood is best preserved, and the copy that carries the dominant disease burden.

The unlabelled tips the tree places in no paralogue clade are left unlabelled in these runs rather than swept into the reference set, because a branch whose paralogue identity is unresolved is not evidence about either side of a contrast.

One implementation note is worth keeping because the failure it causes is silent in the wrong direction. The descriptive model estimates a per-branch rate, and a branch with no synonymous change gets an infinite one, which the program writes as a bare token that is not legal JSON. It is normal output, the analysis succeeds and the parse throws, and on the first run that turned one paralogue's entire result into a blank row.

10.10 What this chapter does not establish

Saturation is not repealed by using a tree. Nearly nine tenths of all within-paralogue pairs exceed the saturation bar. Tree-based models handle it far better than pairwise estimation, but a rate estimated where the synonymous rate is poorly determined is a ratio with a soft denominator.

Fourteen of the coding sequences come from genomic reconstructions with masked codons. The curated subsets show the estimates barely move without them, but those models are also the only evidence for several lineages, so the subset is a control rather than a replacement.

The tree is conditioned on. Every branch test runs on Chapter 6's topology. If the sister arrangement were wrong, the branches marked here would be the wrong branches, which is why that chapter ran a topology test over all three arrangements before this one started, and why the dependency is recorded rather than quietly relied on. A branch test cannot corroborate the topology it is conditioned on.

This is a vertebrate result. The non-vertebrate grade is not in the codon alignment at all, so nothing here bears on the constraint acting on the single-copy receptors Chapter 5 found across the eukaryotes.

Two of the three branch-site effect sizes are not identified, which is not the same as being large.

The whole suite ran unattended in 22.7 hours, and every stage renders whatever has landed and marks an unfinished section as unfinished, because waking up to a partial analysis that says which parts are partial is worth more than waking up to nothing.

11. The channel's shape, its constraint, and the clinical variants

11.1 This chapter asks two questions about the protein rather than the gene

Everything so far has been about genes: where they are, where they came from, and whether they are still there. This chapter is about the protein.

It asks two questions. Does what the census calls an IP_3 receptor fold like one, and can it be told from a ryanodine receptor by shape alone? And which parts of the receptor cannot change, meaning where the constraint is, whether it sits where the structure says the machine works, and whether it explains the clinical variants?

The first question has an answer that is mostly about a database, and it is worth having in advance because it constrains what the second can be built on.

11.2 The structure prediction database does not hold this family

A vertebrate IP_3 receptor subunit is about 2,700 residues, which is where the monomer prediction pipeline [88] stops, and the sister family at about 5,000 is past it outright. So structural coverage is measured before anything is compared, and it is measured three ways that do not agree: the cross-reference in the protein record, an actual probe of the prediction database [89], and model coverage against the length the census holds.

Only the third can see the trap.

The database is keyed on an accession and answers with whatever record it holds, and that record may be an isoform. Asked for the three human paralogues it returns a 2,695-residue model of a 2,758-residue protein, the canonical 2,671-residue ITPR3, and **181 residues of the 2,701-residue ITPR2.** A probe that took the first record and called it a hit would report the family's own reference paralogues as fully modelled.

20.9 % of the census's protein-database records have a usable model, and of the 134 representatives every alignment, tree and selection result in this thesis stands on, **nine** do.

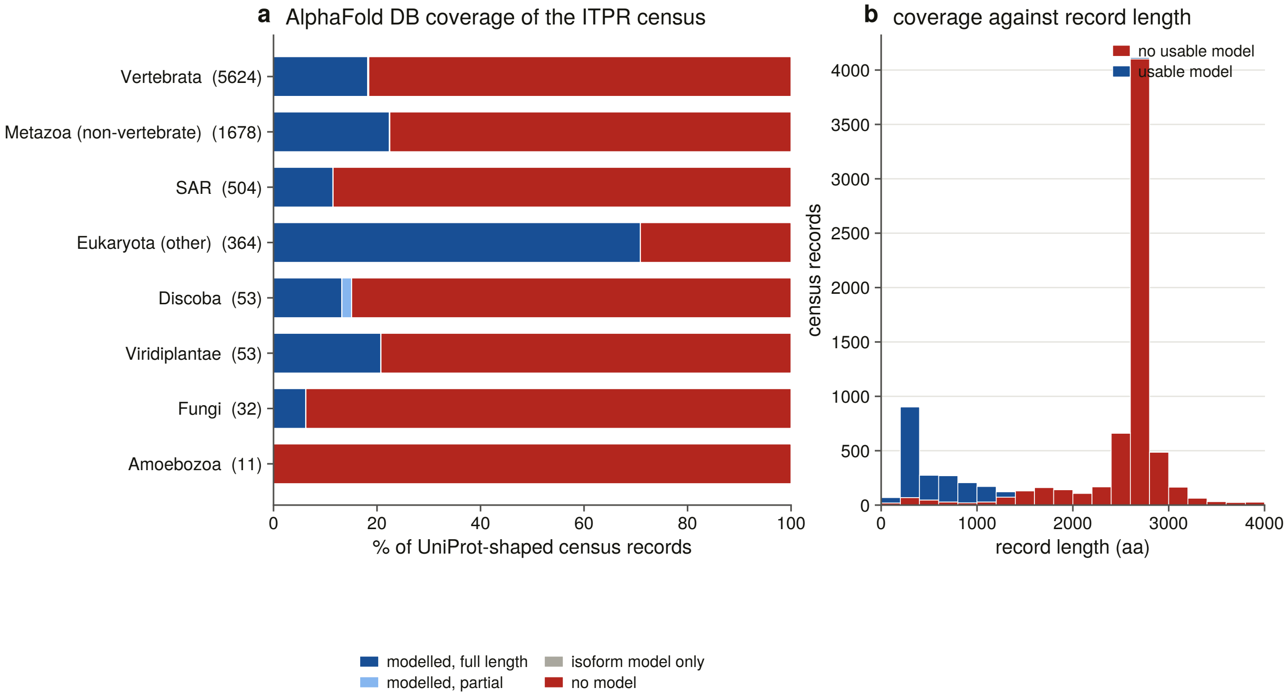


Figure 11.4. Coverage by group, and against record length. The second panel is the result, and it sets the ceiling on every structural argument this family can support: the usable-model mass sits below about 1,300 residues, while the peak at about 2,700, which is a full-length subunit, is almost entirely unmodelled. A reader who assumed the prediction database covers modern proteomes uniformly would expect a receptor of this size to be modelled. Every structural statement in this chapter therefore rests on experimental depositions, and the figure is what makes that a measured constraint rather than a preference.

The median modelled record is 392 residues and the median unmodelled one 2,674. Of the 5,861 census records at or above the family's own length floor, meaning the plausibly full-length ones, **13, or 0.2 %, have a usable model.** The coverage this database offers the family is concentrated on its fragments, which are the records a structural argument can do least with.

This was written down as a guess before it was measured, which is the only reason it is worth reporting as a confirmed prediction rather than as an observation.

11.3 Seven rules build the structural panel, and nothing enters unread

The references are resolved by query rather than from memory, and seven rules choose what goes in, each a positive test.

The family call comes from this project's census on the entity's accession, never from an entry title, because the two families share every diagnostic domain and half their titles. Structures must be cryo-electron microscopy with a recorded resolution and full length in the family's declared band, so a ryanodine receptor cannot enter through the IP_3 band because the band was defined to exclude it. The conformational state must be readable from the title against a stated vocabulary, ordered so that compound states are tested before the words they contain. One reference is taken per paralogue, and a paralogue the census does not name gets no label rather than falling back to the family name. A state panel is built on the paralogue offering the most conformations, because a model scored against one conformation confounds fold with gating. And the negative controls come from Chapter 3's committed decoy panel, one per class, held to the same rules and size-matched.

Enumerating candidates on the **union** of the four family signatures is load-bearing. The structural database's own domain annotation of human ITPR3, which is this project's own IP_3 receptor reference, carries four signatures and **not** the one that names the family, which is the same absence Chapter 3 measured on 2,911 of 15,417 proteins. A query on that signature alone would have missed the reference.

One further instrument is available and is deliberately optional: a structure-based homology search over the whole prediction database [90], which is the one stage in the project needing a multi-gigabyte index and is off by default. What it returns is the shared domain rather than the family, which is the same statement §11.5 makes with coordinates.

Nothing enters the panel unread. Every file is parsed, reduced to **one chain**, because both families are homotetramers and an assembly is about 11,000 residues, so scoring a monomer against a deposited assembly would answer a question about quaternary structure, and then measured. Twenty-nine of thirty structures are usable, and the one rejection is the 181-residue isoform model, caught by a rule rather than by inspection.

Where a taxonomic slot has no representative with a model, the best-covered census record stands in, and the audit records that the slot was filled rather than met. Without that, the panel is six proteins.

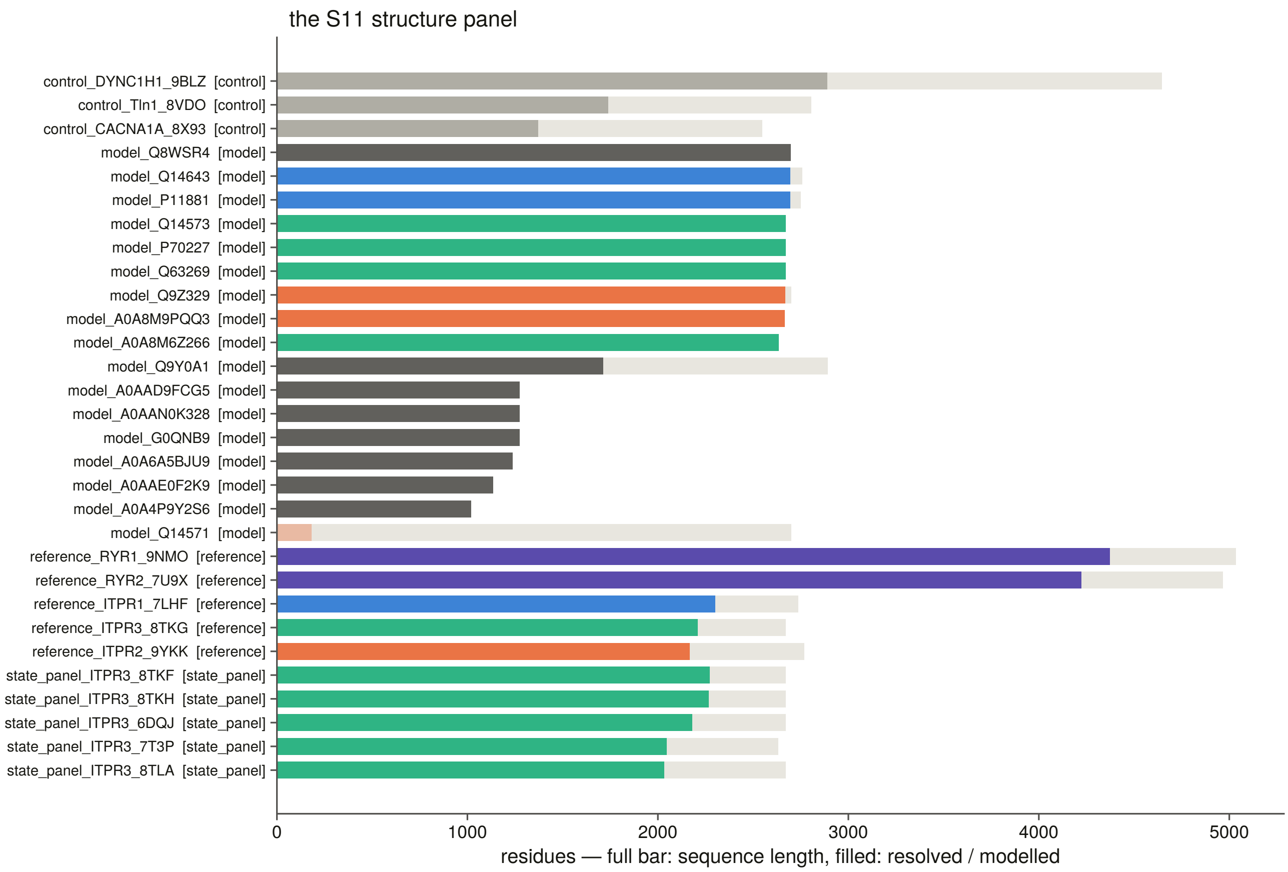


Figure 11.1. Every structure in the panel, where the pale bar is the length the record claims and the filled bar what the structure delivers. The importance of drawing the claimed and delivered lengths as two bars is that the gap between them is what a fold comparison is actually computed on. A structure resolving half its record has half its residues, and a score normalised by the wrong one of those two numbers is a statement about size rather than about shape. The panel lets a reader see, for every entry, how much protein each later comparison had to work with.

11.4 The comparison scale is calibrated on this panel before any structure is called

Before any structure is called, the scale is calibrated on this panel rather than taken from the literature, using five classes of pair, each answering a different question.

The negative controls span 0.09 to 0.25 and none of 81 control pairs reaches the same-fold bar under either normalisation. That is the floor every positive result is read against, measured on proteins chosen as decoys for a sequence scorer and re-used here without reselection.

Conformation is worth 0.22 of the score. Two structures of the same paralogue in different states score a median 0.78, while two different IP_3 receptors score 0.43. So a fold difference of that size or less is not readable on this panel and is not claimed, which is what the state panel exists to establish.

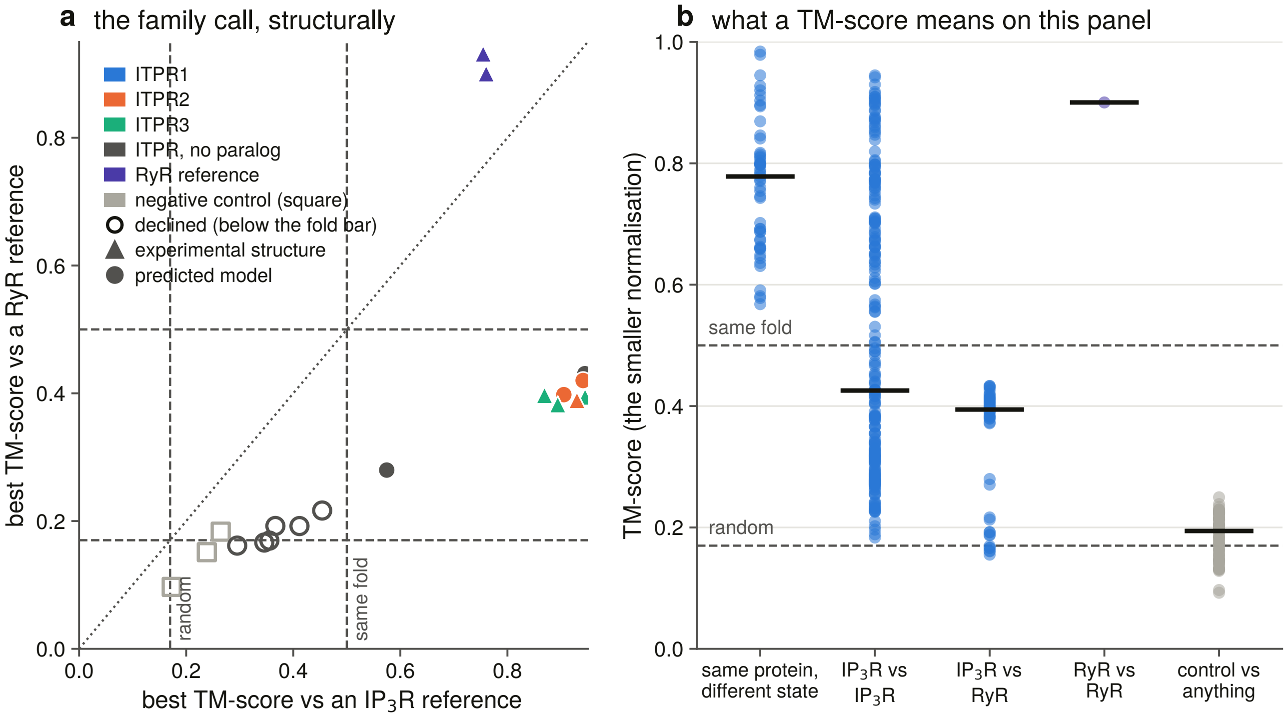


Figure 11.2. The family call made structurally, and the calibration behind it. Both of the alignment method's published bars are drawn, meaning the random-similarity floor and the same-fold bar [91], rather than described. A calibration figure that asked to be believed would not be one. The importance of this figure is that it turns the family separation into a structural result rather than a sequence one. Everything before this chapter told the two families apart on sequence and domain content; here the same separation is recovered from shape alone, against published bars the project did not choose. That matters because the two families share every diagnostic domain, so a reader is entitled to ask whether the distinction survives when domain annotation is removed from the evidence.

Both normalisations are kept, because on this panel they say different things. A subunit resolves to about 2,200 residues and a ryanodine receptor to about 4,300, so **which chain a cross-family score is normalised by decides whether it reads as a fold result or a size one.**

11.5 Shape separates the two families exactly as the census does

This is the sixth instrument in the thesis to separate the two families, and the first that reads no gene symbol, no domain architecture, no alignment score and no tree, but only coordinates.

The test is the best score against the IP_3 references, the best against the ryanodine references, and a **relative** margin between them, with scores normalised by the reference, which is the question being asked: does this structure account for an IP_3 receptor, or for a ryanodine receptor?

A margin is only consulted once the winner clears the same-fold bar, and that gate is not decoration. All three negative controls beat their own runner-up by about 30 % of their score while scoring 0.17 to 0.27 against everything, so a rule gated on the margin alone calls a dynein heavy chain an IP_3 receptor.

Twenty of twenty structures the instrument could call are called the way the census calls them, and none of three negative controls receives a family call at all.

The instrument declines six further census-named structures, and a refusal is not a disagreement. Every one of them still prefers the IP_3 receptors over the ryanodine receptors by roughly two to one. What they cannot do is account for enough of a reference to earn a call.

11.6 The fold test does not reach the deep records

This is where the structural work is worth most. The plant, fungal, protist and non-vertebrate metazoan records were called IP_3 receptor on sequence evidence alone, meaning profile score, architecture, and a per-record contamination chase. None of that is shape.

Two of seven deep models reach the same-fold bar and the rest fall short.

The interesting column is the **ceiling**. A reference-normalised score divides by the reference's length, so a model of n residues scored against an L-residue reference cannot exceed n/L before similarity is considered at all. Most of these models are 1,000 to 1,300 residues, which are the only records the prediction database holds for these groups, against references of about 2,050.

None of them is excluded by arithmetic alone. Every one had the headroom to pass and did not, reaching 0.56 to 0.74 of its own ceiling, which is far above the negative controls and far below a full-length receptor. That is what a genuinely divergent homologue modelled at low confidence should look like.

The honest verdict is that the fold test neither confirms nor contradicts these records. It does not reach them, and the project's vocabulary has a word for a measurement that did not happen.

11.7 Prediction confidence is highest at the ligand core and lowest at the pore

A mean confidence score over a 2,700-residue multi-domain channel averages a well-predicted β-trefoil with hundreds of residues of linker, and the resulting number is high enough to look reassuring while saying nothing about the part any claim rests on. So confidence is reported **per domain**, with the boundaries transferred by pairwise alignment, and a domain landing on too little of its reference span is reported unplaced rather than averaged over whatever aligned.

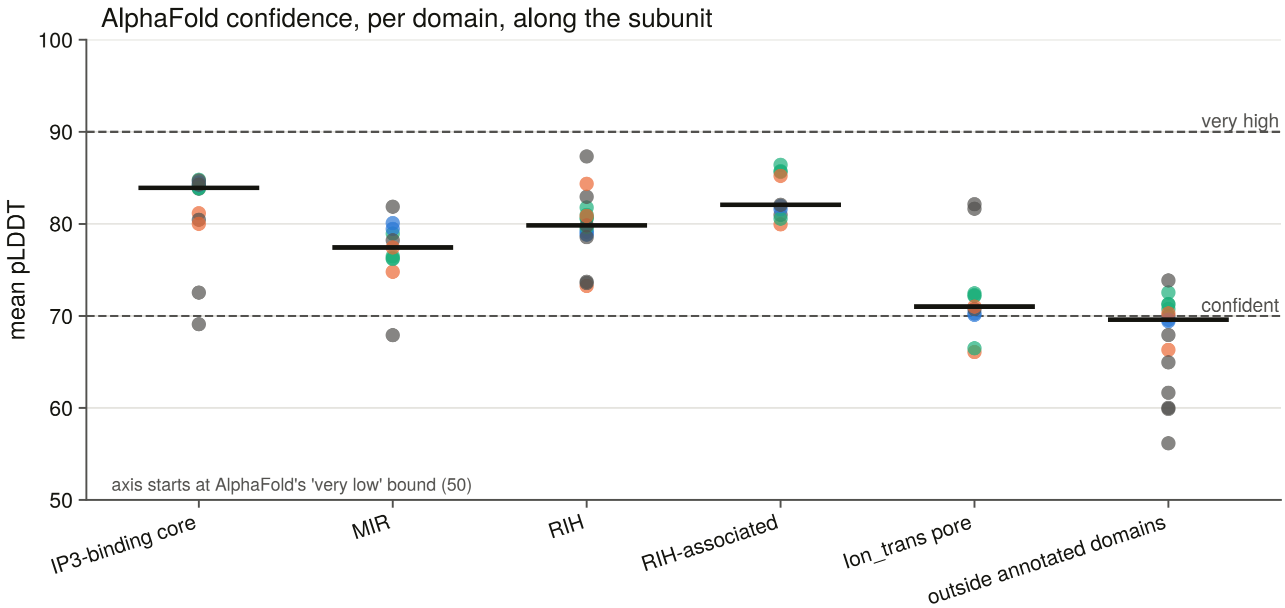


Figure 11.3. Confidence per domain, ordered along the subunit, with the prediction method's own confident and very-high bands drawn. Every model also gets an outside-annotated-domains contrast row, which is what gives a high value something to be high against.

The IP_3-binding core is the best-modelled domain of the receptor at a median of 83.9, against 69.5 outside the annotated domains, and **the pore, which is the part the two families genuinely share as a working channel, is the worst of the named domains** at 71.0. That matters for the next chapter, which is scoped around the ligand site, because whether the prediction is good there decides whether a structural argument about it can stand at all.

11.8 The constraint map needs depth the representative alignment does not have

The rest of this chapter is per-residue conservation, and it needs depth the representative alignment does not have. Thirteen to nineteen tips per paralogue is a phylogeny and it is not a per-site estimate.

So one locus per genome per paralogue was taken out of the 309 sweep outputs, giving **249 to 265 orthologues of each individual paralogue.** Five rules select them, and the fifth is one the sweep's own bars cannot do.

Bait coverage is aligned residues over bait length, so a model that covers the bait and carries two thousand extra residues passes everything, which is exactly what a large intron setting manufactures in the giant genomes. So the fraction of each sequence's own residues landing in reference columns is measured, and the bar is put in that distribution's own gap with both edges committed, refusing to fire at all when nothing sits below the lowest curated record.

It drops one sequence, and that sequence was inflating one paralogue's alignment from 3,380 to 5,676 columns.

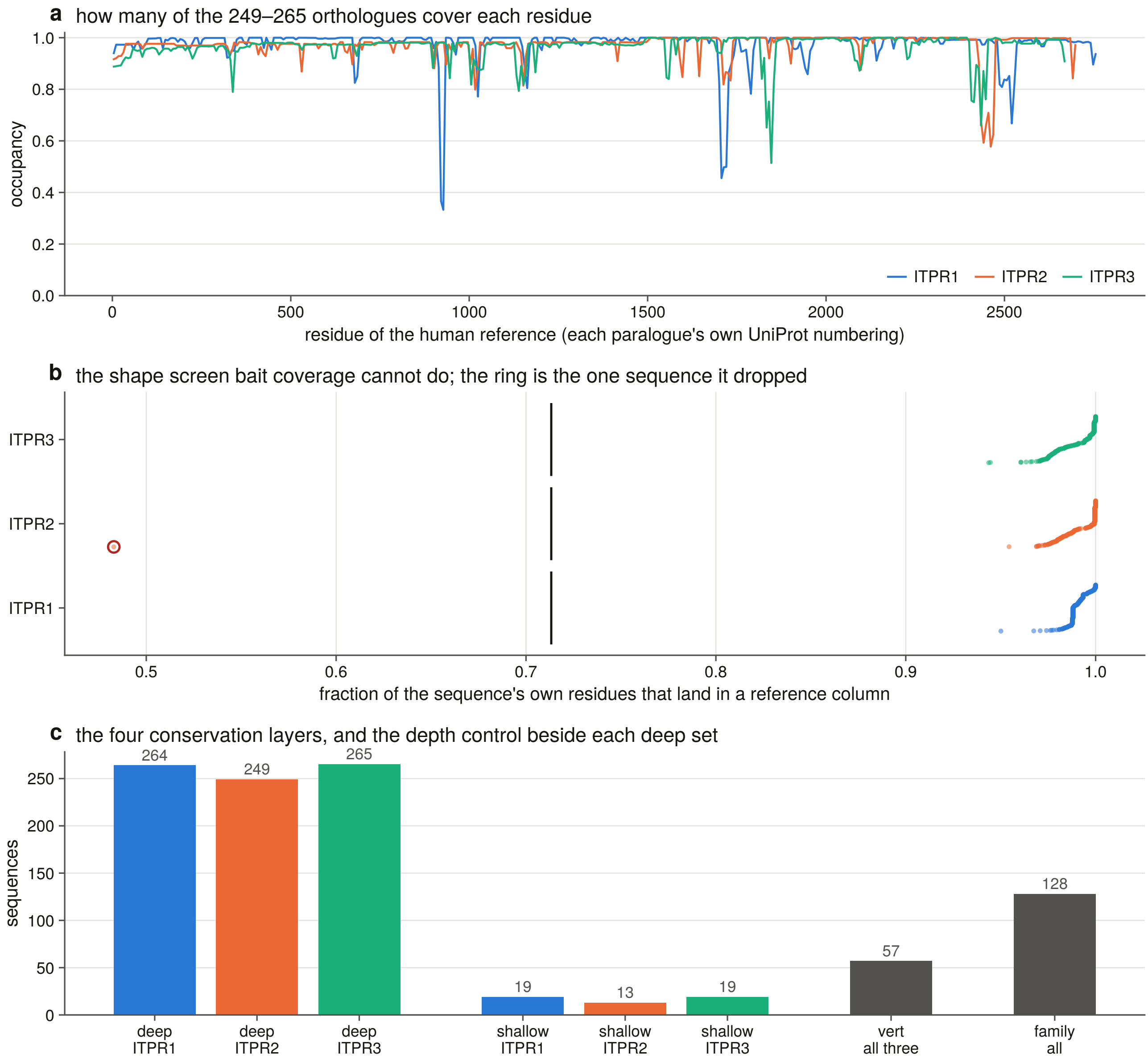


Figure 11.9. The three deep alignments the constraint map is computed on, drawn as occupancy along the human reference rather than along an alignment column, since the three have three widths and no shared coordinate. What the panel shows is the depth behind the map, which is the difference between this resource and one built on a handful of sequences, and where that depth thins out enough that a score there should be trusted less.

Three conventions run through every score. **Conservation is sequence-weighted** [92] before any column statistic, and the column metric is an information-theoretic divergence [93], because ray-finned fish are about a third of the genome scope and unweighted, every teleost-specific residue in a 260-sequence alignment reads as conserved. **Gaps are missing data rather than a 21st state**, and the occupancy each score is conditional on is reported. And the paralogue of each alignment tip is read from Chapter 6's committed table rather than re-derived from the tree, because a third derivation that disagreed would be a silent fork in what this project means by "ITPR2".

11.9 The Pfam signature named after the ligand does not contain the ligand site

Joining the domain coordinates to the structural measurements says something that changes how the rest of this chapter has to be read.

None of the ten measured IP_3 contacts lies in the Pfam signature named *Inositol 1,4,5-trisphosphate/ryanodine receptor*. They sit in MIR and RIH, which are the two domains the family shares with the ryanodine receptors.

So the N-terminal β-trefoil is not the ligand site, and the ligand question is asked throughout of the measured contacts and never of the Pfam label. A report that had taken the label at face value would have measured the suppressor domain and called it the IP_3-binding core.

The structural elements, meaning the filter, the gate and the ten contacts, are in one paralogue's numbering and are transferred, each with an **anchor test that can fail**: the filter must arrive on the same motif, the gate on the same lining residues, and a failed anchor aborts rather than writing a coordinate. The Pfam elements are measured per accession and are not transferred at all, so the whole class of transfer error does not arise for them.

One element no annotation carries is located **by geometry**: the luminal loop, defined as the contiguous run of channel residues whose backbone lies beyond the membrane on the luminal side of the measured axial span. A boundary drawn on the conservation profile would be the profile explaining itself.

Residues outside every named element are given **named linkers** rather than left blank, because a single unassigned bucket would pool the N-terminus, five inter-domain stretches and the whole C-terminal tail, and then use it as the within-protein control.

11.10 The gate and the selectivity filter are the least changeable elements

Every element is tested against the same protein's own linkers rather than against its whole-protein mean, which would contain the element being tested.

The gate and the selectivity filter are the most constrained elements of the protein, on both metrics and in all three paralogues. The RIH-associated domain is the most constrained domain. Every named element except one sits above its own protein's linker mean.

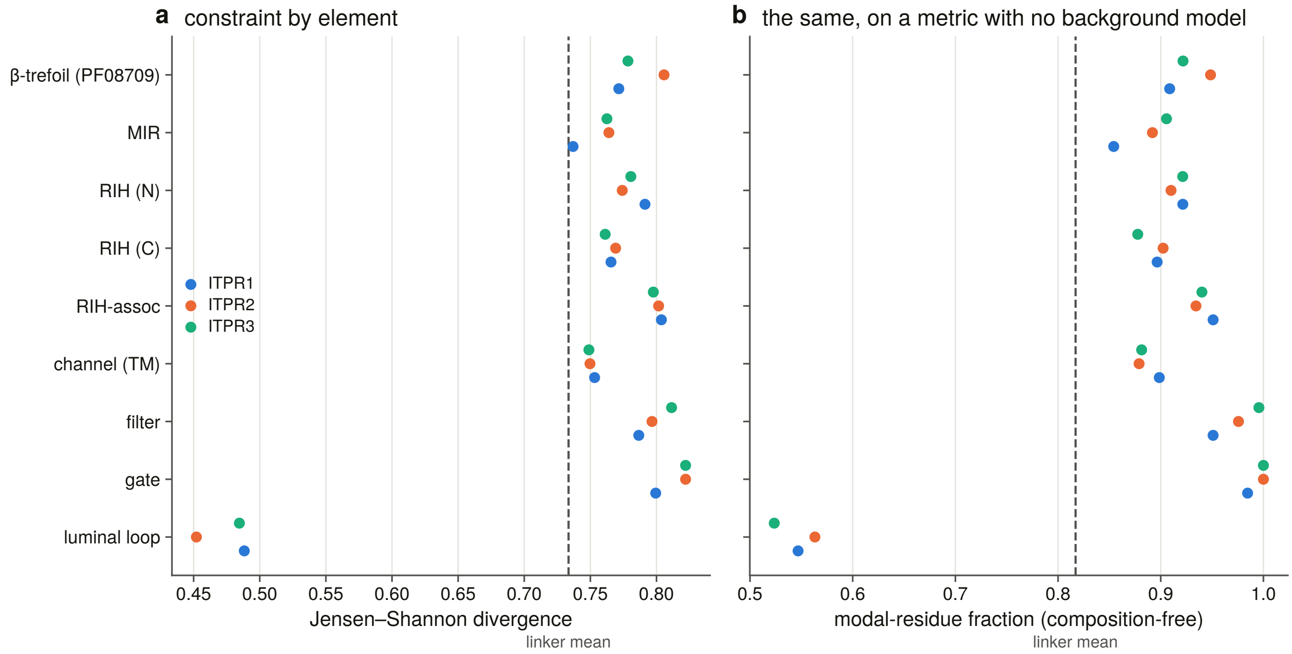


Figure 11.6. Constraint by element, with the composition-free metric drawn beside the divergence metric rather than instead of it. The headline result and its most obvious artefact are therefore on the same axis: the gate and the filter come top on both metrics, so the ranking is not a property of one scoring choice, and a reader can check that agreement rather than take it on assurance. The second metric is there because a divergence from a background frequency table [93] scores a transmembrane element lower at equal conservation, which would have manufactured the very result the chapter reports.

11.11 One element inside the channel is the exception, and finding it changed the result

The luminal loop scores far below every other element and below the linker mean, making it by a wide margin the least conserved sequence in the receptor, on the composition-free metric as well.

Before it was separated out, the pore domain read as the least constrained named element in the receptor, below the linkers. That is a strange thing to report about the pore of an ion channel, and the sort of result that should not be printed without asking what else could produce it. Three explanations were checked and all three were measured rather than argued.

The first candidate explanation was a **metric artefact**, since the divergence measure is computed against a background frequency table and a transmembrane domain is built of common residues. It was ruled out, because the composition-free metric gave the same picture.

The second was a **gene-model artefact**, since the deep alignment is about 95 % genomic translations and the exon-dense transmembrane region is where a mis-placed boundary would do most damage. It was ruled out, because recomputing on the curated subset alone still put the domain below the linkers.

The third was **an unresolved element**, and it was confirmed. The domain's mean was the average of the most conserved stretch in the protein and the least, and a bar of that mean is a number no residue has.

With the loop separated, **the pore module is more constrained than the linkers, and the fifty residues of luminal loop inside it are the most variable sequence in the receptor.** Those two live about fifty residues apart in the same Pfam domain.

This is what the geometric definition was for. Had the loop boundary been drawn on the conservation profile, this section would be circular. Drawn on the membrane's own axial span in the structure, it is an independent prediction that landed on the dip.

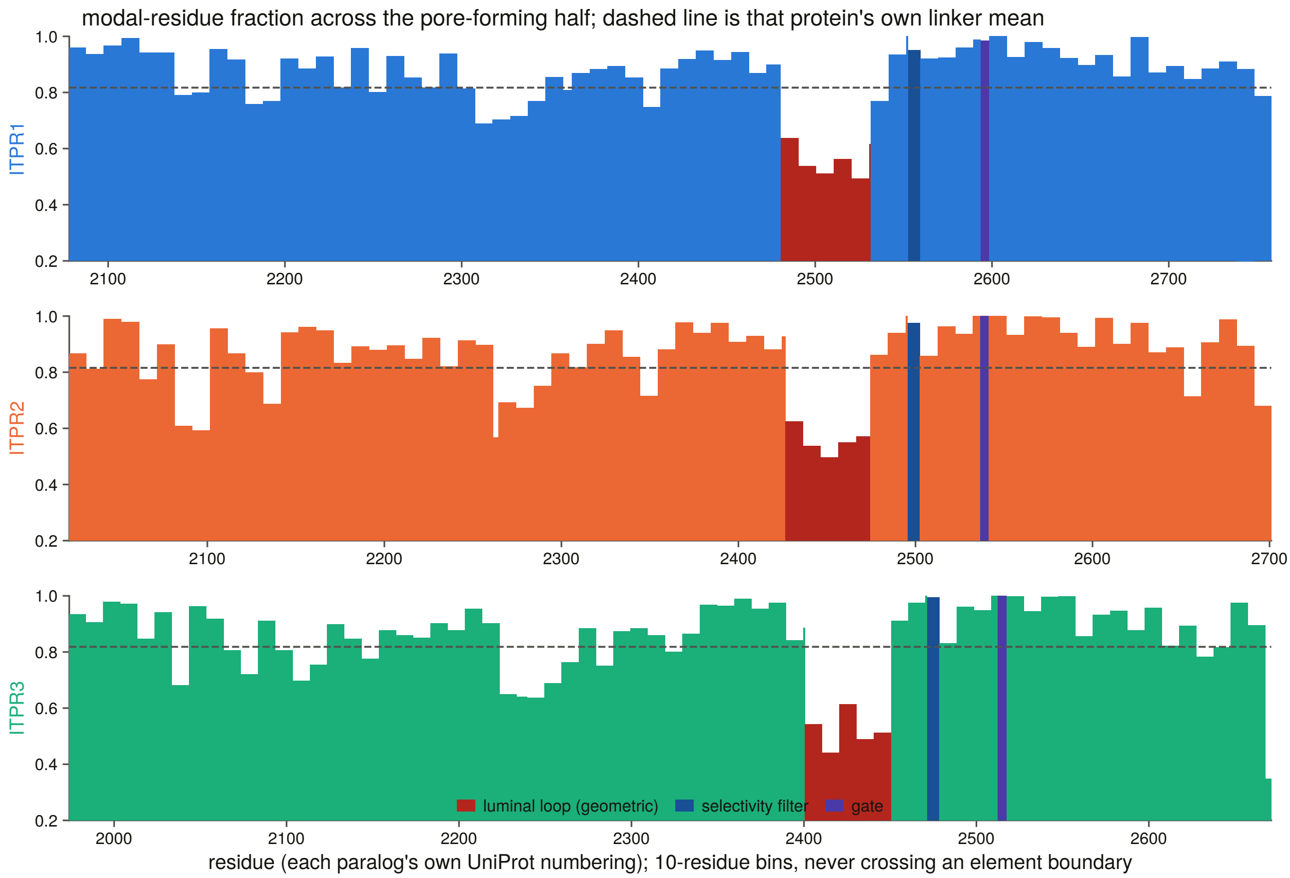


Figure 11.5. Constraint along the channel, binned, with the bins never crossing an element boundary. This is where the chapter's structural claim becomes visible as a shape rather than a table: constraint is not uniform along the channel but peaks on the residues the structure says do the work and collapses in the luminal loop between them. The binning rule matters because a sliding window across that edge would smear the collapse into the gate and erase the contrast the panel exists to show.

11.12 The ligand contacts and the gate are the two extremes of paralogue divergence

Each site class is tested against two controls: against the whole protein, and, the sharper one, against the rest of its own element. A gate residue beating the average residue of a 2,700-residue receptor is nearly guaranteed, and beating the rest of the pore domain is not.

The measured IP_3 contacts are more constrained than the rest of the domains that carry them, in all three paralogues. That is a statement about ten residues and is reported as such. The filter and gate sets are two residues each, and their means are given because the element-level test is where those elements are actually powered, so a p-value on two residues is not offered.

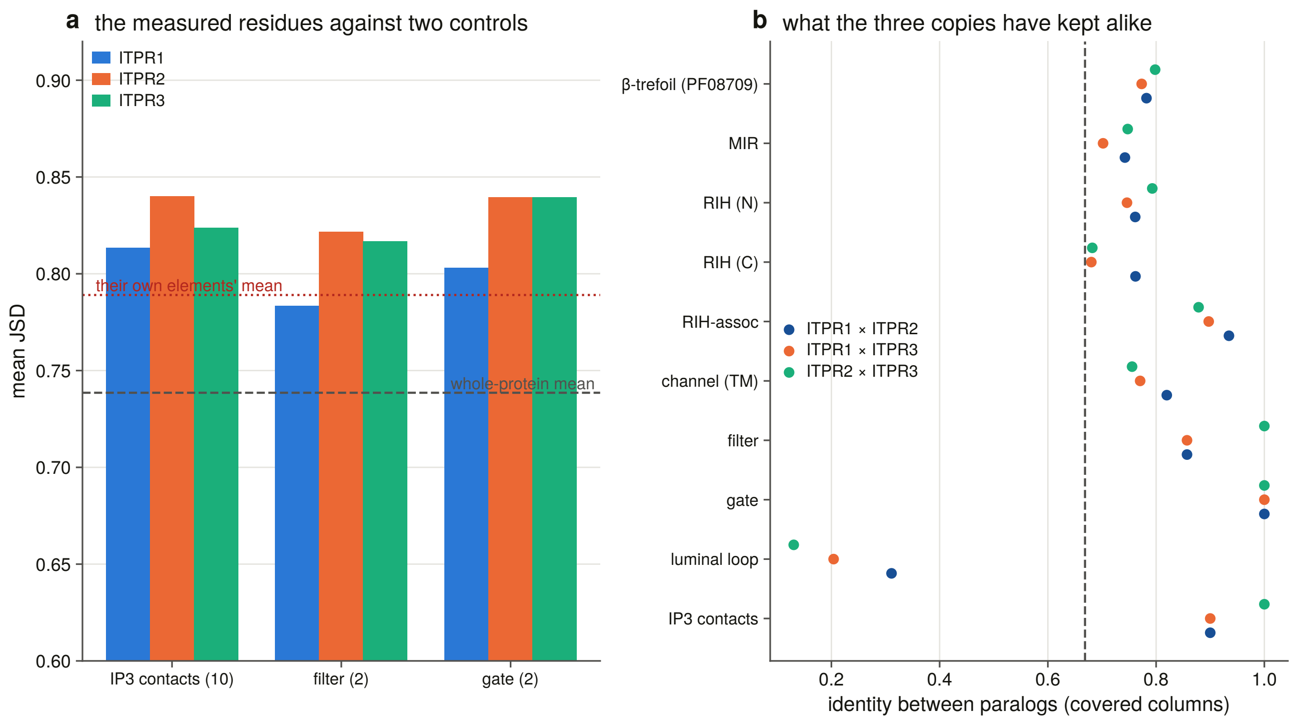


Figure 11.7. The measured functional residues scored twice, against the whole protein and against the rest of the element each sits in. The second comparison is the harder one and the one worth reading. Most of a 2,700-residue receptor is linker, so clearing the first bar is close to guaranteed, whereas clearing the second shows that the structure's annotation picks out something its surrounding domain does not already have. That is the test which makes a constraint map useful for interpreting a new position.

A second and completely independent instrument on the same question is identity between the paralogues, which needs no alignment depth and no conservation metric at all.

The gate is identical in all three pairs. The luminal loop retains 13 to 31 %. The whole protein sits at 0.64 to 0.70.

So the two extremes of paralogue divergence in this receptor are **five residues that have not changed since the duplications** and **fifty that retain a tenth to a third of their identity**, and they sit inside the same domain. Whatever the three copies were free to differ in, it was not the gate.

One number looks like a disagreement with Chapter 6 and is not. Chapter 6 reports between-paralogue identity at 0.741 to 0.791 and this chapter reports 0.64 to 0.70. Neither is wrong, because Chapter 6 measured on the trimmed alignment, which kept 15 % of the columns. So every per-element row here carries the same pair measured Chapter 6's way as well, and it reproduces that chapter's committed matrix exactly.

This chapter cannot use the trimmed alignment, and the reason is the result. Trimming deletes divergent and gappy columns, and the element this section's headline rests on is the most divergent and gappiest in the receptor: 12 of the luminal loop's 51 residues survive trimming, against all of the gate and all of the filter. A per-element identity read off the trimmed alignment would be a measurement of what survived trimming.

11.13 Conservation works as a variant classifier, and the best layer is not the deep one

The harvest returned 1,753 missense records across the three genes, and **none was dropped.**

Numbering is checked rather than assumed. Every cited transcript's own translated coding sequence is fetched, aligned to the canonical, positions transferred through that alignment, and the reference amino acid then required to match. All three genes file their records on a transcript whose translation is the canonical, so the check passes with nothing dropped.

That is a result of the check rather than a reason it was unnecessary. The same code on another channel family found 512 of 773 positions carrying a different amino acid in the canonical sequence, and a resource built without the check would have been wrong everywhere. Here it is right everywhere, and the report can say which.

The harvest carries a positive control it could fail. Chapter 2's curated table localises exactly two variants to a named residue, and both must come back out of a query returning about 1,750 records. Both are recovered.

The clinical record is mostly uncertain, and that is a finding. 1,546 of 1,753 records, or 88 %, are of uncertain significance, and the labelled sets a classifier can be scored on are 49, 1 and 5 pathogenic positions. One paralogue's entire clinical record is a single pathogenic missense variant, so no per-gene classifier score is reported for it.

The four layers are compared on one fixed set of variants. They do not cover the same residues, because the representative-alignment layers lose gappy columns the deep alignment fills and the reverse, so ranking four scores measured on four slightly different variant sets would compare the sets as much as the layers. The comparison is restricted to the 44 pathogenic and 34 benign positions where every layer has a reliable score.

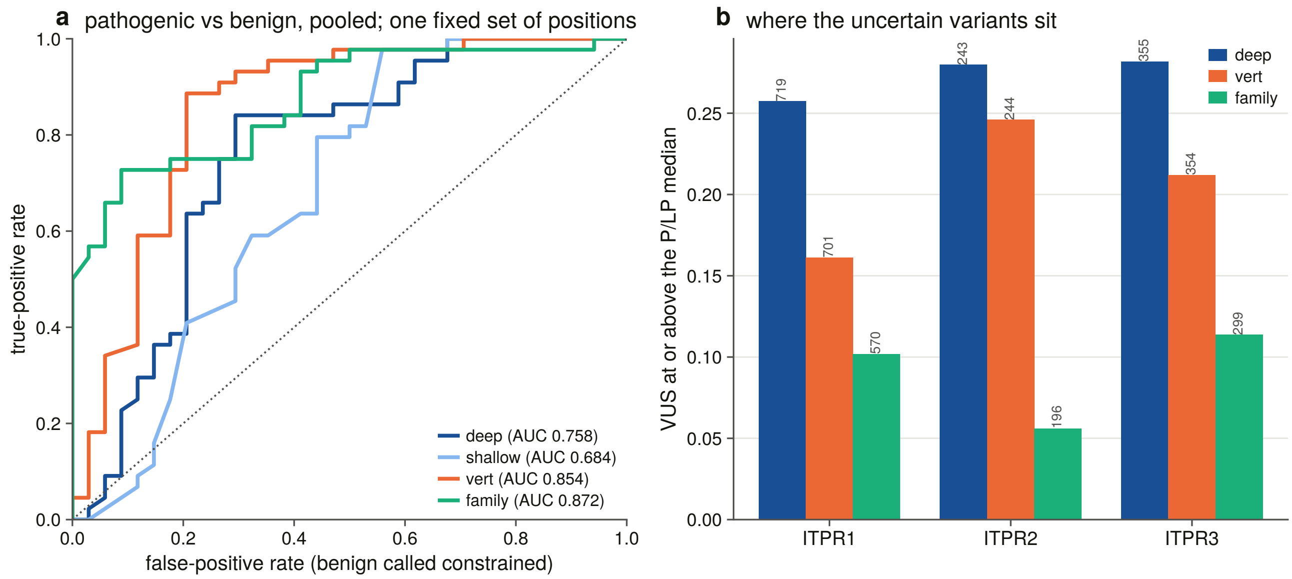


Figure 11.8. Four conservation layers as classifiers, drawn as full curves rather than as a bar of summary scores, because the layers cross. This figure carries a result that runs against the chapter's own design, which is why it is drawn as full curves. The family-wide layer outperforms the deep within-paralogue layer this chapter was built to produce, and the shallow control is worst, so depth was worth building but breadth was worth more. Summary scores would have hidden the crossings that make the ranking readable, and a resource for this family should quote the layer this panel puts on top.

Two things fall out, and one is against this chapter's own design.

The shallow control is the worst layer, so the depth was worth building, because without the deep orthologue sets that is the number this thesis would have had.

But **the best layer is the family-wide one rather than the deep within-paralogue one.** Taxonomic breadth beats within-gene depth here, because a position conserved across 500 million years of paralogue divergence and across the eukaryotes discriminates pathogenic from benign better than a position merely invariant across 260 vertebrate orthologues of the same gene. A resource for this family should quote the family-wide layer, and the instrument this chapter was built around is the second best of four.

The whole-protein control is why the first column cannot be read alone. Every layer also separates pathogenic positions from the average residue, but by less than it separates them from the benign set, so part of the signal is benign calls being enriched in unconstrained positions rather than only pathogenic calls being enriched in constrained ones. Both directions are real and the control is what distinguishes them.

11.14 What a constraint resource can offer an uncertain variant

Eighty-eight per cent of the record is uncertain, and a per-site score is what a resource of this kind can offer such a variant. Each is placed against the labelled distributions of the same gene, on thresholds that are those distributions' own medians, so no cut was chosen to make a count.

This is a stratification rather than a call. It says where a variant sits on an axis the labelled variants separate on, which is a different claim from pathogenicity, and the per-gene numbers inherit the problem above, in that one paralogue's pathogenic median is a single position's score.

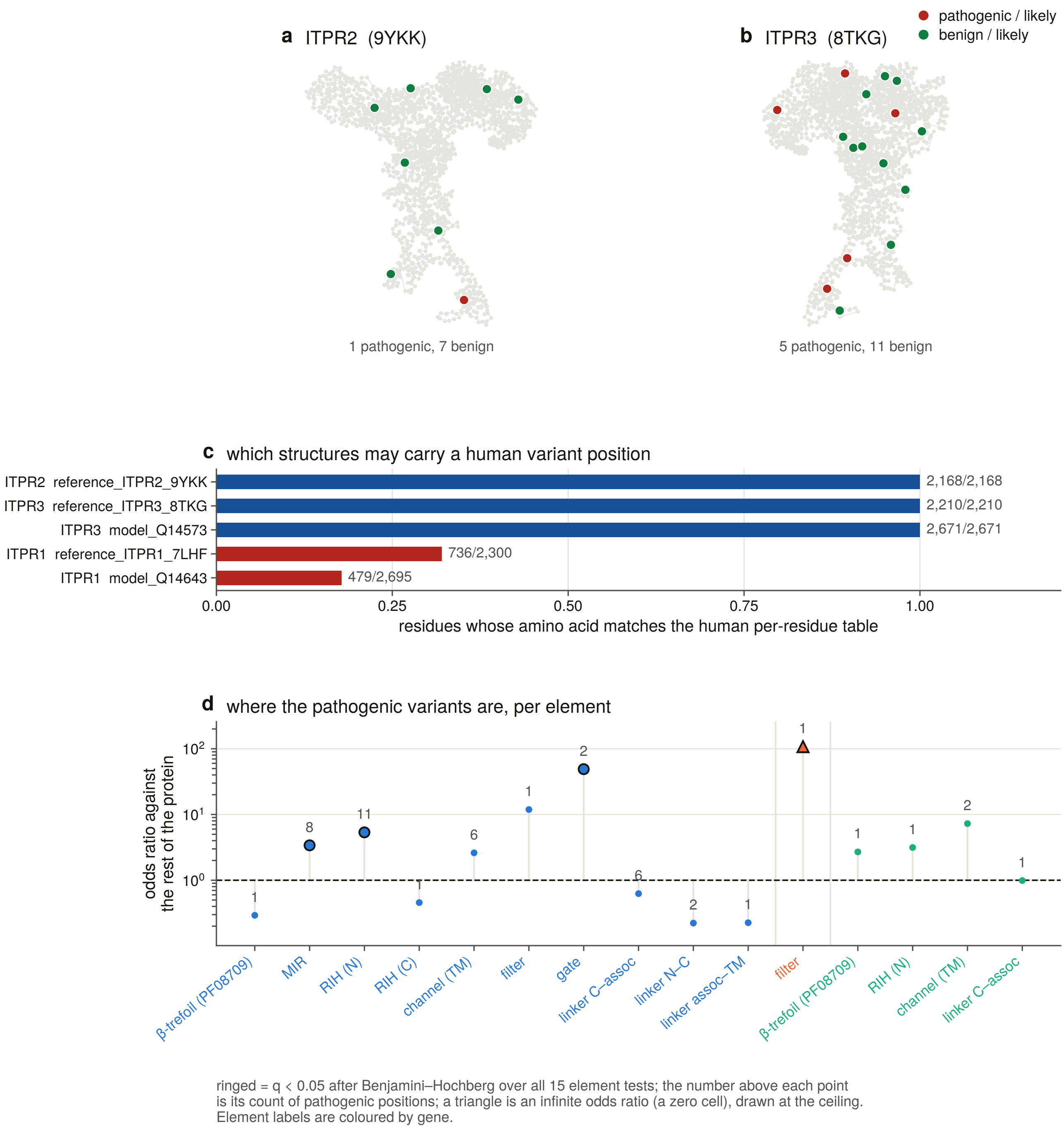


Figure 11.11. Every labelled variant with the per-element enrichment test beside it. An infinite odds ratio is drawn at the ceiling with a marker rather than allowed off the axes, where a significant result would simply vanish. The importance of this figure is that it puts the clinical record on the same coordinates as the constraint map, which is what a variant resource has to do to be usable. Pathogenic positions concentrating in the elements the structure says gate and select is the observation that licenses stratifying the 1,546 uncertain variants at all. Drawing an infinite odds ratio at the ceiling keeps the strongest enrichments on the figure, where otherwise they would silently leave it.

The deliverable is one table per human paralogue, giving every residue in its own numbering, four conservation layers, the occupancy each is conditional on, its element, and whether it is a measured contact, filter or gate residue.

11.15 Per-site selection rates land on the same coordinates

Chapter 10 answered the whole-gene question and not where the constraint sits. A per-site rate model [94] on the same codon alignments gives that, and every site is carried onto the human reference through a **validated transfer**, in which a site whose amino acid disagrees after realignment is written with no residue rather than with a plausible wrong one.

The gate and the filter come back at a non-synonymous rate of zero with 100 % of sites called significantly constrained, in every paralogue where the test has power.

Three cautions are carried forward rather than restated. A per-site ratio is taken only over sites where the synonymous rate is identifiable, with the excluded count as a column. The constrained fraction is read at the method's own significance threshold. And the obvious aggregate, meaning the sum of one rate over the sum of the other, is not usable here, because Chapter 10 measured synonymous saturation on these alignments and the method duly pins the synonymous rate at its bound on a share of sites, so the sum is dominated by sites whose denominator is unidentifiable rather than large.

11.16 The constraint map is painted onto the experimental structures

Both layers are written into the B-factor column of every IP_3 receptor structure in the panel, on one scale read the same way round, so they colour with one command. Sixteen structures are painted at 98 to 100 % of their resolved residues.

Unscored residues are written as (-)1 rather than 0, and that distinction is load-bearing for exactly one region. The luminal loop is both the least conserved element in the receptor and the one the maps resolve worst, and on a coloured structure those two must not look the same. A chain that maps less than half of itself to its paralogue is refused rather than painted with somebody else's profile.

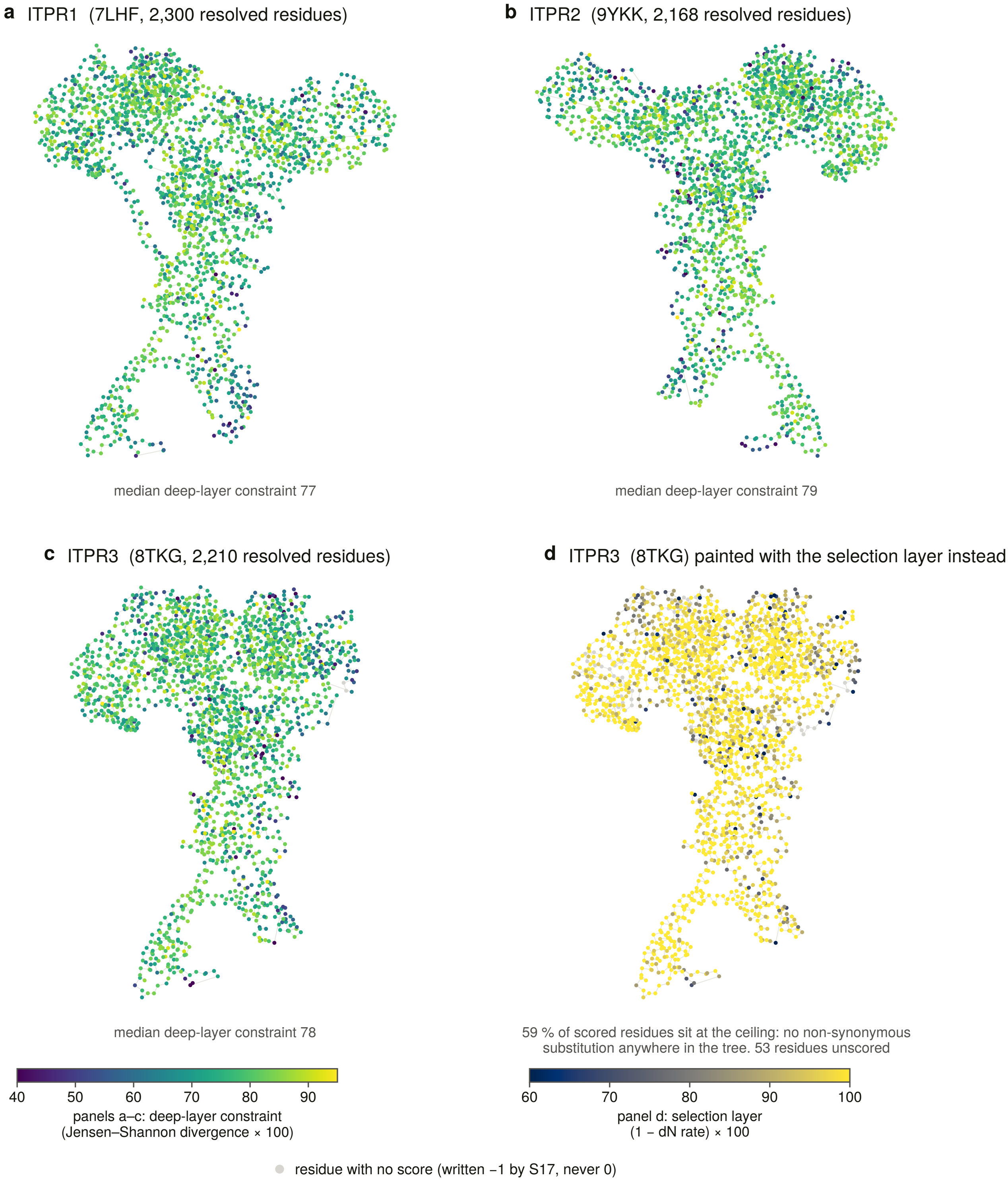


Figure 11.10. The constraint map painted onto the channel, drawn from the file the painting step wrote and coloured from its own B-factor column, so a disagreement with the per-residue tables would be a bug in the painting step. Writing the layer into a structure's own coordinate column is what makes the chapter's numerical claims checkable by a third party in any viewer. Unscored residues are grey rather than at the low end of the scale, which keeps missing data from reading as an absence of constraint, and that matters most in the luminal loop, where the two coincide.

The panel is carried by experimental structures rather than predicted ones. The human ITPR2 prediction is the 181-residue isoform of §11.2 and has no file to paint, so that paralogue's constraint is shown on a cryo-electron microscopy entry. A structural resource for this family cannot be built out of the prediction database.

One further check has to be recorded because it fails on half its candidates. A structure may carry a human variant position only if every residue it shares with the human table carries the same amino acid, and two of four candidates fail, comprising a rat reference and an isoform model. One paralogue's 55 pathogenic positions are therefore not placed on a structure at all, which is a limit stated rather than worked around.

11.17 What this chapter settles about the channel

The gate and the selectivity filter are the least changeable elements of the receptor, on two metrics, in all three paralogues, and the gate is identical between all three. The measured IP_3 contacts are more constrained than the rest of the domains carrying them, and those domains are MIR and RIH rather than the Pfam signature named after the ligand. The pore module is more constrained than the linkers once the fifty residues of luminal loop are separated from it, and that loop is the most variable sequence in the receptor.

Conservation is a usable variant classifier for this family, at a discrimination the shallow control does not reach, and the layer that works best is not the one this chapter was built to produce.

Fourteen constructed negative controls run before anything is written, and one of them found a real bug on its first run: the within-protein control was being selected by a prefix test that swept 225 residues of the N-terminal β-trefoil, which is the element the ligand question is about, into the control set. The suite was mutation-tested on four deliberate rule breakages and caught all four.

What it does not settle is what the ligand site is for evolutionarily, which is the next chapter.

12. The ligand site, which is the one part the ryanodine receptors do not share

12.1 One functional asymmetry makes three questions askable

Every chapter so far has had to separate two families that share a fold, a pore, four domains and half their titles. There is one thing they do not share: the ryanodine receptors do not bind IP_3, and their gating is organised around a different set of ligands entirely [95].

That makes the ligand site the single place where a functional difference between the families is visible in sequence, and it makes three questions askable that are not askable anywhere else. Is the IP_3-binding core under different constraint from the pore? Do the ten residues that touch the ligand behave differently from the rest of the site? And in lineages that have lost the enzyme that makes IP_3, does the receptor's binding site relax?

The answers are, in order: yes and in the opposite direction to the expectation; no, and the reason is informative; and no, with the null bounded.

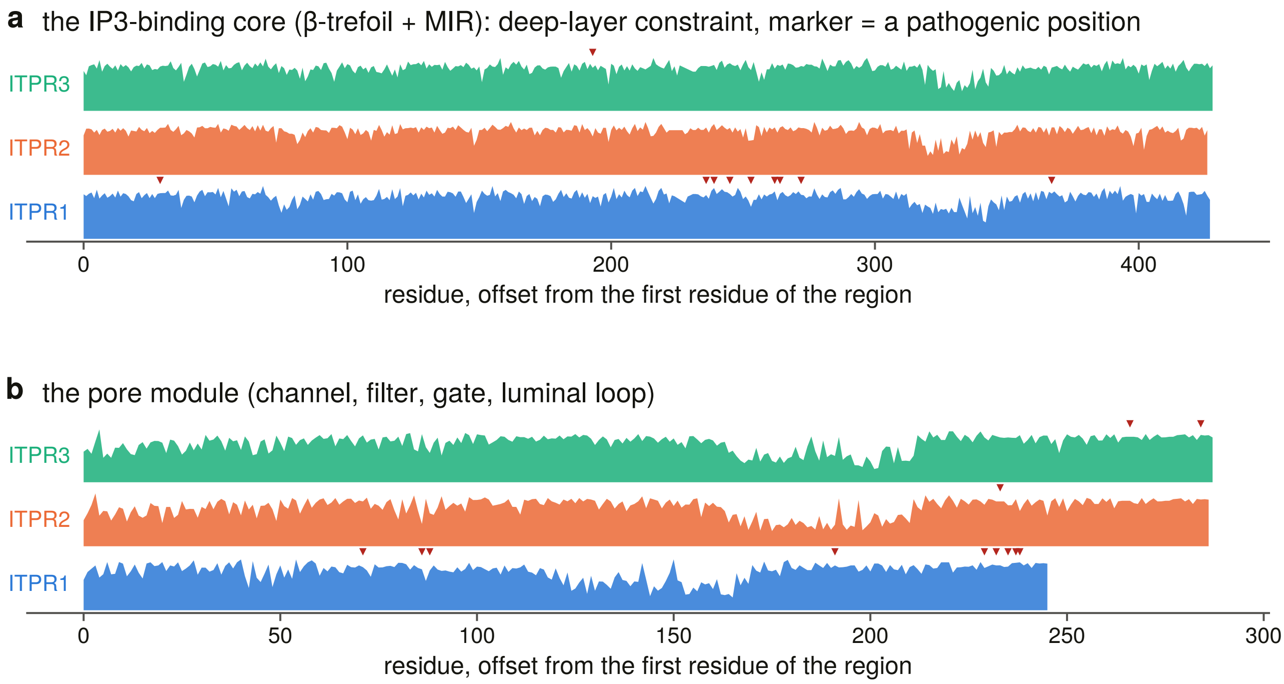
12.2 Neither module can be defined from Pfam, so both are defined by measurement

Neither module is a Pfam domain, and that is the whole reason this stage exists.

Pfam calls the ligand core *MIR* and *RIH_N* and knows nothing about the ligand, because Chapter 11 established that none of the ten measured contacts lies in the signature named after IP_3. And Pfam's pore domain contains fifty residues of luminal loop that Chapter 11 measured as the least conserved sequence in the receptor.

So the **primary ligand core** is defined as the minimal contiguous span containing every measured IP_3 contact, and the **primary pore module** as the Pfam pore domain less the luminal loop. A sensitivity definition is committed beside each and every test is run under all four combinations, using the published binding core [96] transferred through the same routine, and the pore domain as the database draws it.

Every definition is checked against something it does not contain. The core must hold all ten contacts and no pore site, and the pore both filter residues and both gate residues and no contact. A failure raises rather than being written.



c every pathogenic position in the two regions, and the residue each paralogue carries there

		ITPR1/2/3 1	2	3				ITPR1/2/3 1	2	3	
ITPR1 34	β-trefoil (PF08709)	D	D	D	→N	ITPR1 2449	channel (TM)	I	I	I	→V
ITPR3 196	β-trefoil (PF08709)	A	A	A	→T	ITPR1 2451	channel (TM)	L	L	L	→P
ITPR1 241	MIR	R	R	R	→K	ITPR1 2592	channel (TM)	L	L	L	→P
ITPR1 244	MIR	H	H	H	→P	ITPR1 2600	channel (TM)	T	T	T	→P
ITPR1 250	MIR	F	F	F	→L	ITPR1 2601	channel (TM)	F	F	F	→L
ITPR1 258	MIR	K	K	G	→M	ITPR3 2506	channel (TM)	I	I	I	→N
ITPR1 267	MIR	T	T	T	→R	ITPR3 2524	channel (TM)	R	R	R	→H
ITPR1 269	MIR	R	R	R	→L	ITPR1 2554	filter	G	G	G	→R
ITPR1 277	MIR	S	S	S	→I	ITPR2 2498	filter	G	G	G	→S
ITPR1 372	MIR	P	A	P	→L	ITPR1 2595	gate	G	G	G	→A
ITPR1 2434	channel (TM)	T	T	T	→I	ITPR1 2598	gate	I	I	I	→T

red letter = the other paralogue carries a different residue; every letter checked against that paralogue's own per-residue table

Figure 12.5. The two modules at residue resolution across all three paralogues, with every pathogenic position's residue printed. Printing the residues converts the chapter's module definitions from an assertion into something a reader can audit position by position, which matters because both modules are defined here by measurement rather than taken from a domain database. Every letter has been through the join guard described in Chapter 14: each variant's reference amino acid must be the residue its own paralogue's per-residue table holds there, and each aligned partner the residue the other paralogue's table holds.

12.3 The site is measured as a distance to the ligand, not as a contact label

A binary contact-or-not label discards the one thing a structure can say that an alignment cannot. So every residue within 15 Å of the bound ligand was measured, all-atom, in **six independent depositions**, comprising the structure used throughout this thesis [6] and five further IP_3-bound entries from other groups and other gating states [97,98,99].

The reader is deliberately small and all-atom. Chapter 11's structural reader is backbone-only, and a backbone trace puts an arginine 8 Å from a phosphate its side chain is hydrogen-bonded to.

The ten contacts are the positive control and a hard failure: the reader must recover all ten in the reference structure or the stage raises. A residue past the search radius is absent from the table rather than pooled into an open bin, which would be a population defined by the geometry of the search.

12.4 The comparison is paired inside one protein, which controls the taxon sampling

The two modules sit in the same protein, so every comparison uses **one core number and one pore number per orthologue**, and a difference between them cannot be a difference in taxon sampling.

A tip must resolve half of **both** modules to enter, and the dropped ones name the module that lost them, without which an N-terminally truncated gene model enters as an extreme ligand-core divergence.

Forty-four constructed negative controls run before anything is written, split across two suites with one entry point so the driver cannot run half of them, and the suite was mutation-tested on ten deliberate rule breakages, all ten caught. Four of this chapter's rules return a number that looks better when they are wrong. A core drawn without checking it holds the contacts still gives a clean p-value. A paired test admitting a tip that covers the pore and not the core reports a truncated gene model as ligand-core divergence. A permutation that resamples the whole module instead of the label answers a different question significantly. And a panel filing "no reference proteome" as "no enzyme" manufactures its own test set.

12.5 The pore is more conserved than the ligand core, which is not what was expected

The pore module is more conserved than the ligand core rather than less.

Paired per orthologue inside each paralogue's own alignment, across roughly 250 orthologues each, ITPR1 puts the pore ahead by 0.024 identity units with the tips running seven to one, ITPR3 by 0.020 with the tips running five to one, and ITPR2 shows no difference at all.

The module that gives this family its name and its defining signature is the less constrained of the two across a vertebrate orthologue set.

12.6 The core-against-pore answer reverses on one boundary nobody has had to state

All three paralogues flip sign when the luminal loop is left inside the pore module, which is how the domain database draws it and therefore how the comparison would be made by default.

Include the loop and the ligand core wins by about five identity points at overwhelming significance in all three paralogues. Exclude it and the pore wins in two and ties in the third.

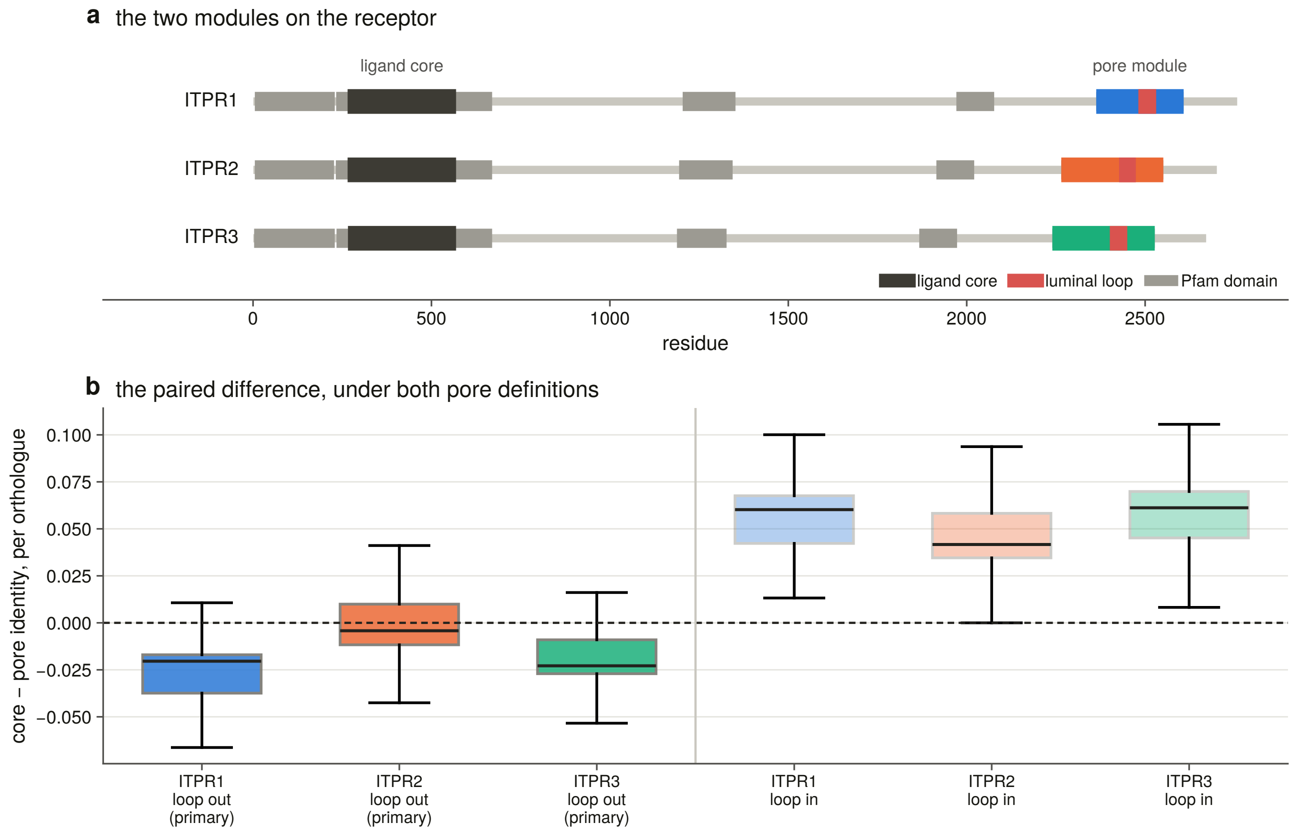


Figure 12.1. The core against the pore under both pore definitions on one axis with zero marked. This is the chapter's central result and its principal caveat in one panel: the pore is more conserved than the ligand core, which is the opposite of what a ligand-gated channel invites one to expect, and the comparison reverses in all three paralogues when fifty residues of luminal loop are left inside the pore. Drawing both definitions shows what the boundary choice costs instead of asserting it in a methods sentence.

Neither answer is wrong about its own module. They are answers about different modules, and the difference between them is one boundary that a comparison drawn on database spans would never have to declare. **Any version of this comparison that does not state the boundary is not interpretable.**

A second disagreement is reported rather than dropped. At column level the divergence metric sees no difference between the modules while the per-orthologue identity does. That is not a contradiction, because the divergence metric is measured against a background amino-acid table [93], so a transmembrane module scores lower than a soluble one at equal conservation. The composition-free column metric agrees with the paired test. The null result is a property of the metric, and it is printed here because a reader coming from Chapter 11's tables would otherwise find two of this project's own numbers pointing different ways with no explanation.

12.7 What selection holds at the ligand site is a pocket, not a contact set

The ten residues that touch IP_3 are more constrained than the rest of the binding core, and not more constrained than the rest of the pocket.

Against the core the contacts win in all three paralogues. Against every other residue the structure places within 15 Å of the ligand they win in none.

The permutation resamples **which positions carry the contact label**, keeping the constraint values where they are, because the hypothesis is about these particular residues.

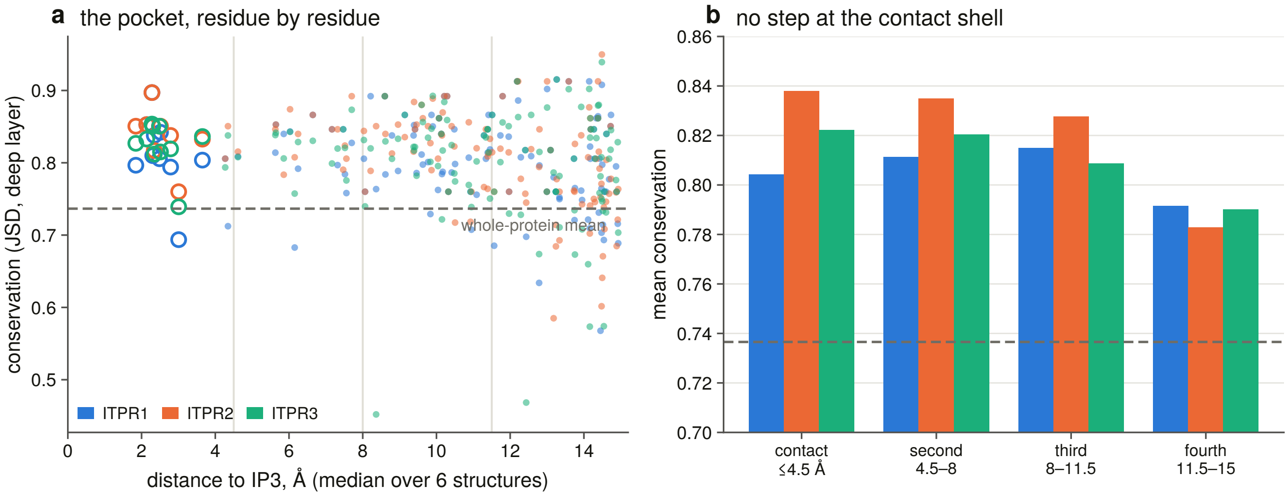


Figure 12.2. Constraint against distance from the ligand, drawn as a scatter rather than as a bar of shells, because the claim is the absence of a step at the contact radius and four bars cannot show an absence. The importance of this figure is the absence it shows. If the constrained unit were the contact set, constraint would step down beyond the contact radius, and it does not: it declines smoothly across the pocket. That is what moves the claim from ten residues that touch the ligand to a pocket about 15 Å across, and it is a claim only a continuous distance axis can support.

Every shell out to 15 Å sits above the whole-protein mean, and the step a contact-driven model predicts at the contact radius is not there. The gradient across the neighbourhood is real and shallow, and reaches significance in one paralogue.

The result replicates on an independent axis. Every contact site in every paralogue is under detectable purifying selection by the per-site rate model [87], and the share of significantly constrained sites falls with distance from the ligand. It is not monotone, because the outermost shell sits a little above the third, so what the data show is a step down from the ligand's first two shells onto a floor rather than a gradient running all the way out, and it is reported as such rather than as a gradient.

Two instruments, two alignments and two different statistics give the same pocket.

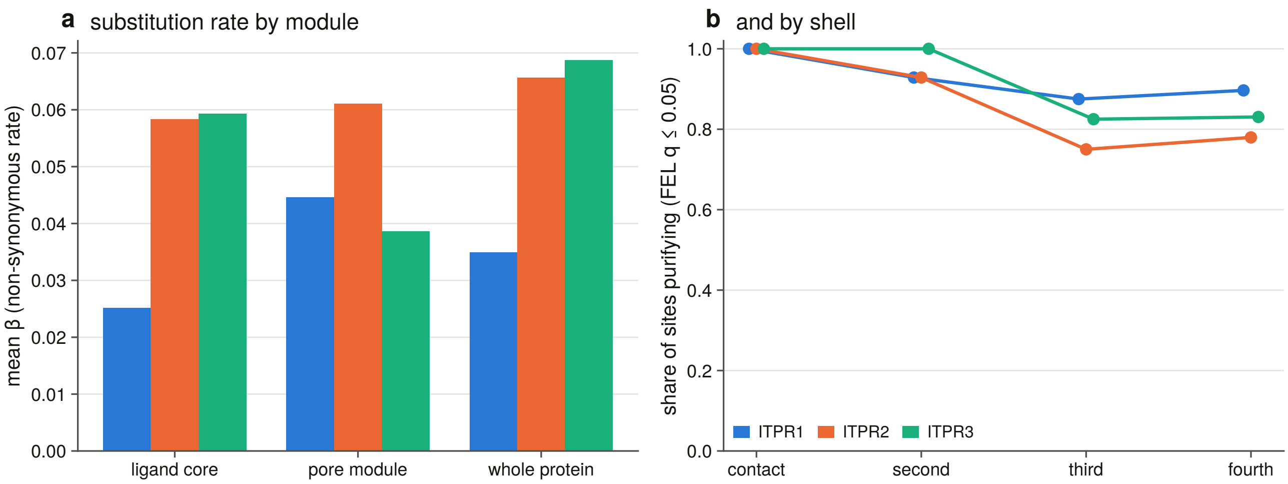


Figure 12.3. Per-site rates by module and by shell. The importance of this figure is that it asks the same

question with a different quantity. A column's dispersion across orthologues and a site's substitution rate on a tree are both called constraint and are not the same measurement, so a result that appears on one axis and not the other is bounded rather than confirmed. The contact result replicates here and the module comparison does not, and the figure is where that distinction is visible.

The module comparison does not replicate on the rate axis, and in one paralogue it points the other way. Both are computed correctly and they are not the same measurement, in that one uses 250 sweep orthologues of one paralogue and the other 57 vertebrate tips, and one is a column's dispersion and the other a rate on a tree. But a reader should have the disagreement in front of them rather than only the half that agrees. With a rate that is exactly zero at most sites and a quarter of the sequences, this is the underpowered version of the same comparison, and the module result is read off the identity axis.

12.8 The upstream enzyme repertoire was measured rather than assumed

The third question needs a list of lineages whose IP_3-producing pathway is reduced or absent. A list taken from reading would be an assumption dressed as a scope, so it is derived from the same reference proteomes the family was swept over.

What counts as a phosphoinositide-specific phospholipase C is a protein carrying both halves of the catalytic barrel in one sequence. Either half alone is not one, because one of them in particular turns up in unrelated proteins and counting it would report a pathway where there is none. Both halves are counted separately, so a proteome scoring one and not the other is visible as such.

The vertebrate group is swept as the positive control for the search, and 760 of 763 vertebrate reference proteomes carry the enzyme. A vertebrate proteome scoring zero would be an instrument failure rather than a result.

Sixty-four proteomes in the whole sweep carry an IP_3 receptor and no phosphoinositide-specific phospholipase C. They are not scattered: they are the oomycetes, the early-diverging fungi, three *Perkinsus* species, two ciliates, four prasinophyte algae and a group of flatworms, with two clades accounting for more than half.

The internal control for that list costs nothing and is strong. Every one of those proteomes carries a full-length IP_3 receptor, meaning a 2,700-residue multi-exon gene the sweep found in it. A gene set complete enough to hold this receptor is complete enough to hold a phospholipase C.

A separate stratum records proteomes with no reference set at all, because filing "we could not look" as "there is none" would manufacture the test set out of missing data.

12.9 The pooled lineage comparison is a clade artefact

Asked of the representative alignment, the test has two tips on one side. That alignment was built to span four kingdoms with 134 sequences, and these taxa are not the taxa a diversity rule picks.

The positive control fires there. The ryanodine receptors, which have the same pore, the same diagnostic domains and no IP_3 site, move the paired difference by about a tenth of an identity unit on six tips.

So the test is repeated on the whole population the question is asked of, meaning all 662 non-vertebrate reference proteomes carrying a receptor, each aligned to the human reference through the same module residues, of which 486 resolve both modules and enter.

Pooled, the enzyme-absent records have a wider gap, and that number should not be believed. The covariate table says why: enzyme-absent proteomes sit at a median pore identity of 0.358 against 0.589 for the rest. They are oomycetes and early-diverging fungi, far further from the human reference than the arthropods and nematodes that dominate the other cell, and the paired difference is itself correlated with divergence. A pooled comparison of absent against present is largely a comparison of distant against near.

Inside a phylum, only three strata have both cells populated at all. And the one that cannot be tested is the most informative: **every early-diverging fungal proteome in this set that carries a receptor lacks the enzyme**, so the phylum has no internal control.

12.10 Matched on divergence the effect disappears, and the null is bounded

Each enzyme-absent record is matched to up to three enzyme-present records within a stated distance of its own pore identity and scored against their median. That is the identity-matched paired design of Chapter 9 applied to the same kind of claim, and it removes the confound rather than arguing with it.

All 36 matched. **The median within-pair difference is (-)0.0064, with a confidence interval from (-)0.0161 to 0.0152, the tips splitting 19 to 17, and p = 0.87.**

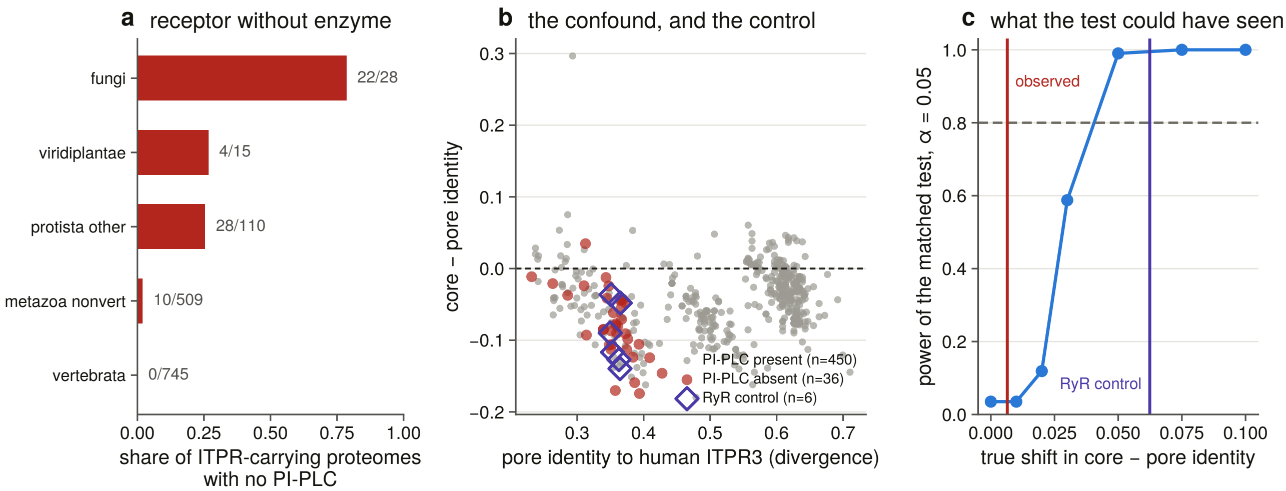


Figure 12.4. The lineage strata with the power curve beside them, which is what makes a negative result readable. Lineages that lost the enzyme making the ligand show no relaxation at the binding site, and without a bound on what the comparison could have detected that would be an untestable statement. The ryanodine receptors are the positive control, because they carry the same pore and no ligand site, so the shift the design can see is measured rather than assumed. The ligand core is drawn in a neutral dark rather than a hue, because the palette reserves its accent colour for that control.

The null is bounded rather than empty. Measured through the same pairwise instrument at the same divergence as the test group, the ryanodine receptors' paired difference is a shift of (-)0.062. The matched test has essentially full power at that shift and 59 % at half of it, and its confidence interval excludes anything larger than about 0.016.

Losing the enzyme that makes IP_3 does not relax the receptor's IP_3-binding core. This is not a case of being unable to tell. The test can see a ligand-free core when there is one, at the same evolutionary distance, and it does not see one here.

12.11 What this chapter settles, and what it does not

The ligand core is not the most constrained part of this receptor. Paired per orthologue against roughly 250 sequences per paralogue, the pore module is ahead in two and level in the third, and that answer depends on one boundary and reverses when fifty residues of luminal loop are counted as pore.

The constrained unit at the ligand site is a pocket about 15 Å across rather than a contact set. The ten measured contacts beat the rest of the core and do not beat the rest of the pocket, and the same shape appears on the rate axis.

Losing the upstream enzyme does not relax the site, with the null bounded by a positive control measured on the same instrument.

What is not settled is whether the pocket's constraint is about IP_3 at all, because everything within 15 Å of the ligand is also within the fold that holds it, and no measurement here separates ligand binding from domain packing. A mutational or binding dataset would, and sequence will not. Nor is it settled what these lineages' receptors are gated by, because a search for one enzyme family says the canonical route to IP_3 is missing rather than that the receptor has no ligand, and the bounded null is consistent with the site being held by something else.

There is one prior this chapter does not corroborate, and it is worth stating plainly. The signature that names the family is the IP_3-binding core, and it is the one diagnostic domain the ryanodine receptors do not carry functionally. It is also the less constrained of the two modules measured here. **Naming a family after its diagnostic domain is a statement about what the domain identifies rather than about what selection holds most tightly**, and this is the first measurement in the project to ask the second question.

13. An archive that cannot find the gene

13.1 Three quarters of these genes are unreachable from any protein database

Chapter 4 measured that **940 of 1,232 demonstrated genes, or 76.3 %, are not reachable by any protein-database search.** They are not hard to find, because no protein record of them exists.

That is a statement about databases rather than about biology, and it is easily the most immediately useful result in the thesis. Automated annotation is known to be imperfect, and the ways in which it fails have been reviewed [77]; what has not been available for this family is a measurement of how often and in which direction. It is also a claim that has to be made carefully, because there are three quite different ways a gene can be missing from a public record, and they have different remedies.

The gene may not be annotated at all. It may be annotated and split across several models, none of which delivers the protein. Or it may be annotated perfectly, named perfectly, and filed as something that emits no protein.

All three happen. This chapter measures how often, audits whether it is worse for this family than for its sister, validates two cases down to the exon, confirms with reads that the lost genes are transcribed, and produces a correction list addressed to somebody else.

13.2 Three rules decide whether the audit is measuring anything

Every gene-scale locus in the 309-genome scope is scored into one of five states, comprising complete, split, fragmentary, non-coding only and unannotated, against each assembly's own submitted gene set [32], under three rules that decide whether the audit is measuring anything.

Only same-strand features count, because an antisense gene overlapping the locus perfectly is not an annotation of it.

Loci are scored against coding blocks rather than gene spans, because a gene span covers its own introns and these genes' introns reach 152 kb, so a passenger gene inside one would score as covering the locus.

The name is read off whichever model the annotation actually places there, deliberately generously, because accusing a database of failing to name a gene it did name is the expensive error.

The name verdict itself has two values that exist because their absence manufactured errors. A name that claims neither family, and a name that claims the family but no paralogue, are recorded as such rather than as wrong. Without those two values the audit produced 66 wrong-family and 6,660 wrong-paralogue errors out of names that claim neither.

The bar for "complete" is validated rather than re-derived. The audit inherits the sweep's coverage threshold and scores it against the distribution it sits in, and every state is re-counted across the whole bar range so the reader can see what the choice costs.

Forty-four constructed negative controls run before anything is written, and the suite was mutation-tested on nine deliberate rule breakages, all nine caught. The measurement is cached and the verdicts are not, because the first version cached both and a later rule change survived into a committed table.

13.3 A quarter of annotated loci are not delivered as one complete model

Of the 932 loci in assemblies that ship a gene set, **73.9 % are delivered as one complete annotated model and 26.1 % are not.** Thirty-eight have no same-strand annotated feature at all, and 42 are held only by a non-coding one, so the database serves no protein for either.

The single largest determinant is not the paralogue but the assembly. Above the contiguity bar the failure rate falls from 26.1 % to **6.7 %**, so two thirds of what looks like an annotation problem is a contig too short to hold a 2,700-residue gene.

By class the spread is wide. Cartilaginous fish and mammals are near-perfect, ray-finned fish good, and **birds are complete at 56 % with 29 % fragmentary**, which is the same class the contiguity bar removes two thirds of.

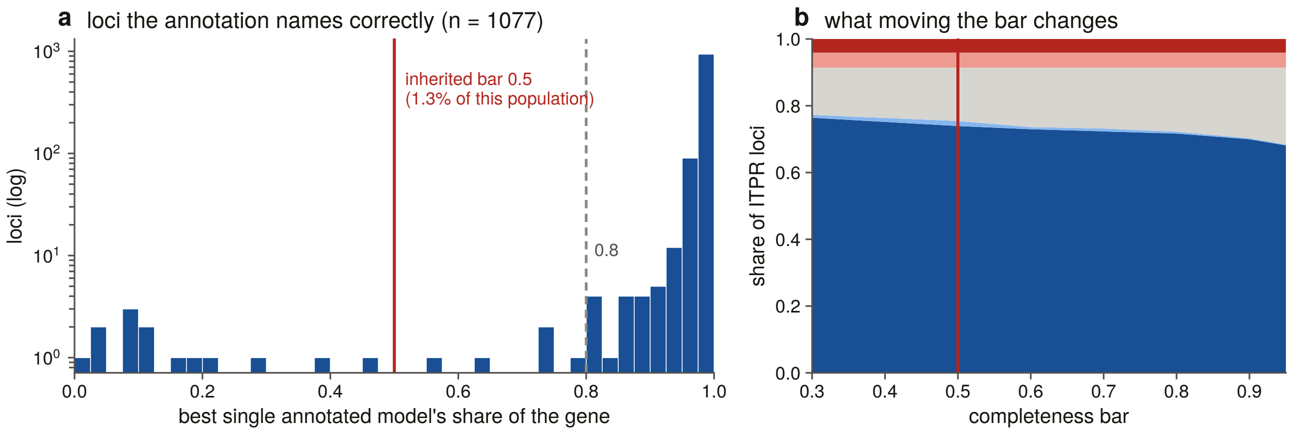


Figure 13.2. The completeness bar drawn inside the distribution it sits in, because a number in a legend cannot show a tail. The importance of drawing the bar inside its own distribution is that the whole audit turns on it. Every locus in this chapter is called complete or not against one coverage threshold, and a threshold quoted as a number in a sentence cannot show whether it sits in a gap or in the middle of a population. The bar was validated rather than replaced here, and this panel is the evidence for that decision.

13.4 Curated and submitter-deposited gene sets differ, and only partly because of assembly quality

A curated reference gene set and a submitter-deposited one are not comparable evidence, and they are not close.

98.8 % of loci in curated annotations are complete, against 37.5 % in submitter-deposited ones.

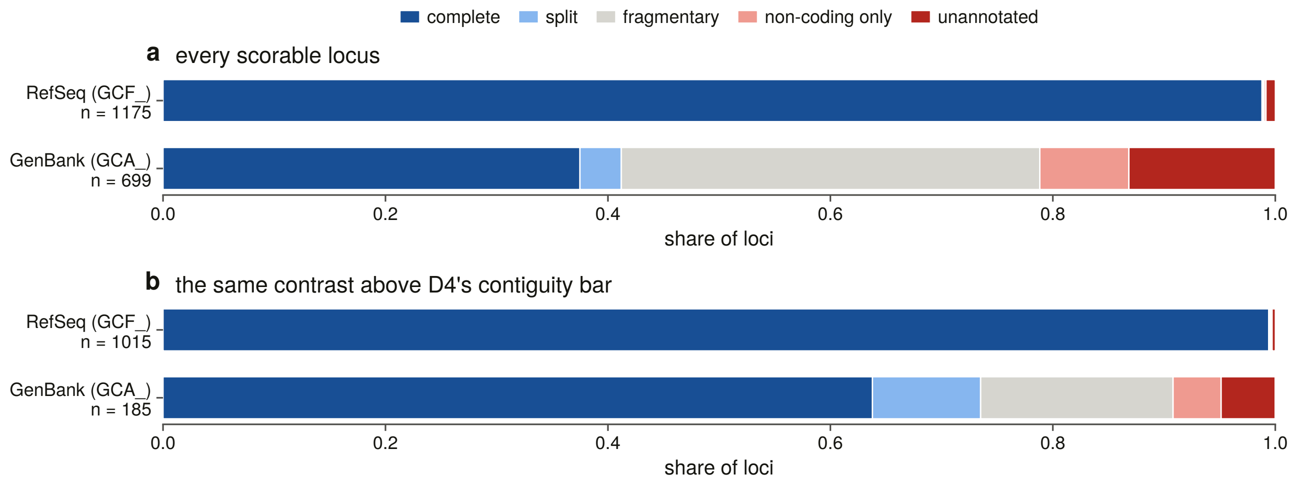


Figure 13.1. Locus state by annotation source, raw and above the contiguity bar. The control is drawn beside the raw contrast rather than instead of it, because that is what separates a database effect from an assembly effect. Curated and submitter-deposited gene sets are not applied to the same assemblies, so a difference between them partly measures which genomes each was run on, and the two bars together show how much of the gap survives the control.

The confounder is obvious and is controlled rather than argued. Submitter assemblies are less contiguous, and a locus on a contig too short to hold the gene cannot be annotated completely by anybody. Re-run over the loci that clear the bar, the contrast **narrows and does not close**, at 99.4 % against 63.8 %. About 42 % of the gap between the two archives is assembly quality and the rest is the gene set.

13.5 The family is not recorded worse than its sister family

The audit's premise was that a 2,700-residue, 58-exon gene with a sister family sharing every diagnostic domain should be badly recorded.

The control says otherwise, and that is the result. The IP_3 receptor cells fail on 26.1 % of loci, and the ryanodine receptors, in the same assemblies through the same pipelines, on 22.1 %. Above the contiguity bar the two are 6.7 % and 5.5 %. **No overall difference survives correction. How this family is recorded is how vertebrate genes of this size are recorded.**

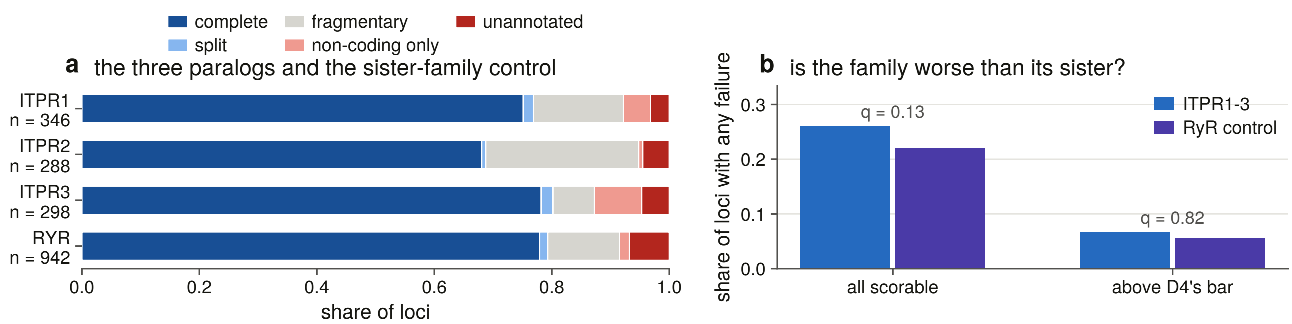


Figure 13.3. This family against its sister in the same assemblies. Without this comparison, a failure rate is not a statement about this family at all. This figure carries the result that overturns the chapter's own premise. The audit was built expecting a large, many-exon gene with a confusable sister family to be recorded badly, and the sister family in the same assemblies through the same pipelines fails at a comparable rate. Without a

control of this kind, a quarter of loci not delivered as one model reads as an indictment of this family; with it, it reads as the failure rate for vertebrate genes of this size.

One state does separate, and it is the family-specific one. An IP_3 receptor locus is 2.7 times more likely than a ryanodine locus to be held only by a non-coding feature, in 42 loci against 16. That is a gene the annotation identifies correctly, names correctly, and files as non-coding, so no protein record is ever created and no name-based search can reach it.

It does not survive the contiguity control. Above the bar the same comparison is 4 against 5, so the effect is confined to assemblies too broken to carry the gene, which is where a submitter has most reason to file a model as non-coding in the first place. The excess is real in the record set, and this measurement cannot say it is a fact about the family rather than about the assemblies its loci happen to sit in.

13.6 Half the protein records carry no usable gene symbol

Each of the 11,402 full-length family protein records held by the protein databases [30] was scored against the committed bait panel, assigned to a family only on a stated margin, then to a paralogue inside the winning family, with the record's own gene symbol and protein name read through the same verdict rule the genome half uses.

The family call is not in dispute. The sequence disagrees with the census call on 4 of 11,402 records, and the databases name the sister family at only 5, every one a non-vertebrate record where the sequence barely separates the families either. The hazard measured at 49 % of records in one query in Chapter 2 does not appear as a naming error in the full-length record set.

The paralogue call is not in dispute either. Fifty-two of 8,306 vertebrate records carry a symbol naming a paralogue the panel assigns elsewhere, and a further 74 claim a paralogue **this panel cannot call**, because the bait table records no bait for it, so those are reported as outside the instrument's reach rather than as errors. Without that guard they would have been 74 database naming errors manufactured by a missing bait.

What is in dispute is whether the records are findable. 3,872 records carry a placeholder gene symbol and 2,395 carry none at all, so **55.0 % of the family's full-length protein records have no usable gene symbol.** On the protein-name side, 66 are named for the superfamily in a way that names the family and its sister together and therefore separates neither.

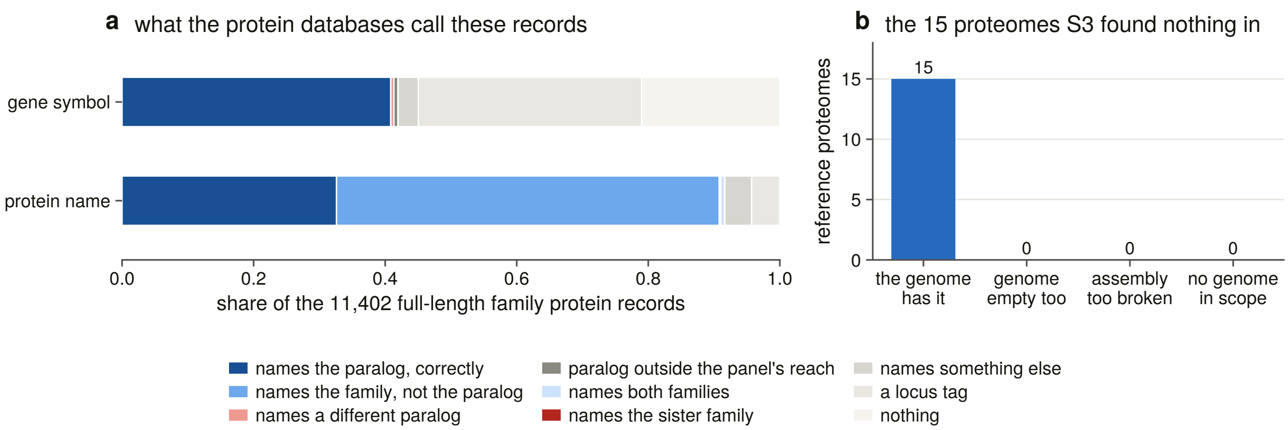


Figure 13.4. The protein records, showing what the name claims against what the sequence is. The importance of separating what a name claims from what a sequence is is that the two failure modes have different

remedies. The family and paralogue calls turn out not to be in dispute, so this is not a record full of misidentified proteins; more than half of it simply carries no usable gene symbol. A record that is correct and unfindable needs a name, not a reannotation, and that distinction is what the panel makes visible.

Would a signature query have found them? The signature that names this family reaches 95.6 % of the records both instruments agree on and 81.2 % of those only one instrument found. It is a good but not complete index of its own family, and the deficit is concentrated outside the vertebrates, falling to 64 % in the amoebozoa and the stramenopiles.

13.7 All fifteen empty proteomes are gene-caller failures

Chapter 3 swept 763 vertebrate reference proteomes and 15 returned no family hit at all. On its own that is uninterpretable, because a receptor-shaped hole in a proteome is either a gene the species lacks or a gene its gene caller did not find.

Each was resolved against an assembly of its own species, searched with an instrument that owes the gene caller nothing.

All 15 are gene-caller failures. Every one of the 15 species has a genome in scope, and in every one the genomic sweep recovers at least one locus at over half the bait's length while the proteome holds none.

The verdict vocabulary carries a fourth value that fires on none of them, and that is the point of having it: a species with no assembly in scope would be recorded as undecidable rather than as an absence. Being able to say it fires on none is worth more than assuming it would.

13.8 Two annotation failures were validated down to the exon

An audit that counts states is a survey. Proving that a particular annotation is wrong about a particular gene needs molecular evidence, and this section takes two cases down to the exon.

Which two cases, and why those. The measurement is annotation loss, meaning the share of a recovered gene's coding footprint that no single annotated model delivers, read straight off the sweep's own coverage rather than re-derived. Five eligibility rules apply, each a positive test, and the fifth is the one that does the work: at least ten annotated protein-coding genes elsewhere in the same genome must be longer than the locus.

Without it, the top of the ranking is loci in assemblies with a genome-wide gene-length ceiling, and validating one would report a property of the whole gene set as a bug at this gene. Measured over the sweep it removes six loci in two genomes, one whose longest annotated gene anywhere is 42 kb and one whose longest is 138 kb against a 726 kb locus, and nothing else.

Three failure modes are labelled and **two are selectable, one case each**, because taking the top two of a single ranking gives two omissions and leaves the fragmentation claim unvalidated. Ties are broken on **control strength**, meaning how many other family loci in the same genome the annotation gets right.

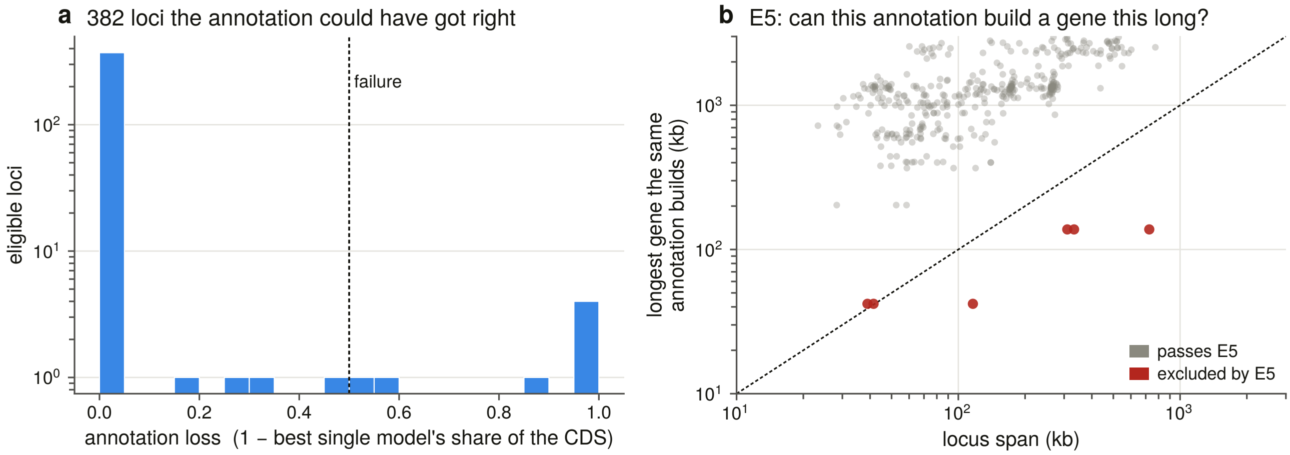


Figure 13.6. Annotation loss across every recovered locus: 880 loci, 382 eligible, and 360 of them at zero loss. The full distribution is drawn because it is the denominator the two validated cases are chosen from, and a case study without its denominator is an anecdote. The failures are a tail rather than a norm, so the two cases taken forward are the extreme of a measured distribution rather than the two examples that happened to be noticed.

Case A is an omission. It is a 52.5 kb locus in a chromosome-level fish assembly whose contigs are 84 times the length of the gene. The gene has 56 coding exons totalling 8,021 bp, and **the annotation places zero gene models over them.** Not a partial model and not a mis-named one: the annotation has nothing there, while working on both sides of the gap, with named genes 4.2 kb downstream and 5.5 kb upstream.

Case B is a fragmentation. It is a 68 kb locus in another chromosome-level fish assembly, with 60 coding exons totalling 8,064 bp, over which the annotation places **three gene models** contributing 38 disjoint coding blocks and covering 58 % of the coding footprint. 3,400 bp in 28 blocks has no coding model at all and reaches no protein.

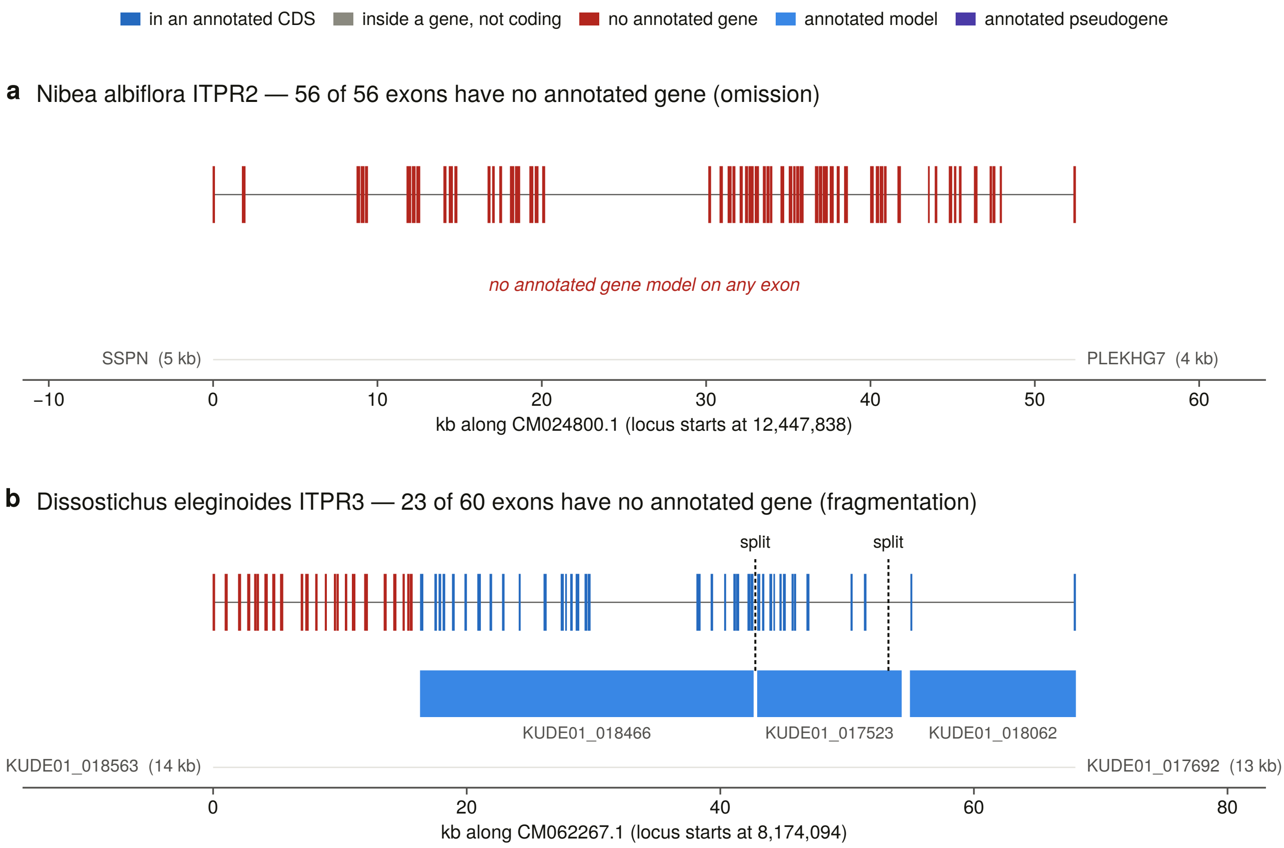


Figure 13.5. The two loci at true genomic width, with exons never widened to be visible. A 56-exon gene over 52 kb averages 143 bp an exon, so fattening the exons would draw a gene whose coding fraction looks like 40 % when it is 15 %. That geometry is the point: what the annotation missed is a few per cent of the locus distributed over dozens of small exons across tens of kilobases, which is why the gene is hard to annotate and why the failure is systematic rather than careless.

13.9 What the DNA says at those two loci, and five checks on it

Case A's locus splices into 2,673 codons with zero internal stops and a terminal stop present. Under neutral drift to its observed divergence, 14.2 stops would be expected, computed from the locus's own codon usage. **Case B gives 2,687 codons, zero internal stops, and 11.4 expected.**

That comparison is the right shape for the claim being made. The annotation's implicit assertion, that this is not a protein-coding gene, is falsifiable, and zero nonsense codons over 2,700 falsifies it. The rule is deliberately one-sided: zero stops falsifies a pseudogene call, and a handful would not establish one.

Five independent checks follow, and they are stated in the order a reader needs them rather than in the order that flatters the conclusion.

Splice sites. Every intron in both loci carries a canonical splice pair. An alignment is not a gene, and an exon structure whose introns are spliceable is evidence that this one is.

Exon boundaries against other genomes' annotations. Thirty-five and 40 swept genomes carry the same paralogue, aligned from the same bait, with their own annotation independently placing one model over the whole alignment. 94.5 % and 100 % of each locus's internal boundaries are shared by a majority of them. This validates the instrument at this gene rather than the individual locus, which is why it is reported alongside the reading frame and the splice sites rather than instead of them.

Transcript evidence at the junction was searched for, and the denominator is the result. Probes across every junction the annotation does not model were searched against every transcript record for the species, and returned nothing. The species has 10 transcript records in total. A search of that many returning nothing has not shown the gene is untranscribed. It has shown the species has almost no deposits.

That criterion has its own negative control, and it needed a second half. Run against the locus's own genomic DNA, where a probe of contiguous spliced sequence cannot span its junction by construction, the probes make 50 hits and none spans. So the zero above is a discriminating zero rather than a test that never fires. On its first run that control returned six false spans, which is how the rule acquired its second half: an anchor requirement alone admits an alignment that has run a dozen bases past the junction into the intron.

The rest of the family in the same genome is intact. Every other family locus in both assemblies also splices into an uninterrupted reading frame. Whatever the annotation is doing here, it is not responding to a damaged gene.

The neighbourhood check is available for one case and not the other. Case A's missing gene sits between the two neighbours its paralogue carries in 197 and 183 of 215 swept vertebrates. For Case B the check is not available at all, because that annotation names its genes by locus tag and there is nothing to match, and that is reported as a limitation of the comparison rather than as a negative result.

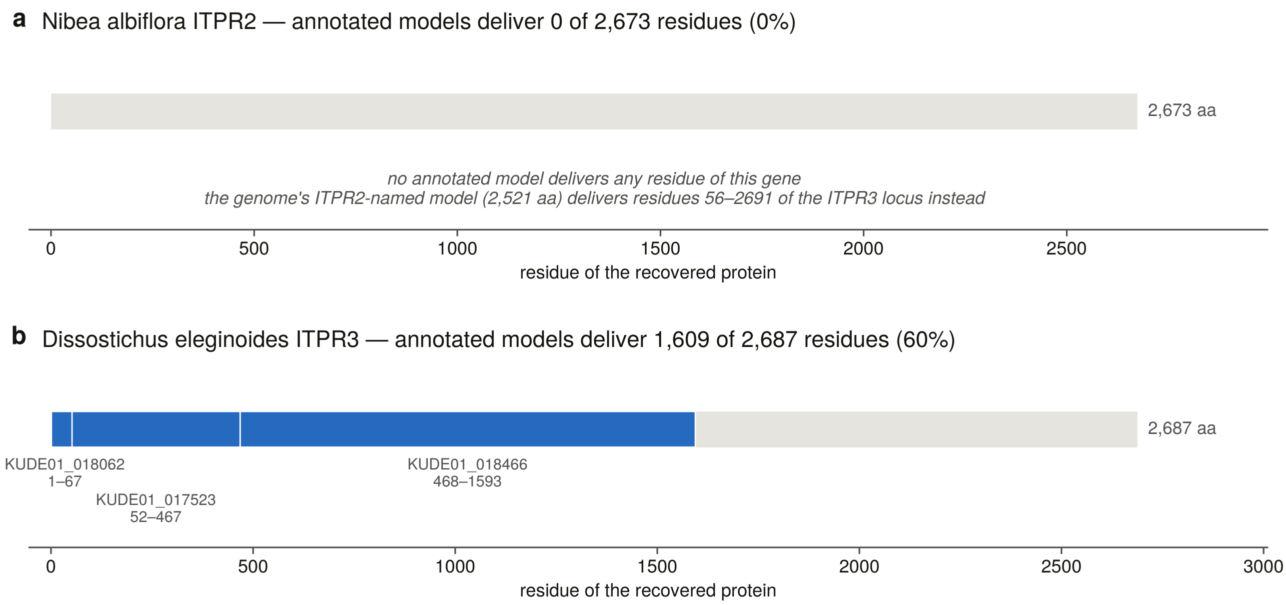


Figure 13.7. The annotated proteins tiled back onto the genome's own recovered loci, which is what says whether the annotation missed a gene or broke it into pieces. The proteins are translated from the assembly's own annotation and genome rather than downloaded, because a locus filed as a pseudogene emits no protein record and those loci are exactly the ones under dispute. The guard that matters is that the subject set is the genome's own loci: a query whose own locus is missing from it lands on its nearest paralogue instead, and the row then reads as a confident naming disagreement, which is exactly what happened before the check existed.

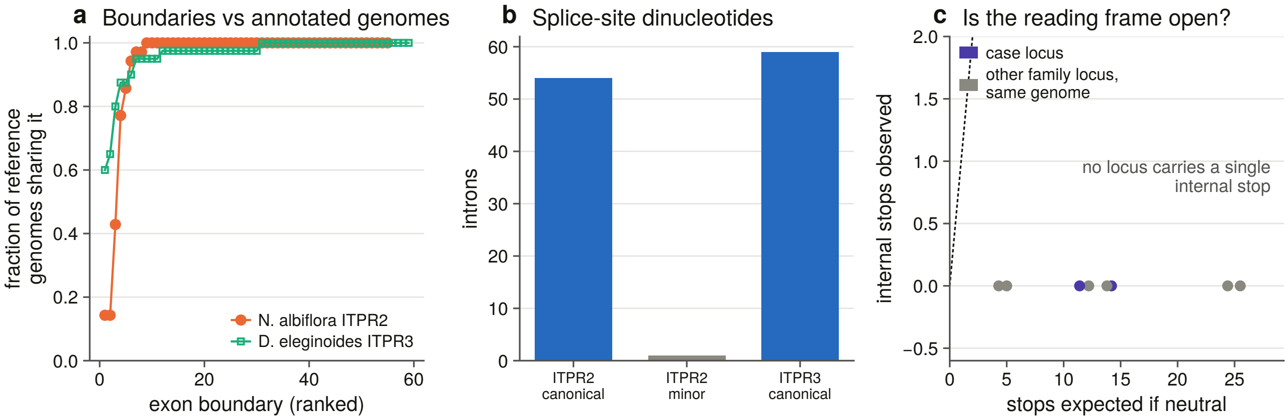


Figure 13.8. The five checks on each case, together. The importance of showing the five checks together is that no single one of them is decisive. Splice dinucleotides, an intact reading frame, junction probes, exon-boundary concordance and the neighbourhood consensus each fail in different circumstances, and they do not fail together. A reader asking whether a recovered gene is an alignment artefact is asking whether every one of these could be wrong at once, which is what this panel is arranged to answer.

One number frames both cases. One of the two assemblies files 7,380 of its 23,345 genes as pseudogenes, which is 31.6 % of the gene set. Its 14 family-named gene models include 8 filed as pseudogenes and 6 emitting a protein.

13.10 Reads show that all seven lost genes are transcribed and spliced

The junction question the case studies could not answer needs reads rather than deposits, and this is where it is answered.

The scope is derived from the ranking rather than chosen. Four rules, each a positive test, apply. Every locus the eligibility rules admit whose annotation loss clears a floor is included, read straight off the committed ranking rather than re-derived. The species must have enough public runs, which is a floor on askability rather than on power. And every other family locus in the same genome joins as an internal control, including the ryanodine locus, whose reads must land on the ryanodine gene.

A control locus carries the loss measured for it or none at all, in that the field is empty rather than zero, because a locus never ranked and a locus measured at zero loss must not share a cell.

Reads come from the public sequence archive [100] and are mapped with a splice-aware aligner [101], run in unspliced mode for the reason given below.

The reference every read is mapped against is built rather than downloaded. It is the spliced genomic coding sequence, deliberately not the frameshift-corrected version, because correcting a frameshift inserts one or two bases that every read crossing it would carry as an indel, costing reads at exactly the difficult sites.

Validation is **colinear block placement**, which has no tuned threshold. Each block is translated in its own phase and searched verbatim in the aligner's own protein from the previous block's match onwards, and a block may fail to place only if the model's own frameshift count explains it. The identity test it replaced was the wrong instrument, and measuring it showed why: a correct reference scores 1.00 with no frameshifts and 0.92 with nine.

Three housekeeping anchors are aligned **by the same aligner and spliced by the same module as the targets** rather than pulled from each annotation, because two of the three annotations name no genes at all, one filing all 29,240 of its models as hypothetical proteins. Building anchor and target alike means a difference between them cannot be a difference in construction. The anchor accessions are **resolved by query against a declared length band** rather than remembered, because the first version hard-coded three and got all three wrong, including a 427-residue protein named for a 335-residue enzyme.

Seven of seven loci that the annotation loses are transcribed and spliced, meeting all three detection criteria in at least one library, against their own reversed decoys at zero.

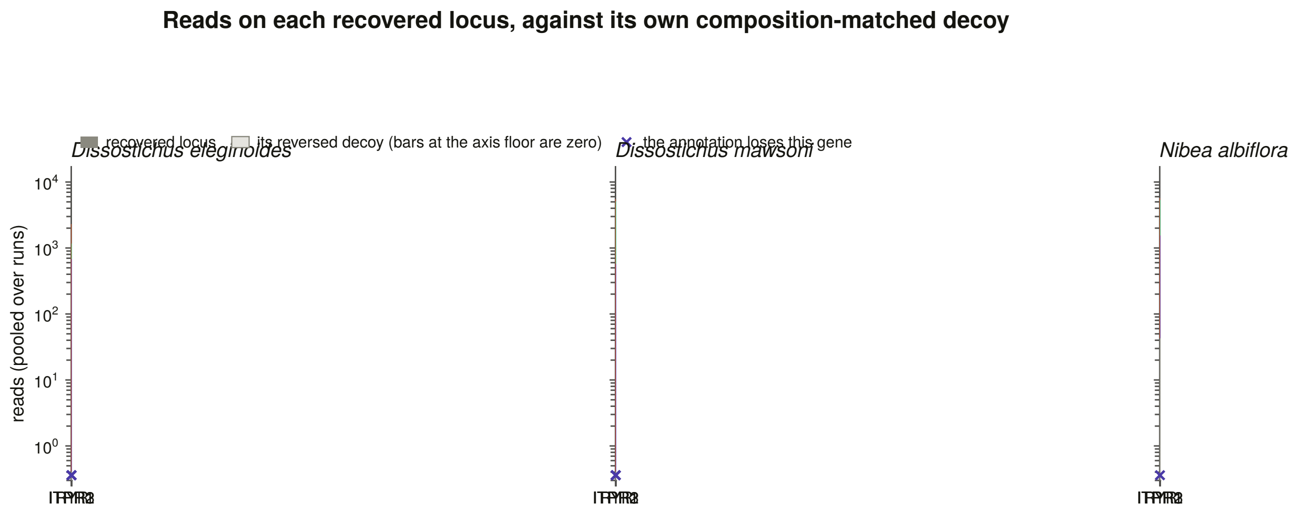


Figure 13.10. Detection per locus against its own reversed decoy, which is the spurious-mapping floor of this particular reference rather than a threshold. The decoy replaces a chosen cut with a measured one: a sequence of identical length and composition and no homology collects whatever this reference collects by accident, so a locus above its own decoy is detected on evidence. Counts are on logarithmic axes because these libraries differ roughly forty-fold in depth, and the comparison that means anything is between a locus and its decoy in the same run.

298 of 314 junctions that no annotated model spans are crossed by reads, which is 94.9 %, against 95.2 % of the annotated junctions in the same genes.

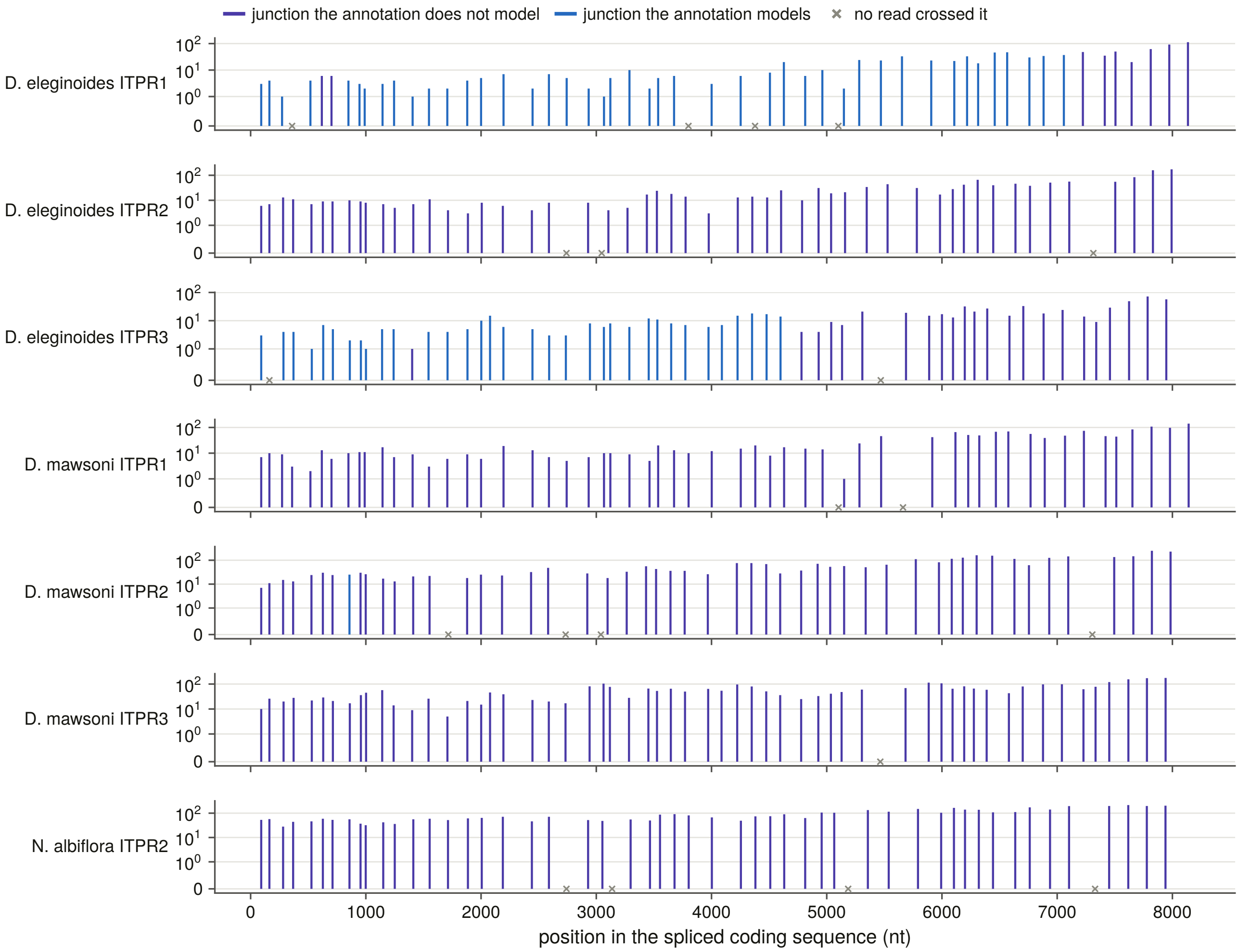


Figure 13.9. Every junction of every reference, crossed or not, scored per junction rather than per gene, with the decoy floor drawn rather than stated. That unit is what makes the read evidence answer the annotation question rather than a weaker one: that a gene is transcribed says little when part of it is already annotated, whereas the particular junctions no model contains being crossed by reads is direct evidence for the exon structure the annotation is missing.

The confound is printed beside the ratio. Several of these libraries are strongly 3′-biased, and in a truncated or fragmented gene the annotated junctions are the ones at the 5′ end, where coverage is thinnest. Comparing the two recovery rates without saying so would report a property of library preparation as a property of the annotation. **The claim this makes is the absolute one**, that these particular junctions are crossed, rather than the ratio.

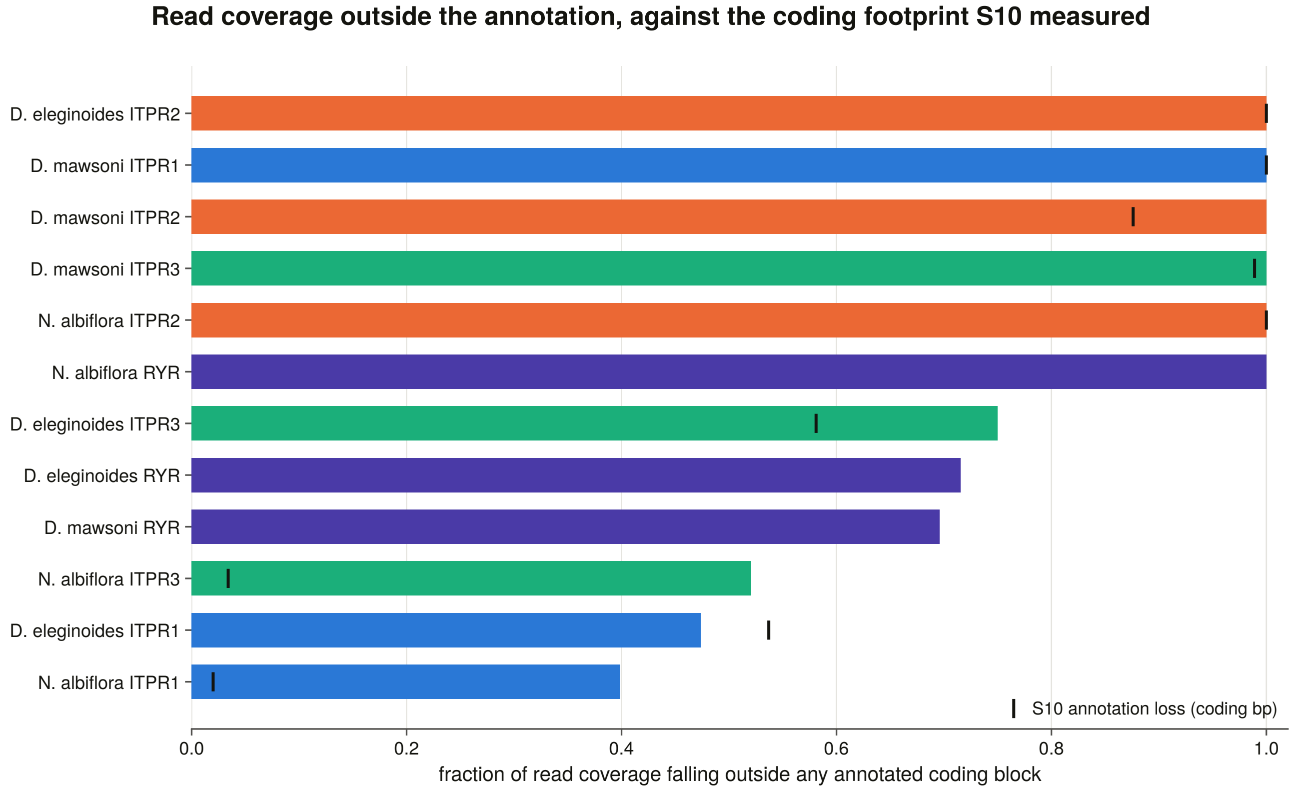


Figure 13.11. Read coverage over the sequence the annotation loses, which is the read-level form of the loss measurement. Two independent instruments, one aligning a protein to a genome and one mapping reads to a spliced reference, are asked about the same missing sequence, and their agreement is what rules out the loss being an artefact of either.

13.11 Four controls make the read evidence readable

A spurious-mapping floor is measured per sequence. Every reference carries a reversed decoy of identical length and composition with no homology, so what it collects is this reference's own floor rather than a number from a paper.

Cross-mapping between paralogues is measured rather than argued away. The reference is a closed set, so if the paralogues were similar enough at nucleotide level for a read to cross between them, the whole exercise would be measuring the wrong gene, and these paralogues are more similar to each other than the family this method was ported from, whose version dismissed the risk in a caveat. Every reference was tiled exhaustively with synthetic reads at fixed steps and mapped back under the same settings. **Zero of 12,500 reads are assigned elsewhere**, and the result is committed whether it is zero or not. It is tiled rather than sampled, so it is exhaustive over positions and deterministic.

Spliced alignment is switched off deliberately. The reference is already coding sequence, so a junction read is contiguous here, and leaving spliced mode on would let the aligner open a gap inside the coding sequence and call an intron that does not exist.

Junctions with no reads are written as zeros rather than omitted, because the denominator is half the result.

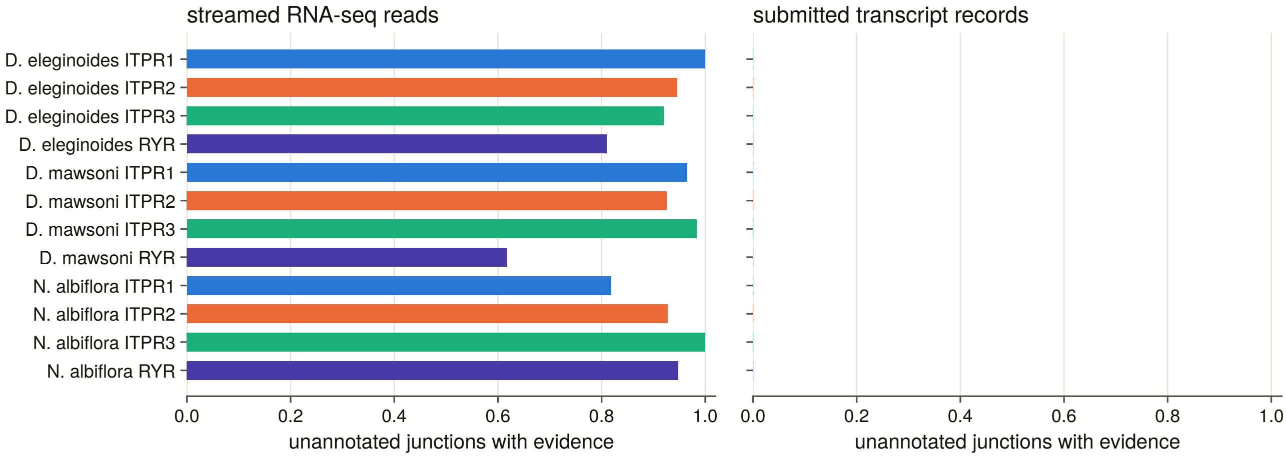


Figure 13.12. The deposit cross-check with its genomic negative control. The same junction-probe test the case studies ran is re-asked on a panel that now includes a species with 37,166 transcript records against the other two species' 43 and 10, and it still cannot answer. A search returning nothing in a species with tens of thousands of deposits and nothing in a species with ten is two very different situations, and the negative control is what shows the probes themselves work. Reporting the result as underpowered is what keeps a database's emptiness from being read as a biological absence.

Eleven constructed negative controls run before anything is written, and were mutation-tested on five deliberate breakages, all five caught. The most instructive requires junction classes to be matched **by genomic coordinate rather than by index**, on a two-intron minus-strand gene, which is the smallest case that can tell a coordinate lookup from an index lookup, because a coordinate-sorted and a target-sorted list have the same length and opposite order on a minus-strand gene. A one-intron test cannot distinguish them, and a mutation test found exactly that mapping slipping through.

13.12 The correction list is 297 items addressed to somebody else

The deliverable of this chapter is addressed to somebody else: **297 correction items, 52 of them high priority, 18 withheld.**

Each row carries the assembly, the coordinates, the current state, the current name, the proposal, an archived evidence file a curator can open, and the numbers the class fired on.

Priority is a rule rather than an impression. High priority needs the gene demonstrably present, the assembly demonstrably able to carry it, and the reading frame intact. A partial recovery or an unscored reading frame is medium. Anything below the contiguity bar is low, because there the annotation's silence may be the assembly's fault rather than the annotator's.

The integrity veto is applied as a column rather than a filter. Where the lesion screen scored a locus as lesion-rich, the audit does not propose resurrecting it, but the row is still written, flagged, with its reason, because a locus this audit declined to correct is evidence about the audit.

A protein-record rename additionally needs an absolute score as well as a relative margin, since the candidates score 90 to 191 bits over roughly 2,700 residues.

13.13 What this chapter settles about the record

The family's public record is substantially worse than the family. Three quarters of the genes this project demonstrated are unreachable from any protein database, more than half of the full-length protein records that do exist carry no usable gene symbol, and a quarter of the loci in annotated assemblies are not delivered as one complete model.

And it is not this family's fault. The sister family, in the same assemblies through the same pipelines, fails at the same rate. What this chapter measures is how vertebrate genes of this size and structure are recorded, for which this family is an unusually well-controlled probe, because it comes with a size-matched, domain-sharing sister to compare against.

Two of the failures are proved at the exon and confirmed by reads. Both are genes with intact reading frames, canonical splice sites, boundaries corroborated by dozens of other genomes' independent annotations, and transcript evidence crossing the junctions no model spans, in chromosome-level assemblies whose contigs are dozens of times the length of the gene.

Two thirds of the apparent annotation problem, however, is assembly quality, and every claim here is made with the contiguity control beside it.

14. Methods, written as the reasoning behind each instrument

14.1 Why the methods are written here as argument rather than as procedure

A paper's methods section says what was done. It has to, because a reader needs to reproduce it. It cannot say why each threshold is the number it is, and for this project that is where most of the work went.

87 methodological decisions were recorded across 35 working sessions, numbered as they were made. Several of them changed an answer. Several were made, tested and overturned. None of them has ever been written out as prose, and this chapter is that: not a list, but the arguments in the order they matter, with the incident that produced each one.

The chapter is organised by what a decision is about, covering thresholds, controls, refusals, reproducibility, and the rules that keep a document from drifting away from its data.

Where the procedures themselves are. Because each instrument is built in the chapter that first depends on it, a reader looking for a procedure rather than for an argument needs a map, and this is it.

Instrument	Built in
Control benchmark, domain-architecture call, profile models	§3.2 to §3.5
Bait panel, spliced-alignment sweep, rescue attribution	§4.3 to §4.6
Contiguity bar	§4.7
False-negative rate, panel ablation, channel contribution	§4.11 to §4.13
Proteome sweep, per-clade positive control, identity floor	§5.2, §5.7, §5.9
Representative selection, alignment, model choice, tree search	§6.2 to §6.6
Neighbourhood windows, matched control, paralogue caller	§7.2, §7.5
Species tree and reconciliation, dated paralogy	§7.7, §7.10
Exon and intron reading, coordinate frame, shared-intron null	§8.2 to §8.7
State vocabulary, reassembly bar, Dollo count, sensitivity grid	§9.2, §9.3, §9.8, §9.9
Coding-sequence validation, codon alignment, selection models	§10.2, §10.3, §10.7
Structural panel, fold calibration, constraint layers	§11.3, §11.4, §11.8
Module definitions, ligand shells, paired test	§12.2 to §12.4
Locus states, name verdicts, read evidence	§13.2, §13.10

Every one of those sections names the committed table its output lives in, and the analysis code is one file per task under `scripts/`, mapped in the repository's `INTERFACE.md`. The exact version of every external program the project shelled out to is recorded in `results/toolchain_manifest.txt`.

14.2 Thresholds are measured rather than chosen, and never ported

A threshold ported from another project must be re-measured before it is used rather than merely restated in this project's units. This project inherited a methodology from a sister project on a different gene family, and the inheritance is deliberate, because the methods were paid for over two dozen sessions and re-deriving them buys nothing.

The boundary of that rule is a number. The attribution margin arrived as a bit-score ratio restated in this project's units, which made it look derived while leaving it tuned to a family whose paralogues are 40 to 50 % identical. These are 61 to 68 % identical. Measured against loci whose identity an independent annotation establishes, the inherited value would have reported every rescue fragment ambiguous and made every absence claim unattributable.

A method ports, and a number that encodes how far apart that family's paralogues sit does not. Neither does one that encodes its gene sizes. The same session found the aligner's maximum-intron parameter needed measuring for the same reason. Where a ported constant cannot be re-measured yet, it is used with its limitation stated in the code and the per-row value recorded, so the cut can be revisited without re-running anything.

A threshold cannot be calibrated from the population it has already filtered. The sweeps deliberately record every candidate down to a floor far below the one they apply, precisely so that the floor is not measured on data it has already removed.

A threshold the data has no population for is not lowered to a round number. One brief asked for a merge rule that would join the pair of coding blocks an aligner emits either side of a frameshift. Measured across all 309 genomes, **no such pair exists**, because every one of 149,148 consecutive block pairs has contiguous query spans and the smallest genomic gap anywhere is 10 bp. So the calibration refuses to derive a bar, since there is nothing on the other side of it, and the floor goes at the smallest gap the genome calls spliceable rather than at the declared fallback, which would have merged ten real junctions. The merge then fires on zero junctions, and a constructed control builds such a pair and requires the branch to fire, which is what makes the zero a measurement rather than a rule that cannot act.

A calibration must be able to refuse. More than one in this project does: it reports its separation statistic beside its threshold and raises rather than handing out a bar its own data does not support. One refused with a Youden index of 0.52 and sent a chapter back to find the population that was contaminating its decoy.

A calibration must not be able to pass vacuously. A smoke-test run over one genome once produced a threshold from two loci, wrote it, and the next sweep read it back and moved three genomes between statuses. Calibrations now carry floors on the number of observations they may be computed from, and one equivalence test refuses to report success on a comparison of fewer than 25 targets after its first run went green on a group that had no hits at either setting.

An operating point is not the same as a separation statistic. Youden's index on a perfectly separated pair of populations lands on the lowest positive, which is the most permissive bar the data allow. Where a gap exists, the operating point goes at its midpoint and **both edges are committed**, so a later run whose populations have drifted into contact is visible in the table rather than only in a verdict.

14.3 Controls are positive, negative and matched

The sister family is this project's positive control rather than its nuisance. The ryanodine receptors carry every diagnostic domain of the IP_3 receptors, are twice their length, and are inside every search this project ran. Rather than filter them out, they are carried through every stage and measured through the identical instrument: as a presence control in the genome sweep, as a decoy in the discovery benchmark, as the outgroup that roots the tree, as a second family the synteny method has to work on, as the family whose triplication makes the teleost singletons interpretable, as the annotation audit's comparison, and as the ligand-site chapter's positive control for a receptor with no IP_3 site.

A positive control does not port across the scope it controls. The vertebrate sweep can write "the ryanodine control fired in all 309 genomes" because every vertebrate has three. Outside the metazoa that sentence is not available. The replacement was a domain-sharing protein family, and that failed too, in the one clade where it matters, because a single profile fixed in advance for every clade is the mistake rather than the choice of profile. The fix is to choose the control per clade by measurement, running six candidates over each clade's own proteomes and taking the one that is actually there.

A within-genome paired test makes a cell and its siblings one observation. Where a comparison is between paralogues inside one genome, the strata from it are not independent corroboration, because an elevated cell makes its own siblings look deficient by construction and three significant rows can be one result.

A confounder that can be matched is matched rather than argued away. Two chapters here found their headline effect disappear under identity matching, and both report the matched result as the answer. A third measured a risk that a sister project had dismissed in a caveat, namely cross-mapping between paralogues, by tiling every reference exhaustively with synthetic reads and mapping them back. **An argument in a caveat and a measurement in a table are not the same evidence, and the second costs minutes.**

A ported control's result is re-measured rather than inherited. The sister project's reversed decoys collected zero reads, and here they collected 42, in two of 469 comparisons, confined to 64 bp of one decoy. It changes no detection call. Writing "zero" because zero was the ported answer is exactly the failure the rule exists to prevent.

14.4 A refusal is not a disagreement

A refusal is not a disagreement. This is the rule that appears in the most instruments. A structural test declines six models, and their agreement column is left **blank rather than zero**, because scoring a refusal as a disagreement would turn "this model is too partial to place" into "shape contradicts the census". The tree's unplaced tips, the sweep's uncontrolled genomes and the census's unassigned records are the same rule in earlier instruments.

The piece that makes a refusal checkable was added late. The report prints the **arithmetic ceiling** each declined structure could have reached, so "too short to pass" becomes a claim that can be falsified. Here it was: every one of the six had the headroom and fell short on similarity instead.

A verdict vocabulary needs a word for a measurement that did not happen. This project's is five-valued, comprising confirmed, contradicted, not corroborated, orthogonal and underpowered, and the last three do the work that stops a comparison being overclaimed. *Not corroborated* is a related property measured cleanly that does not support the prior. *Orthogonal* is a different property of the same object, such as a tree's branching order against a retained flanking ohnologue, a column's dispersion against a rate on a tree, or a search's recovery rate against a gene's evolutionary rate. And *underpowered* is not a polite word for a negative, because a neighbourhood comparison taken where 73 % of the flanking genes are unnamed is a measurement that did not happen.

A ratio is estimated on a tree rather than from a pair, once the pairs are saturated. The counting method a pairwise estimate rests on [102] is used in the application's own exploratory tool and nowhere in this thesis's results, for the reason Chapter 10 measures: nearly nine tenths of within-paralogue pairs are past the saturation bar, so a pairwise matrix is a diagnostic here rather than an estimate.

A significant test is not a claim, and the fitted parameter is. Three of the selection chapter's significant tests mean something other than what their p-values suggest, and in every case the tell was a parameter printed outside the test: an effect size pinned at the optimiser's bound, a rate class at exactly one with a share of zero, or a likelihood-ratio test with no site clearing its own posterior threshold. Every significant test in this thesis is printed beside the parameter it is about.

A likelihood-ratio test whose null sits on a boundary needs the right null distribution, and both are written out rather than one chosen.

14.5 Four things in this project were not reproducible until they were pinned

Anything a committed artefact is built from is aligned single-threaded. The alignment program's [48] iterative refinement combines partial results in whatever order the threads finish, so at automatic thread count it is not reproducible: the same seeds aligned twice gave 8,510 and 8,468 columns, and the profiles built from them 4,933 and 4,908 match states. This was discovered by rebuilding profiles after an unrelated edit and noticing the match-state count had moved. **A profile that changes when it is rebuilt cannot be the profile a committed result was produced with.** Single-threaded costs 87 seconds instead of 15, once.

Thread counts and seeds are pinned for the tree search [51] too, because automatic thread selection reads the machine's current load and the search is reproducible only at a fixed seed and a fixed thread count. The number pinned is what the program's own benchmark returned, so pinning costs nothing.

A database release is pinned like a parameter. The paralogy map is taken from a dated archive rather than the rolling host, because the rolling host follows the release cycle and a re-run would score the same windows against a different tree. The archive's own registry is committed beside the map.

A figure is not reproducible until its file is. Figures were being written with a creation timestamp in the vector output, so the same code on the same data produced different bytes and no checksum recorded against a figure meant anything. The timestamp is now dropped at save time. A mutation test written to catch this initially passed on the broken code, because both saves landed in the same second.

A failed run is never cached, and a running stage owns a lock. An interrupted structural run left its parent process orphaned while its children were killed. It went on spawning work beside the driver that replaced it, and the killed pairs were written to the cache as zero-score results with empty error text, giving ten permanent false negatives that only the stage's own self-test caught. A result that is not marked successful is no longer written, and a stage claims a lock checked against a live process, so a crash does not block the next run for ever.

A cache may hold what a parser found, never what a rule decided. One audit cached both, and a later rule change survived into a committed table.

14.6 Six rules make a report unable to lie

Reports are rendered from the committed tables rather than written by hand alongside them. Every report in this project's results tree is generated, and the numbers in it come from the tables in the same directory. That is why the claims ledger can use a text search against a report as a check that is not weaker than a table lookup: the report carries no hand-written numbers.

Headlines are chosen by the data. Where a chapter tests a prediction an earlier chapter made, the prior is stated in code with where it was said, the new statistic is computed, and the verdict is rendered from the comparison, with both numbers printed either way. A generator written to narrate the expected answer would print it whatever the data said, which is how a pipeline launders an assumption into a result.

A comparison that has to be sayable is the one that goes badly. Several chapters here report a result against their own design. The constraint chapter finds the deep layer it was built around is the second best of four. The ligand-site chapter finds the module that names the family is the less constrained of two. The annotation chapter finds the family is not recorded worse than its sister. Each of those had to be as easy to print as its opposite, and the report generators are written so that it is.

Every load-bearing number is declared with the table it comes from and the operation that recovers it. The manuscript carries 276 such declarations and this thesis carries its own, through the same engine rather than a fork, so a number quoted in both documents is recovered once and cannot disagree between them.

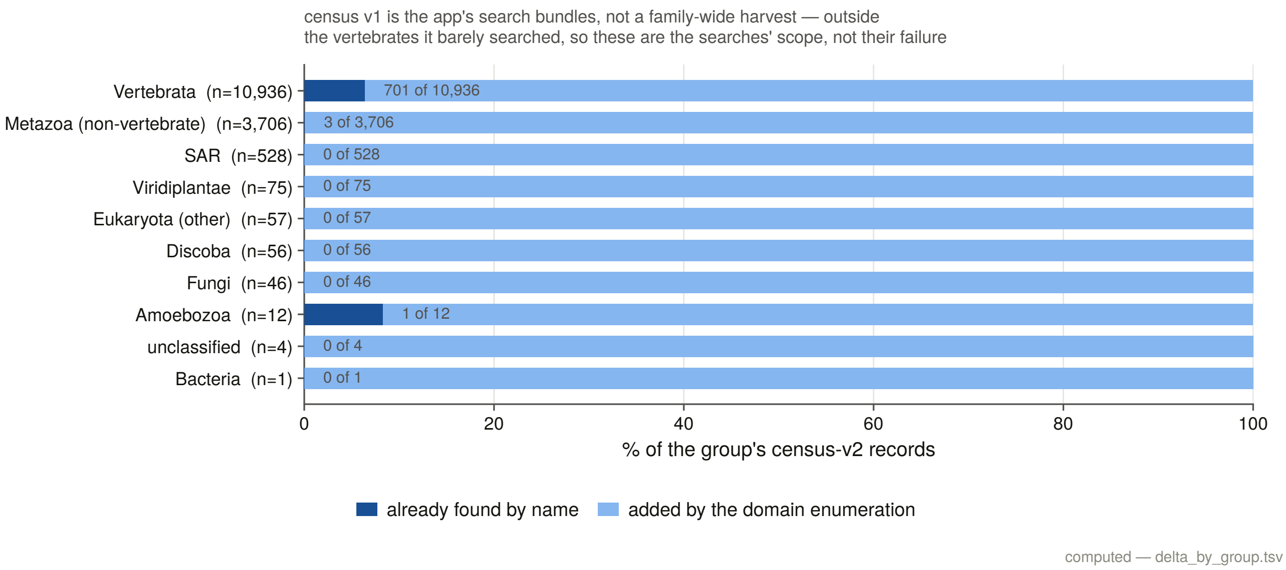


Figure 14.1. How the census grew across its six editions, and on what evidence each addition rests. It is in the methods chapter rather than in a results chapter because it is a statement about the search rather than about the family: each edition is a channel, and the figure is the shape of what each channel was worth. For a reader planning similar work it prices four search strategies against each other on one family. A targeted search of the obvious databases, which is where most projects stop, returned about a tenth of what exhaustive enumeration by domain signature did, and only the two genome sweeps recovered the genes no protein database holds at all.

Citations are keys resolved at build time and the bibliography is rendered rather than typed. No author, title, year or journal is written into a chapter file, and a cited key with no reference row is a build error. This is a decision rather than a convenience, because a bibliography typed beside the text is the one part of a document no table check can reach.

A typeset build is not finished until its own log has been read for dropped glyphs. A missing character is a warning to the typesetter and a hole in the page to a reader. One early build printed the receptor's own name without its subscript because the document font has no subscript glyphs, and nothing in the build said so.

14.7 Four rules are specific to this family

Four of the 87 recorded decisions are about this family and could not have been ported from anywhere.

Separating IP_3 from ryanodine receptors is a positive test at every stage. Never assume a domain hit, a similarity hit or a gene model is a family member. Assign by best profile or by a labelled-bait margin, record the margin, and treat the length band as a filter that supports the call rather than as the call itself.

A paralogue label is only meaningful inside the vertebrates. ITPR1, ITPR2 and ITPR3 are a whole-genome-duplication product, so a protist record whose protein name reads "receptor type 2" carries that number by annotation transfer rather than by descent. Four representatives were affected.

A bait attribution is only a label where it varies. The sweeps record which bait won each locus, and for most clades that discriminates because the three baits win different loci. In the cyclostomes it does not, because each genome carries three full-length loci and the same bait wins all of them. A constant attribution is no information, and filling a paralogue grid cell from it would assert an orthology the data does not contain. The rule counts distinct attributed cells per band, so the exception is found rather than named.

In a family whose members are 61 to 68 % identical, "the same query aligns twice" measures paralogy rather than duplication. Run on raw alignments, a duplication detector fires in essentially every vertebrate genome because every bait aligns at all three genes, at a specificity of 0.16. The fix is the family call applied to a pairwise test, in which the pair must sit inside the cell's own loci, and specificity goes to 0.977 against a copy call the detector never sees. **A geometric signature is only evidence once the family assignment has been made.**

14.8 Two rules govern the scope of an absence claim

An absence claim must pass two bars rather than one: a genome-wide contiguity floor, and a local check that the region itself is present. Residual false negatives are regional rather than random, and a genome missing a whole syntenic block is not evidence of gene loss.

That bar was chosen a priori from gene geometry, meaning the median measured genomic span of the gene, with no error rate anywhere in its derivation, and it has now been scored against a measured false-negative rate for the first time. It gives 0.9 % residual error on the family series and 0.0 % on the sister series, and it **stands**. The cost is reported beside it, because it removes 38.8 % of the scope, disproportionately from the clades a loss survey most wants.

A search's sensitivity is measurable rather than estimable when the family has no losses. That is the design that makes Chapter 4's second half possible, and it needs three rules to be a measurement rather than a restatement. The cells the state rules decline to call are in neither the numerator nor the denominator. The sister cell is carried as an independent replicate using none of those state rules. And a false negative must be constructible, because a control that can only ever return zero misses is not a measurement.

14.9 The constructed negative controls are worth taking as a body of work

The paper reports these in one Methods paragraph and a count. There are more of them than there are paragraphs in the paper's Results, and the ones that mattered mattered a great deal.

There are **437 named constructed checks across 19 suites.** The inventory in Appendix B is derived from the test modules themselves rather than typed, so a control that is deleted disappears from the appendix and one that is added appears in it. The unit counted is a named check, meaning one assertion reported by name as the suite runs, and where a suite groups its checks into named test functions instead, that is what is counted and the table says so, because the two units are not the same size and adding them together silently would be worse than either.

Every suite runs **before anything is written**, and a failure refuses the build.

What the controls are checks on

Almost none of them checks that a routine returns the right answer. They check that it **refuses**, and that it **can act**.

That emphasis is not stylistic. In this project almost every rule returns a plausible number when it is wrong. An interval routine that scores against gene spans instead of coding blocks credits an annotation with sequence it never called. A minus-strand intron read without reverse-complementing both ends is non-canonical, and getting that wrong makes every minus-strand gene in the sweep look broken, which is a systematic signal that would discredit exactly the alignments a chapter is defending. A codon model will fit a frame-shifted alignment and return a perfectly plausible rate. A duplication detector that ignores the family call measures paralogy at a specificity of 0.16 and a sensitivity of 0.997, which looks like an excellent instrument.

A rule that fires on nothing is indistinguishable from a rule that cannot fire. Several results in this thesis are zeros: no locus encodes the same part of the protein twice, no block pair is a frameshift pair, no cell reaches the absence state, the merge fires on none of 2,146 loci, and the chimera screen rejects none of 57 candidates. Each of those zeros has a constructed control that builds the case the rule is supposed to catch and requires it caught, which is the only thing that makes the zero a measurement.

Six controls changed something

The chimera screen's own control failed on its first run, and the test was wrong. It asked the domain-envelope rule to reject a truncation, which the envelope cannot and should not do, because a protein truncated to 40 % of its length aligns 100 % of itself to the profile in one clean segment. Truncation is the length band's job and fusion is the envelope's.

A reassignment rule's positive and negative halves both failed, and the rule was wrong rather than the test. It climbed the tree from a mislabelled tip until some group dominated, which hands a tip to whichever clade is largest three nodes up.

A ranked-table order-invariance check caught real hash-seeded nondeterminism in a committed table.

A within-protein control was being selected by a name-prefix test that swept 225 residues of the element the ligand question is about into the control set.

A self-test damaged the artefact it tested. One suite called the real routine on constructed rounds and wrote its two synthetic rows over a committed table, after which the report read two runs where there are seven, and nothing failed. Two rules follow: any routine a control exercises takes a write flag and the control passes it false, and the suite checks the checksum of every committed table before and after it runs.

A mutation test passed on broken code, twice, in one suite. A duplicate-column case has to be built where the two columns hold the same residues, or an earlier guard catches it and the one being tested is never exercised. And a byte-identity check on a saved figure passes vacuously whenever both saves land in the same second, so the property is now tested directly.

Mutation testing is the check on the checks

Where a suite's own report records it, the suite was **mutation-tested**, meaning the rule it guards was deliberately broken and the suite required to catch it. Across the tasks that recorded this, every deliberate breakage was caught by the check responsible, giving a total of dozens of mutations across the analysis chapters, with each one named in its task's committed statistics rather than summarised.

That is the check on the checks. A suite that has never been shown to fail is a suite whose passing means nothing.

14.10 One rule cannot be automated: look at the figure

One decision in this project cannot be automated, and it is stated as an instruction rather than a rule: **look at the figure.**

Every main and supplementary figure was opened and read against its own legend, because errors of the kind "the legend says four and the figure draws three" are invisible to every table check in the project. A figure is generated from a committed table, so its data cannot drift. Its description can, and nothing downstream notices.

Twenty-three figures were inspected and 26 findings recorded, comprising 16 legends corrected and 10 figures corrected. They are recorded as data, one row per finding, with what the legend said, what the figure shows, the committed table the correction was re-derived from, and which of the two was changed.

The findings are worth characterising because they are all of one kind. One key described two colours where the figure draws four. One legend said "lineages grouped by kingdom" where three of the groups are not kingdoms. Another named the class with the most non-blue area where the figure shows a different class ahead on the fraction it plots. One figure carried a bold panel letter with no second panel to go with it. And one legend called four labels the extended paralogue clades where the figure carries five nodes. None of these is a data error, and none would have been caught by re-reading a table.

The mechanical half of that inspection is now automated and runs on every build. Every figure must have a legend, every legend a figure, and each multi-panel figure's legend letters must match its panel files. A later pass added the other mechanical half, requiring every figure to be cited by a sentence of the body and the citations to run in ascending order, because before it two figures were cited by nothing at all and the methods figure was numbered last and first mentioned in the third results section.

14.11 Two joins behind the supplementary figures are checked in code

A supplementary figure exists so a reader can check a join, so both joins were checked first, in code, as hard failures.

The trimming column map requires that every trimmed column be the input column the map names, for every sequence, across 1,797 columns and 134 sequences, walked exhaustively. The trimming program writes no column map of its own. One was recovered from a reporting flag, and it is **not trusted**, because an off-by-one would have no other symptom: every column would still map to a column, and every residue claim downstream would be renumbered with nothing to show for it.

The residue at the column requires that every variant's reference amino acid be the residue its own paralogue's per-residue table holds there, and every aligned partner the residue the other paralogue's table holds, across 1,780 variants and 2,699 partners.

Both passed, which is why a residue-level figure in this thesis prints letters.

A structure may carry a position from another numbering only if it earns it. A structure carries a human variant position only when every residue it shares with the human table carries the same amino acid, and two of four candidates fail, so one paralogue's pathogenic positions are simply not drawn on coordinates that are not theirs.

14.12 How this document is built, in eight stages

Running `python scripts/s25_assemble.py` runs eight stages in dependency order and exits non-zero on any failure.

The **guards** stage breaks every build guard on purpose and checks that each fires, and §14.13 and Appendix E describe it.

The **assign** stage commits the chapter grouping and fails on a results directory that is neither assigned to a chapter nor explicitly excluded, which is the no-leftovers rule enforced rather than asserted, and which means a future task's results cannot be silently left out of this thesis.

The **controls** stage derives the negative-control inventory from the test modules themselves.

The **refs** stage audits every reference added for this document, and §14.14 is that audit.

The **figures** stage copies every placed figure from the results directory that committed it, in both formats, and fails on a missing file, a slug used twice, or a committed figure that the thesis neither places nor explicitly excludes.

The **claims** stage re-verifies every load-bearing number against its committed table through the manuscript's own engine, and additionally requires each declared number to **appear in the chapter that declares it**, which is what stops a ledger being padded with checks the text never makes.

The **stitch** stage concatenates the chapters, numbers the figures per chapter in order of first placement, resolves the citation keys, and fails on a missing chapter, a placed figure the map does not declare, a declared figure no chapter places, a figure placed in a chapter other than its own, or a figure reference that resolves to nothing.

The **pdf** stage typesets the result and reads the typesetter's own log for dropped glyphs.

14.13 Every guard is broken on purpose, on every build

The requirement was that each guard be broken on purpose once. Doing that by hand proves it for one afternoon, so it is instead a build stage that runs first, and it carries two properties that make it worth the file.

Each case declares a fragment the guard's own message must contain, so a case that starts failing for a different reason is a failure rather than a pass. The suite's own first run had four cases returning non-zero for the wrong reason: two path bugs in the harness, one guard reading a module-level constant the sandbox could not redirect, and one mutation applied to a list the driver had already copied.

It runs against a sandboxed copy of the thesis directory and checks the SHA-256 of every committed file before and after, because this project has already had a self-test overwrite the committed table it was testing.

All 18 fire, and `thesis/guard_check.tsv` records what each said.

14.14 The reference audit caught nine citations that named the wrong paper

The literature review carries 137 references, audited when it was built. This document needed 58 more, almost all of them methods and tools, and the rule for them is the review's rule unchanged: **a reference added and not audited is worse than no reference, because it launders an assumption into a citation.**

So nothing bibliographic is typed. Each new reference is declared by an identifier alone, meaning a PubMed identifier or a digital object identifier where the work predates indexing, together with a distinctive phrase that must appear in the title that identifier resolves to, and a note of what the thesis rests on it for. The build resolves each identifier against a live record, admits it **only if the title carries the declared phrase**, and then writes the bibliographic row from the fetched record.

That check caught nine of 76. Nine identifiers written from memory resolved perfectly well, to entirely different papers. A gene-duplication inference algorithm's identifier returned a Bayesian phylogenetics program. A reconciliation method's returned a paper on real-time gene expression. A morphological-character likelihood model's returned a paper on species names in phylogenetic nomenclature. A vertebrate ancestral-genome reconstruction's returned a paper on structured RNAs.

Every one of those would have entered a bibliography looking entirely normal, with the right journal era, a plausible author and a number nobody checks. **A bibliography assembled by hand has no way to notice**, and that is the whole argument for the rule.

The audit is committed as a table with one row per new reference, giving the source database, the identifier queried, the phrase required, the title returned, and the verdict. All 58 now read *verified*.

15. General discussion

15.1 What is now known about the family that was not known before

Before this work, the IP_3 receptor family was described as animal machinery with scattered occurrences elsewhere, three vertebrate paralogues of unresolved relationship, and a well-studied protein whose gene had never been counted.

Six things are now measured that were not. Each is stated below with what it changes, because a census is only worth the questions it makes askable, and several of these results bear directly on literatures that have been arguing without the relevant numbers.

The family now has a measured range, and every absence names the space it was measured in

The receptor is present across the metazoa, in several protist lineages, in the green algae and in the early-diverging fungi. It is absent from land plants, from the yeasts and moulds, from apicomplexans, microsporidia, red algae, diatoms, tapeworms and every archaeal and bacterial proteome swept. Each of those absences is a claim about a declared space, and each of the 35 clade-level ones has been taken to the genome in assemblies where a measured positive control recovers a comparably long, deeply conserved gene.

This changes the status of the family's distribution from an impression to a measurement. The receptor's occurrence outside the animals has been reported from whichever genomes were available when each survey was written, and the resulting picture is a list of sightings rather than a range [37,38,39,47]. A sighting list cannot support an absence, because the absence of a report and the report of an absence look identical in it. What is different here is the denominator: 7,691 reference proteomes and 194 non-vertebrate genomes, with a control in every genome that carries the claim.

The land-plant absence answers a question that has been open in the plant literature for two decades. Plant cells release calcium in response to IP_3 in physiological assays, and no plant receptor gene has ever been identified, so the standing review of the subject asks in its title whether the receptor is real [36]. This work gives the genomic half of that answer in the strong form: land plants do not have the gene, and the green algae they descend from do [42,41]. That relocates the question rather than closing it. The physiology still needs a carrier, and the plant calcium toolkit is built from channel families that animals use differently or not at all [43,44], so the useful next experiment is about what performs the function in a lineage that demonstrably lost the receptor.

The absences in the reduced parasite genomes are the least surprising and the most useful as controls. Microsporidia and apicomplexans have undergone severe genome reduction [46], and the tapeworms have lost gene families across their entire biology as an adaptation to parasitism [45]. A method that failed to find the receptor in those lineages when it is there would show the same pattern, which is why each of those claims is carried by a positive control drawn from the same clade rather than by the search returning nothing. The apicomplexan calcium literature has had to work around exactly this uncertainty [47].

A measured range turns the receptor into an instrument for other questions. Co-occurrence across genomes is one of the oldest ways of assigning function without an experiment [103], and it requires a range on both sides of the comparison. Chapter 12 uses this one: the receptor's distribution is compared against the distribution of the enzyme that makes its ligand, across the same 7,691 proteomes, and the comparison is only possible because both were swept rather than recalled.

The distribution has a shape, and the shape is repeated independent loss

The family is present in the early-diverging lineage of both green plants and fungi and gone from the derived one in each. Combined with its presence across the metazoa and in several protist groups, that makes it ancestrally eukaryotic and lost repeatedly rather than invented more than once.

Loss is the expected fate of a gene family and is still the harder thing to demonstrate. Gene loss is now understood as an ordinary evolutionary mechanism rather than as degradation [25], but a loss claim needs a denominator and a control, which is what most reports of a missing channel lack. The pattern here is worth stating because it is convergent: the same receptor is retained by the chlorophyte algae and absent from land plants [42,41], and retained by the early-diverging fungi and absent from the Dikarya [40]. Two kingdoms lost the same signalling component on the two branches where each made its largest change in cell biology, and calcium signalling in both is now carried by other machinery [43].

That is a comparison a physiologist can act on. It identifies the pairs of lineages between which the substitution happened, and it names assemblies on both sides of each pair.

The three vertebrate paralogues now have a dated order of duplication

ITPR2 and ITPR3 are sisters, with both alternatives outside the confidence set of topologies. The split that separated ITPR1 from the other two sits on the vertebrate stem and the split between them on the gnathostome stem, so the two duplications are not on the same branch and ITPR1 is the earlier-diverging copy. The neighbourhood agrees that the three blocks are paralogous and that the surviving links run through ITPR1, with one of the two dated to the vertebrate radiation and the other far older.

The placement matters because it puts this family on the two branches the genome-scale reconstructions assign to the two vertebrate duplications. Reconstructions of the proto-vertebrate, proto-cyclostome and proto-gnathostome genomes place the first round before the cyclostome and gnathostome split and the second after it [104], which is the arrangement this family's two splits recover from an independent gene tree and its own neighbourhood evidence [24,69,56]. A family whose duplication history matches the genome-scale account needs no lineage-specific explanation, and one that did not would have been a problem for either the family or the account.

It also changes how the functional literature should be read. The three receptors are usually discussed as three coequal subtypes with different regulation, distribution and physiology [105,106,107]. The tree says they are not symmetric: ITPR2 and ITPR3 are a younger pair and ITPR1 is the outgroup to both. A functional difference shared by ITPR2 and ITPR3 and absent from ITPR1 is therefore the cheaper hypothesis, because it needs one change on one branch rather than two.

The teleost genome duplication kept one paralogue doubled and discarded the other two

The teleost genome duplication doubled ITPR1 and only ITPR1, in 97 % of teleost genomes, while doubling all three ryanodine receptors in the same genomes.

Which duplicate copies survive is one of the few things about whole-genome duplication that is predictable, and this result fits the prediction and sharpens it. Retention after a genome duplication is biased towards dosage-sensitive genes, whose products work in complexes where the ratio of partners matters [108], and the genes that stay duplicated are a small and non-random remainder of an otherwise rapid return to single copy [55,70]. The receptor is a homotetramer that also forms heterotetramers, so it is a dosage-sensitive gene by that argument, and the interesting part of the result is not that a copy was retained but that only one paralogue's was. The ryanodine receptors in the same genomes, under the same duplication and the same subsequent loss, kept all three.

The same paralogue is dosage-sensitive in humans. Ohnologues from the vertebrate duplications are overrepresented among the genes whose copy-number changes cause disease [109], and ITPR1 is one of them: heterozygous deletion of ITPR1 causes spinocerebellar ataxia in humans [110,111,112] and the gene is among the most constrained in the human population record [113]. A copy kept doubled for 300 million years in one clade and intolerant of hemizygosity in another is the same statement about dosage made twice.

The gene has one architecture that is conserved and one that is not

Each paralogue has fifty-seven to fifty-eight coding exons, of which 46 to 49 positions are shared between any two paralogues in every genome carrying both, which is an enrichment of nearly two orders of magnitude over a null drawn from the alignment itself. The ryanodine receptors, which carry every diagnostic domain of the family, share one. What differs between the paralogues is intron length, by a factor of four.

Shared intron positions are among the most durable characters a gene has, conserved across kingdoms and lost or gained at rates low enough to be informative about deep relationships [75,76]. Reporting 46 to 49 shared positions therefore says something the sequence alignment does not: the three paralogues did not merely diverge from a common protein, they inherited a common gene, intron by intron, and have kept the boundaries through 500 million years of sequence divergence.

The ryanodine receptor control is what makes that a statement about descent. Two proteins with the same domain content and the same fold could share intron positions because the splicing machinery prefers the same places in that kind of sequence, and the way to test it is to ask the same question of a family with the same domains and a separate origin. One shared position against 46 to 49 is the answer, and it is the difference between an observation about splicing and an observation about ancestry.

The four-fold difference in intron length is the other half of the result, and it is a live question rather than a settled one. Intron length is under selection, with shorter introns in highly expressed genes [73], although the energetic explanation usually offered for that has itself been contested [74]. Three paralogues that agree on every coding boundary and disagree four-fold on the DNA between them are a controlled comparison for whatever sets intron length, because everything a comparison of unrelated genes would have to control for is already matched.

No vertebrate genome in the scope has lost the gene

Across 927 genome by paralogue cells in 309 vertebrate genomes, none reaches the state this project defines as an absence, and parsimony places no loss anywhere on the tree.

A zero is the least interesting result to report and the most demanding to defend, which is why §15.3 is about the defence rather than the number. What makes it worth reporting at all is the contrast with the expectation. Gene loss is common enough to be a normal evolutionary mechanism [25], and this family has demonstrably used it repeatedly outside the vertebrates, so a clade in which none of three paralogues has been lost in any of 309 genomes is a statement about that clade rather than about the family's general behaviour.

Every paralogue has a loss-of-function phenotype, which is the mechanism the retention result points at. Deleting ITPR1 in mice produces ataxia and seizures [114], the human heterozygous deletion is a recognised ataxia [110,111], loss of ITPR2 function abolishes sweat secretion in humans and mice [115] and ITPR3 variants cause a peripheral neuropathy [116]. Retention across a whole vertebrate clade, an ohnologue kept doubled in the teleosts, and dosage intolerance in humans are three independent observations pointing at the same conclusion, and none of them establishes it on its own.

15.2 Five independent measurements put ITPR1 on the same side

Five independent measurements point the same way, and they were made by methods that share almost nothing.

ITPR1 is the paralogue whose neighbourhood retained a dated ohnologue with each of the other two, while they retained none with each other. It is the copy the teleost genome duplication kept doubled. It is the most constrained of the three by coding-sequence rate, at roughly twice the purifying selection of its sisters, and the only one whose selection is intensifying relative to them. It is the earlier-diverging copy on the reconciliation. And it carries the family's dominant missense disease burden.

It is also the one whose stem branch carries the single branch-site result this thesis reports: a class of sites evolving several times faster than neutrally, on the interval between the duplication and the first surviving split, while the rest of the tree sits near a rate of 0.03.

That is a coherent story and it is not a demonstrated one. The obvious reading, in which ITPR1 retained or acquired the ancestral dosage requirement and the other two were free to specialise, is consistent with every measurement above and is not tested by any of them. What can be said is that whatever distinguishes ITPR1 from its sisters is visible in the genome's arrangement, in the coding sequence's rate, in what a whole-genome duplication kept, and in the clinic, and that a hypothesis with that many independent footprints is worth a targeted experiment.

15.3 What it takes to defend a count of zero losses

The retention result is the one this thesis worked hardest for and the one that would be easiest to disbelieve.

A count of zero losses is what a broken pipeline produces, what an over-conservative rule produces, and what an insensitive search produces. The defence is not a claim about care. It is four measurements.

The search's own false-negative rate is measured rather than estimated, at 15.2 % of control cells over the whole scope and 0.9 % above a contiguity bar set from the gene's own geometry. That measurement was only possible because the count is zero, since if no gene in the scope is missing then every unfound cell whose gene is independently known present is a miss with a known right answer, and it is carried on two independent series, one of which uses none of the state rules the other depends on.

The state vocabulary was built so that exactly one state can be counted as a loss, and every alternative explanation must be positively excluded first. Two of its eight rules exist because of specific incidents in which a naive count would have scored losses that never happened.

The threshold that decides the hardest cells was calibrated against a decoy population, refused to separate on its first attempt, and was fixed by identifying the population that was contaminating the decoy, comprising 20 regions that are pieces of real genes in genomes where two paralogues are both shattered.

The sensitivity of the count is reported as a grid rather than as a robustness claim, covering thirty-two settings, on two codings, under three branch-length schemes. The family-level count manufactures a loss in 2 of 32 settings and the paralogue-resolved one in 18, and moving the calibrated bar across the whole gap its own calibration measured changes nothing in 927 cells.

The strongest form in which a zero can be stated is the one that names its own escape routes, and this thesis names them: refuse all reconstruction evidence and ignore assembly contiguity, and a loss appears. That is a description of a setting rather than a defence of it, and it is in a committed table rather than in a sentence.

15.4 The public record is the weakest thing about this family

Three quarters of the genes demonstrated here are unreachable from any protein database. Fifty-five per cent of the full-length protein records that do exist carry no usable gene symbol. A quarter of the loci in annotated assemblies are not delivered as one complete model.

And it is not this family's fault. The sister family, in the same assemblies through the same pipelines, fails at the same rate. What is being measured is how vertebrate genes of this size and structure are recorded, for which this family is an unusually well-controlled probe because it comes with a size-matched, domain-sharing sister to compare against.

Two thirds of the apparent problem is assembly quality, since the failure rate falls from 26 % to 7 % above the contiguity bar, and the remaining third splits between the two archives, with 99 % of loci in curated gene sets complete against 64 % in submitter-deposited ones above the same bar.

The parts that are not assembly quality are worth naming because they are fixable. A gene the annotation identifies correctly, names correctly, and files as non-coding emits no protein and cannot be reached by name. A gene with a locus tag instead of a symbol is invisible to every symbol-based query. A gene delivered as three models covering 58 % of its coding footprint is a gene nobody can retrieve.

Two of these failures are proved here at the exon and confirmed by reads. They are genes with intact reading frames over 2,700 codons, canonical splice sites, boundaries corroborated by dozens of other genomes' independent annotations, and transcript evidence crossing the junctions no model spans, in chromosome-level assemblies whose contigs are dozens of times the length of the gene. The deliverable addressed to somebody else is a list of 297 corrections with coordinates, evidence and a priority rule.

15.5 What the protein measurements say, and what they cannot

Constraint mapped onto the channel puts the gate and the selectivity filter at the top, as the least changeable elements of the receptor on two metrics in all three paralogues, with the gate identical between all three since the duplications. The pore module as a whole is more constrained than the linkers, once fifty residues of luminal loop are separated from it, and that loop is the most variable sequence in the receptor, sitting about fifty residues from the most conserved.

Two results at the ligand site were not what the question expected.

The IP_3-binding core is the less constrained of the two functional modules, paired per orthologue across roughly 250 sequences per paralogue, and that answer reverses on whether the luminal loop counts as pore. Any version of that comparison which does not declare the boundary is not interpretable, which is a methodological result as much as a biological one.

The constrained unit at the ligand site is a pocket rather than a contact set. The ten measured contacts beat the rest of the binding core and do not beat the rest of the 15 Å neighbourhood, and the same shape appears independently in per-site substitution rates. What selection holds is a neighbourhood, and the step a contact-driven model predicts is not there.

Losing the enzyme that makes the ligand does not relax the site. That null is bounded rather than empty, because the test can see a ligand-free core when there is one, in the ryanodine receptors, at the same evolutionary distance, on the same instrument, and it does not see one in the 36 lineages that have lost the upstream enzyme.

Conservation is a usable variant classifier for this family, and the layer that works best is not the one built for it, because taxonomic breadth beats within-gene depth. That is worth stating because building the deep layers took most of a session and the honest result is that the family-wide layer, which was already available, discriminates better.

15.6 Five results here are about method rather than about receptors

Some of what this thesis found is not about IP_3 receptors at all.

A genome-scale orthologue sweep misses about one cell in seven, and every miss is an assembly. A survey that reports absences without a contiguity floor is reporting assembly quality.

Phylogenetic breadth in a protein bait panel is nearly worthless within the vertebrates, and paralogue coverage is everything. Four human baits recover 782 of the 783 cells a 38-bait panel recovers, while removing one paralogue's own baits costs about a quarter of the recovery.

A family profile over reference proteomes is not a discovery method for whole genes. At gene scale it returns what domain annotation already returns, adding one extra record in 3,135 in the vertebrates. Its gain is fragments, and gene-scale records only where the annotation is worst.

The obvious rule for catching iterative-search drift measures the wrong thing. Sister-family contamination is the quantity this project and its predecessor both chose to monitor, and it is diluted by the drift it is meant to detect. Scored as classifiers of an outcome measured on the finished model, the sister-family rule has a sensitivity of zero on seven runs including all three that drifted, while the off-family share separates them perfectly.

A reference added without an audit is worse than no reference. Nine of 76 identifiers written from memory resolved to entirely different papers, each of which would have entered a bibliography looking completely normal.

15.7 Six things this thesis does not settle

Which of the two duplication rounds made which split. The dated ohnologue link is dated to the vertebrate radiation, which is both rounds, and separating them needs the cyclostome side of the quartet, which the neighbourhood evidence cannot reach, because 27 to 29 % of those genomes' coding genes carry a symbol at all.

Whether the paralogue quartet had a fourth slot. With one dated ohnologue pair surviving between the blocks that do carry a gene, a block that lost the gene too has nothing left to be recognised by.

Why ITPR1 alone kept its teleost duplicate. The observation is clean and the cause is not in this evidence.

Whether the ligand pocket's constraint is about IP_3. Everything within 15 Å of the ligand is also within the fold that holds it, and no sequence measurement separates ligand binding from domain packing.

What the receptors of the enzyme-less lineages are gated by. A search for one enzyme family says the canonical route to IP_3 is missing rather than that the receptor has no ligand.

The bird lesion result is named but not explained. ITPR3 carries an excess of disabling indels against its own genome's identity-matched sibling loci, and stratified it is entirely a bird result, at 25 genomes to 2. Twenty-one of the 27 informative pairs are in assemblies below the contiguity bar, which is where the test has its power, and the six above it all point the same way without being enough to test. The lineage is named and the mechanism is not, and what would settle it is a different sample, meaning birds at chromosome level, rather than a different statistic.

15.8 The general shape of the argument is a stance about negatives

If there is one thing this project would offer somebody starting a comparable census, it is not a pipeline. It is a stance about negatives.

Every interesting result here is a negative or rests on one. The family is absent from land plants. No vertebrate lineage has lost a paralogue. The ryanodine receptors share one intron position with the IP_3 receptors. Losing the upstream enzyme does not relax the binding site. No locus encodes the same part of the protein twice. The family is not recorded worse than its sister.

A negative is only worth stating when three things are true of it: the search space is declared, a positive control demonstrates the search could have found the thing, and the rule that produced the zero has been shown able to produce something else. The third is the one most often missing, and it is cheap, requiring only a constructed case that the rule is supposed to catch, required to be caught.

Almost every methodological decision in Chapter 14 is a specialisation of that stance. Measure the threshold rather than choosing it, because a chosen threshold has no error bar. Carry the sister family through every stage, because a control that rides along costs nothing and makes every number readable. Report a refusal as a refusal, because a blank and a zero are different claims. Print the parameter beside the p-value, because a significant test with an unidentified effect size is a different sentence. Render the report from the table, because a number typed beside its data will eventually stop matching it.

None of that is specific to calcium channels. What is specific to this family is the reason it was all necessary: a sister family that carries every diagnostic domain, sits in every genome in triplicate at twice the length, and is returned by every query. Half the rules in this thesis exist because of the ryanodine receptors, and the other half exist because 309 genomes contain enough broken assemblies to manufacture any answer somebody wants.

16. How this project was carried out with Claude Code

16.1 What Claude Code did, and what it did not

This project was carried out by Claude Code, which is Anthropic's Claude running as an agent in a terminal with access to the file system, the shell and the network. It reads and writes files, runs commands, inspects their output, and decides what to do next.

Claude Code did the following. It designed the project plan and wrote it down as a task ledger with dependencies. It wrote all 373 Python files under `scripts/`, which run to over a hundred thousand lines, together with the 9,243 lines of the search application under `src/`. It audited the literature the project started from, claim by claim, against primary sources. It decided which genomes and proteomes to download, wrote the manifests that declare them, and fetched 624 GB of assemblies and 60 GB of proteomes. It chose every threshold in the work and, in most cases, measured it rather than choosing it. It ran every search, alignment, tree, selection test and structural comparison. It produced all 444 committed result tables and all 111 committed figures. It recorded 83 numbered methodological decisions as it made them. And it wrote the literature review, the manuscript, this thesis and every one of the 28 rendered task reports that stand behind them, which together run to 109,250 words.

Every figure in that list is measured rather than recalled. The build re-derives them from the repository on every run and commits them as `thesis/production_stats.tsv`, so they move as the project does and this chapter cannot describe a repository that no longer exists.

The human collaborator did the following. He stated the goal, which was a publication-grade genome-scale census of one gene family. He provided the machine, the external storage and the network access. He ran the small number of commands that require an interactive login or elevated privileges. He read the drafts and corrected them, including two corrections to this document that are recorded in the decisions log. And at the start of each working session he said, in effect, continue.

He did not write the code, choose the genomes, select the statistical tests, set the thresholds or draft the text.

What follows is a description of that arrangement rather than a defence of it. An agent that runs for hours unattended can produce a great deal of plausible output, and the interesting question is not whether it can but how anybody, including the agent, can tell whether the output is true. Almost every methodological rule in this thesis is an answer to that question, and §16.9 sets out what the answer looks like in practice.

16.2 The work was organised as a ledger of one-task sessions

The organising artefact is a single file, `PUBLICATION_ROADMAP.md`, which Claude Code wrote in the first session and has amended in every session since. It contains four things: a statement of the goal and how the claim is scoped; a session protocol; a task ledger; and a decisions log.

The session protocol is a fixed procedure that starts and ends every working session. At the start it opens a progress dashboard, pulls the repository, verifies that the external storage is attached and stops if it is not, then picks exactly one task, which is the single row already in progress, or otherwise the topmost pending row whose dependencies are all complete. It announces that task and works only that task until its completion criteria pass. At the end it updates the ledger and the results column, appends to the session log, appends a plain-language entry to a findings document, refreshes the repository's README, commits and pushes.

That protocol exists because of a specific failure mode. An agent with a long list of things to do will do several of them partially, and a project of this size dies of half-finished tasks rather than of hard ones. **Working one task to its stated completion criteria, and splitting a task in the ledger when it proves too large, is the rule that kept it finishable.** Several tasks were split this way, and each split is visible in the ledger as two rows with the reason recorded.

The ledger is 36 tasks, of which 34 are complete. Each row carries the task, its dependencies, its status with a date, and a results column holding the load-bearing numbers and the paths they came from. The two incomplete rows are the human-gated deposit, which needs a person to create a public archive record, and the paper series, which is the next task.

Alongside the ledger, a briefs document holds a detailed specification for every row: the goal, the steps, the completion criteria, the outputs, and, for the later rows, what would make that task a failure. Claude Code wrote those briefs in advance of doing the work, which matters because a brief written after the fact describes what happened rather than what was intended.

35 sessions were logged across 9 working days, the first on 2026-08-18 and the most recent on 2026-09-09. The session log is a running technical record of what ran, what resulted and what is next. It is written for the next session rather than for a reader, and it is the mechanism by which an agent with no memory between sessions resumes work that is already in progress.

16.3 The literature was surveyed and then audited claim by claim

The first substantive task was not a search. It was to write down what the project was taking as known, and then to check it.

Claude Code assembled a background document in which every statement carries a tag: `[db]` if it can be re-derived from a database with a stated query, `[lit]` if it rests on a published source, and `[open]` if it is a question the project answers. It then extracted nineteen atomic claims from the `[lit]`-tagged statements and checked each against its primary sources, and re-queried every `[db]` number against the live database it came from.

Chapter 2 reports the outcome. Twelve claims were verified, three were qualified, two were corrected and one was demoted to an open question. The demoted one mattered most: the background asserted that the IP_3 and ryanodine receptor families had independently expanded to three paralogues each, no primary source establishes it, and leaving it in place would have made three later chapters into a confirmation of something already assumed.

Claude Code then wrote a full literature review, running to 137 references and twelve generated figures, as a separate document. That review is not decoration. It is where the project's claim about what is already known lives, and it is built by a script from numbered source files with citations resolved against one reference table, so its text and its bibliography cannot drift apart.

Two habits from that stage run through everything after it. **A number is quoted with the query that produces it**, and **a claim is tagged by how far it can be trusted**, so that a later chapter can tell the difference between a fact it inherited and a fact it measured.

16.4 What to download was decided by rule and committed as a manifest

The project needed 503 genome assemblies and 7,691 reference proteomes. Which ones was not a judgement call, because a judgement call cannot be audited.

The denominator is declared before the search runs, and it is derived from the data rather than hand-listed. For the vertebrate scope, Claude Code wrote two rules: one best assembly per vertebrate order, ranked by a stated function, and a margin rule with four positive tests that reads the census to find species whose protein records are missing, fragmentary or absent. The union is 309 genomes. For the non-vertebrate scope it wrote five rules under one principle, which is to sample most finely where the negative claim is, and one of those rules re-derives its own target list from a committed presence table. That rule found three absences the earlier summary had never named, including a metazoan clade with no record at all.

Both manifests are committed tables with a reason column on every row, so the scope rebuilds itself if the census changes, and a reader can see why any given genome is in the study.

The downloads themselves were run by a script Claude Code wrote for the purpose, which verifies every archive against its published checksums, retries download, extraction and verification together because a truncated archive surfaces as a checksum failure rather than a download error, and can operate in a fetch, search and purge cycle so that a 624 GB scope can be processed at one genome of peak storage.

16.5 What to analyse was decided by a menu written before the analyses ran

Between building the search and running the analyses, Claude Code wrote a document listing what the harvested data could answer. Each entry names its inputs, its method, its deliverable and the caveat that would invalidate it, and the entries became ledger rows.

Writing that menu before doing the work is the difference between a study and a fishing expedition. **An analysis chosen after seeing the data is chosen partly because of what the data shows**, and the menu is the record that these were not.

The menu is also where the project's scope discipline is visible. Several entries were written and then answered as underpowered rather than dropped, because a measurement that did not reach its question is a result about the evidence and an omission is indistinguishable from a section nobody ran.

16.6 The analyses ran unattended, and each one is a driver with stages

Every analysis task is a set of small modules and one driver script with ordered, resumable stages. The pattern is the same throughout: a driver with `--only` and `--from` flags, stages that are individually re-runnable, a negative-control suite that runs before anything is written and refuses the build if it fails, and a report and figures rendered at the end from the tables the stages committed.

Three properties made unattended operation possible.

Stages are resumable, so a run that is interrupted resumes at the stage that failed rather than from the beginning. The genome sweep is resumable per genome, and the selection suite is resumable per job.

A failed stage does not stop the stages after it that do not depend on it, and the report marks an unfinished section as unfinished. Waking up to a partial analysis that says which parts are partial is worth more than waking up to nothing.

And expensive work is cached at the level that makes a re-run cheap, with the rule that **a cache may hold what a parser found and never what a rule decided**, so that changing a rule invalidates the verdicts without invalidating the parse.

The largest single run was the selection suite, which ran for 22.7 hours unattended. The genome sweep processed 309 vertebrate assemblies and then 194 non-vertebrate ones. **15.2 compute-hours are recorded across the eight search channels**, measured on clean recomputes rather than on cached re-runs, because a cached run makes a budget look free.

During long runs the drivers write a small progress file that a dashboard reads, so the human collaborator could see what was happening without interrupting it.

16.7 Thresholds were measured rather than chosen, which is where the agent's judgement went

The part of this project that most needed judgement was not which analysis to run. It was where to put each threshold, and the answer Claude Code arrived at early and applied throughout is that **a threshold should be measured against a population the threshold has not already filtered, and reported with its separation**.

Chapter 14 sets out the rules in full. What is worth saying here is that this is where the agent's own reasoning is most visible, and it is visible because it was written down at the time. **87 numbered decisions** are recorded in the roadmap, each with the incident that produced it. Several changed an answer. An attribution margin inherited from a sister project was measured and overturned, because it came from a family whose paralogues are half as similar to each other as this one's. An intron-length parameter was measured across eleven species because it is not a performance setting but a threshold on whether a gene reads as a fragment. A calibration refused to separate its two populations and sent a chapter back to find the contamination in its decoy.

The decisions log is also where the agent recorded being wrong. A kill criterion written for the project's central hazard turned out to have a sensitivity of zero when it was finally scored. A duplication detector reached a specificity of 0.16 before it was scoped correctly. A self-test overwrote the committed table it was testing. Each is in the log with the fix and the rule that generalises it.

16.8 Results, tables and figures are generated, and nothing is typed twice

Every number this project reports lives in one of 444 committed result tables, and every document that quotes one reads it from there. That rule has three parts, and Claude Code applied all three from the first analysis onward.

Reports are rendered from the committed tables rather than written beside them. All 28 task reports are generated by a script from the tables in their own directory. None contains a hand-written number. That is why a text search against a report is a legitimate check in the claims ledger: the report cannot contain a number its tables do not.

Headlines are chosen by the data. Where a task tests a prediction an earlier task made, the prior is stated in code together with where it was said, the new statistic is computed, and the verdict is rendered from the comparison, with both numbers printed either way. A report generator written to narrate the expected answer would print it whatever the data said. Several reports consequently say that an earlier task's finding was not corroborated, and one says that the task's own premise was wrong.

Figures are drawn only from committed tables, by one style module, at the width they will be printed at. There are 111 committed figures. A single module holds the page geometry, the validated palette and a save function that refuses to write a figure whose layout runs off the canvas or whose text needs a glyph the font lacks. Figures are never re-plotted downstream: the paper and this thesis copy the file the analysis committed, and record its checksum.

Several figure decisions are recorded because they are substantive rather than cosmetic. A contiguity panel bins on the threshold rather than across it, because a sliding window straddling the bar reports a value no genome in it has. An exon track draws exons at true genomic width and never widens them to be visible, because widening them would draw a gene whose coding fraction looks like 40 % when it is 15 %. A saturation panel is drawn on log-log axes, because on a log-linear plot the neutral diagonal is not a line and the first draft left the saturation bar as the only line on the figure, which reads as neutrality.

One check on figures cannot be automated, and Claude Code did it by hand. Every main and supplementary figure was opened and read against its own legend, which found 26 discrepancies of the kind a table check cannot see: a key describing two colours where the figure draws four, a legend naming the wrong class as the largest, a bold panel letter with no second panel. Sixteen legends and ten figures were corrected, and each finding is recorded as a row of data with the table the correction was re-derived from. The mechanical half of that inspection now runs on every build.

16.9 The final documents are built by script, and every guard is broken on purpose

Three documents are produced from the same committed evidence: a 137-reference literature review, a manuscript, and this thesis.

Each is assembled by a driver with ordered stages, and each exits non-zero on a defect rather than producing a document with a hole in it. The manuscript's build collects figures, re-verifies its numbers, stitches its sections, typesets and builds a deposit manifest. The thesis's build adds three stages of its own: it commits the chapter grouping and fails if a results directory is neither assigned nor explicitly excluded, it derives the negative-control inventory by parsing the test modules, and it measures the repository to produce the statistics this chapter quotes.

Citations are keys resolved at build time and the bibliography is rendered rather than typed. No author, title, year or journal appears in any source file of any of the three documents. A cited key with no reference row is a build error.

Every load-bearing number is declared with the table it comes from and the operation that recovers it. The manuscript declares 276 and this thesis declares 220, through one shared engine so that a number quoted in both is recovered once and cannot disagree between them. The thesis adds a second condition: a declared number must also appear in the chapter that declares it, which is what stops a ledger being padded with checks the text never makes.

Every guard in the thesis build is broken on purpose on every build. A suite runs first, in a sandboxed copy, breaking each of the fifteen guards in turn and requiring each to fire with the message it is supposed to, and then verifying that it altered no committed file. That last condition exists because an earlier self-test in this project overwrote the committed table it was testing, after which a report read two runs where there were seven and nothing failed.

The bibliography is audited on entry, and the audit caught nine references in fifty-eight. Nothing bibliographic is typed: a new reference is declared by an identifier and a phrase its title must carry, resolved against a live record, and written from what comes back. Nine identifiers written from memory resolved to entirely different papers, each of which would have looked completely normal in a reference list. This is the clearest single example in the project of a check catching the agent rather than the data.

16.10 What the human collaborator contributed

The division of labour is worth stating precisely, because "written by an agent" is easy to overstate in either direction.

The goal and the standard came from the human. The instruction was a publication-grade census of one gene family, to the evidence standard of a molecular-evolution journal, with an exhaustive search space, a completeness argument, support values, synteny-backed orthology, model-based selection tests, molecular validation of every annotation-error claim, and one-command reproducibility. That specification shaped everything.

The infrastructure came from the human: the machine, the external drive that holds 700 GB of bulk data, the network access, and the credentials for the interactive logins that an agent cannot perform.

Continuation came from the human. Each session began with an instruction to continue. The agent chose the task, but a person decided that there would be a session.

Correction came from the human, and it mattered. The literature review was expanded to a full illustrated document at his request, outside the ledger. The prose of this thesis was rewritten after he read it and reported that sentence fragments and overused dashes made it confusing, which produced two recorded decisions and a full rewrite of all 27 chapter files. The title and the authorship of this document were corrected the same way.

And every commit carries both. Every commit on the main branch is authored by the human collaborator and carries a co-author trailer naming Claude, which is an accurate record of how they were made: the agent wrote the change and the message, and the human's account and machine committed it.

16.11 What this arrangement is good at, and where it needs watching

Three things this way of working did well.

It sustained a long, dependency-ordered plan. 36 tasks with declared dependencies, worked one at a time to stated completion criteria across 35 sessions, with the ledger and the session log carrying the state between them. An agent with no memory between sessions can do this only if the state lives in the repository, and the protocol is what puts it there.

It made the methodology explicit. 87 numbered decisions with the incident that produced each is a level of methodological record-keeping that is uncommon in a paper, and it exists because the agent had to write down why it did something in order to be able to follow the same rule three weeks later.

That count is itself an example of the arrangement working, and it has demonstrated it twice. Writing this chapter added a decision to the log, and the editorial pass that followed added another. On both occasions the build re-measured the repository and the claims ledger refused the document, naming the number the chapter still carried and the number the log now held. The figure in the sentence above was corrected each time because a check declined to pass it, which is the whole of what §16.9 describes, applied to the chapter that describes it.

It made checking cheap enough to do everywhere. 437 constructed negative controls, 496 declared claims across two documents, guards that are broken on every build, and reports that cannot contain a number their tables do not. Each of those is tedious for a person and almost free for an agent, so they were applied uniformly rather than where somebody remembered.

Three things that need watching, stated because they are the risks of the arrangement rather than incidental faults.

A fluent agent produces plausible prose about work it has not checked. The reference audit is the clearest case: nine of seventy-six citations were confidently wrong, and no amount of care in the writing would have caught them. The only thing that caught them was a check that resolved each identifier against a live record. **The general lesson is that an agent's confidence is not evidence, and the fix is a mechanical check rather than more care.**

A rule that fires on nothing looks exactly like a rule that cannot fire. Several of this project's results are zeros, and an agent generating both the rule and the result has an obvious way to produce a satisfying zero by accident. Every such zero here has a constructed control that builds the case the rule is supposed to catch and requires it caught, and that is the only thing that makes the zero a measurement.

Style rules are applied where somebody notices unless they are checked. The first prose pass through this thesis fixed the confusing headings and removed every long dash, and a systematic check afterwards found thirty more headings with no verb and fifteen fragmentary paragraph openings. The rule that came out of it is the same one the science follows: **a pass that has not been checked mechanically is a pass that fixed the instances somebody happened to see.**

Appendix A. The correction list

This appendix is the one deliverable of the thesis addressed to somebody else. It is a list of specific, coordinate-level proposals to the public archives, each with the evidence a curator would need to act on it, and it is committed in full as `results/annotation_audit/corrections.tsv`.

There are **297 items, comprising 52 at high priority, 19 at medium and 226 at low, with 18 vetoed.**

A.1 What each row contains

Every row carries the assembly accession, the annotation source, the organism and its vertebrate class, the paralogue cell, the contig with start, end and strand, the current annotation state, the current name and biotype, the proposal, the numbers the class fired on, the priority with the rule that assigned it, and an archived evidence file a curator can open.

A.2 The correction list has six classes

Twenty items are unannotated. No same-strand annotated feature overlaps the recovered gene at all, and the proposal is a coding model over the aligned exons.

Fifty-four items are held only by a non-coding feature. The annotation places a feature there and files it as something that emits no protein, most often a pseudogene. This is the class where the family differs from its sister, and where the failure is most invisible to a user, because the gene is identified, frequently named correctly, and simply produces no protein record, so no name-based or sequence-based protein search can ever reach it.

Two hundred and seventeen items are incomplete. One or more coding models are present and between them deliver a fraction of the gene. The proposal names how many models are there, what fraction they deliver together, what the best single one delivers, and asks for extension or merging to the aligned exons. The typical row delivers a tenth of the gene across two models.

One item carries a wrong paralogue name. Its annotation names a paralogue the alignment assigns elsewhere, in a genome that also carries a separate locus for the paralogue the annotation names, so the two cannot both be that gene.

Five items are protein records naming the wrong family. Each is a full-length protein record whose sequence the bait panel assigns to one family and whose name claims the other. All five are non-vertebrate records where the sequence barely separates the families either, and each additionally required an absolute score as well as a relative margin before it was written, because these candidates score 90 to 191 bits over roughly 2,700 residues.

Two hundred and ninety-two of the items concern a genome annotation and five a protein record.

A.3 Priority is assigned by a rule

High priority requires three things simultaneously: the gene demonstrably present, the assembly demonstrably able to carry it, and the reading frame intact. Only 52 items meet all three, and they are concentrated in ray-finned fish, with a handful in birds, mammals and amphibians.

Medium priority covers a partial recovery or an unscored reading frame.

Low priority covers anything below the contiguity bar, because there the annotation's silence may be the assembly's fault rather than the annotator's, and a correction proposed on that basis would be asking a curator to annotate a gene the assembly cannot represent.

A.4 The integrity veto is a column rather than a filter

Eighteen items are vetoed. Where the lesion screen scored a locus as lesion-rich, this audit does not propose resurrecting it.

The row is still written, flagged, with its reason. A locus the audit declined to correct is evidence about the audit, and filtering those rows out would leave a list that looks cleaner than the evidence behind it.

A.5 What the list is not

It is not a claim that these annotations are wrong in a sense a curator must accept. It is a set of specific, checkable proposals with coordinates and evidence, produced by one instrument, about genes that instrument recovered. Two of them, one omission and one fragmentation, are validated in Chapter 13 down to the exon and confirmed by junction-spanning reads. The other 295 rest on the same instrument that produced those two, applied at scale.

Appendix B. The constructed negative controls

This appendix lists the constructed negative controls of this project. It is derived from the test modules rather than counted by hand and is committed as `thesis/control_inventory.tsv`.

There are **437 named constructed checks across 19 suites.**

B.1 What is counted, and what a unit is

Two conventions are in use across the project and both are counted, because it grew them at different times and neither is wrong. Some suites group their checks into named test functions, while others call a helper once per assertion, naming each check as it runs. **A named check is the finer unit, so it is preferred where a module has any, and the table records which unit each row counts**, because the two are not the same size and adding them together silently would be worse than either.

The count is derived by parsing each module. Any call to one of the project's check helpers whose first argument is a string literal is one named control, and any top-level function whose name matches the project's test-function convention is one grouped control. A control deleted from a module disappears from this appendix, and one added appears in it.

B.2 Where the controls are

suite	unit	controls
`s6_test_selection.py`	named check	24
`s7_test_tree.py`	named check	18
`s8_test_flanks.py`	named check	49
`s9_test_codon.py`	named check	14
`s10_test_evidence.py`	named check	58
`s11_test_structures.py`	named check	24
`s12_test_expression.py`	named check	11
`s13_test_recon.py`	test function	14
`s15_test_loss.py`	test function	15
`s15b_test_counts.py`	named check	21
`s16_test_dup.py`	named check	21
`s17_test_constraint.py`	test function	14
`s18_test_audit.py`	named check	42
`s19_test_methods.py`	named check	33

suite	unit	controls
`s21_test_arch.py`	named check	9
`s21_test_claims.py`	named check	12
`s22_test_ligand.py`	named check	23
`s22_test_lineage.py`	named check	18
`s24_test_supp.py`	named check	17

Further controls run inside the analysis modules themselves rather than in a test module, comprising the bait screen's three synthetic failures, the chunked-genome equivalence test, the resume test and the one-search-two-sensitivities equivalence test. They are not in the table because they are not separately addressable checks.

B.3 What the controls are checks on

Almost none of them checks that a routine returns the right answer. They check that it **refuses**, and that it **can act**. Chapter 14 gives the argument. The short form is that in this project almost every rule returns a plausible number when it is wrong, and several of the thesis's results are zeros that would be indistinguishable from a rule that cannot fire.

Every suite runs before its task writes anything, and a failure refuses the build.

B.4 Six of the controls found something

Six are described in Chapter 14: the chimera screen's control that failed because the test was wrong rather than the rule; a reassignment rule whose positive and negative halves both failed and whose rule was wrong; an order-invariance check that caught real hash-seeded nondeterminism in a committed table; a within-protein control selected by a name prefix that swept the element under test into the control set; a self-test that overwrote the committed table it was testing; and two mutation tests that passed on broken code until the case was rebuilt to exercise the guard it was aimed at.

Appendix C. The sensitivity tables

Four results in this thesis rest on a threshold that somebody chose, and in each case the count was recomputed across the whole range of that threshold and committed. This appendix says where those tables are, what each axis is a knob on, and what moving it costs.

The principle behind all four is the same, and it is worth stating once. **A threshold-dependent result must say which settings manufacture it, and it must walk the threshold across its own measured uncertainty before any invented value.** A robustness claim asserted in prose is not checkable, and a grid is.

C.1 The loss count is walked across four axes

The table is `results/loss_counts/sensitivity_matrix.tsv`, with 384 rows.

It has four axes. **Coding** contrasts the family-level presence character with the paralogue-resolved one. **Evidence** is an ordered eight-rung ladder, each rung named after what it refuses, whose first three rungs are the reconstruction calibration's own measured gap edges and midpoint rather than round numbers. **The cyclostome rule** is switched on and off, and so is **the contiguity bar**. Every cell is computed under three branch-length schemes.

The count is zero at the operating point on both codings. The family-level coding manufactures a loss in 2 of 32 settings and the paralogue-resolved one in 18, up to 45 loss edges.

The companion table `manufactured_losses.tsv`, with 144 rows, says the same thing at cell resolution, giving every genome by cell that ever reads absent under any setting, the state and rule that held it at the operating point, and the loosest setting that releases it. That table is what makes the sensitivity claim checkable one cell at a time rather than one count at a time.

The branch-length axis is carried and reported as invariant. Parsimony counts edges and cannot read a length, so no scheme changes any count, and a matrix that quietly dropped that axis would be indistinguishable from one that had tested it and found nothing.

C.2 Copy number is recomputed at seven coverage bars

The table is `results/duplication/copy_sensitivity.tsv`, with 56 rows.

It recomputes every copy count in the duplication chapter at seven coverage bars. A duplication claim that survives only one bar is a claim about the bar, and the teleost result is the one this table exists to protect, because it holds across the range.

C.3 Gene architecture is recomputed at six coverage bars

The table is `results/gene_architecture/architecture_sensitivity.tsv`, with 24 rows.

It recomputes exon counts, spans and intron statistics at six coverage bars. The result the table protects is that the exon count is conserved and the span is not, which is a comparison between two quantities measured on the same loci and therefore robust to the bar by construction. But the bar decides which loci, and the table says so.

C.4 Annotation states are recomputed across the completeness bar

The table is `results/annotation_audit/state_sensitivity.tsv`, with 21 rows.

It re-counts every locus state across the whole range of the completeness bar. The audit inherits that bar from the sweep rather than deriving a second one, so this table is what validates the inherited value: it shows where each state boundary sits in the distribution and what a different bar would have done to the headline.

C.5 Two thresholds have no sensitivity table, and here is why

The reconstruction bar has none, because the evidence ladder in C.1 is its sensitivity analysis. Its first three rungs are the two edges of the gap the calibration measured and the midpoint between them, and moving across the whole gap changes no cell in 927.

The identity floor for what counts as a locus outside the vertebrates has none, because the calibration that produced it reported that neither identity nor coverage separates its two populations cleanly, and a sensitivity grid around a threshold whose calibration refused to separate would be a grid around nothing. What was done instead was to add a second, independent axis, in which every recorded cluster is scored against both family profiles, and to report both statistics.

Appendix D. The reference audit

The audit table is `thesis/reference_audit.tsv`, with one row per reference added for this document, giving the source database, the identifier queried, the phrase the title had to carry, the title returned, the year, the journal, the first author, the verdict, and what the thesis rests on the reference for.

76 references were added and all 76 verified. Nine required a corrected identifier before they could be.

D.1 What the rule for a new reference requires

The literature review's 137 references were audited when it was built, and this document inherits them as a frozen baseline, committed as `thesis/reference_baseline.txt` before any new reference was added. Every key cited by this thesis that is not in that baseline is a new reference, and the build fails on a new reference with no verified audit row.

The rule for a new reference is the review's rule unchanged: it is **audited on entry**. What that means in practice is that nothing bibliographic is typed. Each new reference is declared by an identifier alone, together with a distinctive phrase that must appear in the title that identifier resolves to. The build resolves the identifier against a live record, using PubMed for anything indexed there and Crossref otherwise, admits it only if the title carries the phrase, and then writes the bibliographic row from the fetched record.

So a mistyped identifier fails the audit instead of quietly putting a different paper in the bibliography, and no author, title, year or journal can be wrong, because none of them is entered by hand.

Responses are archived, so the audit re-runs offline.

D.2 What the audit caught

Nine of the 76 identifiers resolved to entirely different papers.

A gene-duplication inference algorithm's identifier returned a Bayesian phylogenetics program. A reconciliation method's returned a paper on statistical challenges in real-time gene expression. A morphological-character likelihood model's returned a paper on species names in phylogenetic nomenclature. A vertebrate ancestral-genome reconstruction's returned a paper on structured RNAs in the human genome. A zebrafish gene-cluster paper's returned a paper on a poly(ADP-ribose) polymerase at telomeres. Two timetree papers, a gene-loss review and a Dollo-parsimony paper resolved elsewhere or not at all.

One of the nine was a different failure and is worth separating: the identifier was right and the phrase was wrong. That one was corrected by adjusting the phrase, because the record the identifier returned was the intended work.

Every one of the other eight would have entered a bibliography looking entirely normal, with the right journal era, a plausible author list, and a number nobody checks. That is the argument for the rule in one sentence: **a bibliography assembled by hand has no way to notice.**

D.3 What the new references are

By audit class, there are 12 tools this project actually ran, whose versions are recorded in the toolchain manifest; 21 methods, models or statistical procedures implemented or applied; 10 databases, cited at the release the project used; and 33 substantive biological or evolutionary claims.

The tool and method references are the bulk of the addition and the reason it was needed at all. The manuscript cites 29 references and none of them is a tool, because a paper can name its software in a Methods section and be understood. A chapter that explains why an alignment was run single-threaded, or why a branch-site likelihood ratio is tested against a mixture, is citing a method and must say whose.

D.4 What is not audited here

The 137 inherited references are not re-audited. They were audited once, against their primary sources, in the session that built the literature review, and 19 atomic claims resting on them were scored, giving 12 verified, 3 qualified, 2 struck and 1 downgraded to an open question. Chapter 2 reports that audit and what it changed.

Re-auditing them would be a second session's work for no new information. What this appendix adds is that the new references have been through a check the inherited ones were not: their identifiers have been resolved against a live record and required to return the paper the declaration names.

Appendix E. How the results were assigned to chapters, and the build guards

E.1 What the assignment table records

The assignment is committed as `thesis/chapter_assignment.tsv`, with one row per entry under `results/`, giving the chapter it is primary in, the rule that placed it, what it is, and the chapter it might plausibly have gone to instead with the reason it did not.

There are **37 results directories assigned across 13 chapters, with one excluded.**

The rules are written out in `thesis/chapter_rules.md`, which was committed before any chapter prose was written, and four of the seven are enforced by the build rather than asserted.

Rule T4 says a results directory is primary in exactly one chapter. It is enforced by construction, because the assignment is a mapping.

Rule T6 says a grouping that only holds leftovers is an appendix. It is enforced in the strong direction: every entry under `results/` is either assigned to a chapter with a rule and a reason, or named in an exclusion list saying why it is not a result. A directory that exists and is unassigned fails the build, which also means a future task's results cannot be silently left out of this thesis. One entry is excluded, namely the dashboard's progress panel, which is session state.

Rule T7 says every figure is copied from a committed results directory and never re-plotted. It is enforced, in that a missing file, a slug used twice, or a committed figure the thesis neither places nor explicitly excludes all fail the build. **The thesis places 103 figures**, every one copied from the results directory that committed it in both formats, with the source and the SHA-256 of each recorded in `thesis/figure_manifest.tsv`. Eight figures are explicitly excluded, all of them the search application's own bundle plots from smoke-test runs, which are drawn from whatever one search returned rather than from a committed analysis table.

Rule T3 allows four to twelve figures per results chapter. It is checked by inspection against the figure manifest rather than by the build. The introduction, the methods chapter and the discussion are exempt as exposition, and the methods chapter carries one figure.

E.2 The paper series should expect one departure from this assignment

This task ran before S26, so this table is the one S26 starts from. There is one place where a paper series will have to choose differently, and it is recorded here as a disagreement rather than resolved.

The methods results, meaning the measurement of what the search was worth, sit entirely in Chapter 4 with the search they measure. S26's own starting proposal splits them across two papers: a retention paper taking the false-negative rate, and an archive paper taking the contribution half.

Both are right about their own format, and the reason they differ is not a matter of taste.

The thesis's reason. The false-negative rate is a property of the search, in that 15.2 % of cells whose gene is independently known present were not found by the ledger. A document that explained the search in one chapter and then withheld the number saying what that search was worth until five chapters later would be hiding the instrument's error bar from the chapter that builds the instrument.

The series' reason. A retention claim without its own false-negative rate is a paper whose central control lives somewhere else, which is exactly what S26's second rule forbids, since every control a paper's claims rest on must be measured inside that paper.

The rule that forces the choice is the one both documents share, namely that a result is primary in exactly one place, and it is shared deliberately so that a disagreement between the two groupings is visible rather than invisible.

What a reader holding both documents should take from it. The thesis and the series are not the same object cut two ways. A chapter can be a stage in a longer construction and a paper cannot. Where the two groupings differ, the difference is a fact about the two formats, and this appendix is where it is recorded rather than smoothed over.

E.3 Every guard is broken on purpose, and this is what each one said

Every guard in the build is broken on purpose on **every** build, by `scripts/s25_test_guards.py`, which runs as the first stage and writes `thesis/guard_check.tsv`. **All 18 fire with the message they are supposed to**, and the suite verifies that it altered no committed file while doing it, because a self-test that damages what it tests is a failure mode this project has already had once.

Each case declares a fragment the guard's own message must contain, so a case that starts failing for a different reason is a failure rather than a pass.

guard	how it was broken	what it said
the assignment (T6)	a results directory removed from the assignment	results/<name> is neither assigned to a chapter nor listed in ASSIGNMENT_EXCLUDED (rule T6)
the assignment (existence)	an assignment pointing at a directory that does not exist	results/<name> is assigned but does not exist
a missing chapter	a chapter file moved aside	missing chapter file thesis/<name>
a missing figure	a declared figure's source file moved aside	<slug>: missing <path>.png
a duplicated slug	one figure declared twice	slug '<slug>' is declared more than once
an unexcluded figure	a committed figure removed from the exclusion list	<path>.png is committed under a chapter's results tree but is neither placed in the thesis nor listed in UNPLACED

guard	how it was broken	what it said
an unplaced figure	a declared figure's placement deleted from its chapter	`<slug> is declared for chapter N but no chapter places it`
a borrowed figure	a figure placed in a chapter other than its own	`thesis/<file> places <slug>, which is declared for chapter N`
a dangling figure reference	a reference to a figure that is never placed	`{fig:<slug>} is referenced but the figure is never placed`
an undeclared reference	a citation key with no row in the reference table	`cited keys with no row in references.tsv: <key>`
an unaudited reference	a new key cited with its audit row removed	`cited keys added after the frozen baseline with no verified audit row: <key>`
a decorative reference	a key's single citation deleted	`references added for the thesis and never cited: <key>`
a failed claim	a claim's expected value altered	`[FAIL] <id> <claim>: expected X, found Y`
a padded claim	a claim declared for a chapter whose text does not state its value	`chapter N does not state 'X', so either the number is missing from the text or the claim is padding`
a dropped glyph	a character the document font cannot set	`N distinct missing characters, so the document font lacks a glyph the text uses`
an em-dash	an em-dash reintroduced into a chapter source	`N em-dash(es) in a chapter source`
a legend repeating itself	a sentence duplicated inside one figure legend	`legend <slug> says the same thing twice`
a legend repeating its neighbour	a legend given the text of the paragraph beside it	`legend <slug> restates the paragraph after it`

Four of these are the ones this thesis added over the manuscript's build: the padded-claim guard and the three prose guards at the foot of the table. The padded-claim guard closes the ledger in the direction the manuscript's cannot, because it fails not only when the document states a number no table produces, but when the ledger declares a check the document never makes. The three prose guards exist because the two editorial passes on this document each introduced a defect that every other guard passed over: an em-dash carrying a clause a reader needed, and a figure legend saying the same thing twice.

References

Every entry below is rendered from `results/s0_baseline/references.tsv`, which is the one reference table this project keeps, and numbered in order of first appearance in the text. No author, title, year or journal is typed into a chapter file, and a cited key with no row in that table fails the build.

The 137 keys inherited from the literature review were assembled and audited in the session that built it. The 76 added for this document were each resolved against a live record, using PubMed for anything indexed there and Crossref otherwise, and admitted only if the title the identifier returned carried the phrase the declaration said it should. Appendix D is that audit, and it reports what it caught.

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